

A N I N T R O D U C T I O N T O

MICRO- ECONOMICS

K U T U S H C A T U S H

AN INTRODUCTION TO MICRO-ECONOMICS

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An Introduction to Micro-Economics

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C O N T E N T S

I	WHAT ECONOMICS IS ABOUT	1
	The Problem of Wanting More	2
	Choice and Opportunity Cost	3
	What Economics Is	4
	<i>Wealth: Adam Smith, 1776</i>	5
	<i>Well-being: Alfred Marshall, 1890</i>	5
	<i>Scarcity and Choice: Lionel Robbins, 1932</i>	5
	<i>The Central Theme</i>	6
	The Three Basic Problems	6
	A Worked Case: The Owner's Choice	7
	Summary	8
	Think It Through	8
	Where People Go Wrong	8
	In History	8
	Sources	8
	Review Questions and Practice	9
	<i>Review questions</i>	9
	<i>Practice: from a household to a bakery</i>	9
II	HOW ECONOMISTS THINK	11
	Two Ways of Looking	12
	Scope, Importance and Purposes	13
	Models and Economic Laws	14
	Positive and Normative; Science or Art	15
	Microeconomics in Bangladesh	15
	A Worked Case: One Firm, One Economy	16
	Summary	17
	Think It Through	17
	Where People Go Wrong	17
	In History	17
	Sources	18
	Review Questions and Practice	18
	<i>Review questions</i>	18
	<i>Practice: sorting statements</i>	18
III	UTILITY AND CONSUMPTION	19
	The Idea of Utility	20
	Can Satisfaction Be Measured?	21
	The Law of Diminishing Marginal Utility	21
	Limits of the Law	23
	Consumption: How a Buyer Spends	23
	A Worked Case: The Customer's Bundle	24
	Summary	25
	Think It Through	25
	Where People Go Wrong	25
	In History	25
	Sources	26
	Review Questions and Practice	26
	<i>Review questions</i>	26
	<i>Practice: from a table to a bundle</i>	26
IV	DEMAND	27
	What Demand Means	28
	The Law of Demand	29
	What Moves the Curve	30
	Exceptions: Giffen, Snob and Bandwagon	32
	A Worked Case: Three Groups of Buyers	32
	Summary	33
	Think It Through	33

C O N T E N T S (continued)

	Where People Go Wrong	33
	In History	34
	Sources	34
	Review Questions and Practice	34
	<i>Review questions</i>	34
	<i>Practice: reading, filling, adding and shifting</i>	34
V	ELASTICITY OF DEMAND	35
	Why Measure Reaction?	36
	Price Elasticity of Demand	37
	<i>The Five Types</i>	37
	<i>Point and Arc Methods</i>	38
	<i>Elasticity Along a Straight Line</i>	38
	<i>Worked Examples</i>	39
	What Makes Demand Elastic	40
	Income and Cross Elasticity	41
	Price and Revenue	42
	Summary	43
	Think It Through	43
	Where People Go Wrong	43
	In History	43
	Sources	44
	Review Questions and Practice	44
VI	CONSUMERS' SURPLUS AND INDIFFERENCE CURVES	45
	Consumers' Surplus	46
	Indifference Curves	48
	The Budget Line and Choice	49
	Measuring Consumers' Surplus with Indifference Curves	51
	Summary	51
	Think It Through	52
	Where People Go Wrong	52
	In History	52
	Sources	52
	Review Questions and Practice	52
VII	SUPPLY AND ITS ELASTICITY	55
	What Supply Means	56
	The Law of Supply	57
	Determinants of Supply	58
	Exceptional Supply Curves	59
	Elasticity of Supply	59
	Summary	61
	Think It Through	61
	Where People Go Wrong	62
	In History	62
	Review Questions and Practice	62
VIII	MARKETS AND EQUILIBRIUM	63
	What a Market Is	64
	Equilibrium Price and Quantity	64
	<i>Balance in a Table</i>	64
	<i>Balance on a Graph</i>	65
	<i>Balance in Algebra</i>	66
	Forces Above and Below Equilibrium	66
	Shifts and New Equilibria	67
	Gains From Trade	68
	Summary	70
	Think It Through	70
	Where People Go Wrong	70

C O N T E N T S *(continued)*

	In History	70
	Sources	71
	Review Questions and Practice	71
IX	PRODUCTION	73
	Production and the Production Function	74
	Product in the Short Run	75
	The Law of Diminishing Returns	76
	Returns to Scale	78
	Least-Cost Combination	79
	Summary	81
	Think It Through	81
	Where People Go Wrong	81
	In History	81
	Sources	81
	Review Questions and Practice	81
X	COST	83
	Cost and Opportunity Cost	84
	Kinds of Cost	85
	Average and Marginal Cost	86
	Short-Run and Long-Run Cost Curves	88
	Economies and Diseconomies of Scale	88
	Summary	89
	Think It Through	89
	Where People Go Wrong	89
	In History	90
	Sources	90
	Review Questions and Practice	90
XI	MARKET STRUCTURES AND THE GOAL OF THE FIRM	91
	What Market Structure Means	92
	The Main Types	92
	Revenue	93
	The Profit Rule and Other Goals	95
	How Structure Shapes Pricing	96
	Summary	97
	Think It Through	97
	Where People Go Wrong	97
	In History	97
	Sources	97
	Review Questions and Practice	97
XII	PERFECT COMPETITION	99
	Features	100
	Price Determination	100
	Short-Run Equilibrium of the Firm	101
	Short-Run Equilibrium of the Industry	102
	Long-Run Equilibrium	103
	Summary	103
	Think It Through	104
	Where People Go Wrong	104
	In History	104
	Review Questions and Practice	104
XIII	MONOPOLY	105
	Nature and Kinds	106
	Demand, Revenue and Equilibrium	107
	Compared with Competition	110
	Price Discrimination	111

C O N T E N T S *(continued)*

	Controlling Monopoly	112
	Summary	113
	Think It Through	113
	Where People Go Wrong	113
	In History	113
	Review Questions and Practice	114
XIV	MONOPOLISTIC COMPETITION AND OLIGOPOLY	115
	The Idea	116
	Product Differentiation and Selling Costs	117
	The Theory	117
	Oligopoly	120
	Price Determination Compared	121
	Summary	122
	Think It Through	122
	Where People Go Wrong	122
	In History	122
	Sources	123
	Review Questions and Practice	123
XV	RENT AND WAGES	125
	15.1 Pricing the Factors	126
	15.2 Rent	127
	15.3 Wages: Meaning	128
	15.4 How Wages Are Determined	129
	15.5 Why Wages Differ	129
	Worked Case: The Bakery Hires	130
	Summary	130
	Think It Through	131
	Where People Go Wrong	131
	In History	131
	Sources	131
	Review Questions and Practice	131
XVI	INTEREST AND PROFIT	133
	16.1 Interest	134
	16.2 How Interest Rates Are Determined	134
	16.3 Profit	135
	16.4 Why Profit Occurs	136
	16.5 The Whole Picture	137
	Worked Case: The Second Oven	138
	Summary	138
	Think It Through	138
	Where People Go Wrong	138
	In History	138
	Sources	139
	Review Questions and Practice	139
	GLOSSARY OF KEY TERMS	141

TO THE READER

This book assumes that you have never studied economics. Each idea is built from ordinary things, such as a family budget, a shop or a bun, and each new word is explained where it first appears. A few notes on how the page is arranged will help you to read it easily.

The wide column carries the main text. Beside it, in the narrow side column, are short notes in italic. Each one sums up the paragraph next to it in a line or two, so that if you are short of time you can read the side notes alone and see the shape of the chapter.

A word set in bold in the main text is a key term, and it is printed in bold only where it first appears. The side column gives its meaning in one plain line. The glossary at the back gives a fuller definition, with the chapter and the page where the term was first explained.

Small raised numbers point to footnotes at the foot of the page. A side note that says *Go slowly here* marks a passage that deserves a second reading, and one that says *Advanced* marks a passage that you may leave for a later day. Paragraphs headed *Case file*, *illustrative* follow an imagined example through the chapter; the people and figures in them are invented to make the reasoning clear.

Most chapters end in the same way: a summary, a question to think through, a list of common mistakes, a short note from history, and review questions with practice. Try the practice before you look at anything else. The work you do yourself is the part that stays with you.

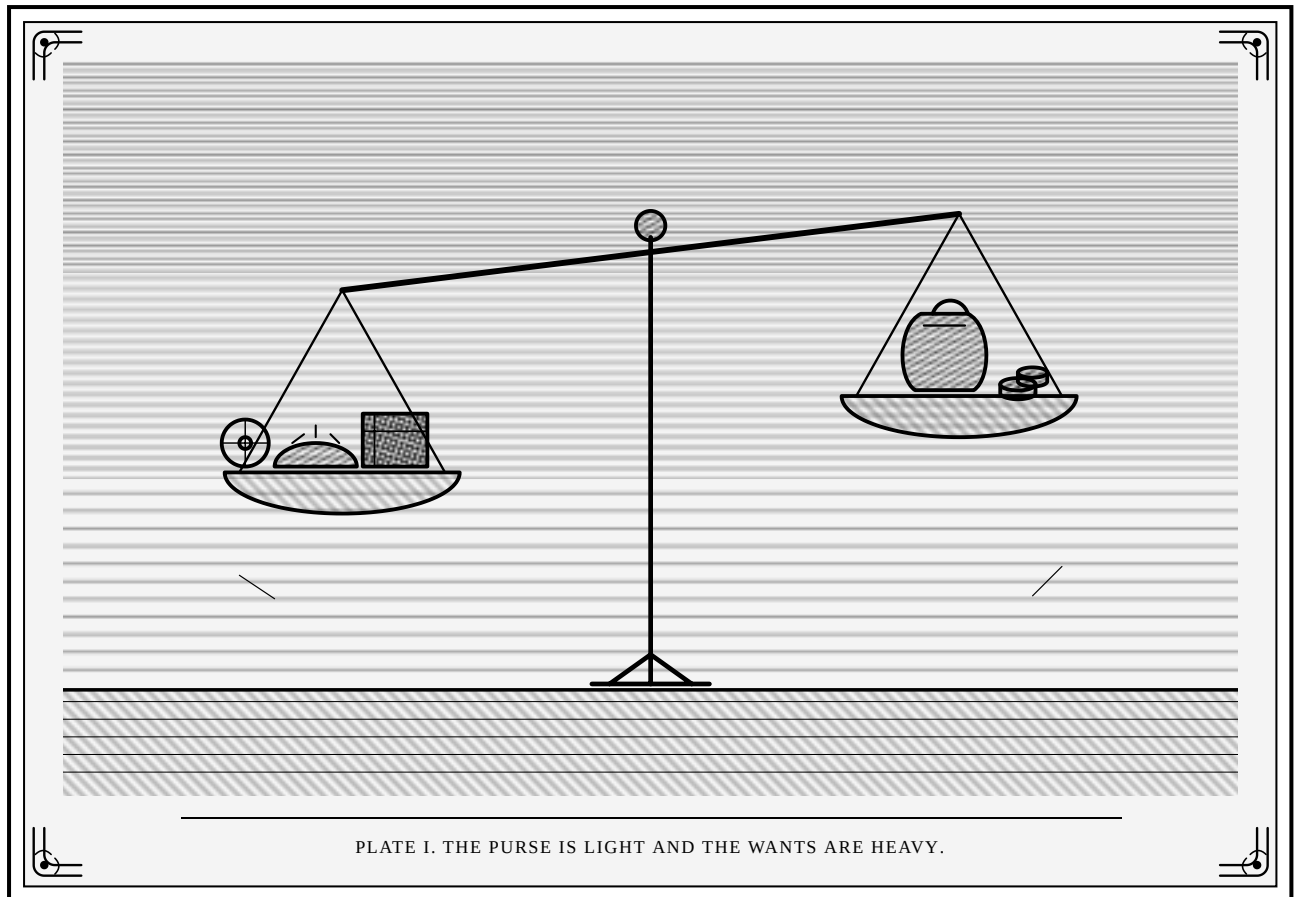
P R E F A C E

Microeconomics is the study of how single buyers, sellers and firms make choices, and of how those choices meet in markets. It is often taught as a heap of curves and formulas. This book tries to teach it as a way of thinking, one idea at a time.

The plan follows the order in which the ideas lean on one another. The first two chapters explain what economics is and how economists think. Chapters III to VI deal with the buyer: utility, demand, elasticity, consumers' surplus and indifference curves. Chapters VII and VIII turn to the seller and bring buyer and seller together in the market. Chapters IX and X study production and cost. Chapters XI to XIV take up the market structures: perfect competition, monopoly, monopolistic competition and oligopoly. The last two chapters ask how the owners of land, labour, capital and enterprise are paid, through rent, wages, interest and profit.

The examples come from everyday life in Bangladesh, and prices are given in taka. A bakery and its customers return again and again, so that you can watch one small business meet each new idea in turn. The arithmetic is kept simple, with a little algebra where it is needed.

The book is meant to be read with a pencil in hand. Work through the cases, answer the questions at the end of each chapter, and return to the earlier chapters whenever a later one seems hard, because in this subject the later ideas are built from the earlier ones.



CHAPTER I

What Economics Is About

Why can no one have everything?



This chapter starts from a plain fact of daily life: wishes are many and means are few. It shows why every person, family and country must choose, and what each choice costs. It then explains what economics is, and the three questions every society must answer.

A small family budget shows the problem that the whole book studies.

The same problem faces farms, shops and governments.

The chapter asks why choice is forced on us and what a choice really costs.

GO SLOWLY HERE.

A student's small purse shows the gap between wishes and money.

Want. A wish for something.

A want is any wish, wise or not, and new wants keep arriving.

Resource. Anything used to make goods.

Resources are the things used to make goods, and they are limited.

Good. In daily life, good means nice or kind. In economics, a good is a thing that satisfies a want.

Service. Work done for others, such as teaching.

A good satisfies a want; a service is work done for others.

Factors of production. Land, labour, capital and entrepreneur.

Land. Gifts of nature, such as soil and water.

Labour. Human work of the body or the mind.

Resources fall into four groups; the first two are land and labour.

Imagine a family with Tk 20,000 to spend this month. The family wants rice, fish, school books, a bicycle for the eldest child, new roof sheets, and a journey to visit relatives. Added together, these wishes cost far more than Tk 20,000. The family cannot buy everything. Someone must decide what to buy now and what to leave for later.

Now widen the picture. A farmer has one field. It can grow rice or jute, but not both in the same season. A government has one budget. It can build a road or a school, but it cannot use the same taka for both. A baker has one oven and twelve working hours in a day. Each of them wants more than they can have. Each of them must choose.

This chapter asks three plain questions. Why is choice forced on everyone, even on rich people and rich countries? What does a choice cost, when no money is paid? And what is the subject that studies these questions? You need no earlier knowledge to begin. Everyday experience of spending, saving and choosing is enough.

THE PROBLEM OF WANTING MORE

Begin with a student who has Tk 100 to last the week. The student would like a notebook for Tk 60, a lunch for Tk 70, a bus ticket for Tk 40 and a cinema seat for Tk 80. Add the four prices and the total is Tk 250. The purse holds Tk 100. The student can have some of these things, but not all of them. This gap between what is wished and what is possible is the root of the whole subject.

We need a word for the wishes in this picture. The word is **want**. A want means a wish for something. The student's wish for a cinema seat is one want. A family's wish for a better roof is another. A country's wish for a new bridge is a third. Economists do not judge whether a want is wise or foolish. They note only that wants exist and that they are very many. When one want is met, another often takes its place. A person who has a bicycle may soon want a motorcycle.

Wants are met by goods, and goods must be made. Making them needs materials, tools and people. Each such thing is a **resource**. A resource means anything used to make goods. Soil, water, human work and machines are all resources. A baker uses flour, an oven and hours of work to make bread. If the flour runs out or the hours end, no more bread can be made.

What do resources make? They make things that satisfy wants. We call such a thing a **good**. Rice, a bicycle and a pair of shoes are goods. Some wants are met by work done for others, and not by a thing that can be held. A teacher's lesson and a barber's haircut are examples. This book calls such work a **service**. To save words, the book often writes goods for both, and it says so when the difference matters.

Economists sort resources into four groups. Together the groups are called the **factors of production**. The first is **land**, which means all gifts of nature, such as soil, water, forests, minerals and fish.¹ The second is **labour**, which means human work, whether of the body or of the mind. A farmer who ploughs and a clerk who keeps accounts both supply labour.

¹ Economists use land in a wide sense. A river and the fish in it, and the oil under the sea bed, are all land.

The third is **capital**, which means made things that are used to produce other goods, such as ovens, tools and buildings. In this book capital does not mean money. The fourth is the **entrepreneur**. An entrepreneur is a person who starts and runs a business and bears its risk. If the business fails, the entrepreneur loses the most.

The table below gives one example of each factor, taken from a bakery. Read each row from left to right: the factor, its meaning, and the example.

THE FOUR FACTORS OF PRODUCTION

FACTOR	MEANING	EXAMPLE IN A BAKERY
Land	Gifts of nature	The wheat grown in the soil; the water for the dough
Labour	Human work	The baker who kneads and bakes
Capital	Made things used in production	The oven and the baking trays
Entrepreneur	The person who starts the business and bears its risk	The owner who decides to open the shop

Every factor earns a reward when it is used. Later chapters study these rewards and how they are set.² For now, remember only that all four are resources, and that all four are limited.

We can now name the gap seen in the student's purse. **Scarcity** means that too little exists to satisfy all wants. Resources are limited, and wants are not, so some wants must go unmet. The student has Tk 100 against wants worth Tk 250. A country has a limited amount of land, labour and capital, against a long list of wants. Scarcity is the starting point of economics.

Two confusions should be removed at once. First, scarcity is not the same as poverty. A rich family also has limited money and limited time, and it too must choose. Second, scarcity is not the same as a shortage. A shortage is a short-lived lack of one good, such as no rice in the shop today because the lorry is late. A shortage may end next week. Scarcity applies to nearly all goods, and it does not end.

Is anything not scarce? A **free good** means a good that exists in unlimited amount. Sea air on an empty beach is an example. Anyone may breathe as much as they like, and no one's use leaves less for another person. Free goods are rare, and they raise no problem of choice. Notice that the word free here is not about price. A lesson in a government school costs the student nothing, yet it is not a free good, because teachers, buildings and books were used to provide it.

Before moving on. Wants are many. Resources are few, and they come in four groups. The gap between them is scarcity. A free good is the rare thing that is not scarce. *You should now be able to say:* wants are many, resources are limited, and this gap is called scarcity.

Pause and check. A friend says that a shop in the town has no sugar this week. Is this scarcity or a shortage? Give a reason.

Capital. Made things used to make other goods.

Entrepreneur. A person who starts and runs a business and bears its risk.

The other two groups are capital and the entrepreneur.

A table gives one bakery example of each group.

Each factor earns a reward, which later chapters will study.

Scarcity. Too little exists to satisfy all wants.

Scarcity means resources are too few to meet all wants.

Scarcity is not poverty, and it differs from a shortage.

Free good. A good that exists in unlimited amount.

A free good exists in unlimited amount, so no one must choose.

Recap: wants are many, resources are few, and the gap is scarcity.

A short question tests the idea of scarcity.

CHOICE AND OPPORTUNITY COST

Return to the student, now with an evening free on Friday. A cousin has asked for help at the family shop and will pay Tk 200 for the evening. A friend has invited the student to a film, and the ticket is free. The student cannot do both,

GO SLOWLY HERE.

Every choice gives something up, even a free evening.

² Chapters XV and XVI study the rewards of the four factors.

Choice. Picking one use of limited resources.

A choice means picking one use of limited resources.

Opportunity cost. The best alternative given up.

Opportunity cost is the best alternative given up.

Only the single best alternative counts, not all of them.

Recap: the cost of a choice is the best thing given up.

A short question tests the idea of opportunity cost.

Opportunity cost can be counted in money, in time or in goods.

The bakery owner meets the same cost, in larger amounts.

Opportunity cost needs a real alternative to exist.

Thinkers have answered the question "what is economics?" in different ways.

because both take the same evening. The student chooses the film. What did the film cost? No money was paid, so we might say that it cost nothing. But the student gave up Tk 200. That is the real cost of the evening.

The student made a **choice**. A choice means picking one use of limited resources, such as money or time, and leaving the others. Choice exists only because of scarcity. If resources were unlimited, the student could help at the shop and see the film as well. No one would need to choose.

Economists give the cost of a choice a special name. The **opportunity cost** of a choice is the best alternative given up. In our example the film evening has an opportunity cost of Tk 200, because helping at the shop was the best alternative. The word opportunity is a reminder: the cost is an opportunity that was lost.

The word best matters. Take a farmer with one field. Planted with rice, the field would give a net income of Tk 40,000 this season. Net income means income after all costs are paid. Planted with jute, it would give Tk 30,000. Planted with pulses, it would give Tk 25,000. The farmer plants rice. The opportunity cost is Tk 30,000, the net income from jute. It is not Tk 55,000, the sum of jute and pulses, because the field can hold only one of them. The farmer gives up the best of the other choices, and that one counts.

Before moving on. Choice follows scarcity. The cost of a choice is the best alternative given up, even when no money is paid. Only one alternative counts: the best of those not chosen. *You should now be able to say:* the cost of a choice is the best thing I give up to make it.

Pause and check. A student has one hour. The student can tutor a child for Tk 150, or sell snacks for Tk 100. The student chooses tutoring. What is the opportunity cost of the hour?

Opportunity cost is not always money. For a student, an hour on a phone game is an hour not spent on study. For a country, a taka spent on a road is a taka not spent on a school. Money gives a convenient way to measure the cost, but the cost itself is the alternative that was lost.

Now view the same idea from the owner's seat. A person who opens a bakery pays money for rent, flour and wages. These payments are easy to see. But the owner also gives up things that no one is paid for: the wage that a job would have brought, and the interest that the savings would have earned in a bank. These forgone rewards are real costs. The worked case at the end of this chapter measures them. Chapter X returns to the idea and shows how a firm counts such costs.³

Some limits keep the idea clear. If there is truly no alternative, nothing is given up, and the opportunity cost is zero. The cost is judged at the time of the choice, by what could have been done then. And opportunity cost is not regret. A choice may turn out well even when its opportunity cost was high.

WHAT ECONOMICS IS

Ask what economics is, and the answer depends on whom you ask and when. Over two hundred years, three well-known answers were given. Each was a step forward, and each left something out. Looking at them in order shows why the modern answer rests on the two ideas you have met: scarcity and choice.

³ Chapter X, section 10.1, measures opportunity cost and separates costs paid in money from costs not paid.

WEALTH: ADAM SMITH, 1776

Adam Smith was a Scottish thinker who taught at the University of Glasgow. In 1776 he published a book about the nature and causes of the wealth of nations. Many readers took from it a plain definition: economics is the study of wealth, meaning how a nation produces goods of value and grows richer. This is called the wealth definition.

The wealth definition has two limits. It places goods at the centre and the people who need them at the side. It also makes the subject seem a study of money-making only, which many readers found too narrow. People want wealth for what it does for their lives.

The earliest famous answer says economics studies wealth.

The wealth definition is too narrow, because it puts goods before people.

WELL-BEING: ALFRED MARSHALL, 1890

Alfred Marshall was a professor at Cambridge, in England. In 1890 he published *Principles of Economics*. He described economics as a study of people in the ordinary business of life. He held that wealth matters because it serves human well-being, and he spoke of material well-being, meaning the well-being that comes from goods that can be bought and sold. This is called the welfare definition.

The welfare definition put people first, yet it too has limits. It seems to count only material things, so the work of a teacher or a doctor sat uneasily inside it. It also says little about choice, which is the activity that economics actually studies. Many lives mix material and other aims in one decision.

Marshall moved people to the centre and treated wealth as a means.

The welfare definition has limits: it divides life into material and other parts.

SCARCITY AND CHOICE: LIONEL ROBBINS, 1932

Lionel Robbins taught at the London School of Economics. In 1932 he published *An Essay on the Nature and Significance of Economic Science*. He said that economics studies human behaviour as a relation between wants and scarce resources that have alternative uses.⁴ In plain words, economics studies choice under scarcity. This is the scarcity and choice definition.

This definition fits any choice, in any society, whether or not money is used. The student's evening, the farmer's field and a government's budget all fall inside it. Its weakness is that it says little about why a whole economy may produce less, or why many jobs may vanish. Later writers treated those questions as a separate branch, as Chapter II shows.

The table sets the three answers side by side. Read it row by row. Each row gives one thinker, the central idea, one strength and one limit.

Robbins defined economics by scarcity and choice.

The scarcity definition fits every choice, but it says little about whole economies.

A table sets the three definitions side by side.

THREE DEFINITIONS COMPARED

THINKER AND DATE	CENTRAL IDEA	STRENGTH	LIMIT
Adam Smith, 1776	Wealth: how nations grow rich	Focus on producing goods	Puts goods before people
Alfred Marshall, 1890	Well-being: wealth serves people	People come first	Seems to count only material things
Lionel Robbins, 1932	Scarcity and choice	Fits any choice	Says little about whole economies

⁴ Robbins wrote ends where this book writes wants, and means where it writes resources. The meaning is the same.

Economics. The study of choice under scarcity.

Modern books join scarcity, choice and well-being in one definition.

THE CENTRAL THEME

Today most textbooks keep the idea of Robbins as the core and keep the concern of Marshall for people. This book therefore uses a working definition. **Economics** means the study of choice under scarcity. It asks how individuals, firms and governments use limited resources to meet wants, and what follows for the well-being of people.

IN SHORT

Economics is the study of how people and societies choose among uses of scarce resources. Its theme is scarcity and choice.

Everything in the book grows from one chain of ideas.

Economics explains, predicts and guides choice.

Recap: three definitions led to one working definition.

The central theme can be written as a chain. Wants are many. Resources are limited. So there is scarcity. Scarcity forces choice. Every choice has an opportunity cost. Each later chapter takes one link of this chain and studies it closely: how buyers choose, how sellers choose, and how markets bring the two together.

Knowing what economics studies, we can ask what it is for. It does three jobs. It *explains*: why did the price of onions rise in one month? It *predicts*: if a tax is placed on fuel, buyers will probably use less. And it *guides* choice: it shows a family, a firm or a government what each option costs, so that the choice is made with open eyes. Economics cannot choose for anyone, because the final choice depends on values. Chapter II returns to this point.

Before moving on. Smith looked at wealth, Marshall at well-being, and Robbins at scarcity and choice. This book defines economics as the study of choice under scarcity. Its three jobs are to explain, to predict and to guide choice.

THE THREE BASIC PROBLEMS

Because of scarcity, every society must answer three questions.

What to produce: which goods, and how much of each.

How to produce: which method and which mix of resources.

For whom to produce: who receives the goods that are made.

Think of a village after a poor harvest. There is not enough rice for everyone. The villagers must decide what to plant next year, how to work the fields, and who will receive the rice that exists. These questions are not special to villages. Every society, large or small, poor or rich, meets the same three problems, because every society has scarce resources and many wants.

The first problem is *what* to produce. Land, labour and capital can make rice or jute, schools or roads, shirts or mobile phones. Using them for one purpose takes them from another. The society must decide which goods to make, and how much of each. This is the choice of the farmer's field, now made for a whole country.

The second problem is *how* to produce. The same output can be made in different ways. Rice can be harvested by many workers with sickles, or by a few workers with a machine. The first method uses much labour and little capital. The second uses much capital and little labour. The society must choose a method, and the choice depends on which resources it has in plenty and which are scarce.

The third problem is *for whom* to produce. When goods are made, someone receives them. They may go to those who pay the most, to those in greatest need, or in equal shares. This question is about sharing the output of a society among its people.

A society's answers to the three problems form its **economic system**. An economic system means the way a society answers what, how and for whom. Some writers call it an economic organisation. This book uses the term economic system throughout.

Real economies differ in who decides. In a traditional system, custom decides: sons farm as their fathers did. In a market system, prices and the free choices of buyers and sellers decide. In a command system, the government decides by plan. In a mixed system, markets and government share the work. Most countries today, Bangladesh among them, are mixed. The table shows the main answers. Read each row as one kind of system.

HOW FOUR SYSTEMS ANSWER THE THREE PROBLEMS

SYSTEM	WHAT TO PRODUCE	HOW TO PRODUCE	FOR WHOM
Traditional	What custom has always made	By inherited methods	By family and custom
Market	What buyers will pay for	By methods that cost least	To those who can pay
Command	By government plan	By government orders	By government rule or ration
Mixed	Partly by prices, partly by government	Mostly by firms, under rules	By prices, with taxes and public services

No system removes scarcity. Each handles the three problems in its own way, and each has strengths and costs. Later chapters study the market system closely, because prices and markets are the subject of microeconomics. Notice also that the three problems are choices, and so each carries an opportunity cost: more of one good means less of another.

Before moving on. Every society must decide what to produce, how to produce it, and for whom. The pattern of answers is its economic system. The four kinds are traditional, market, command and mixed.



A WORKED CASE: THE OWNER'S CHOICE

Case file, illustrative. Situation. On 2 February 2026 an owner has savings of Tk 600,000 and a job that pays Tk 40,000 a month. There are two options. Option A is to keep the job and deposit the savings in a bank, at an assumed 6% a year. Option B is to open a bakery.

Case file, illustrative. Question. Apart from money paid to others, what does Option B cost the owner each month?

Case file, illustrative. Reasoning. Option A gives two rewards. The wage is Tk 40,000 a month. The interest is 6% of Tk 600,000, which is Tk 36,000 a year, or Tk 3,000 a month. Choosing the bakery gives up both. The opportunity cost of the bakery is therefore Tk 40,000 + Tk 3,000 = Tk 43,000 a month. Rent comes on top of this, because it is a money payment. On 1 March 2026 the bakery opens, with shop rent of Tk 25,000 a month and a loaf price of Tk 60.

Case file, illustrative. The three problems. What to produce: loaves and buns. How to produce: with one oven and hired bakers. For whom: the households of the neighbourhood.

Economic system. The way a society answers what, how and for whom.

An economic system is the way a society answers the three problems.

Four broad kinds of system answer the problems differently.

No system removes scarcity; each handles the problems in its own way.

Recap: scarcity forces every society to answer what, how and for whom.

The case file begins on 2 February 2026 with a choice.

The question is what Option B costs besides money paid out.

Reasoning: Option A would pay interest of Tk 3,000 a month, plus the wage.

The owner must also settle the three basic problems.

Conclusion and check: the bakery is worth opening only if it earns more than Tk 43,000 a month above its running costs.

Case file, illustrative. Conclusion. The bakery is a good choice only if it can earn more than Tk 43,000 a month above what it pays out in running costs. Later chapters test this. Check. 6% of Tk 600,000 is 36,000, and 36,000 divided by 12 is 3,000. Then 40,000 + 3,000 is 43,000.



SUMMARY

Scarcity forces choice, and choice has a cost.

Wants are many and resources are limited, so scarcity exists. Resources are land, labour, capital and the entrepreneur. Scarcity forces choice, and the cost of a choice is the best alternative given up. Economics is the study of choice under scarcity, and it explains, predicts and guides choice. Every society must decide what to produce, how, and for whom. Its answers form its economic system.

THINK IT THROUGH

A small choice with two uses of Tk 500.

A student has Tk 500 left at the end of the month. She can buy a pair of shoes for Tk 500. Or she can keep the money towards a course fee of Tk 5,000, which she hopes to pay in ten months. State the opportunity cost of buying the shoes. Then say what facts you would want before advising her, and whether your advice would change if she had Tk 5,000 instead of Tk 500.

WHERE PEOPLE GO WRONG

Scarcity does not mean poverty or a shortage.

Many students write that scarcity means poor people have too little. It does not. Scarcity means that wants exceed resources, and it applies to the richest family and the richest country. A shortage is a temporary lack of one good. Scarcity is lasting.

Opportunity cost is not the money price, and it is not the sum of alternatives.

Many students give the price paid as the cost of a choice. The price is only part of it, and for a free film it is zero while the cost is not. Others add up every alternative given up. Only the single best alternative counts.

Free in price is not the same as a free good.

A good that costs the user nothing, such as a government school lesson, is not a free good. Resources were used to provide it, and they could have been used elsewhere. A free good exists in unlimited amount, with no resources used up.

IN HISTORY

Wartime food rationing in Britain answered the question "for whom".

During the Second World War, Britain was short of imported food. Rationing began on 8 January 1940, first with bacon, butter and sugar, and later it covered many other foods. The last food controls ended in 1954. Each person received a ration book that gave a fixed weekly share. The government answered the question of *for whom* by shares, and not by price and money. The rule did not remove scarcity. It changed who received the scarce food, and it showed that every society must answer the question in some way.

SOURCES

The three definitions come from three landmark books.

Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), set out how nations produce wealth. Alfred Marshall, *Principles of Economics* (1890), made well-being the aim of the subject. Lionel Robbins, *An Essay on the Nature and*

Significance of Economic Science (1932), defined the subject by scarcity and choice.

REVIEW QUESTIONS AND PRACTICE

REVIEW QUESTIONS

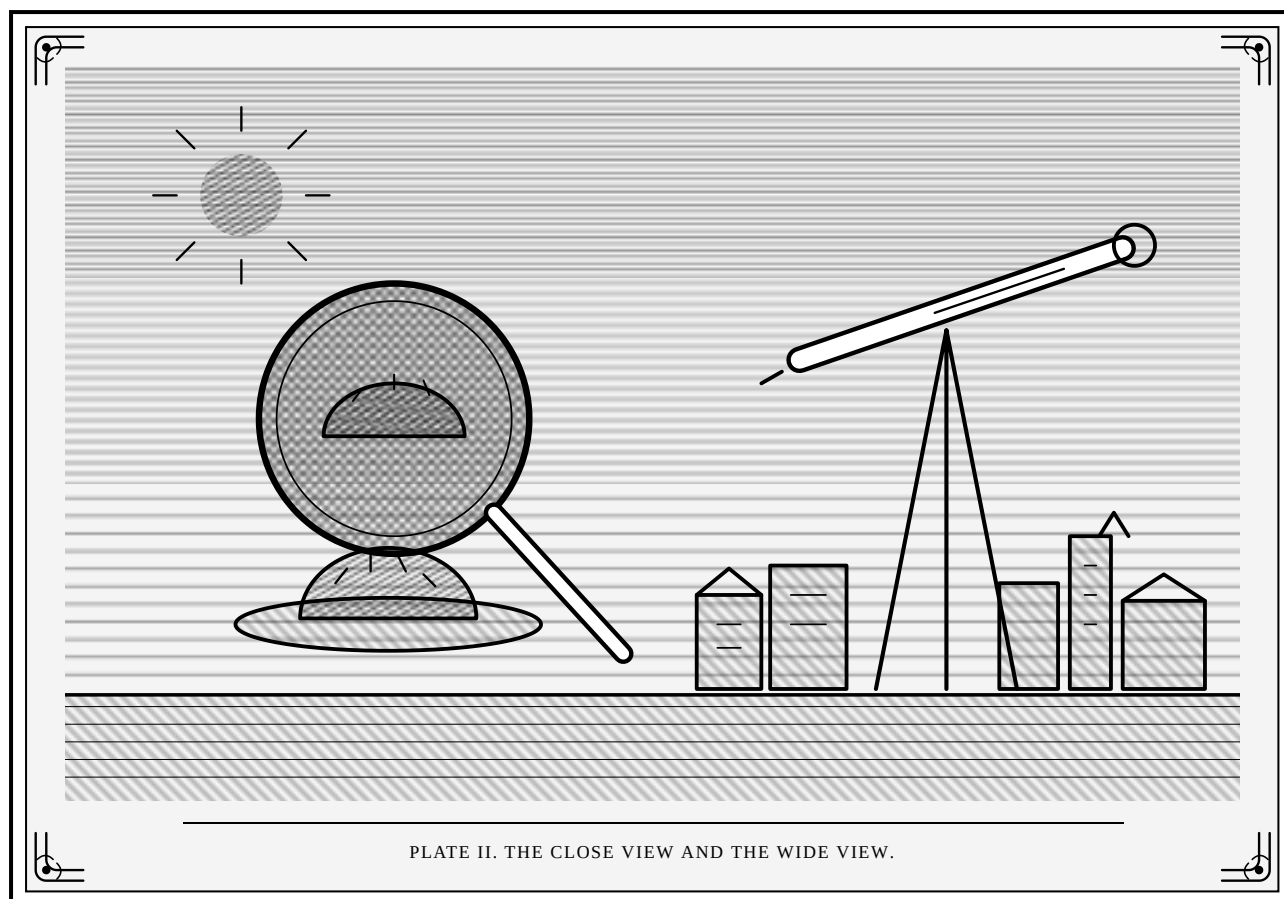
1. Explain the difference between scarcity and a shortage, with one example of each. 2. Give two goods that are free goods and two that are not, and say why. 3. Name the three basic problems of an economic system and give one example of each. 4. List the four factors of production and give one example of each from a farm.

Four review questions test the main ideas of the chapter.

PRACTICE: FROM A HOUSEHOLD TO A BAKERY

Item 1: a household. A family has Tk 7,500. It would like to buy a bicycle for Tk 7,000, pay school books and fees of Tk 7,000, or buy a mobile phone for Tk 6,500. The family ranks them in that order: bicycle first, books second, phone third. It buys the books. What is the opportunity cost? *Item 2: a market stall.* A stall holder has Tk 3,000 to buy stock. Vegetables would sell for Tk 3,900, dried fish for Tk 3,600 and lentils for Tk 3,300. The stall holder buys vegetables. State the expected gain from each stock, and the opportunity cost of the choice. *Item 3: a government.* A district council has Tk 50 lakh. It can repair a bridge, build a clinic or repair schools. The council ranks them in that order. It repairs the bridge. State the opportunity cost, and say which basic problem the choice answers. *Item 4: a bakery.* A person has savings of Tk 400,000 and a job at Tk 30,000 a month. A bank pays 6% a year. The person plans to open a bakery. Find the opportunity cost each month, and name the three choices the person must make.

Four practice items move from a familiar case to a combined one.



CHAPTER II

How Economists Think

One event, seen from two places.



This chapter separates the close-up view of economics from the wide view, and shows how each depends on the other. It explains how economists build simple pictures and state laws, how facts differ from values, and where the subject is used in Bangladesh.

A rise in bus fares can be seen from two places.

The two pictures answer different questions, and both are true.

This chapter builds on scarcity and choice from Chapter I.

Economics has a close-up branch and a wide branch.

Microeconomics. The study of single households, firms and markets.

Microeconomics studies the choices of single households, firms and markets.

Household. People who live together and spend together.

Firm. An organisation that makes goods for sale.

Households and firms are the two main decision makers in micro study.

Macroeconomics. The study of the whole economy.

Macroeconomics studies the whole economy.

A table sets the two branches side by side.

On a Monday in March, bus fares in a city rise by Tk 5 a ride. A student at a bus stop does the sum: the two daily rides now cost Tk 10 more. The student decides to walk the shorter half of the journey and ride the longer half. At the same moment, a watcher in a government office looks at the whole city. The watcher sees that fares, food prices and house rents have all risen together, and that the number of passengers across the city has fallen.

Both pictures are true. The student's picture is small and close: one person, one choice, one price. The watcher's picture is wide: the whole city, many prices, total travel. Economists use both. This chapter shows how the two differ, and how they depend on each other. It then shows how economists build simple pictures of the world, how they separate facts from opinions, and where the close-up picture is used in Bangladesh.

You will need two ideas from Chapter I: scarcity, and choice with its opportunity cost. Both appear here again, in new settings.

TWO WAYS OF LOOKING

Look again at the two viewers in the opening story. The student studied one household's choice. The watcher studied a whole economy. Economics has a name for each way of looking. The two names come from Greek words: *mikros*, which means small, and *makros*, which means large.

Microeconomics means the study of the choices of single households, firms and markets, and of the prices that result. A family choosing what to eat is a micro subject. So is a tailor choosing the price of a shirt, and so is the market for fish in one town. The unit is small, and the questions are about choice and price.

Two words need clear meaning. A **household** means people who live together and spend together, such as a family sharing one kitchen. A **firm** means an organisation that makes goods for sale, such as a bakery, a garment factory or a bus company. Households buy goods and supply labour. Firms hire labour and make goods. Chapter III onwards studies how each chooses.

Macroeconomics means the study of the whole economy: its total output, the general level of prices, and the total number of jobs. The watcher in the government office was asking macro questions. Why did the total output of the country grow more slowly this year? Why did prices in general rise? The unit is the whole, and the questions are about totals and averages.

The table below compares the two branches. Read down a column to see what one branch studies. Read across a row to compare the branches on one feature.

MICROECONOMICS AND MACROECONOMICS COMPARED

FEATURE	MICROECONOMICS	MACROECONOMICS
Unit of study	A household, a firm, a market	The whole economy
Typical question	Why did the price of onions rise?	Why did prices in general rise?
Example of a subject	The price a bakery sets for a loaf	The total number of jobs in the country
Main idea	Choice and price in one market	Totals and averages over all markets
Viewpoint	Close up	Wide

The two branches are not rivals. They depend on each other. The total output of a country is built from the output of its firms, so macro results rest on micro choices. Yet the whole can behave in a way that no single part shows. Suppose every family decides to save more and spend less. Each family acts sensibly. But shops then sell less, some firms cut jobs, and family incomes fall. This result can be seen only from the wide view. In the other direction, general conditions, such as the level of prices, form the surroundings in which each firm chooses.

Before moving on. Microeconomics studies single households, firms and markets. Macroeconomics studies the whole economy. Each depends on the other. This book is about microeconomics.

The two branches depend on each other.

Recap: micro is close up, macro is wide, and each needs the other.

SCOPE, IMPORTANCE AND PURPOSES

What does microeconomics cover? Its scope has three parts. The first is the pricing of products: how demand, supply and cost decide the price of bread, cloth or bus rides. Chapters III to XIV study this. The second is the pricing of resources: what rent, wages, interest and profit are paid, and why. Chapters XV and XVI study this.

The third part is **welfare**. Welfare means the well-being of people. Here we ask whether the way markets work serves people well, and who gains or loses when something changes. A new tax may raise the money a government needs. It may also leave some households with less to spend. Welfare study keeps both effects in view.

Why does it matter? For consumers, it explains how prices arise and how to spend limited money with care. For firms, it shows how costs, demand and rivals affect the price and output to choose. For governments, it shows who gains and who loses from a tax, a price limit or a subsidy. A government that sets a price limit on rice without knowing how buyers and sellers react may bring about the very shortage that it fears.

Chapter I named three jobs of economics. Microeconomics does each job for single units. It explains why a price is what it is. It predicts what may happen when a price, a tax or an income changes. It guides choice by showing what each option costs. In this way it serves the student, the shopkeeper and the official alike.

A student of microeconomics also gains a habit of asking three questions. What is scarce here? What is the best alternative? Who bears the cost? These questions help in running a shop, in reading a news report, and in judging a public plan.

Microeconomics covers three large areas.

Welfare. The well-being of people.

The third part is welfare, the well-being of people.

Microeconomics helps consumers, firms and governments choose.

The purposes are to explain, to predict and to guide choice.

The student also gains a habit of questions.

Recap: three areas, three kinds of user, three purposes.

GO SLOWLY HERE.

A bus route map is wrong in detail and still useful.

Economic model. A simple picture of how something works.

A model is a simple picture that keeps only what matters.

A model is judged by whether it helps, not by whether it is complete.

Ceteris paribus. All other things held unchanged.

Ceteris paribus means all other things held unchanged.

Economic law. A statement of a regular tendency.

An economic law is a statement of a regular tendency.

Economic laws have five features.

A table compares economic laws with laws of nature.

A law does not fail where its conditions are absent.

Before moving on. The scope is product pricing, resource pricing and welfare. The users are consumers, firms and governments. The purposes are to explain, to predict and to guide choice.

MODELS AND ECONOMIC LAWS

Think of the route map pinned inside a bus. It shows lines and stops. It leaves out the shape of the roads, the shops, the trees and the hills. A map with all of these would be too crowded to read. The route map is useful because it leaves things out. It shows only what a traveller needs in order to find the way.

An **economic model** is the same kind of thing for the economy. An economic model means a simple picture of how something works. It keeps a few important features and sets the rest aside. A model of the market for tea may keep only the price of tea and the number of cups bought, and ignore the colour of the cups. Because the model is simple, we can reason through it step by step.

No model is complete. A model is judged by a plain test: does it help us explain what happens and predict what may happen? A model that fails the test is changed or dropped.

To use a model we must change one thing at a time. Suppose the price of tea rises, and in the same month the weather turns hot and incomes fall. The number of cups bought changes, and we cannot tell which cause did what. So economists use an old Latin phrase, **ceteris paribus**, which means all other things held unchanged. We then ask: if only the price of tea changes, what happens to the cups bought? This is a tool of thought, like testing one switch at a time. It does not claim that the real world stands still.

Models help us find regular patterns. When a pattern holds in most cases, under stated conditions, we call it an **economic law**. An economic law means a statement of a regular tendency. One example, met in Chapter IV, is that buyers tend to buy less of a good when its price rises, other things unchanged. The word law here does not mean a rule made by a government. No one is punished for breaking an economic law.

Five features mark an economic law. First, it is a statement of tendency and not of certainty: most buyers, most of the time. Second, it holds under conditions, usually ceteris paribus. Third, it rests on the observed behaviour of people. Fourth, it can be tested by facts, though not in a laboratory where everything is controlled. Fifth, it is less exact than a law of nature, because people differ and change their minds.

A law of nature, such as the law that a dropped stone falls, is a useful comparison. The table below sets the two kinds of law side by side. Read each row to see one point of difference.

LAWS OF NATURE AND ECONOMIC LAWS

FEATURE	LAW OF NATURE	ECONOMIC LAW
Subject	Matter and forces	Human choices
Statement	Holds exactly	Holds as a tendency
Test	Controlled experiment	Observation of events
Exactness	High	Lower, because people vary
Example	A dropped stone falls	A price rise lowers purchases

If the stated conditions are absent, the law does not apply. Most laws of economics come with exceptions, and later chapters name and explain them. An

exception is not a defeat for the law. It tells us where the conditions have changed.

Before moving on. A model keeps what matters. *Ceteris paribus* holds other things unchanged. An economic law states a tendency under conditions. *You should now be able to say:* an economic law tells what usually happens when other things stay the same.

Pause and check. A friend says: "The law of demand is false, because I bought more tea last month even though the price rose." What has the friend left out?

Recap: models simplify, ceteris paribus isolates one cause, and laws state tendencies.

A short question tests the idea of an economic law.

POSITIVE AND NORMATIVE; SCIENCE OR ART

Compare two statements. One: the price of lentils was higher in March than in February. Two: the price of lentils should be lowered for poor families. The first can be checked by looking at prices in the two months. The second cannot be checked in that way, because it expresses a view about what is right.

A **positive statement** means a statement that can be tested by facts. It says what is, what was, or what will be. It may be true or false, but the way to settle it is by evidence. "A rise in bus fares reduces the number of passengers" is a positive statement. Evidence may show that it is wrong, and it is still positive.

A **normative statement** means a statement of what ought to be. It rests on values, such as fairness. "Bus fares should be kept low for students" is a normative statement. Evidence about costs may help the debate, but evidence alone cannot settle it, because people hold different values.

When two people disagree, ask which kind of disagreement it is. If they disagree about a positive matter, such as whether a price limit causes shortages, they can look at evidence. If they disagree about a normative matter, such as whether poor families should receive cheap rice, they must debate values. Much confusion comes from mixing the two. Economics helps with the first kind directly. It helps with the second kind by showing what each choice costs.

Is economics a science or an art? A science is organised knowledge built by observation and reasoning. It tells us what is. An art is the skilled use of knowledge to reach an aim. It tells us how to do something. In 1891 John Neville Keynes, an English economist, described three parts of the subject: a positive science, a normative science of ideals, and an art of rules for reaching goals.¹ By this view, economics is a science when it explains, and an art when it advises a government or a firm. It also uses normative judgements when it asks what ought to be done. The honest answer is that it is both.

Before moving on. A positive statement can be tested by facts. A normative statement says what ought to be. Economics is a science when it explains and an art when it advises.

Some statements can be tested by facts, and some cannot.

Positive statement. A statement that can be tested by facts.

A positive statement can be tested by facts.

Normative statement. A statement of what ought to be.

A normative statement says what ought to be.

Disagreements about facts and about values are different disagreements.

Economics is both a science and an art.

Recap: positive means testable, normative means ought, and economics is science and art.

MICROECONOMICS IN BANGLADESH

Microeconomics is not only a classroom subject. Many ordinary events in Bangladesh are micro questions. Why does the price of vegetables change between a good harvest and a flood season? How does a farmer decide between boro rice and another crop? Why do garment factories in one district compete for the same workers, and what does this do to wages? What rules stop sellers from agreeing together to fix a price?

Microeconomic questions appear in daily life in Bangladesh.

¹ John Neville Keynes was the father of John Maynard Keynes, who appears in the history section of this chapter.

The subject has growing uses in business, farming, public work and research.

Several kinds of institution teach, study and apply microeconomics.

These questions point to the prospects of the subject. Firms need people who can judge costs, prices and demand. Farmers and farm advisers choose crops and methods. Public bodies set rules for markets. Researchers use data to study households and firms. There are also difficulties: many markets are informal, and good data on small firms is hard to obtain. A student who learns the method can help with both the use and the difficulty.

Several kinds of institution teach, research and apply the subject. The table lists six, with one example of each and its main role. The names and duties should be checked against current official sources, because public bodies are renamed and reorganised.²

INSTITUTIONS THAT TEACH, RESEARCH AND APPLY MICROECONOMICS

KIND	EXAMPLE	MAIN ROLE
University	Department of Economics, University of Dhaka	Teaches the subject and trains researchers
Research institute	Bangladesh Institute of Development Studies	Studies households, farms, prices and public plans
Central bank	Bangladesh Bank	Regulates banks and money; its work affects interest rates
Statistics office	Bangladesh Bureau of Statistics	Collects data on prices, farm output and households
Planning body	Planning Commission	Prepares national plans and weighs public spending
Competition commission	Bangladesh Competition Commission	Applies the Competition Act 2012 to rules on price fixing and abuse of market power

Some of these bodies return later in the book.

Some of these bodies return later in the book. The statistics office supplies the price and output data on which the market chapters rely. The central bank appears when we study interest in Chapter XVI. The competition commission appears in Chapter XIII, where we study sellers who have power over price. Do not confuse microeconomics with microfinance. Microeconomics is a way of studying choices and prices. Microfinance means small loans to poor borrowers, and Chapter XVI treats it briefly.

Recap: the subject applies to prices, farms, wages and market rules, and has public institutions.

Before moving on. Microeconomic questions arise in prices, farm output, wages and competition rules. Universities, research institutes, the central bank, the statistics office, the planning body and the competition commission teach, study or apply the subject.



A WORKED CASE: ONE FIRM, ONE ECONOMY

The bakery is a micro unit, and a general price rise is a macro event.

Case file, illustrative. Situation. The bakery opened on 1 March 2026 and sells loaves at Tk 60. It is one firm in the neighbourhood market for bread. Suppose that the price of flour rises as part of a general rise in prices across the economy.

The question is which of the bakery's concerns are micro and which are macro.

Case file, illustrative. Question. Which concerns of the owner belong to microeconomics, and which to macroeconomics?

² The names, dates and duties in this table are given as general descriptions and should be confirmed before use.

Case file, illustrative. Reasoning. The owner decides the price of a loaf, the number of bakers to hire and the hours of work. These are choices of one firm in one market, and so they are micro decisions. The general rise in prices comes from outside the bakery. It affects all firms and households, and it is measured for the whole economy, so it is a macro event. The owner cannot change it, but must respond to it.

Case file, illustrative. Conclusion. The price the bakery sets is micro. The general price rise is macro. Check. Ask of any question: is it about one household, firm or market, or about totals for the whole economy? No new figures are added in this chapter.

Reasoning: choices inside the market are micro; the general price rise is macro.

Conclusion and check: ask whether the question is about one unit or the whole economy.



SUMMARY

Microeconomics studies single households, firms and markets. Macroeconomics studies the whole economy, and each depends on the other. Microeconomics covers product pricing, resource pricing and welfare. Economists use simple models and the idea of *ceteris paribus* to state laws of tendency. Positive statements can be tested by facts, and normative statements say what ought to be. Economics is both a science and an art, and it is applied in Bangladesh in many ways.

Economics has two views and uses models and laws.

THINK IT THROUGH

Bus fares in a city rise by 10%. Describe two questions that a micro economist would ask about this event, and two that a macro economist would ask. Then write one positive and one normative statement about the rise.

A rise in bus fares seen from two seats.

WHERE PEOPLE GO WRONG

Many students think microeconomics studies small firms and macroeconomics studies large ones. The difference is in the question, not in the size. A large company's price is a micro matter. The total jobs in a country is a macro matter.

Micro does not mean only small businesses.

Many students think *ceteris paribus* means that nothing else changes in the real world. It is a way of thinking. We hold other things unchanged in the model, in order to see one cause at work.

Ceteris paribus does not claim that nothing else changes.

A positive statement can be false. It is called positive because it can be tested. An economic law is a statement of tendency. It is not a rule that anyone enforces.

A positive statement is not a true statement, and a law is not a rule of government.

IN HISTORY

In the early 1930s many countries suffered a long fall in output and a great rise in unemployment, called the Great Depression. Thinking about single markets did not explain why so many jobs vanished at once. In 1936 John Maynard Keynes published *The General Theory of Employment, Interest and Money*, which argued that total spending in the economy decides total output and jobs. The wide view became a branch of its own. The Norwegian economist Ragnar Frisch is often credited with using the words micro and macro for the two views in a paper of 1933.

The Great Depression of the 1930s showed the need for a wide view.

SOURCES

The ideas of this chapter come from two writers on method.

John Neville Keynes, *The Scope and Method of Political Economy* (1891), separated the positive, the normative and the art. Milton Friedman, *The Methodology of Positive Economics* (1953), argued that a model should be judged by the quality of its predictions.

REVIEW QUESTIONS AND PRACTICE

Four review questions test the main ideas of the chapter.

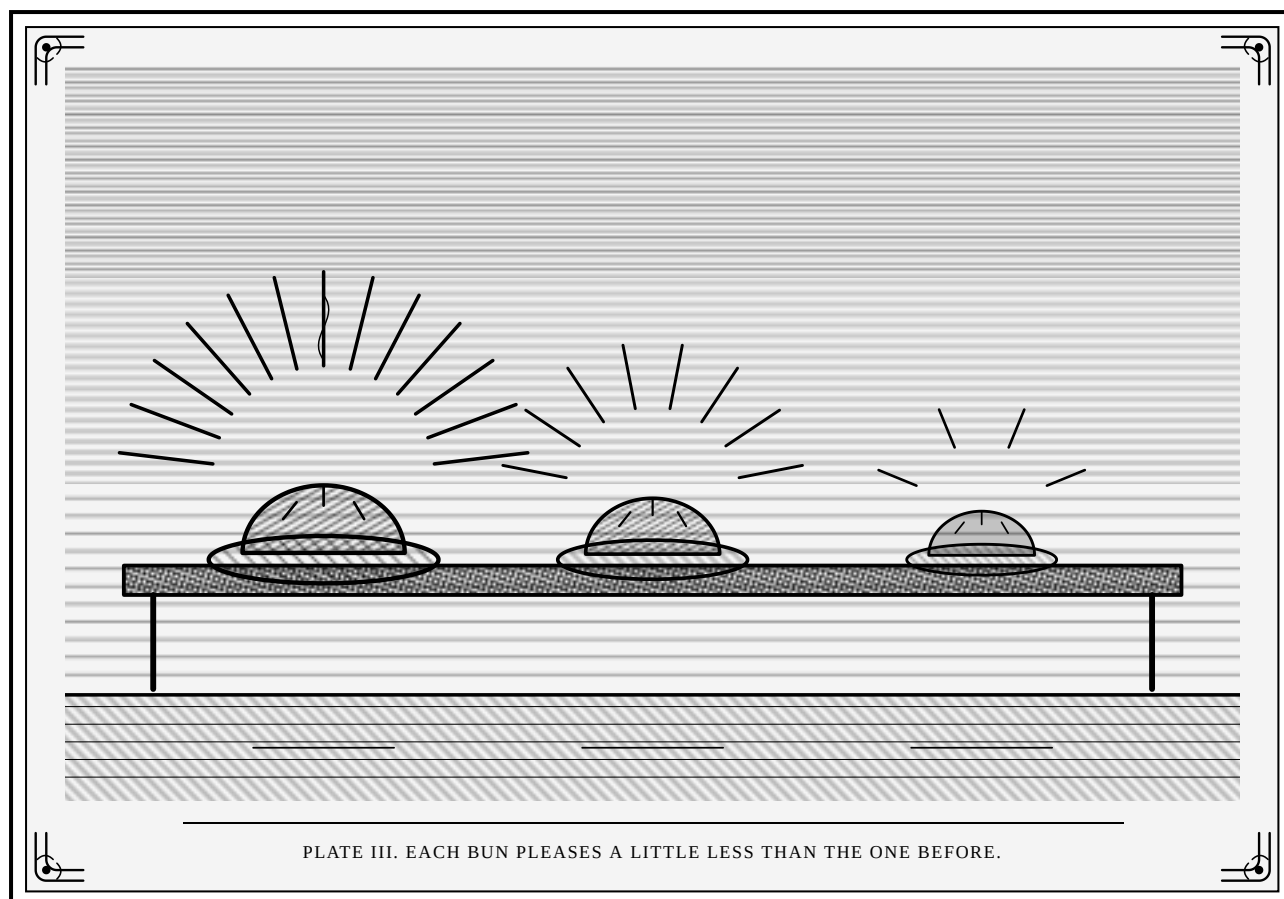
REVIEW QUESTIONS

1. Explain the difference between microeconomics and macroeconomics, and say how each depends on the other. 2. State what microeconomics covers and why it matters to a consumer, a firm and a government. 3. Write a short note on *ceteris paribus*. 4. Describe five features of an economic law, and say how it differs from a law of nature.

Two practice items ask for sorting and for a short note.

PRACTICE: SORTING STATEMENTS

Item 1: positive or normative. Sort these ten statements. (a) The price of lentils was higher in March than in February. (b) The government should lower the price of lentils. (c) A rise in bus fares reduces the number of passengers. (d) Everyone ought to be able to afford a doctor. (e) Garment workers earn more than they did ten years ago. (f) Rich families should pay a higher rate of tax. (g) Most students prefer a hostel near the campus. (h) Cheap rice is better than cheap mobile phones. (i) If the price of tea rises, buyers will take less. (j) Competition between shops is good for the town. *Item 2: micro or macro.* Sort these six. (a) A tailor raises the price of a shirt. (b) Total jobs in the country fell this year. (c) The price of fish in one bazaar fell. (d) Prices in general rose by 6%. (e) A family chooses a school for its child. (f) The total output of the country grew. *Item 3: a short note.* Write four lines on *ceteris paribus*, with one example of your own.



CHAPTER III

Utility and Consumption

Why does the third bun please less than the first?



This chapter explains why a buyer values each extra unit of a good a little less than the one before. It shows how this simple fact can be written as a table and a law, where the law seems to fail, and how a buyer with a fixed budget decides what to buy.

A hungry customer finds each bun less welcome than the last.

The puzzle is how anyone decides how many to buy.

The chapter builds on choice, opportunity cost and ceteris paribus.

GO SLOWLY HERE.

A hungry student's samosas show satisfaction falling with each one.

Utility. In daily life, the usefulness of a tool. In economics, the satisfaction a person gets from a good.

Utility is the satisfaction a person gets from a good.

Total utility. All the satisfaction from the units consumed.

Marginal utility. The extra satisfaction from one more unit.

Total utility is all the satisfaction; marginal utility is the extra from one more unit.

Margin. The edge: the next or last unit.

Util. A made-up unit of satisfaction.

Margin means the edge; a util is a made-up unit of satisfaction.

The customer in the case file gives the schedule used through this book.

We find total utility by adding the marginal utilities one by one.

A customer walks into a bakery, hungry, and eats a bun. It is delicious. A second bun is still good. By the third, the pleasure is weaker. By the fifth, the customer leaves half of it on the plate. Yet each bun cost the same Tk 10. The buns did not change. Something in the customer did.

This raises a puzzle. If each extra bun is worth less to the customer, how does the customer decide when to stop? And if the customer has money for other things too, such as a slice of cake, how is the money shared between the two? This chapter answers both questions, step by step.

You will need three earlier ideas: scarcity and choice (Chapter I), opportunity cost (Chapter I), and ceteris paribus (Chapter II). All three return here.

THE IDEA OF UTILITY

Begin with a hungry student at a tea stall. The first samosa is a great relief. The second is still welcome. The third is pleasant, but the hunger is gone. A fourth would be too much. Each samosa was the same size and the same price. The pleasure from each one fell as the number eaten rose. Notice that the student never stopped to count this pleasure. It is felt, not calculated.

Economists need a word for this pleasure. The word is **utility**. Utility means the satisfaction a person gets from using a good. A samosa gives utility to a hungry person. A bus ride gives utility to a person late for work. A good gives utility if someone wants it, whether or not it is good for that person.

Two kinds of utility must be kept apart. **Total utility** means all the satisfaction from all the units consumed. **Marginal utility** means the extra satisfaction from one more unit. If three samosas give a total of 21 units of pleasure and four samosas give 22, then the marginal utility of the fourth samosa is $22 - 21 = 1$.

The word **margin** means the edge, or the next step. Marginal means at the edge: the one more unit that is added, or the last unit taken. To put numbers on satisfaction we use a made-up unit, the **util**. A util is a made-up unit of satisfaction. It has no existence outside the book, as a metre has outside a ruler. It lets us write a feeling as a number.

Case file, illustrative. A customer at the bakery eats buns. We count one util as worth Tk 1. The marginal utility of buns 1 to 5 is 20, 16, 12, 8 and 4 utils. The price of a bun at opening is Tk 10. The table gives these figures, and adds total utility.

UTILITY FROM BUNS (CUSTOMER IN THE CASE FILE)

BUNS EATEN	MARGINAL UTILITY (UTILS)	TOTAL UTILITY (UTILS)
1	20	20
2	16	36
3	12	48
4	8	56
5	4	60

Given. Marginal utility of buns 1 to 5: 20, 16, 12, 8, 4 utils. *Find.* The total utility after each bun. *Formula.* Total utility of n units = total utility of $(n - 1)$ units + marginal utility of the n th unit. In symbols,

$TU(n) = TU(n - 1) + MU(n)$, where TU is total utility and MU is marginal utility.

Steps. After 1 bun: 20. After 2 buns: $20 + 16 = 36$. After 3: $36 + 12 = 48$. After 4: $48 + 8 = 56$. After 5: $56 + 4 = 60$. *Answer.* The totals are 20, 36, 48, 56 and 60 utils. *Check.* Going backwards, $60 - 56 = 4$, which is the marginal utility of the fifth bun. Notice that the total keeps rising, though more slowly, while the marginal utility falls.

Common slips. Students often confuse the two columns. Total utility is the running sum. Marginal utility is the step added at each line. A fall in marginal utility does not mean that total utility falls.

Before moving on. Utility is satisfaction. Total utility is all of it. Marginal utility is the extra from one more unit. *You should now be able to say:* total utility is the whole, marginal utility is the last step added.

Pause and check. A buyer's total utility from 4 mangoes is 30 utils and from 5 mangoes is 33 utils. What is the marginal utility of the fifth mango?

The steps show total utility rising while marginal utility falls.

A common slip is to confuse total utility with marginal utility.

Recap: total is all the satisfaction, and marginal is the extra.

A short question tests marginal and total utility.

CAN SATISFACTION BE MEASURED?

Satisfaction is a feeling inside a person. No instrument can read it. How, then, can we write it as numbers, as the table did? Economists have two answers, and the book will use both.

The first answer is **cardinal** measurement. Cardinal means measured in numbers, such as 20 utils and 16 utils. It tells us how much more satisfaction one thing gives than another. The second answer is **ordinal** measurement. Ordinal means ranked as first, second and third. It says only that a bun is preferred to a biscuit, without saying by how much.

Most of this chapter uses the cardinal method, with a money measure. We take one util as worth Tk 1. A buyer who will pay at most Tk 20 for the first bun is saying that it gives 20 utils. Money is useful because it is a number that we can see. The method assumes that a taka means the same to the buyer at each step. This is reasonable when a purchase is small compared with the buyer's whole income.

The cardinal method has limits. Satisfaction differs from person to person, so one person's utils cannot be added to another's. And people can rank things more easily than they can score them. For these reasons Chapter VI introduces the ordinal method, which needs only ranking. The two methods lead to the same main results.

Satisfaction cannot be seen, so we must decide how to count it.

Cardinal. Measured in numbers.

Ordinal. Ranked as first, second, third.

Cardinal measures satisfaction in numbers; ordinal only ranks it.

The money measure is used first, because money is a number we can observe.

The method has limits, which is why ranking returns in Chapter VI.

THE LAW OF DIMINISHING MARGINAL UTILITY

Return to the customer. The fifth bun gave 4 utils, which is worth Tk 4. The price of the bun is Tk 10. A customer who weighs this will not buy the fifth bun. The first bun gave 20 utils, worth Tk 20, and was a bargain at Tk 10. What happened in between? The answer is one of the oldest laws of economics.

The **law of diminishing marginal utility** states that, as a person consumes more units of a good one after another, the marginal utility of each extra unit becomes smaller, other things unchanged. Put in plain words, each extra unit adds less satisfaction than the one before. The bun schedule shows the law: 20, 16, 12, 8, 4.

The customer's fifth bun is worth only 4 utils against a price of Tk 10.

Law of diminishing marginal utility. Each extra unit adds less satisfaction than the one before.

The law states that each extra unit adds less satisfaction.

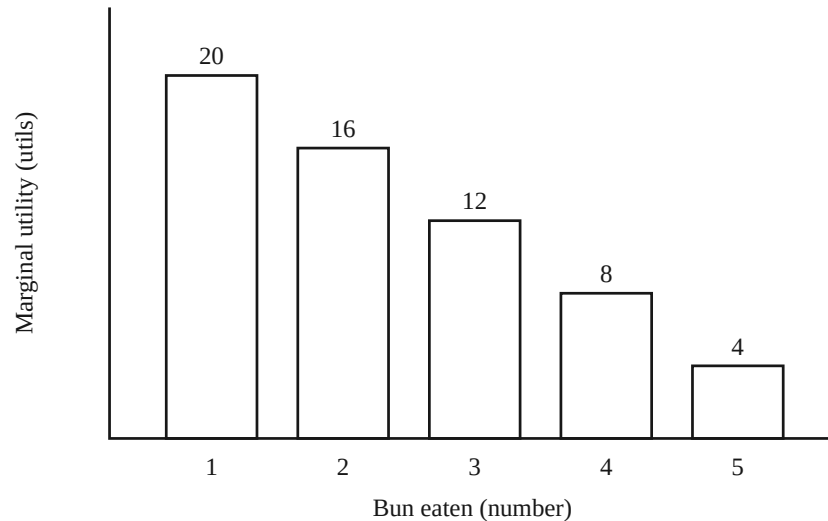


FIG. 1. Marginal utility of each bun for the customer in the case file

The diagram shows the law as a staircase that goes down.

A second example shows how to find marginal utility from totals.

The steps show marginal utility reaching zero and then turning negative.

Total utility is greatest where marginal utility is zero.

The law has three reasons behind it.

The law holds only under stated conditions.

Recap: marginal utility falls as more units are consumed, under stated conditions.

Read Figure 1 as follows. The bottom line shows the number of the bun. The side line shows the marginal utility in utils. Each bar is one bun, and its height is the satisfaction that bun adds. The bars become shorter from left to right. This staircase going down is the law in a picture.

Example. Given. A thirsty cyclist drinks glasses of sugarcane juice. The total utility after 1 to 7 glasses is 14, 26, 35, 41, 44, 44 and 41 utils. *Find.* The marginal utility of each glass. *Formula.* $MU(n) = TU(n) - TU(n - 1)$, with $TU(0) = 0$.

Steps. Glass 1: $14 - 0 = 14$. Glass 2: $26 - 14 = 12$. Glass 3: $35 - 26 = 9$. Glass 4: $41 - 35 = 6$. Glass 5: $44 - 41 = 3$. Glass 6: $44 - 44 = 0$. Glass 7: $41 - 44 = -3$. *Answer.* The marginal utilities are 14, 12, 9, 6, 3, 0 and -3. *Check.* Adding them again, $14 + 12 + 9 + 6 + 3 + 0 - 3 = 41$, the last total.

The example shows two more points. When marginal utility is zero, as at the sixth glass, total utility stops rising and has reached its highest value. Beyond that point marginal utility is negative, and the extra glass brings discomfort. A buyer who thinks carefully never goes past the point of zero marginal utility, and usually stops sooner, because goods have a price.

Why does marginal utility fall? There are three reasons. First, each want has a limit. A person who is hungry has a limited need for food, and each unit meets part of that need. Second, the person uses the first unit for the most important purpose. The second unit goes to a less important use. Water, for example, serves first to drink, then to wash, and only later to water plants. Third, units of the same good are partial substitutes for one another, so the more one has, the less any single unit matters.

Like every economic law, this one comes with conditions. The units must be of the same size and quality. They must be consumed one after another within a short time, not spread over years. The buyer's tastes must not change. The quantities of other goods the buyer has must not change. And the money measure assumes that a taka means the same at each step. When these conditions hold, the law holds. When they are broken, it seems to fail, as the next section shows.

Before moving on. The law of diminishing marginal utility says that each extra unit adds less satisfaction than the one before. It rests on limited wants and on using the best units first. It holds when units, tastes and other goods stay the

same. *You should now be able to say:* each extra unit adds less than the one before.

Pause and check. A buyer's marginal utility from tea is 10, 8, 6, 4 and 2 utils for cups 1 to 5. What is the total utility from 3 cups?

A short question tests the shape of the law.

LIMITS OF THE LAW

Does the law hold for everything? Several cases look like exceptions. In each, a closer look shows that one of the conditions of the law has been broken. This is the pattern for every exception in economics: ask which condition has changed.

Some cases seem to break the law, though each has an explanation.

The first case is the hobby or the collection. A stamp collector may feel more pleasure with each new stamp, because the collection is nearer to completion. The explanation is that the collector is not buying separate units of one good. The unit of the good is the whole collection, and the last stamp completes it. For the collector, the whole set is one unit.

A collector's pleasure may rise with each new item.

The second case is money. A person who loves money seems always to want more. But money is the means of buying many goods, and the marginal utility of each extra taka falls more slowly than for a single good, since there are always more uses. For a person with a very large amount, each extra taka does add little.

The wish for money may seem to have no limit.

The third case is the very small unit. A thirsty person given a single drop of water does not feel more satisfaction from a second drop. The units are too small to matter. The explanation is that the useful unit is a glass, not a drop, and the law applies to the useful unit.

Very small units give too little to feel.

The fourth case is the habit. A person who drinks a harmful drink may want more with each glass, because the habit has changed the person's tastes. The law assumes that tastes stay the same. Here they have changed, so the condition is broken.

A habit can change tastes while the units are consumed.

Before moving on. The apparent exceptions are collections, money, very small units and habits. Each breaks one condition: the unit, the measure, the size of the unit, or unchanged tastes. The law remains true under its conditions.

Recap: each exception breaks one condition of the law.

CONSUMPTION: HOW A BUYER SPENDS

Think of a customer at a tea stall with Tk 8. A cup of tea costs Tk 2. A biscuit costs Tk 1. How should the customer share the Tk 8 between them? Eight biscuits would be too many. Four cups of tea would also be too many. The best choice lies somewhere between, and we now find a rule to locate it.

A customer with Tk 8 must split it between tea and biscuits.

Consumption means using a good to meet a want. A **consumer** is a person who uses a good. Eating a samosa, riding a bus and attending a lesson are all consumption. A consumer has wants, a limited budget, and goods with prices. The consumer's task is to turn the budget into the greatest satisfaction.

Consumption. Using a good to meet a want.

Consumer. A person who uses a good.

Consumption is using a good to meet a want; a consumer does it.

Comparing units can mislead, because goods have different prices. A cup of tea may give more utility than a biscuit, but it also costs more. The fair comparison is the marginal utility per taka spent: the marginal utility of the unit divided by its price. The buyer should spend the next taka where it brings the most satisfaction.

A buyer should compare the satisfaction from each taka, not from each unit.

Equilibrium. A state of balance with no reason to change.

Equi-marginal rule. Spend so that marginal utility per taka is equal across goods.

The buyer is in balance when each good gives equal satisfaction per taka.

A worked example finds the best bundle for the tea-stall customer.

The steps list the utility per taka and buy in order, best first.

The answer is 3 cups and 2 biscuits, and the check compares other bundles.

The rule has limits: a fixed budget, given prices and independent goods.

Recap: a buyer is in equilibrium when each good gives equal utility per taka.

Suppose the buyer spends the next taka on the good with the highest marginal utility per taka. As more of that good is bought, its marginal utility falls. At some point the other good becomes the better use. The buyer is then in **equilibrium**. Equilibrium means a state of balance with no reason to change. The rule that gives this balance is the **equi-marginal rule**: spend so that marginal utility per taka is equal across goods, until the budget is used.

Example. Given. Budget Tk 8. Tea costs Tk 2 a cup, and its marginal utility for cups 1 to 4 is 12, 10, 8 and 6. A biscuit costs Tk 1, and its marginal utility for biscuits 1 to 4 is 5, 4, 3 and 2. *Find.* The bundle that gives the greatest total utility. *Formula.* Marginal utility per taka = marginal utility ÷ price.

Steps. Tea per taka: $12 \div 2 = 6$, $10 \div 2 = 5$, $8 \div 2 = 4$, $6 \div 2 = 3$. Biscuit per taka: 5, 4, 3, 2. Buy in order of the highest figure, as long as the budget lasts: tea 1 (6) costs Tk 2, total Tk 2. Tea 2 (5) costs Tk 2, total Tk 4. Biscuit 1 (5) costs Tk 1, total Tk 5. Tea 3 (4) costs Tk 2, total Tk 7. Biscuit 2 (4) costs Tk 1, total Tk 8. The budget is spent.

Answer. The customer buys 3 cups of tea and 2 biscuits. The marginal utility per taka is 4 for the last cup and 4 for the last biscuit, which is equal. Total utility is $12 + 10 + 8 + 5 + 4 = 39$ utils. *Check.* Try 2 cups and 4 biscuits, which also cost Tk 8: $22 + 14 = 36$ utils. Try 4 cups and no biscuits: 36 utils. Both are lower than 39.

The rule has limits. The budget must be fixed and fully spent. Prices must be given, since one buyer cannot change them. The marginal utility of one good must not depend on how much of another is bought. And it must be possible to buy goods in small steps. Chapter IV builds the demand curve from this idea.

Before moving on. A consumer spends a budget on several goods. The equi-marginal rule says to equalise marginal utility per taka across goods. At that point the consumer is in equilibrium and has no reason to change.



A WORKED CASE: THE CUSTOMER'S BUNDLE

The case file adds a second good, cake, and a budget.

The question is which bundle of buns and cake gives most satisfaction.

We first compute marginal utility per taka for each unit.

Case file, illustrative. Situation. The customer of the case file has a budget of Tk 70 at the bakery. Buns cost Tk 10 each, with marginal utility 20, 16, 12, 8 and 4 utils. A slice of cake costs Tk 20, with marginal utility 40, 24, 16, 8 and 4 utils. One util is worth Tk 1.

Case file, illustrative. Question. How many buns and how many slices of cake should the customer buy with Tk 70?

Case file, illustrative. Reasoning. The table gives marginal utility per taka for each unit: the marginal utility divided by the price.

MARGINAL UTILITY PER TAKA

UNIT	BUN (TK 10)	CAKE SLICE (TK 20)
1	2.0	2.0
2	1.6	1.2
3	1.2	0.8
4	0.8	0.4
5	0.4	0.2

Case file, illustrative. Buy in order of the highest figure. Cake 1 (2.0) costs Tk 20, total Tk 20. Bun 1 (2.0) costs Tk 10, total Tk 30. Bun 2 (1.6) costs Tk 10, total Tk 40. Bun 3 (1.2) costs Tk 10, total Tk 50. Cake 2 (1.2) costs Tk 20, total Tk 70. The budget is spent. The customer buys 3 buns and 2 slices of cake, and the last bun and the last slice each give 1.2 utils per taka.

Case file, illustrative. Conclusion. Total utility is $20 + 16 + 12$ for the buns, which is 48, plus $40 + 24$ for the cake, which is 64. The total is 112 utils. Check. All three bundles below cost Tk 70. The chosen bundle gives the most. At the price of Tk 10 the customer buys 3 buns, a figure used again in Chapter VI.

Buying in order of utility per taka spends exactly Tk 70.

The conclusion and check compare three bundles that all cost Tk 70.

THREE BUNDLES COMPARED (BUDGET TK 70)

BUNS	CAKE SLICES	COST (TK)	TOTAL UTILITY (UTILS)
5	1	70	$60 + 40 = 100$
3	2	70	$48 + 64 = 112$
1	3	70	$20 + 80 = 100$



SUMMARY

Utility is the satisfaction from a good. Total utility is all of it, and marginal utility is the extra from one more unit. Marginal utility falls as more units are consumed, under stated conditions, and this is the law of diminishing marginal utility. Apparent exceptions break one of the conditions. A consumer with a budget is in equilibrium when marginal utility per taka is equal across goods.

Satisfaction falls with each extra unit, and buyers spend by equalising utility per taka.

THINK IT THROUGH

A family has bought five kilograms of rice this week, and the shop offers a sixth at the same price. The family declines, though rice is a necessity. Use marginal utility to explain. Then say what would have to change for the family to buy the sixth kilogram.

A family declines a sixth kilogram of rice at the same price.

WHERE PEOPLE GO WRONG

Many students think that when marginal utility falls, total utility falls too. It does not. Total utility rises as long as marginal utility is positive. It falls only when marginal utility becomes negative.

Total utility keeps rising while marginal utility falls.

Utility is the satisfaction a person feels. A good that harms a person can still give utility, since the person wants it. The word does not judge whether a want is wise.

Utility does not mean usefulness or goodness.

Many students apply the law to the whole of a person's wealth, or over a long time. It is about successive units of one good, bought one after another, with tastes and other things unchanged.

The law is about one good, one buyer, a short time and unchanged tastes.

IN HISTORY

In 1776 Adam Smith pointed to a puzzle: water is very useful but costs little, while diamonds are of little use but cost much. For nearly a century the puzzle stood. In the early 1870s, William Stanley Jevons in England and Carl Menger in Austria, and soon after Léon Walras in Switzerland, each showed that the

The puzzle of water and diamonds was solved by marginal thinking.

value of a good depends on its marginal utility, not on its total usefulness. Water is plentiful, so the marginal utility of one more glass is small. Diamonds are scarce, so the marginal utility of one more diamond is large. Hermann Gossen, a German, had stated the law earlier, in 1854, but his book was little noticed.

SOURCES

The law has a short and well-documented history.

Hermann Gossen, *Entwicklung der Gesetze des menschlichen Verkehrs* (1854), stated the law first. William Stanley Jevons, *The Theory of Political Economy* (1871), and Carl Menger, *Grundsätze der Volkswirtschaftslehre* (1871), made it the base of the subject. Léon Walras, *Éléments d'économie politique pure* (1874), set it in a wider system. Alfred Marshall, *Principles of Economics* (1890), gave it the form used in this book.

REVIEW QUESTIONS AND PRACTICE

REVIEW QUESTIONS

Four review questions test the main ideas of the chapter.

1. Define utility, total utility and marginal utility, and give an example of each.
2. Explain whether satisfaction can be measured, and name the two methods.
3. State the law of diminishing marginal utility, show it with a table and diagram, and give its reasons and conditions.
4. Name four cases that seem to break the law and explain each.

PRACTICE: FROM A TABLE TO A BUNDLE

Four practice items move from a clean table to a choice between two goods.

Item 1: a clean table. The marginal utility of dosas for a customer is 18, 14, 10, 6 and 2 utils. Find the total utility after each dosa. *Item 2: a messier table.* The total utility from cups of tea in a morning is 12, 22, 30, 36, 40, 42 and 42 utils for cups 1 to 7. Find the marginal utility of each cup. At which cup does marginal utility reach zero? If one util is worth Tk 1 and a cup costs Tk 6, how many cups does the customer buy? *Item 3: the shape.* Draw the marginal utility of Item 2 as bars and explain why the bars become shorter. *Item 4: two goods.* A buyer has Tk 100. Good A costs Tk 10, with marginal utility 30, 24, 18, 12 and 6. Good B costs Tk 20, with marginal utility 70, 46, 24, 16 and 8. Find the bundle that gives the greatest total utility, and its total utility.

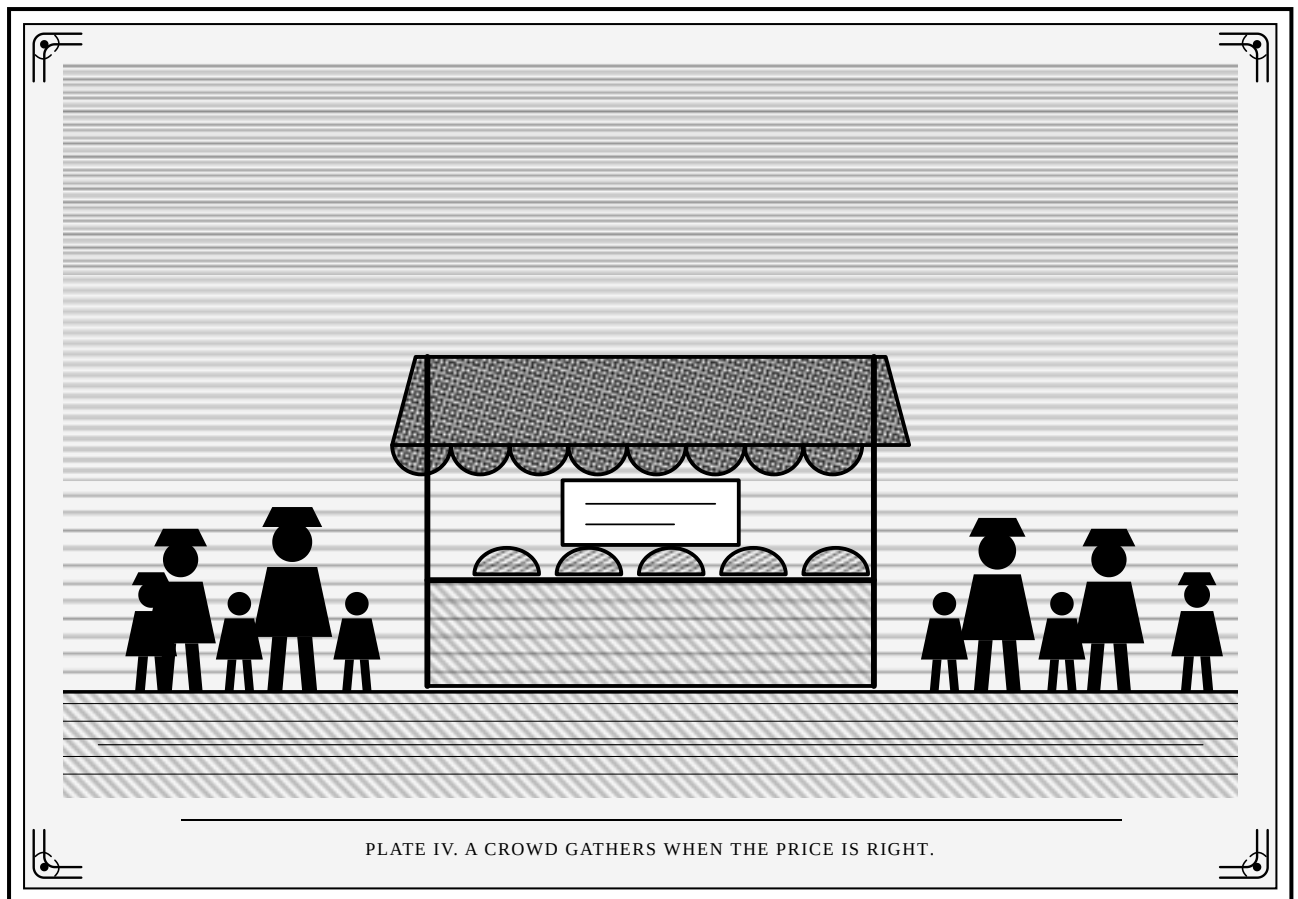


PLATE IV. A CROWD GATHERS WHEN THE PRICE IS RIGHT.

CHAPTER IV

Demand

What makes a crowd buy more?



This chapter shows how the price of a good and the quantity bought move together, and why. It then separates two kinds of change: a move along the demand curve and a shift of the whole curve. It ends with the rare cases where the usual pattern fails.

A bakery lowers its bun price, and a festival raises sales.

The puzzle is to tell a price effect from other causes.

The chapter builds on marginal utility and on ceteris paribus.

GO SLOWLY HERE.

A wish without money does not make demand.

Demand. In daily life, a strong request or order. In economics, the quantity buyers are able and willing to buy at each price.

Demand is the quantity buyers are able and willing to buy at each price.

Quantity demanded. The amount bought at one price.

Quantity demanded is the amount bought at one price.

Demand schedule. A table of prices and quantities.

Demand curve. The line drawn from a demand schedule.

A demand schedule is a table; a demand curve is its line.

The case file gives the neighbourhood bun market.

A bakery lowers the price of a bun from Tk 12 to Tk 10, and sales rise. Some weeks later a festival arrives, and sales rise again, though the price is back at Tk 12. Both events brought more buyers. Yet the two are not the same kind of change, and mixing them up leads to wrong conclusions.

How can we tell a rise in sales caused by a lower price from a rise caused by something else? The answer needs a clear word for what buyers plan to buy, a picture of how that plan depends on price, and a list of the other things that can change it. This chapter provides all three.

You will need marginal utility (Chapter III) and ceteris paribus (Chapter II). Both are used on almost every page.

WHAT DEMAND MEANS

Begin with a student who would love to own a motorcycle. The wish is strong. But the student has no money, and so no shop will ever record a sale. Now picture a family who can afford a motorcycle but has no wish for one. There is no sale here either. A sale needs three things together: the wish, the ability to pay, and the willingness to pay.

Economists give the name **demand** to this combination. Demand means the quantity of a good that buyers are able and willing to buy at each price, in a given time. Notice the words at each price. Demand is not one number. It is a whole list of prices, each with a quantity, and it needs a time such as one day.

Keep demand apart from **quantity demanded**. Quantity demanded means the amount bought at one particular price. If the neighbourhood buys 240 buns a day at Tk 10, the quantity demanded at Tk 10 is 240. Demand is the whole list. The quantity demanded is one line of it.

A **demand schedule** is a table of prices and quantities demanded. A **demand curve** is the line that draws the schedule on a graph. The schedule and the curve say the same thing in two forms.

Case file, illustrative. All buyers of buns in the neighbourhood together demand the quantities in the table, per day. The rule behind the table is $Q_d = 540 - 30P$, where Q_d is the quantity demanded and P is the price in taka.

NEIGHBOURHOOD DEMAND FOR BUNS, PER DAY

PRICE (TK)	QUANTITY DEMANDED
8	300
10	240
12	180
14	120
16	60
18	0

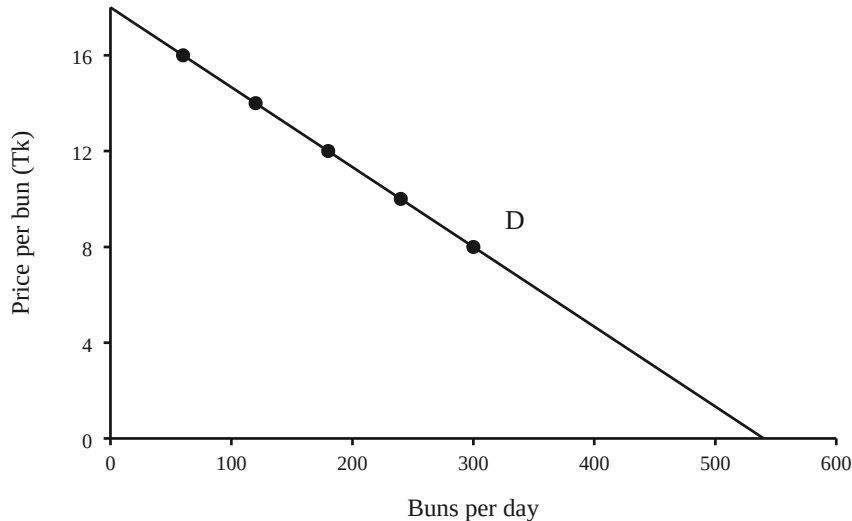


FIG. 2. The demand curve for buns in the neighbourhood, per day

Read Figure 2 as follows. The side line shows the price of a bun in taka. The bottom line shows the number of buns demanded per day. Each dot is one row of the table. The dot at price 10 and quantity 240 means that at Tk 10 buyers will take 240 buns. The straight line through the dots is the demand curve. Starting from any price, move across to the curve and then down to read the quantity.

Individual demand means the demand of one buyer. **Market demand** means the sum of all buyers' demand. To find market demand, add the quantities of all buyers at each price. Never add prices. Prices are the same for all buyers, and quantities are what differ.

Example. Given. At Tk 4 a cup, buyer A takes 5 cups and buyer B takes 4. At Tk 6, A takes 3 and B takes 2. At Tk 8, A takes 1 and B takes none. *Find.* The market demand at each price. *Formula.* Market quantity at a price = quantity of A + quantity of B at that price.

Steps. At Tk 4: $5 + 4 = 9$. At Tk 6: $3 + 2 = 5$. At Tk 8: $1 + 0 = 1$. *Answer.* The market demand is 9, 5 and 1 cups at Tk 4, 6 and 8. *Check.* The quantities fall as the price rises for each buyer, and the totals fall too, which is what we expect.

Before moving on. Demand needs wish, ability and willingness, and it is a schedule of prices and quantities. Quantity demanded is the amount at one price. Market demand adds the quantities of all buyers. *You should now be able to say:* demand is the whole schedule, and quantity demanded is one point on it.

Pause and check. The price of tea falls from Tk 6 to Tk 4, and buyers take 9 cups instead of 5. Has the demand for tea changed, or the quantity demanded?

To read the graph, find a price on the side line and go across.

Individual demand. The demand of one buyer.

Market demand. The sum of all buyers' demand.

Market demand is found by adding the demand of each buyer.

A small example adds two buyers' demand for tea.

The steps add at each price and the check compares totals.

Recap: demand is the whole schedule; quantity demanded is one point.

A short question tests the difference between demand and quantity demanded.

THE LAW OF DEMAND

Look again at the table. As the price falls from Tk 18 to Tk 8, the quantity rises from 0 to 300. The two figures move in opposite directions. This is true for most goods and most buyers, and it is the first law of the market.

The table shows that a lower price brings a larger quantity.

Law of demand. A higher price brings a smaller quantity demanded, other things equal.

The law of demand says price and quantity demanded move in opposite directions.

A worked example fills in a schedule from the rule.

Each Tk 1 rise in price cuts quantity demanded by 30 buns.

There are three reasons why the curve slopes downward.

Income effect. The change in buying power after a price change.

Substitution effect. Switching to a cheaper good.

The income effect and the substitution effect add two further reasons.

The law holds only under stated conditions.

Recap: price and quantity move in opposite directions for three reasons.

A price change moves us along the curve; other changes move the curve itself.

Change in quantity demanded. A move along the demand curve, caused by a change in the price.

Change in demand. A shift of the whole demand curve, caused by something other than price.

Only a change in price causes a move along the curve.

Determinants of demand. The things other than price that decide demand.

The determinants of demand are the things other than price that decide demand.

The **law of demand** states that, other things equal, a higher price of a good brings a smaller quantity demanded, and a lower price brings a larger quantity demanded. The demand curve therefore slopes downward from left to right. Other things equal is the *ceteris paribus* of Chapter II: income, tastes and the prices of other goods stay unchanged.

Example. Given. The market rule is $Q_d = 540 - 30P$. *Find.* The quantity demanded at Tk 9, Tk 11 and Tk 15. *Formula.* Quantity demanded = $540 - 30 \times \text{price}$. *Steps.* At Tk 9: $540 - 30 \times 9 = 540 - 270 = 270$. At Tk 11: $540 - 330 = 210$. At Tk 15: $540 - 450 = 90$. *Answer.* The quantities are 270, 210 and 90 buns. *Check.* Each lies between the neighbouring rows of the table: 270 between 300 and 240, 210 between 240 and 180, 90 between 120 and 60.

The rule also tells us a simple fact: each rise of Tk 1 in the price lowers the quantity demanded by 30 buns. Check this against the table. From Tk 10 to Tk 12 is a rise of Tk 2, and the quantity falls from 240 to 180, a fall of 60.

Why does the curve slope downward? There are three reasons. The first comes from Chapter III. Marginal utility falls as a buyer takes more. A buyer will pay for each extra bun only what it is worth, so a larger quantity is taken only at a lower price. The customer of Chapter III, with a price of Tk 10, took 3 buns, because the fourth was worth only Tk 8.

The second and third reasons are two effects of a price change. When a price falls, the buyer's money buys more than before, as if income had risen. This is the **income effect**: the change in buying power after a price change. Also, the good becomes cheaper than its rivals, so buyers switch to it. This is the **substitution effect**: switching to a cheaper good. Both effects work in the same direction for most goods.

The law holds under conditions. Income, tastes, expectations, the number of buyers and the prices of other goods must stay unchanged. If any of them changes, the whole curve may move, as the next section shows. Section 4.4 describes the rare cases in which the law seems to fail.

Before moving on. The law of demand says that a higher price brings a smaller quantity demanded, other things equal. The reasons are falling marginal utility, the income effect and the substitution effect. *You should now be able to say:* a lower price brings a larger quantity, because each extra unit is worth less.

WHAT MOVES THE CURVE

Return to the opening puzzle. A lower bun price moves buyers along the same curve, to a larger quantity. A festival moves the whole curve, because it changes something other than the price. We need two precise terms to keep these apart.

A **change in quantity demanded** means a move along the demand curve, caused by a change in the price of the good itself. A **change in demand** means a shift of the whole demand curve, caused by something other than the price of the good. The two names look alike, and many students mix them. Remember: price moves us along, anything else shifts the curve.

The **determinants of demand** are the things other than price that decide demand. There are five main ones. The first is income: buyers with more income can buy more. The second is tastes: a fashion for a good raises demand. The third is the number of buyers: a growing town buys more. The fourth is expectations: buyers who expect a higher price next month buy

more now. The fifth is the price of related goods, which has its own paragraph below.

Income acts in two ways. For a **normal good**, demand rises when income rises. Most goods are normal: for example, fish and clothes. For an **inferior good**, demand falls when income rises, because buyers switch to something better. Coarse rice or the cheapest cloth may be inferior for some households. Whether a good is inferior depends on the buyer, not on the quality of the good.

The price of related goods works in two ways. **Substitute goods** are goods used in place of each other, such as tea and coffee. If the price of coffee rises, buyers switch to tea, and the demand for tea rises. **Complementary goods** are goods used together, such as tea and sugar. If the price of sugar rises, buyers take less tea, and the demand for tea falls.

Here is an application. Many vehicles in Bangladeshi cities can run on petrol or on compressed natural gas, called CNG. Petrol and CNG are substitutes for such vehicles. If the price of petrol rises, some owners switch to CNG, and the demand for CNG rises. Petrol and the vehicles that use it are complements. If petrol becomes dear, fewer people buy vehicles that run only on petrol. The price of one good thus moves the demand for another. Chapter V gives this a number, the cross elasticity.

Case file, illustrative. Suppose a festival raises the neighbourhood demand for buns by 60 at every price. The rule changes from $Q_d = 540 - 30P$ to $Q_d = 600 - 30P$. The table compares the old and new demand. At Tk 10 the quantity demanded rises from 240 to 300, though the price has not changed. This is a change in demand. The whole curve moves to the right, as in Figure 3.

A FESTIVAL SHIFTS DEMAND (BUNS PER DAY)

PRICE (TK)	OLD QUANTITY	NEW QUANTITY
8	300	360
10	240	300
12	180	240

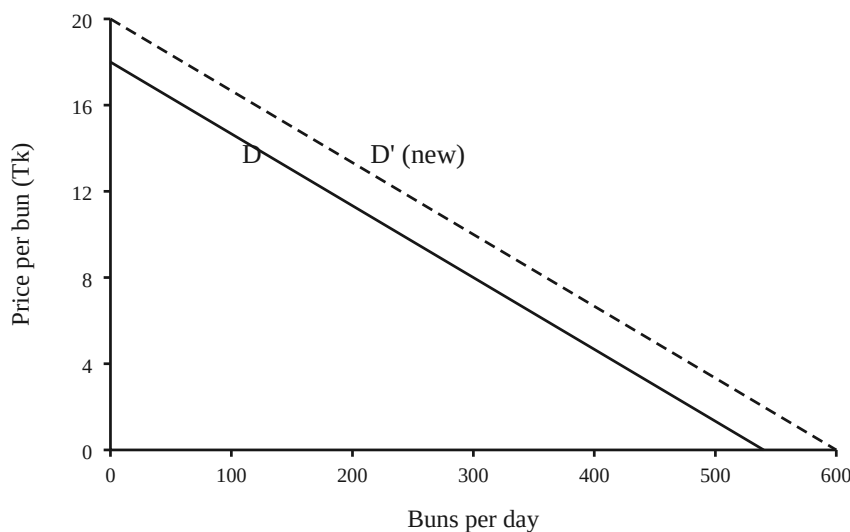


FIG. 3. A rise in demand shifts the curve to the right

Normal good. A good whose demand rises with income.

Inferior good. A good whose demand falls as income rises.

Income raises the demand for normal goods and lowers it for inferior goods.

Substitute goods. Goods used in place of each other.

Complementary goods. Goods used together.

Related goods affect demand: substitutes raise it, complements lower it.

A fuel example shows a cross effect between related goods.

The case file uses a festival that raises demand by 60 buns at every price.

Recap: price moves us along the curve; other determinants shift the curve.

In a few cases a lower price brings a smaller quantity.

Giffen good. A good whose demand falls when its price falls.

A Giffen good is an inferior good that takes a large part of a poor household's budget.

Snob effect. Demand falls as more people buy.

The snob effect makes a good valued for being rare.

Bandwagon effect. Demand rises as more people buy.

The bandwagon effect makes a good valued because others buy it.

These cases are rare and each breaks a condition of the law.

Recap: three exceptions, each with a reason.

The neighbourhood demand is the sum of three customer groups.

Before moving on. A change in price causes a change in quantity demanded, which is a move along the curve. A change in income, tastes, the number of buyers, expectations or related prices causes a change in demand, which is a shift of the curve. Substitutes and complements, normal goods and inferior goods behave differently.

EXCEPTIONS: GIFFEN, SNOB AND BANDWAGON

Does the law of demand ever fail? In a few cases it seems to. In each case, as the last chapter taught us to expect, one of the conditions of the law has been broken. Three cases are well known.

The first is the **Giffen good**, named after the Scottish economist Robert Giffen. A Giffen good is a good whose demand falls when its price falls. It is an inferior good that takes a very large part of a poor household's budget, for example a basic staple food. When its price falls, the household feels richer. Its income effect is strong, and it uses the saved money to buy better food and eats less of the staple. The income effect overcomes the substitution effect. The curve slopes upward over that range.

The second is the **snob effect**. The snob effect means that demand falls as more people buy. Some buyers want a good because few others have it, such as a rare watch or a limited edition. When the price falls and many more people buy, the good loses its appeal, and the demand of snobs falls.

The third is the **bandwagon effect**. The bandwagon effect means that demand rises as more people buy. A fashionable clothing style or a popular mobile phone is wanted because others have it. A fall in price brings more buyers, and that attracts still more. The effect can make the demand curve flatter, or even bend upward.

How likely are these cases? They are rare. Giffen behaviour needs very poor households and a staple with no good cheap rival. The snob and bandwagon effects need buyers whose tastes depend on what others do. In each case the condition of unchanged tastes, or of a small income effect, has been broken. For most goods and most buyers, the law of demand holds.

Before moving on. A Giffen good is a staple whose strong income effect makes demand fall as the price falls. The snob effect lowers demand as more people buy. The bandwagon effect raises demand as more people buy. All three are rare.



A WORKED CASE: THREE GROUPS OF BUYERS

Case file, illustrative. Situation. The neighbourhood bun market has three groups of buyers: students, office workers and families. Their daily demand at each price is in the table.

DEMAND BY GROUP, BUNS PER DAY

PRICE (TK)	STUDENTS	OFFICE WORKERS	FAMILIES	MARKET
8	100	80	120	300
10	80	60	100	240
12	60	40	80	180
14	40	20	60	120
16	20	0	40	60
18	0	0	0	0

Case file, illustrative. Question. Do the three groups add to the neighbourhood demand, $Q_d = 540 - 30P$, and what happens to each group if the price rises?

Case file, illustrative. Reasoning. Add the quantities at each price. At Tk 8: $100 + 80 + 120 = 300$. At Tk 10: $80 + 60 + 100 = 240$. At Tk 12: $60 + 40 + 80 = 180$. At Tk 14: $40 + 20 + 60 = 120$. At Tk 16: $20 + 0 + 40 = 60$. At Tk 18: 0.

Case file, illustrative. Conclusion. The three groups add exactly to the market demand. Each group's demand falls as the price rises, in line with the law of demand. Office workers drop out first, at Tk 16, and all groups drop out at Tk 18. Check. At Tk 10 the rule gives $540 - 30 \times 10 = 240$, the same as the sum.



SUMMARY

Demand is the quantity buyers are able and willing to buy at each price. The law of demand says that a higher price brings a smaller quantity demanded, because of falling marginal utility, the income effect and the substitution effect. A change in price moves us along the curve. A change in income, tastes, the number of buyers, expectations or related prices shifts the curve. Giffen, snob and bandwagon effects are rare exceptions.

THINK IT THROUGH

The price of tea falls. Say what happens to the quantity of tea demanded, and what happens to the demand for sugar. Say whether each change is a move along a curve or a shift of a curve, and why. Then say what would happen to the demand for coffee.

WHERE PEOPLE GO WRONG

Many students write that demand rises when the price falls. It is the quantity demanded that rises. Demand rises only when the whole curve shifts, and a price change cannot shift it.

Many students draw a new curve when the price changes. The price of the good moves us along the same curve. A new curve needs a change in income, tastes, numbers, expectations or the price of another good.

Many students use the Giffen good to dispute the law of demand. It is a rare case that needs special conditions. The law of demand remains true for almost all goods.

The question is whether the groups add to the market demand in the case file.

Reasoning: add the quantities of the groups at each price.

The conclusion and check test the sum against the rule.

Demand is a schedule; price moves along it and other things shift it.

A fall in the price of tea and the demand for sugar.

Demand is not the same as quantity demanded.

The price of the good itself never shifts its own demand curve.

The Giffen good is not common.

A study in China found Giffen behaviour among very poor households.

The ideas of this chapter come from three writers.

Four review questions test the main ideas of the chapter.

Four practice items and ten cases test the chapter.

Ten cases ask whether each event is a move along the curve or a shift.

IN HISTORY

For over a century, Giffen goods were more a puzzle than a fact. In 2008 the economists Robert Jensen and Nolan Miller reported an experiment in Hunan province in China. They gave very poor households a subsidy on rice, the staple of their diet, which lowered its price. Some households ate less rice, not more, and bought other food with the money saved. The study is cited as evidence that Giffen behaviour can occur when households are very poor and the staple takes a large part of the budget.

SOURCES

Alfred Marshall, *Principles of Economics* (1890), drew the demand curve and stated the law. Thorstein Veblen, *The Theory of the Leisure Class* (1899), described buying to show wealth. Harvey Leibenstein, in a paper of 1950, named the bandwagon and snob effects.

REVIEW QUESTIONS AND PRACTICE

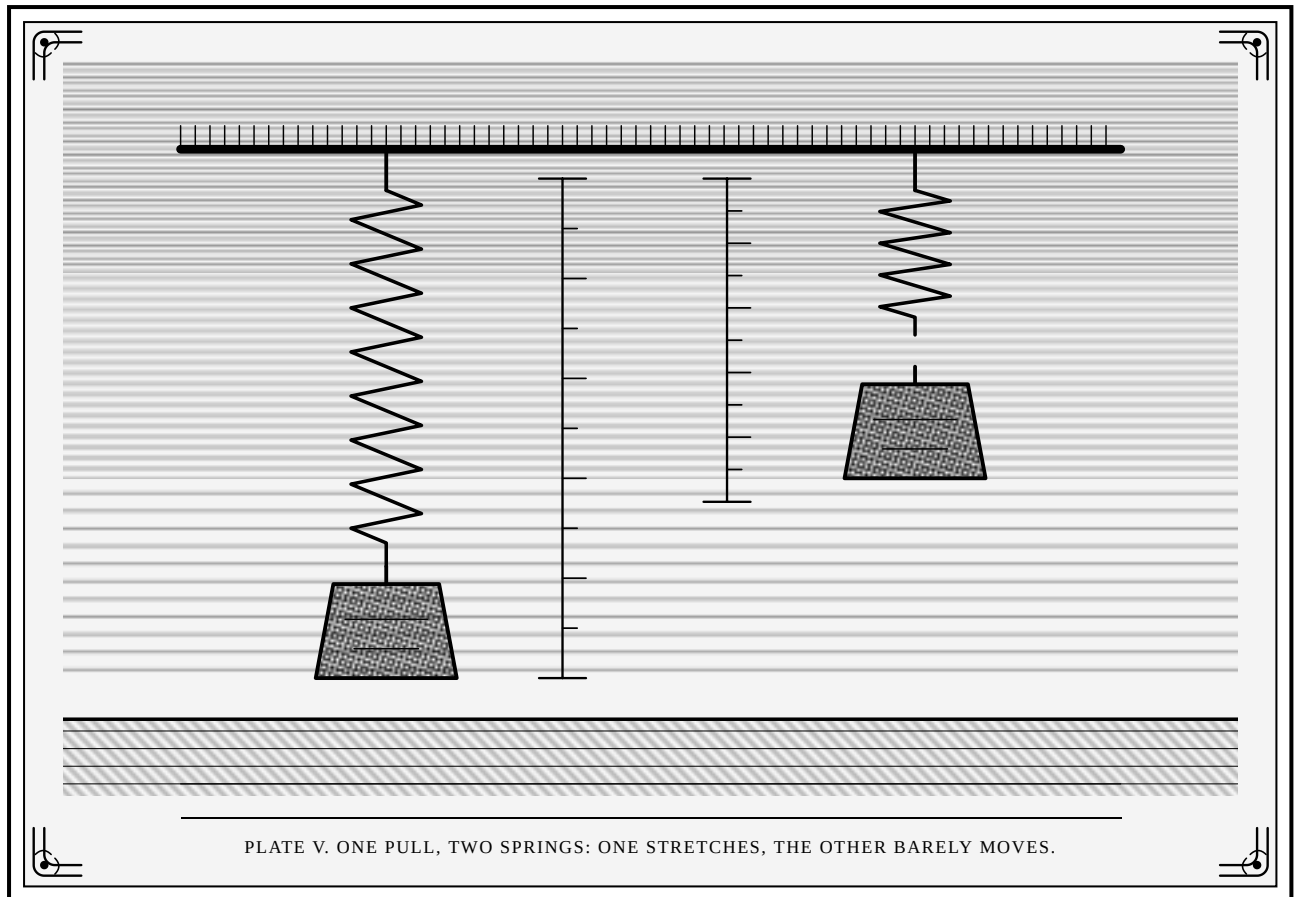
REVIEW QUESTIONS

1. Define demand, and explain the difference between demand and quantity demanded. 2. State the law of demand, draw a demand curve, and explain why it slopes downward. 3. List the factors that change demand, and explain the effect of substitutes and complements. 4. Describe Giffen, snob and bandwagon effects and say why each is an exception.

PRACTICE: READING, FILLING, ADDING AND SHIFTING

Item 1: read a point. The demand for rice in a market is 800, 700, 600, 500 and 400 kg at Tk 40, 45, 50, 55 and 60 a kg. Read the quantity at Tk 50 and the price at which 500 kg are demanded. *Item 2: fill from a rule.* The demand for fish is $Q_d = 900 - 10P$. Find the quantity at Tk 50, 60, 70 and 80, and the price at which demand falls to zero. *Item 3: add schedules.* Three groups demand 90, 60 and 30 units at Tk 20, 30 and 40 (group one), 70, 50 and 30 (group two), and 40, 30 and 20 (group three). Find the market demand. *Item 4: shift by 20%.* A rise in income raises the market demand of Item 3 by 20% at every price. Find the new schedule.

Shift or movement? For each case, say whether it is a change in quantity demanded or a change in demand. (a) The price of mangoes falls and more are bought. (b) Incomes rise and more mobile phones are bought at every price. (c) The price of tea rises and less tea is bought. (d) The price of petrol rises and the demand for CNG rises. (e) A cold wave raises the demand for blankets. (f) The price of sugar falls and more sugar is bought. (g) The price of tea falls and the demand for sugar rises. (h) The number of families in a town doubles and the demand for rice rises. (i) Buyers expect a higher price of rice next month and buy more now. (j) A tax raises the price of cigarettes and fewer are bought.



CHAPTER V

Elasticity of Demand

How hard do buyers push back when a price moves?



The law of demand says that a higher price brings smaller purchases. It does not say how much smaller. This chapter builds a number that measures the reaction, applies it to price, income and the prices of other goods, and shows why every seller, farmer and tax collector needs it.

Two tea stalls raise their prices by the same amount and meet very different results.

The law of demand gives only the direction of the change, not its size.

This chapter rests on Chapter IV and on percentages.

A household reacts to a price rise in salt very differently from one in mangoes.

GO SLOWLY HERE.

To compare reactions, count them as percentages, not as units.

Percentage change. The change divided by the starting value, times 100.

Percentage change: change over starting value, times 100.

Salt: price up 10 per cent, quantity down 1 per cent. Mangoes: quantity down 20 per cent.

Elasticity. How strongly one thing reacts when another thing changes.

Price elasticity of demand. Percentage change in quantity demanded divided by percentage change in price.

Two new terms: elasticity and price elasticity of demand.

Formula: elasticity equals % change in quantity divided by % change in price.

Two tea stalls stand side by side near a bus station in a small Bangladeshi town. On the same morning each raises the price of a cup by Tk 2. By evening the first stall has lost almost half of its customers. The second stall has lost hardly any, and its owner finds more money in the cash box than before.

The law of demand, met in Chapter IV, predicts that both stalls will sell fewer cups, and both did. But the law speaks only of direction. It says that quantity falls. It does not say whether the fall is large or small, and a seller who must choose a price needs to know which. To tell the two stalls apart, we need a number. This chapter builds that number and uses it to answer four questions. Why did the stalls differ? Why do governments tax some goods heavily and others lightly? Why can a good harvest hurt the farmer? When does a price cut help a seller, and when does it not?

You will need the law of demand and the demand curve (Chapter IV), and percentage change, which the appendix *Maths you will use* explains.

WHY MEASURE REACTION?

Start with a household. In one month the price of salt rises by 10 per cent, and the price of mangoes also rises by 10 per cent. The family buys almost the same amount of salt as before, because food without salt is hard to eat. But the family buys far fewer mangoes, because many other fruits are available. The two prices moved by the same amount. The two reactions were very different.

Now put numbers on it. Salt rises from Tk 20 to Tk 22 a kilogram, and the family's purchase falls from 100 kg to 99 kg. Mangoes rise from Tk 100 to Tk 110 a kilogram, and the purchase falls from 20 kg to 16 kg. The raw changes are 1 kg of salt and 4 kg of mangoes. We cannot compare them. A kilogram of salt is not a kilogram of mangoes, and the family buys far more salt. We need a measure that does not depend on the unit. A percentage is such a measure.

A **percentage change** means the change divided by the starting value, multiplied by 100. A price that moves from Tk 20 to Tk 22 changes by Tk 2. Divide by the starting value: $2 \div 20 = 0.10$. Multiply by 100: the price rose by 10 per cent. Always divide by the value you started from, not the value you ended with.

Apply this to both goods. The price of each rose by 10 per cent. The quantity of salt fell by $1 \div 100 \times 100 = 1$ per cent. The quantity of mangoes fell by $4 \div 20 \times 100 = 20$ per cent. Now the reactions can be placed side by side. For salt, a 10 per cent price rise brought a 1 per cent fall in quantity. For mangoes, the same rise brought a 20 per cent fall. Mangoes reacted twenty times as strongly.

Economists compress this comparison into one number. **Elasticity** means how strongly one thing reacts when another thing changes. A rubber band is elastic: a small pull stretches it a long way. **Price elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in price. From now on this book writes "elasticity" for price elasticity of demand, and says so whenever another kind is meant.

Here is the formula, first in words and then in symbols. Elasticity = percentage change in quantity demanded $\div$ percentage change in price. In symbols, $E_d = \% \Delta Q \div \% \Delta P$. Here E_d is elasticity, Q is quantity demanded, P is price, and Δ (the Greek letter delta) means "change in". For salt, $E_d = 1 \div 10 = 0.1$. For mangoes,

$$E_d = \frac{20}{10} =$$

2.0.¹ Salt has the small number and mangoes the large one, as the household story suggested.

Pause and check. The price of a school bag rises by 5 per cent and the quantity demanded falls by 10 per cent. What is the elasticity? The answer is at the back of the book.

Before moving on. Elasticity compares two percentages, so it has no units and can be used for any good. *You should now be able to say:* "Elasticity is the percentage change in quantity demanded divided by the percentage change in price, and a larger number means a stronger reaction."

PRICE ELASTICITY OF DEMAND

This section is the centre of the chapter. It sorts demand into five types, shows two ways of measuring elasticity, explains why one straight line has many elasticities, and ends with worked examples. Take it one subsection at a time.

THE FIVE TYPES

The number 1 divides strong reactions from weak ones. When quantity changes by a larger percentage than price, buyers react strongly, and the number is above 1. When quantity changes by a smaller percentage, the number is below 1. Economists name five cases. Figure 4 shows the five curves side by side. Price is on the vertical axis and quantity on the horizontal axis, as in Chapter IV.

Demand is **elastic** when the quantity demanded changes by a larger percentage than the price, so elasticity is above 1. Mangoes, with 2.0, are an example. Demand is **inelastic** when the quantity changes by a smaller percentage, so elasticity is below 1. Salt, with 0.1, is an example. Demand is **unit elastic** when the two percentages are equal, so elasticity is exactly 1. A 10 per cent price rise then brings a 10 per cent fall in quantity.

Two extreme cases complete the list. Demand is **perfectly elastic** when buyers will take any amount at one price and nothing at a higher price. Elasticity is then infinite, and the curve is a horizontal line. A small seller in a market where thousands of others set the price meets this demand (Chapter XII). Demand is **perfectly inelastic** when quantity does not change at all, so elasticity is 0 and the curve is a vertical line. It is rare. It fits only a need with no substitute, such as a life-saving medicine for one patient.

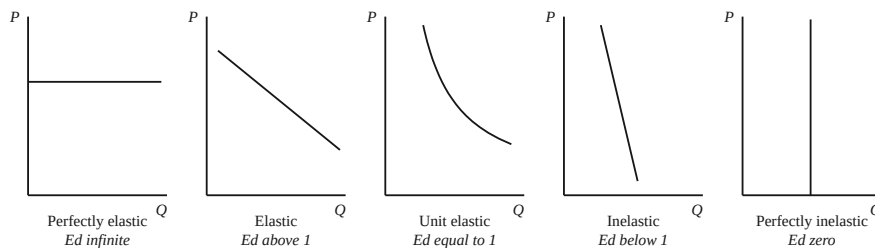


FIG. 4. Five types of price elasticity of demand. In each panel, price is on the vertical axis and quantity on the horizontal axis.

¹ A rise in price brings a fall in quantity, so the true ratio is negative: $-20 \div 10 = -2.0$. For an ordinary demand curve the sign is always negative, so economists drop it and quote the size. This book does the same for price elasticity. For income and cross elasticity, later in this chapter, the sign carries meaning and is kept.

A short test of the formula.

Elasticity is a ratio of two percentages, so it has no units.

This is the longest section of the chapter, and it repays slow reading.

GO SLOWLY HERE.

The number 1 is the dividing line between strong and weak reaction.

Elastic. Quantity changes by a larger percentage than price.

Inelastic. Quantity changes by a smaller percentage than price.

Unit elastic. Quantity and price change by equal percentages.

Elastic, inelastic and unit elastic are the three main types.

Perfectly elastic. Buyers take any amount at one price and nothing above it.

Perfectly inelastic. Quantity demanded does not change at all when price changes.

The two extremes are horizontal and vertical curves.

The table gathers the five types for comparison.

The table below gathers the five types in one place. Read each row from left to right: the name, the value of elasticity, how quantity reacts, and the shape of the curve.

THE FIVE TYPES OF PRICE ELASTICITY OF DEMAND

TYPE	ELASTICITY	REACTION OF QUANTITY	CURVE
Perfectly elastic	Infinite	Any amount at one price	Horizontal line
Elastic	Above 1	Larger percentage than price	Flat
Unit elastic	Exactly 1	Same percentage as price	Bends in toward both axes
Inelastic	Below 1	Smaller percentage than price	Steep
Perfectly inelastic	0	No change	Vertical line

"Elastic" is about percentages, not about the number of units.

Many beginners read "elastic" to mean that buyers change by many units. That is wrong. The test is always a comparison of two percentages. A seller whose customers fall from 1,000 to 990 may still face elastic demand, if the price rose by only 0.5 per cent. Then the quantity fell by 1 per cent, and $1 \div 0.5 = 2.0$.

POINT AND ARC METHODS

Over a large change, the percentage method gives two answers, one for each direction.

The percentage formula has a weakness. Take the neighbourhood bun market of the case file, where demand is $Q_d = 540 - 30P$ a day. If the price rises from Tk 8 to Tk 10, the quantity falls from 300 to 240. The price changed by $2 \div 8 \times 100 = 25$ per cent, and the quantity by $60 \div 300 \times 100 = 20$ per cent. Elasticity is $20 \div 25 = 0.8$. Now travel the other way, from Tk 10 to Tk 8. The price falls by $2 \div 10 \times 100 = 20$ per cent, and the quantity rises by $60 \div 240 \times 100 = 25$ per cent. Elasticity is $25 \div 20 = 1.25$. The same two points give two answers, because the starting value changed.

Point elasticity. Elasticity at one point of the demand curve, for a very small price change.

Point elasticity measures one point, using the slope.

Two remedies exist. The first is to measure at one point, for a very small change. **Point elasticity** means elasticity measured at a single point of a demand curve. On a straight demand curve, quantity changes by the same amount for each Tk 1 of price, so we can use that amount. The formula in words: elasticity = (change in quantity $\div$ change in price) $\times$ (price $\div$ quantity). In symbols, $E_d = (\Delta Q \div \Delta P) \times (P \div Q)$, where P and Q are the price and quantity at the point. This is the percentage formula rearranged, because $(\Delta Q \div Q) \div (\Delta P \div P)$ is the same as $(\Delta Q \div \Delta P) \times (P \div Q)$.

Arc elasticity. Elasticity between two points, using average price and average quantity.

Arc elasticity uses average price and average quantity.

The second remedy suits a real change between two points. **Arc elasticity** means elasticity between two points, found by using the average of the two prices and the average of the two quantities as the starting values. The formula is $E_d = (\Delta Q \div \Delta P) \times (\text{average } P \div \text{average } Q)$. For Tk 8 to Tk 10: $\Delta Q = 60$, $\Delta P = 2$, average P = 9 and average Q = 270. So $E_d = (60 \div 2) \times (9 \div 270) = 30 \times 9 \div 270 = 270 \div 270 = 1.0$. The answer is the same in both directions, and it lies between the two earlier answers, 0.8 and 1.25.

ELASTICITY ALONG A STRAIGHT LINE

One straight line has the same slope everywhere but a different elasticity at each point.

Look again at the bun demand curve. Its slope is the same everywhere: every Tk 1 added to the price removes 30 buns. Yet the elasticity is not the same everywhere. The point formula shows why. The slope part, 30, stays fixed, but the part $P \div Q$ changes. High on the curve the price is large and the quantity small, so $P \div Q$ is large. Low on the curve the price is small and the quantity large, so P

$\div Q$ is small. The table gives six rows. To read a row, take the price, find the quantity from $Q_d = 540 - 30P$, and then compute $30 \times P \div Q$.

ELASTICITY ALONG THE BUN DEMAND CURVE, $Q_d = 540 - 30P$

PRICE (TK)	QUANTITY	ELASTICITY	TYPE
2	480	0.125	Inelastic
8	300	0.8	Inelastic
9	270	1.0	Unit elastic
10	240	1.25	Elastic
14	120	3.5	Elastic
16	60	8.0	Elastic

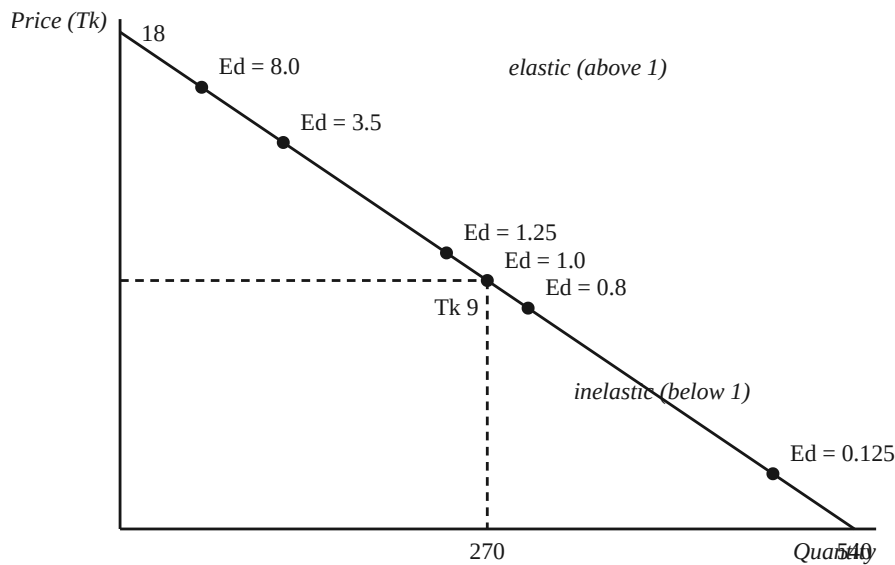


FIG. 5. Elasticity changes along a straight demand curve. It is unit elastic at the middle point.

The curve is elastic in its upper half, unit elastic at the middle point (price Tk 9, quantity 270), and inelastic in its lower half. Many students think that a steep curve means inelastic demand and a flat curve means elastic demand. That is true only when we compare two curves at the same point. A single straight line shows both kinds.

Case file, illustrative. In the neighbourhood bun market, demand is $Q_d = 540 - 30P$ a day. At Tk 10 the quantity is 240 and elasticity is $30 \times 10 \div 240 = 1.25$. At Tk 8 it is 0.8, and at Tk 14 it is 3.5. The arc elasticity between Tk 8 and Tk 10 is 1.0. For a straight demand curve, the percentage method from a starting price equals the point elasticity at that price, which is why 0.8 and 1.25 match the earlier results.

WORKED EXAMPLES

Example 1. Given: the price of an exercise book rises from Tk 20 to Tk 24, and monthly purchases fall from 500 to 450. Find: the elasticity. Formula: elasticity = percentage change in quantity $\div$ percentage change in price, or $Ed = \% \Delta Q \div \% \Delta P$.

Steps. Change in price = $24 - 20 = \text{Tk } 4$. Percentage change in price = $4 \div 20 \times 100 = 20$ per cent. Change in quantity = $500 - 450 = 50$. Percentage change in quantity = $50 \div 500 \times 100 = 10$ per cent. $Ed = 10 \div 20 = 0.5$. Answer: elasticity

Steepness is not the test: the upper half is elastic, the lower half inelastic.

Case file: bun elasticities at Tk 8, 10 and 14.

First example: small clean numbers, so the method is the only new thing.

The steps: two percentage changes, then one division.

Second example: the point formula applied to the bun market.

Six slips cause most wrong answers.

A second test: use the point formula at the top of the curve.

Elasticity sorts demand into five types and has two measuring methods.

Four things decide how strongly buyers react: substitutes, need, share of income and time.

Necessity. A good people need and cannot easily do without.

Luxury. A good people can do without when money is short.

Necessities have inelastic demand; luxuries elastic.

Large shares of income and long time spans make buyers more careful.

The two stalls differ in substitutes, not in prices.

A seller who knows elasticity knows whether a price rise will bring more money or less.

is 0.5, so demand is inelastic: quantity falls by half a per cent for each 1 per cent rise in price. *Check:* the quantity fell by a smaller percentage than the price (10 against 20), so the number must be below 1, and it is.

Example 2. Given: demand is $Q_d = 540 - 30P$, and the price is Tk 10. *Find:* the point elasticity at this price. *Formula:* $E_d = (\Delta Q \div \Delta P) \times (P \div Q)$. *Steps.* Quantity at Tk 10 = $540 - 30 \times 10 = 240$. The change in quantity per Tk 1 of price is 30 (size only). Then $P \div Q = 10 \div 240$. So $E_d = 30 \times 10 \div 240 = 300 \div 240 = 1.25$. *Answer:* elasticity at Tk 10 is 1.25, so demand is elastic there. *Check:* Tk 10 lies above the middle point (Tk 9), where elasticity is 1, so the value must exceed 1. It does.

Common slips. First, dividing by the new value instead of the starting value. Second, comparing units instead of percentages. Third, putting price on top: quantity always goes on top. Fourth, calling a steep line inelastic without checking the point. Fifth, using arc elasticity when the question asks for a point, or the reverse. Sixth, mixing periods, such as a weekly price with a monthly quantity.

Pause and check. Using $Q_d = 540 - 30P$, find the point elasticity at a price of Tk 16. The answer is at the back of the book.

Before moving on. The five types are perfectly elastic, elastic, unit elastic, inelastic and perfectly inelastic. Use the point method for a tiny change and the arc method for a visible change. *You should now be able to say:* "On a straight demand curve, elasticity falls steadily from infinity at the top to zero at the bottom, and equals 1 at the middle."

WHAT MAKES DEMAND ELASTIC

Four main things decide whether demand is elastic or inelastic. The first and strongest is the number of close substitutes. If a buyer can switch easily to another good, a price rise sends buyers away quickly. Tea from one particular stall has many substitutes, namely the other stalls. Tea as a drink has fewer. The narrower the good is described, the more elastic its demand.

The second is how much the buyer needs the good. A **necessity** is a good that people need and cannot easily do without, such as salt or medicine. A **luxury** is a good that people can do without when money is short, such as a restaurant meal. Demand for necessities tends to be inelastic and demand for luxuries elastic. The line is not fixed. A good that is a necessity to one family may be a luxury to another.

The third is the share of income the good takes. A price rise in matches costs a family almost nothing, so it does not notice. A rise in the rent of a flat takes a large part of income, so it reacts. The fourth is time. In the first days after a price rise buyers have little chance to change habits, so demand is inelastic. Over months they find substitutes, change their equipment and move their homes, and demand becomes more elastic. Chapter VII shows that the same idea applies to sellers.

Return to the two tea stalls. The first stall sits among several rivals, so a customer who dislikes the new price walks a few metres to another cup. Its demand is elastic, and the price rise lost it custom. The second stall serves people who wait for a bus and have no time to go elsewhere. Its demand is inelastic, and the price rise raised its takings.

The number matters to three kinds of people. A seller needs it before changing a price (the last section of this chapter shows how). A government needs it before laying a tax. A tax on a good with inelastic demand raises a large sum,

because buyers go on buying. A tax on a good with elastic demand raises less, because buyers switch away. Salt would be a better source of revenue than mangoes for this reason, though other concerns, such as fairness to poor families, also count.

A farmer needs the number too. Suppose the demand for potatoes has an elasticity of 0.5. A bumper harvest lowers the price by 20 per cent, and quantity bought rises by only 10 per cent. Revenue is then $0.8 \times 1.10 = 0.88$ of its former level, a fall of 12 per cent. The farmer harvested more and earned less. This is why farmers' groups sometimes ask the government to buy the extra crop, and why a poor harvest can raise a farmer's income.

Before moving on. Demand is more elastic when there are close substitutes, when the good is a luxury, when it takes a large share of income, and when buyers have more time. *You should now be able to say:* "Demand for a good is inelastic when buyers cannot easily switch away from it."

A good harvest can leave the farmer worse off if demand is inelastic.

Four factors, one number, three users.

INCOME AND CROSS ELASTICITY

Price is not the only thing that moves demand. Chapter IV listed income and the prices of related goods among the determinants. The same percentage method measures their effects, with one change: the sign is now kept, because it carries meaning.

Income elasticity means the percentage change in quantity demanded divided by the percentage change in consumers' income. In words: income elasticity = percentage change in quantity demanded ÷ percentage change in income. A positive value means a normal good, because demand rises with income. A negative value means an inferior good, because demand falls as income rises. A positive value above 1 marks a good whose demand grows faster than income, which is what is meant when economists call it a luxury. A positive value between 0 and 1 marks a necessity.

Cross elasticity means the percentage change in the quantity demanded of one good divided by the percentage change in the price of another good. In words: cross elasticity = percentage change in quantity of good A ÷ percentage change in price of good B. A positive value means the goods are substitutes: when the price of B rises, buyers move to A. A negative value means the goods are complements: when the price of B rises, buyers buy less of both. A value near zero means the goods are unrelated.

Example 3. Given: a family's income rises from Tk 20,000 to Tk 22,000 a month. Restaurant meals rise from 40 to 50 a month, and coarse rice falls from 30 kg to 27 kg. Separately, the price of petrol rises by 10 per cent and demand for CNG rises by 8 per cent. *Find:* the income elasticities and the cross elasticity. *Formulas:* as stated above. *Steps.* Income change = $2,000 \div 20,000 \times 100 = 10$ per cent. Meals: $10 \div 40 \times 100 = 25$ per cent, so $25 \div 10 = 2.5$. Rice: $-3 \div 30 \times 100 = -10$ per cent, so $-10 \div 10 = -1.0$. Cross: $8 \div 10 = 0.8$. *Answer:* restaurant meals have income elasticity 2.5 (a luxury); coarse rice -1.0 (an inferior good); CNG has cross elasticity 0.8 with petrol (substitutes). *Check:* each sign agrees with the story. More income brings more meals and less coarse rice, and a dearer fuel sends buyers to the other fuel.

A common slip is to drop the sign, as we do for price elasticity. Here the sign tells us whether a good is normal or inferior and whether two goods are substitutes or complements. Keep it.

Demand reacts to income and to other prices as well, and the same method measures both.

Income elasticity. Percentage change in quantity demanded divided by percentage change in income.

Income elasticity shows normal, inferior and luxury goods.

Cross elasticity. Percentage change in demand for one good divided by percentage change in the price of another.

Cross elasticity shows substitutes and complements.

Worked example: income and cross elasticity with new numbers.

Common slip: dropping the sign here loses the meaning.

Case file: cake income elasticity and bun cross elasticity.

Income and cross elasticity use the same method with the sign kept.

A seller wants to know what a price change does to the money received.

Total revenue. Price multiplied by the quantity sold.

Elasticity predicts the effect of a price change on revenue.

The bun market shows revenue rising, peaking at the unit-elastic point, then falling.

The bun table shows revenue peaking at the unit-elastic price.

Case file: the bun revenue test.

A lower price is not always better for a seller.

Case file, illustrative. When customers' income rises by 10 per cent, cake sales rise by 15 per cent: income elasticity 1.5, a luxury. When a rival bakery raises its bun price by 10 per cent, demand for this bakery's buns rises by 5 per cent: cross elasticity 0.5, so the buns are substitutes.

Before moving on. Income elasticity tells us about normal and inferior goods; cross elasticity tells us about substitutes and complements. *You should now be able to say:* "A negative cross elasticity means that the two goods are used together."

PRICE AND REVENUE

A seller who changes a price wants to know what happens to the money received. **Total revenue** means the price multiplied by the quantity sold. In symbols, $TR = P \times Q$. If a bakery sells 240 buns at Tk 10, its total revenue is $240 \times 10 = \text{Tk } 2,400$.²

Elasticity tells us the result. When the price changes, two things happen at once. The price per unit moves, and the quantity sold moves the other way. If demand is elastic, the quantity changes by the larger percentage, so the quantity effect wins. If demand is inelastic, the price effect wins. If demand is unit elastic, the two cancel exactly. The table sets out the rule.

ELASTICITY AND TOTAL REVENUE

DEMAND IS	WHEN PRICE RISES	WHEN PRICE FALLS
Elastic	Revenue falls	Revenue rises
Unit elastic	No change	No change
Inelastic	Revenue rises	Revenue falls

TOTAL REVENUE IN THE NEIGHBOURHOOD BUN MARKET, PER DAY

PRICE (TK)	QUANTITY	TOTAL REVENUE (TK)	ELASTICITY
8	300	2,400	0.8
9	270	2,430	1.0
10	240	2,400	1.25
12	180	2,160	2.0
14	120	1,680	3.5
16	60	960	8.0

The table confirms the rule. Between Tk 16 and Tk 12 demand is elastic, and each cut in price raises revenue, from Tk 960 to Tk 2,160. Revenue reaches its highest value, Tk 2,430, at Tk 9, where elasticity is exactly 1. Below Tk 9 demand is inelastic, and a further cut lowers revenue. Between Tk 8 and Tk 10 the arc elasticity is 1.0, so revenue is the same, Tk 2,400, at both ends.

Case file, illustrative. The revenue test for the bun market gives Tk 2,400 at Tk 8, 2,400 at Tk 10, 2,160 at Tk 12, 1,680 at Tk 14 and 960 at Tk 16, with a peak of Tk 2,430 at Tk 9 (derived). Revenue is not profit: the cost of making buns is counted in Chapter X.

Is a lower price always better for a seller? It is not, for four reasons. First, when demand is inelastic, a price cut lowers revenue and raises the number of buns to be baked. Second, revenue is not profit, because each extra unit costs

² Some books say *sales* or *turnover* for the same thing. This book uses *revenue* throughout.

money to make. Third, a very low price may signal poor quality, so that some buyers stay away. Fourth, rivals may match the cut, so that the extra buyers never arrive. Chapter XI turns this into a rule for profit, and Chapter XIV shows what rival replies can do.

Pause and check. In the table, the price falls from Tk 14 to Tk 12. By how much does revenue change? Is this what the rule predicts? The answer is at the back of the book.

Before moving on. A price cut raises revenue when demand is elastic and lowers it when demand is inelastic. *You should now be able to say:* "At the point of unit elasticity, a small change in price leaves revenue almost unchanged."

Revenue rises with price cuts only in the elastic range.

Elasticity and revenue move together by a simple rule.

SUMMARY

Elasticity measures how strongly quantity demanded reacts to a change in price. It equals the percentage change in quantity divided by the percentage change in price, and the size is quoted without sign. Demand is elastic above 1, inelastic below 1, and unit elastic at 1, with perfectly elastic and perfectly inelastic as the extremes. Use the point method for a tiny change and the arc method for a visible one. A straight demand curve is elastic above its middle and inelastic below it. Substitutes, need, share of income and time decide elasticity. Income elasticity and cross elasticity, with signs kept, show normal and inferior goods and substitutes and complements. A price cut raises revenue only where demand is elastic.

The chapter in a few lines.

THINK IT THROUGH

A tea seller charges Tk 10 a cup and sells 200 cups a day. She raises the price to Tk 12, and sales fall to 150 cups. Find the arc elasticity. Find her revenue before and after. Say what type of demand this is. Then suppose each cup costs her Tk 6 in tea, milk and sugar. Compare her daily gain before and after the change (revenue minus these costs). Which figure, revenue or this gain, is the better guide to her decision? The answer is at the back of the book.

A tea seller considers a price rise.

WHERE PEOPLE GO WRONG

Many students believe that a steep curve always means inelastic demand. Slope and elasticity are different things. Slope depends on the units of the two axes. Elasticity uses percentages and changes along a straight line.

Steepness is not elasticity.

Another mistake is to read an elasticity of 2 as a fall of 2 units. It means a fall of 2 per cent for each 1 per cent rise in price.

An elasticity of 2 does not mean that quantity falls by 2 units.

A third mistake is to read "inelastic" as "buyers do not react". Buyers react in every case but the extreme one. They react less than in proportion. A seller facing inelastic demand also should not raise price without limit, because the rule of Chapter XI sets a stopping point.

Inelastic does not mean that buyers ignore the price.

IN HISTORY

In October 1973, several oil-producing states cut supplies to some countries and raised prices. Within months the price of crude oil roughly quadrupled. At first, buyers cut their use only a little. Their cars, homes and factories were built to use the fuel, so demand was inelastic. Over the following years they bought smaller cars, insulated buildings and changed fuels, and demand grew more

A fuel shock in 1973 showed that elasticity grows with time.

elastic. The episode taught economists and governments that the same price rise can bring a small reaction in a month and a large one in a decade.

SOURCES

Alfred Marshall defined the idea.

Alfred Marshall's *Principles of Economics* (1890) introduced the measure of the reaction of demand to price, which this chapter calls elasticity. He valued it because it compared reactions without regard to the units of goods. His work is the root of the five types and of the link between elasticity and revenue.

REVIEW QUESTIONS AND PRACTICE

Seven review questions test the chapter in new words.

Review questions. (1) Why is a percentage better than a count of units for comparing two goods? (2) State the formula for elasticity, and say what a value of 0.4 tells a seller. (3) Name the five types of price elasticity and give the value for each. (4) Why does a straight-line demand curve show different elasticities at different points? (5) Give three things that make demand more elastic, each with an example. (6) State what happens to revenue when price rises and demand is elastic. (7) What sign has cross elasticity for substitutes and for complements?

Four practice items climb from clean numbers to a mixed set.

Practice. Item 1. The price of a notebook rises from Tk 50 to Tk 55, and the quantity demanded falls from 200 to 140. Find the elasticity and name the type.

Items 2 to 4 use messier numbers and new goods.

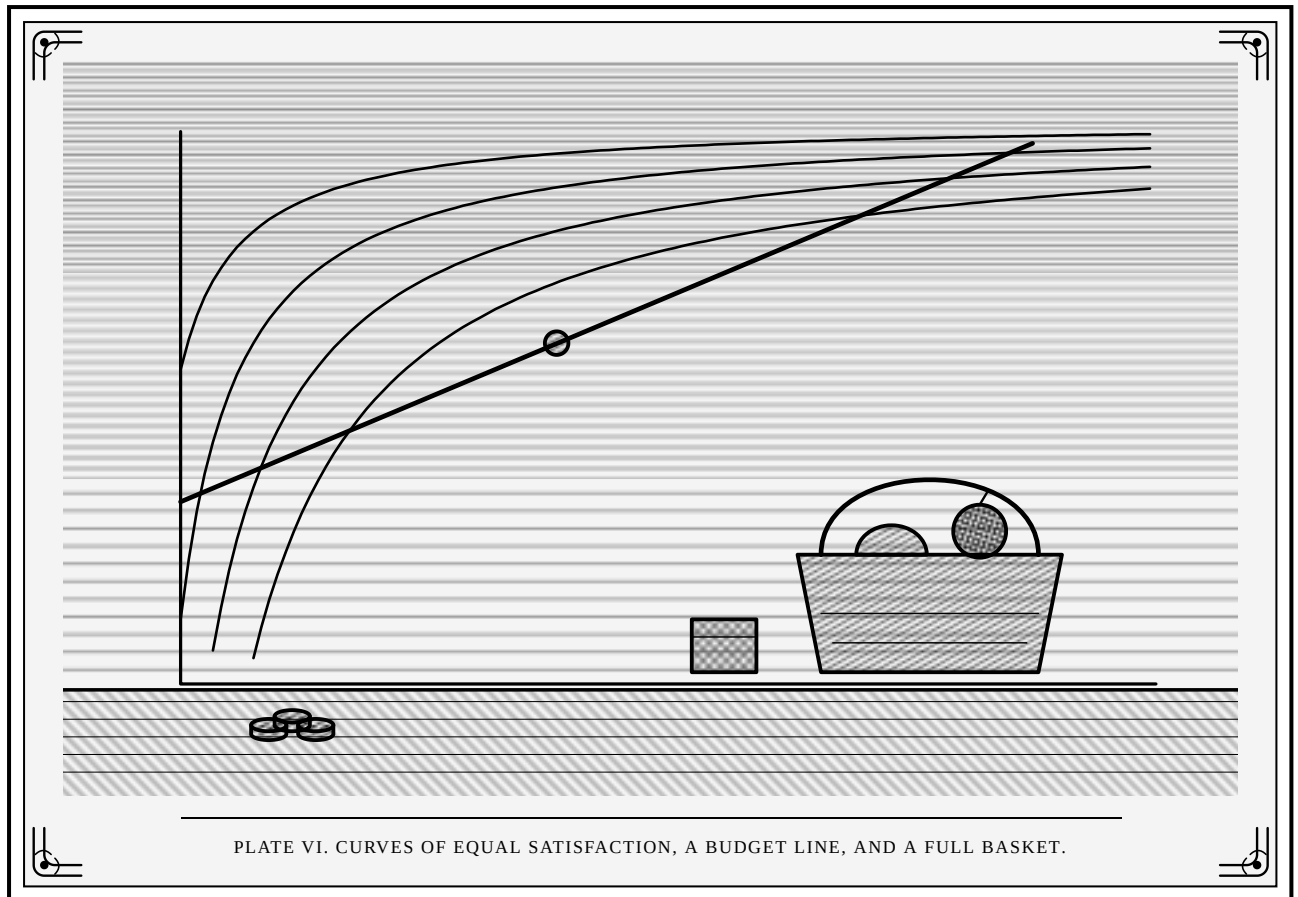
Item 2. Three goods each rise in price, with these results. Find each elasticity and classify it. (a) Rice: price Tk 50 to Tk 56 a kilogram, quantity 1,000 kg to 970 kg. (b) A mobile data pack: price Tk 200 to Tk 220, quantity 500 to 400. (c) A restaurant meal: price Tk 160 to Tk 176, quantity 600 to 540.

Item 3 uses the arc method.

Item 3. The price of a school uniform falls from Tk 500 to Tk 450 and the quantity sold rises from 800 to 880 a month. Use the arc method to find the elasticity, then check whether total revenue rises or falls.

Item 4 covers income and cross elasticity.

Item 4. (a) Income rises by 8 per cent. Demand for packaged juice rises by 12 per cent and demand for coarse rice falls by 4 per cent. Find each income elasticity and name the kind of good. (b) The price of motorcycles rises by 10 per cent and demand for helmets falls by 7 per cent. Find the cross elasticity and say whether the goods are substitutes or complements.



CHAPTER VI

Consumers' Surplus and Indifference Curves

What is a good bargain worth, and how does anyone choose?



A buyer pays one price for every bun, yet values the first bun far more than the last. This chapter measures the gain a buyer keeps beyond the price paid. It then offers a second way to explain choice, one that needs no counting of utils: curves of equal liking and a line that shows what a budget can buy.

A shopper feels that a purchase was a bargain, though everyone paid the same price.

Two questions: how big is the gain, and how can choice be explained without utils?

This chapter rests on utility, marginal utility and the demand curve.

The first bun is worth more to the buyer than the last bun.

GO SLOWLY HERE.

Willingness to pay is the most a buyer will give for one more unit.

Willingness to pay. The most a buyer will give for one unit of a good.

On each bun bought, the buyer keeps the gap between worth and price.

Consumers' surplus. What a buyer would be willing to pay, minus what is actually paid.

Think of a time when you bought something and felt that you had done well. Perhaps the price was Tk 100, and you would have paid Tk 150 if you had to. The shopkeeper took Tk 100 from you, and Tk 100 from every other buyer. Where did your feeling of a good bargain come from? The price shows what you paid. It does not show what the thing was worth to you, and the difference between the two is a real gain.

This chapter answers two questions. First, how can that gain be measured, and what is it used for? Second, Chapter III explained choice by counting utils, which assumes that satisfaction can be measured in numbers. Can choice be explained when buyers can only rank one basket above another? The second method, called indifference curve analysis, shows that it can, and it reaches the same answer.

You will need total and marginal utility (Chapter III), the demand curve and the law of demand (Chapter IV), and the idea of an equilibrium in which a buyer has no reason to change (Chapter III).

CONSUMERS' SURPLUS

Begin with a single customer at the bakery. In Chapter III her utility from buns fell as she ate more: the first bun was worth 20 utils, the second 16, the third 12, the fourth 8 and the fifth 4. We take one util to be worth Tk 1. The first bun is then worth Tk 20 to her, the second Tk 16, and so on. She does not pay these sums. She pays the market price, which is Tk 10 for every bun.

Willingness to pay means the most a buyer will give for one unit of a good. For the first bun it is Tk 20. A buyer buys a unit when the willingness to pay is at least the price. At Tk 10 the customer buys the first bun (worth 20), the second (worth 16) and the third (worth 12). She does not buy the fourth, because it is worth Tk 8, and that is less than the price. She stops at three buns, which is where Chapter III said she would stop.

Now count what she gains on each bun. On the first, she would pay Tk 20 and pays Tk 10, so she keeps Tk 10. On the second she keeps $16 - 10 = \text{Tk } 6$. On the third she keeps $12 - 10 = \text{Tk } 2$. The total is Tk 18. We call this sum **consumers' surplus**, which means what a buyer would be willing to pay for the units bought, minus what is actually paid. Here she would pay $20 + 16 + 12 = \text{Tk } 48$ and pays $3 \times 10 = \text{Tk } 30$, so the surplus is $48 - 30 = \text{Tk } 18$.

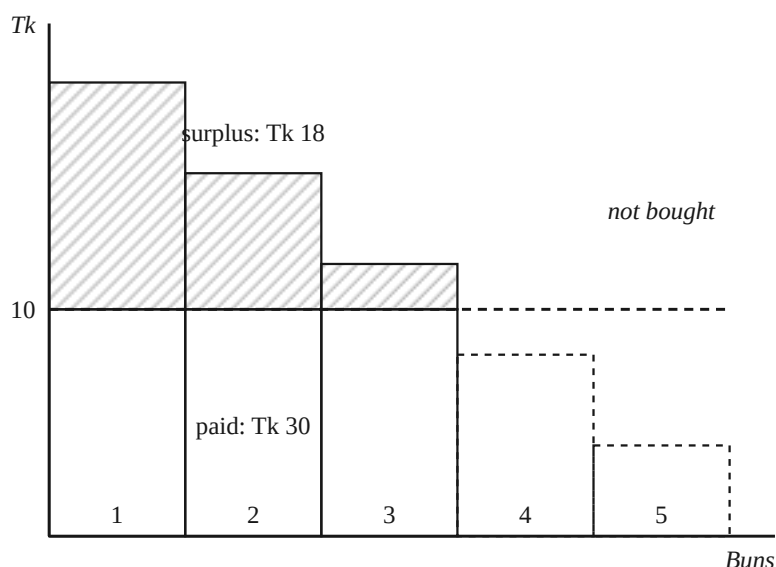


FIG. 6. Consumers' surplus of one customer at a bun price of Tk 10. The shaded parts above the price line sum to Tk 18.

Figure 6 shows the same facts. Each bar is the willingness to pay for one bun, and the dashed line is the price. Read the first bar: it reaches Tk 20, and the price line cuts it at Tk 10, so Tk 10 of it is shaded. The shaded parts of the first three bars add up to Tk 18. The fourth and fifth bars fall below the price line, so those buns are not bought.

Case file, illustrative. At a bun price of Tk 10 the customer of Chapter III buys 3 buns. Consumers' surplus is $(20 - 10) + (16 - 10) + (12 - 10) = \text{Tk } 18$.

The same idea works for a whole market. In the neighbourhood bun market of Chapter IV, demand is $Q_d = 540 - 30P$, and buyers would pay up to Tk 18 for the very first bun. If the price were Tk 12, the market would buy 180 buns a day. The surplus is the triangle between the demand curve and the price line, with base 180 and height $18 - 12 = 6$. Its area is $\frac{1}{2} \times 180 \times 6 = \text{Tk } 540$ a day. Chapter VIII uses this figure again.

Two warnings follow. First, consumers' surplus is a gain in well-being, not money received, and it is not a profit. Second, the measure rests on an assumption that a taka means the same to the buyer whether she is rich or poor, and that the buyer's other purchases do not change.¹ Rich and poor buyers do not value a taka equally, and so the figure is an estimate, not an exact count.

Despite its limits, the idea is useful. A tax raises the price, and buyers lose part of their surplus. A road or a bridge may charge a toll that is lower than what users would pay, so users gain surplus beyond the toll, and that gain counts as a benefit of the project. A seller who understands surplus sees that buyers keep a gain at a single price, and that some sellers try to capture it by charging different buyers different prices (Chapter XIII).

Pause and check. The customer's willingness to pay is Tk 20, 16, 12, 8 and 4 for buns one to five. If the price were Tk 14, how many buns would she buy, and what would her consumers' surplus be? The answer is at the back of the book.

Before moving on. Consumers' surplus is the sum of (willingness to pay – price) over the units bought. You should now be able to say: "A

Figure 6 shows the surplus as the gap between the bars and the price line.

Case file: surplus of Tk 18 at a bun price of Tk 10.

A whole market has a surplus too, shown as a triangle under the demand curve.

Surplus is not profit, and it is not a perfect measure.

Surplus guides decisions on taxes, public projects and prices.

Test the idea on a higher price.

Surplus is willingness to pay minus price, summed over the units bought.

¹ Alfred Marshall made this assumption explicit in 1890. It holds well for goods that take a small part of income, such as buns, and less well for goods that take a large part.

buyer who pays one price for every unit keeps a gain on every unit except the last."

A buyer can say which basket is better without saying by how many utils.

GO SLOWLY HERE.

An indifference schedule lists baskets that the buyer likes equally well.

Indifference schedule. A table of baskets of two goods that give equal satisfaction.

Plotted on a graph, the baskets lie on a curve that slopes down and bends inward.

Indifference curve. A line joining baskets of two goods that give equal satisfaction.

INDIFFERENCE CURVES

Start with an everyday case. A student has a weekly basket of fruit and likes two fruits, mangoes and guavas. Ask her: "Would you rather have 12 mangoes and 1 guava, or 6 mangoes and 2 guavas?" She may answer, "I like them equally well." She cannot say how many utils either basket gives. She can say only that neither is better. This is enough to build a theory of choice.

The student gives these answers, and we record them. An **indifference schedule** means a table of baskets of two goods that a buyer likes equally well. Read the table row by row: each row is a basket, and the buyer is equally happy with all of them. As mangoes fall, the number of guavas needed to keep her equally happy rises.

BASKETS THAT THE STUDENT LIKES EQUALLY WELL

BASKET	MANGOES	GUAVAS	MANGOES GIVEN UP PER EXTRA GUAVA
A	12	1	—
B	6	2	6
C	4	3	2
D	3	4	1
E	2	6	0.5

Plot the baskets on a graph, with mangoes on the horizontal axis and guavas on the vertical axis. Join the points. The line is an **indifference curve**, which means a line joining all the baskets that give a buyer equal satisfaction. The curve in Figure 7, marked II, passes through A to E. Every point on it is as good as every other point.

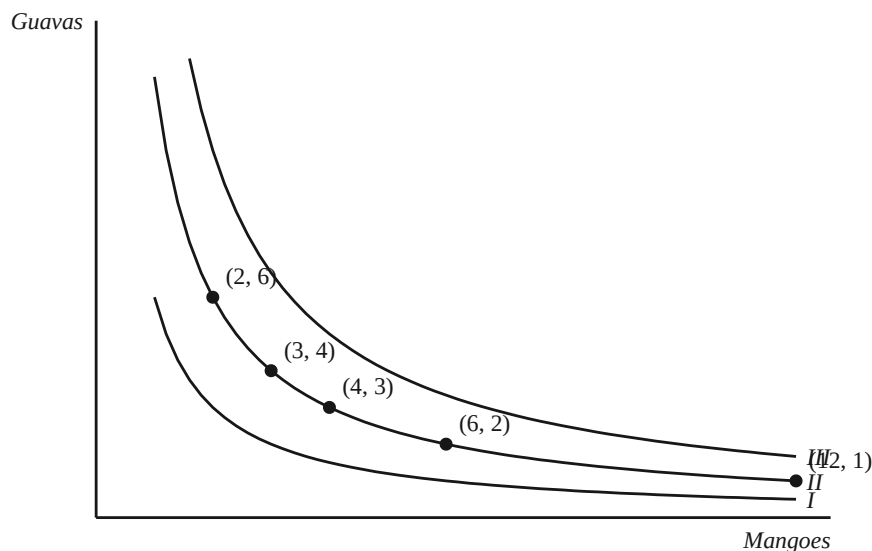


FIG. 7. An indifference map. Every point on one curve is equally liked. Curve III is better than curve II, and curve II is better than curve I.

The student has many such curves. A basket with more of both fruits is better, so it lies on a higher curve. A set of curves of this kind is an **indifference map**. In Figure 7, curve III shows baskets better than any on curve II, and curve I shows baskets worse. The map is like the contour lines on a hill map, except that the height is the buyer's well-being.

The last column of the table needs a name. Moving from A to B, the student gives up 6 mangoes to gain 1 guava. The **marginal rate of substitution** means the number of units of one good a buyer gives up to gain one more unit of the other, while staying equally happy. Here it is 6, then 2, then 1, then 0.5. It falls as we move down the curve. This is the law of diminishing marginal utility (Chapter III) in a new form: when guavas are scarce, one more guava is worth many mangoes, and when guavas are plentiful it is worth few.

The curves have four properties. First, they slope downward, because a buyer who gets more of one good must give up some of the other to stay equally happy. Second, they bend inward toward the origin, because the marginal rate of substitution falls. Third, they never cross. If two curves crossed at a basket, that one basket would be on two levels of well-being at once, which is impossible. Fourth, a higher curve is better than a lower one, because it holds more of at least one good.

Pause and check. Can two indifference curves of the same buyer cross each other? Give the reason. The answer is at the back of the book.

Before moving on. An indifference curve joins baskets of equal well-being, and the marginal rate of substitution measures the rate of exchange between the two goods. *You should now be able to say:* "A higher indifference curve is better, and the curve bends inward because each extra unit of a good is worth less."

THE BUDGET LINE AND CHOICE

The map shows what the buyer likes. It does not show what she can afford. For that we need her money and the prices.

Income here means the money a buyer has to spend in a period. A **budget line** means a line joining all the baskets of two goods that cost exactly the whole income. Take the case of the Chapter III customer: income Tk 70, a bun at Tk 10, a cake slice at Tk 20. If she buys only buns, she can buy $70 \div 10 = 7$. If she buys only cake slices, she can buy $70 \div 20 = 3.5$. Between these points lie other baskets, such as 5 buns and 1 slice ($50 + 20 = 70$), 3 buns and 2 slices ($30 + 40 = 70$), and 1 bun and 3 slices ($10 + 60 = 70$).

The line has a slope that tells us something. One cake slice costs Tk 20, which is the price of two buns. So in the market, giving up 1 slice frees enough money for 2 buns. Baskets inside the line cost less than Tk 70 and are affordable. Baskets outside the line cost more and are not.

Case file, illustrative. The customer has an income of Tk 70, a bun at Tk 10 and a cake slice at Tk 20. The budget line runs from 7 buns to 3.5 slices. The basket of Chapter III, 3 buns and 2 slices, costs $30 + 40 = \text{Tk } 70$, so it lies on the line.

Now put the map and the line together in Figure 8. The buyer wants the highest curve that she can reach. Curve III lies out of reach, because no point of it is on or inside the budget line. Curve I can be reached, but it is low, and she could do better. Curve II touches the budget line at point C, with 3 buns and 2 slices. This is the buyer's equilibrium, and a touching point of this kind is called a tangent point.

A set of curves, each better than the one below, forms an indifference map.

Indifference map. A set of indifference curves, each higher curve preferred to the lower.

The last column of the table gives the rate at which one fruit replaces the other.

Marginal rate of substitution. The units of one good given up for one more unit of the other.

Four properties follow: downward slope, inward bend, no crossing, higher is better.

Test the map.

Indifference curves rank baskets, and the marginal rate of substitution falls along them.

A buyer's wants are unlimited, but the money to spend is not.

Income (of a consumer). The money a buyer has to spend in a period.

Budget line. A line of all the baskets that cost exactly the whole income.

Income and the budget line fix what a buyer can afford.

The slope of the budget line is the price ratio: one slice costs two buns.

Case file: the customer's budget line.

The best affordable basket lies where the budget line touches an indifference curve.

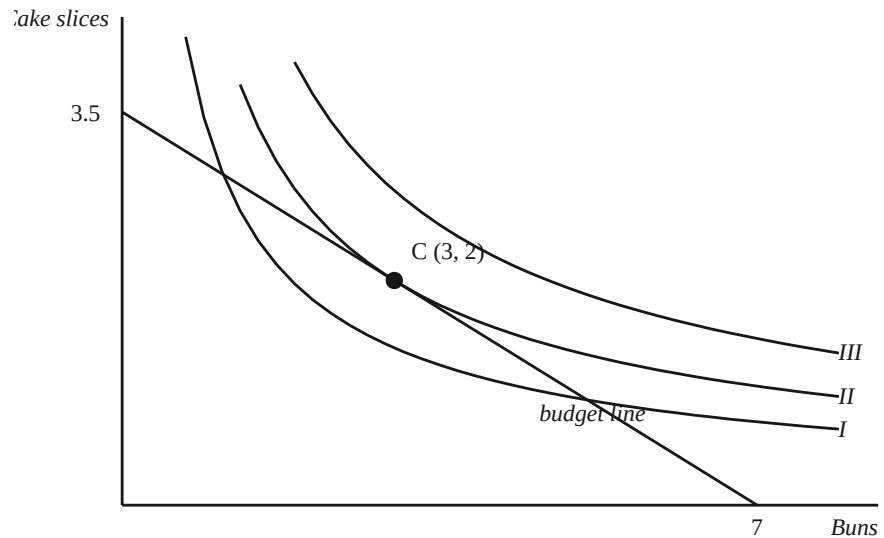


FIG. 8. The choice of the customer. Curve II touches the budget line at C, 3 buns and 2 cake slices. The curves are drawn to be consistent with Chapter III.

At the touching point, what the buyer will swap equals what the market asks her to swap.

The two methods agree: one counts utils, the other ranks baskets.

Price effect. The change in a buyer's best basket when the price of a good changes.

A fall in price turns the budget line outward and raises purchases.

Worked example: a budget line from new prices.

The steps: find each end, then test baskets.

The budget line moves in two ways: income shifts it, a price turns it.

At C the slope of the curve and the slope of the line are equal. The slope of the curve is the marginal rate of substitution. The slope of the line is the price ratio. So at the best basket, the buyer's willingness to swap equals the market's rate of exchange. Check this with Chapter III. The second cake slice gives 24 utils and the third bun gives 12, so she values a slice at about 2 buns. The market also values a slice at 2 buns, since $20 \div 10 = 2$. The two agree. In the language of Chapter III, utility per taka is also equal: $12 \div 10 = 24 \div 20 = 1.2$.

This shows that the two methods are one idea. Counting utils gave the rule that utility per taka is equal across goods. Ranking baskets gave the rule that the marginal rate of substitution equals the price ratio. Both describe the same choice, and the ranking method needs weaker assumptions.

Now let the price of buns fall from Tk 10 to Tk 5. If she spent all her income on buns she could now buy $70 \div 5 = 14$. The budget line swings outward along the bun axis, while the cake end stays at 3.5. She can reach a higher curve and, in the usual case, buys more buns. This change is the **price effect**, which means the change in the best basket when one price changes. If we record the number of buns she buys at each bun price, we obtain her demand curve of Chapter IV.

Example 1. Given: income Tk 120, a notebook at Tk 20 and a pen at Tk 10. Find: the two ends of the budget line and three baskets on it. Formula: price of notebooks $\times$ notebooks + price of pens $\times$ pens = income, or $20N + 10P = 120$.

Steps. All income on notebooks: $N = 120 \div 20 = 6$, with no pens. All income on pens: $P = 120 \div 10 = 12$, with no notebooks. Test 2 notebooks and 8 pens: $20 \times 2 + 10 \times 8 = 40 + 80 = 120$. Test 4 notebooks and 4 pens: $80 + 40 = 120$. Test 5 notebooks and 3 pens: $100 + 30 = 130$, which is more than 120, so it lies outside. Answer: the line runs from 6 notebooks to 12 pens, and the baskets (2, 8) and (4, 4) lie on it. Check: the slope is 12 pens for 6 notebooks, so 1 notebook is worth 2 pens, which equals the price ratio $20 \div 10 = 2$.

A common slip is to confuse the two ways the line can move. A change in income moves the whole line in parallel, outward if income rises. A change in the price of one good turns the line around the other end, as the bun price fall did above.

Pause and check. Could the customer, with Tk 70, buy 4 buns and 2 cake slices? The answer is at the back of the book.

Before moving on. The budget line shows what can be bought and its slope is the price ratio. The best basket is at a tangent point, where the marginal rate of substitution equals the price ratio. *You should now be able to say:* "A buyer chooses the basket on the highest indifference curve that the budget allows."

A third test.

The best basket is where the line touches the highest reachable curve.

MEASURING CONSUMERS' SURPLUS WITH INDIFFERENCE CURVES

This section shows that the surplus of the first section can be found from indifference curves. It is harder than the rest of the chapter, and the chapters that follow do not depend on it. Read it now if you wish, or return to it later.

The method uses a simple trick: let money be the second good. Take the customer with an income of Tk 70 and a bun price of Tk 10. A basket is now a number of buns and the money left over. Because one util is worth Tk 1, a basket is worth the utility of the buns plus the money left, in taka. The table gives the worth of each basket she can buy.

BASKETS FOR THE CUSTOMER AT A BUN PRICE OF TK 10

BUNS	MONEY LEFT (TK)	UTILITY OF BUNS (TK)	WORTH OF BASKET (TK)
0	70	0	70
1	60	20	80
2	50	36	86
3	40	48	88
4	30	56	86
5	20	60	80

Read the last column. The basket with 3 buns is worth Tk 88, which is higher than any other, so she buys 3 buns. The basket with no buns is worth Tk 70. All baskets of equal worth lie on one indifference curve. The curve through her best basket therefore meets the money axis at Tk 88. This means that she is as well off with 3 buns and Tk 40 as with no buns and Tk 88.

She actually has Tk 70 with no buns. So the extra money that would make her give up all buns and stay equally well off is $88 - 70 = \text{Tk } 18$. This is the consumers' surplus found in the first section. The two methods agree because money was used as the measure of worth.

Case file, illustrative. For the customer of Chapter III at a bun price of Tk 10, the indifference-curve method gives the same surplus as the first method: Tk 18.

Before moving on. The surplus is the extra money that would leave the buyer as well off as before without the good. *You should now be able to say:* "The indifference-curve method and the bar method both measure the gain from buying at one price."

This section is advanced and may be left for a second reading.

ADVANCED: YOU MAY DEFER THIS SECTION.

Put money on one axis, and baskets become buns plus money left over.

The best basket is 3 buns, worth Tk 88, against Tk 70 with no buns.

The gap between Tk 88 and Tk 70 is the consumers' surplus.

Case file: the second method gives Tk 18 again.

Two methods, one answer.

SUMMARY

Consumers' surplus is willingness to pay minus the price paid, summed over the units bought. For the customer in the case it is Tk 18 at a bun price of Tk 10. It is useful in judging taxes, public projects and prices, but it is an estimate. An indifference curve joins baskets of equal well-being. It slopes down, bends

The chapter in a few lines.

inward, never crosses another, and a higher curve is better. The marginal rate of substitution falls along it. The budget line shows the baskets that cost the whole income, and its slope is the price ratio. The best basket lies where the line touches the highest reachable curve, where the marginal rate of substitution equals the price ratio. The same method also measures consumers' surplus.

THINK IT THROUGH

A town considers removing the toll on a bridge.

A bridge charges a toll of Tk 6 a crossing, and 400 crossings are made each day. The demand for crossings follows a straight line: at a toll of Tk 0 there would be 1,000 crossings, and at a toll of Tk 10 there would be none. Find the consumers' surplus of users at the toll of Tk 6. Find the surplus if the toll is removed. Find the toll revenue given up. Does the town gain or lose by removing the toll, counting the surplus and the revenue together? Name one cost that this count leaves out. The answer is at the back of the book.

WHERE PEOPLE GO WRONG

Consumers' surplus is not money in the buyer's pocket.

Many students treat consumers' surplus as a refund or a profit. It is neither. It is a measure of the gain in well-being, and no one pays it.

Two indifference curves of one buyer cannot cross.

Some students draw indifference curves that cross, or curves that slope upward. Neither is allowed, for the reasons given in the second section.

The budget line does not show what the buyer prefers.

A third mistake is to think that the budget line shows what the buyer likes. It shows what the buyer can afford. The map shows what she likes. Only the two together fix the choice.

IN HISTORY

A French engineer measured the gain from a bridge in 1844.

In 1844 Jules Dupuit, a French engineer, asked how a government should judge the worth of a bridge or a road. He argued that the toll paid is not the worth of the bridge to its users, because many users would pay more than the toll. The worth is the whole area under the demand curve for crossings, not the toll revenue alone. His argument is the root of the idea of consumers' surplus. Alfred Marshall gave the idea its name and its place in the theory of demand in 1890, and cost-benefit studies of public projects still use it.

SOURCES

Three works shaped the two ideas in this chapter.

Francis Edgeworth, in *Mathematical Psychics* (1881), introduced the indifference curve. John Hicks and R. G. D. Allen, in a paper of 1934, rebuilt the theory of demand on ranking alone, without counting utils, which is the method of the third section. Alfred Marshall's *Principles of Economics* (1890) named consumers' surplus.

REVIEW QUESTIONS AND PRACTICE

Six review questions test the chapter in new words.

Review questions. (1) Define willingness to pay, and say how it differs from the price paid. (2) How is consumers' surplus found for one buyer who buys several units? (3) Give two practical uses of consumers' surplus, and one reason why the measure is only an estimate. (4) State the four properties of indifference curves. (5) What is the marginal rate of substitution, and why does it fall as we move down a curve? (6) Say where the buyer's best

basket lies, and what is true of the marginal rate of substitution and the price ratio there.

Practice. Item 1. A buyer's willingness to pay for cups of tea is Tk 12, 9, 6 and 3 for cups one to four. The price is Tk 5. How many cups does she buy, and what is her consumers' surplus?

Item 2. A buyer likes these baskets of tea cups and biscuits equally well: (2 cups, 12 biscuits), (3, 8), (4, 6), (6, 4) and (8, 3). Find the marginal rate of substitution (biscuits given up per extra cup) between each pair of neighbouring baskets. Does it fall as cups increase?

Item 3. A family has Tk 200 a week to spend on rice at Tk 50 a kilogram and lentils at Tk 100 a kilogram. Find the two ends of the budget line and one basket in between. Then the price of lentils falls to Tk 50. Find the new ends. Say which way the line has moved.

Item 4. Market demand is $Q_d = 100 - 10P$. At a price of Tk 4, find the quantity bought and the consumers' surplus (use a triangle). Then find the surplus if the price falls to Tk 2.

Four practice items: a new schedule, a map, a budget line and a market.

Item 2 reads a map; Item 3 builds a budget line.

Item 3 builds a budget line and moves it.

Item 4 uses a straight-line market demand.

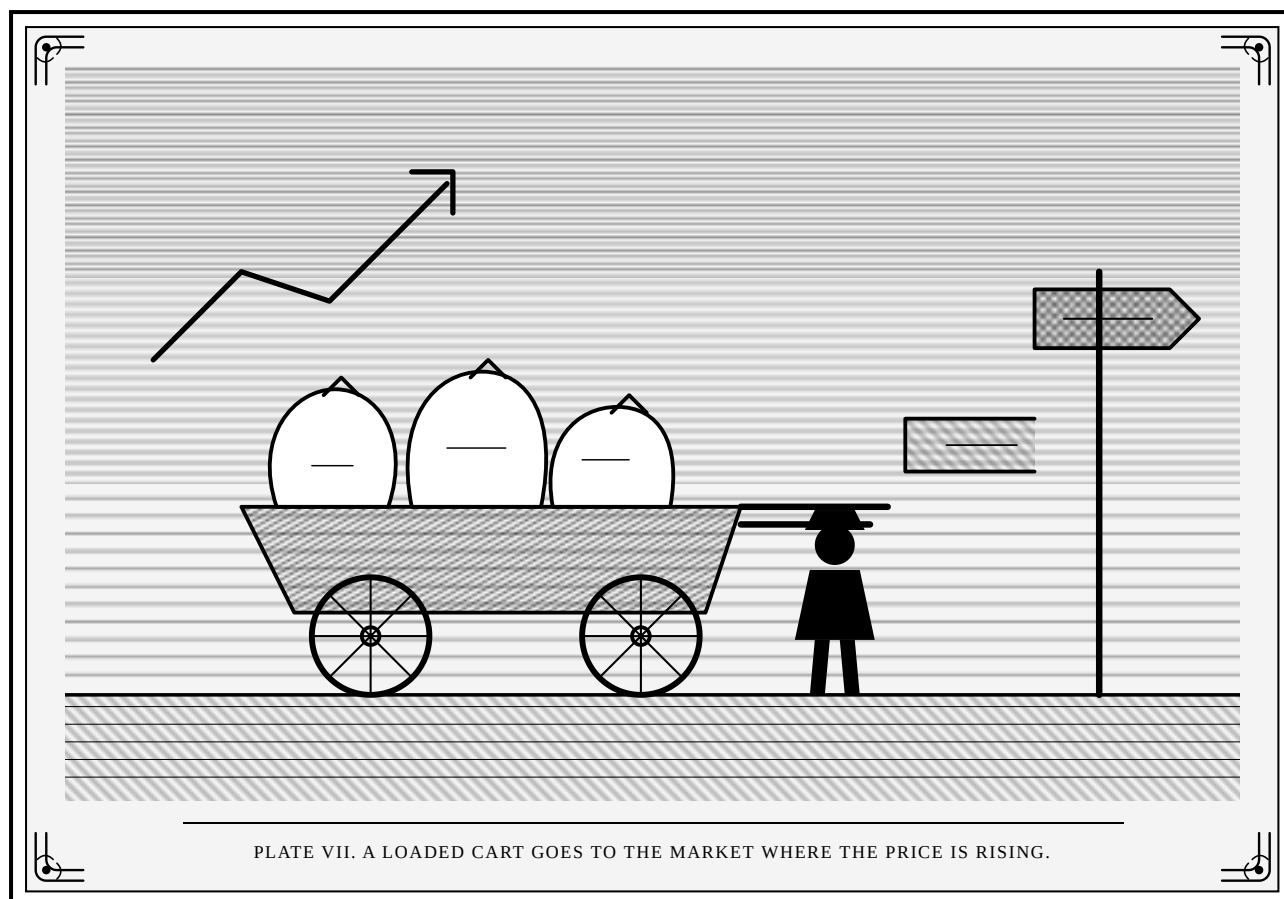


PLATE VII. A LOADED CART GOES TO THE MARKET WHERE THE PRICE IS RISING.

CHAPTER VII

Supply and Its Elasticity

Who sells more when the price rises, and how fast?



Chapters IV and V looked at the buyer. This chapter looks at the seller. It shows why a higher price brings more goods to market, what moves the whole supply curve, where the rule fails, and how to measure the seller's reaction with a number that mirrors the elasticity of demand.

A fisherman and a baker face the same price rise, and only one can respond today.

Buyers were measured in Chapters IV and V; sellers now follow the same path.

This chapter rests on demand and on elasticity.

A family with cows decides how much milk to sell at each price.

GO SLOWLY HERE.

Supply. The quantity sellers are able and willing to sell at each price.

Quantity supplied. The amount sellers offer for sale at one price.

Supply is the quantity sellers can and will sell at each price.

Supply schedule. A table of prices and the quantities offered at each.

Supply curve. The line that plots a supply schedule.

A schedule is the table; the curve is the line.

The market supply is the sum of what every seller offers at each price.

Individual supply. The supply of one seller.

Market supply. The sum of the supplies of all sellers.

On the same morning the price of fish and the price of bread each rise by 10 per cent. A fisherman has landed his catch at dawn. The fish are on the table, and no more can be caught before evening. A baker has an oven standing idle and a sack of flour in store. By afternoon she has baked an extra batch. The price signal was the same for both, but the amount each could offer was very different.

Chapter IV asked how buyers react to a price, and Chapter V measured the reaction. This chapter asks the same two questions of sellers. How much will sellers offer at each price? And how strongly does that amount react when the price moves? The answers explain why the fisherman and the baker differ, and they prepare the ground for Chapter VIII, where buyers and sellers meet.

You will need the demand curve, the idea of a shift (Chapter IV), and elasticity with its five types and two methods (Chapter V).

WHAT SUPPLY MEANS

Begin with a village family that keeps cows. At Tk 50 a litre, the family finds it worth selling 20 litres a day. At Tk 60 it sells 30 litres, because the higher price makes it worth feeding the cows better and milking longer. At Tk 70 it sells 40 litres, and at Tk 80, 50 litres. The family decides how much to offer for each possible price. That list of decisions is the beginning of supply.

Supply means the quantity of a good that sellers are able and willing to sell at each price. Both words matter. A seller who is willing but has no goods or inputs cannot supply. A seller who has goods but will not sell at that price does not supply. **Quantity supplied** means the amount offered at one price. At Tk 60 the family's quantity supplied is 30 litres a day.¹

A table that lists prices and the quantities offered is a **supply schedule**. If we plot it with price on the vertical axis and quantity on the horizontal axis, the line through the points is the **supply curve**. The family's schedule gives four points, and the curve rises from left to right. The word *schedule* names the table, and the word *curve* names the line. Both carry the same information.

One seller's supply is **individual supply**. **Market supply** is the sum of the individual supplies at each price. In the neighbourhood bun market of the case file, three bakeries bake buns. The table shows how their schedules add. Read a row: at Tk 12, Bakery A offers 60 buns, B offers 30, C offers 90, and the market offers 180.

MARKET SUPPLY OF BUNS FROM THREE BAKERIES, PER DAY

PRICE (TK)	BAKERY A	BAKERY B	BAKERY C	MARKET
8	20	10	30	60
10	40	20	60	120
12	60	30	90	180
14	80	40	120	240
16	100	50	150	300

The market schedule is a rule: $Q_s = 30P - 180$, zero below Tk 6.

The market column follows a rule: quantity supplied $Q_s = 30P - 180$, where P is the price in taka. At Tk 12 the rule gives $30 \times 12 - 180 = 360 - 180 = 180$, as in

¹ Supply is a flow, measured over a period such as a day or a month. It is not the stock that a seller holds in a warehouse. A seller may hold a large stock and yet supply little at a low price.

the table. At Tk 6 it gives 0, and below Tk 6 no bakery offers any bun, because the price does not cover the cost of making one.

Case file, illustrative. In the neighbourhood bun market, supply per day is $Q_s = 30P - 180$, with nothing offered below Tk 6. At Tk 8, 10, 12, 14 and 16 the quantity supplied is 60, 120, 180, 240 and 300. Three bakeries, A, B and C, make up this supply, with schedules $10P - 60$, $5P - 30$ and $15P - 90$.

Before moving on. Supply is a list of quantities at each price, and market supply is the sum of the individual lists. *You should now be able to say:* "Supply means how much sellers will offer at each possible price, not how much they happen to hold."

Case file: bun supply and three bakeries.

A supply schedule is a table of what sellers will offer, one row for each price.

THE LAW OF SUPPLY

The pattern in the schedules above is the same in most markets. When the price of a good rises, sellers offer more. When it falls, they offer less. We call this the law of supply.

Higher price, larger quantity offered: the rule mirrors the law of demand.

The **law of supply** states that, other things remaining the same, a higher price brings a larger quantity supplied, and a lower price brings a smaller one. The phrase "other things remaining the same" is the *ceteris paribus* of Chapter II. We hold costs, technology and taxes still while we watch price and quantity. On the graph the supply curve slopes upward from left to right.

Law of supply. A higher price brings a larger quantity supplied, other things equal.

The law of supply, with ceteris paribus.

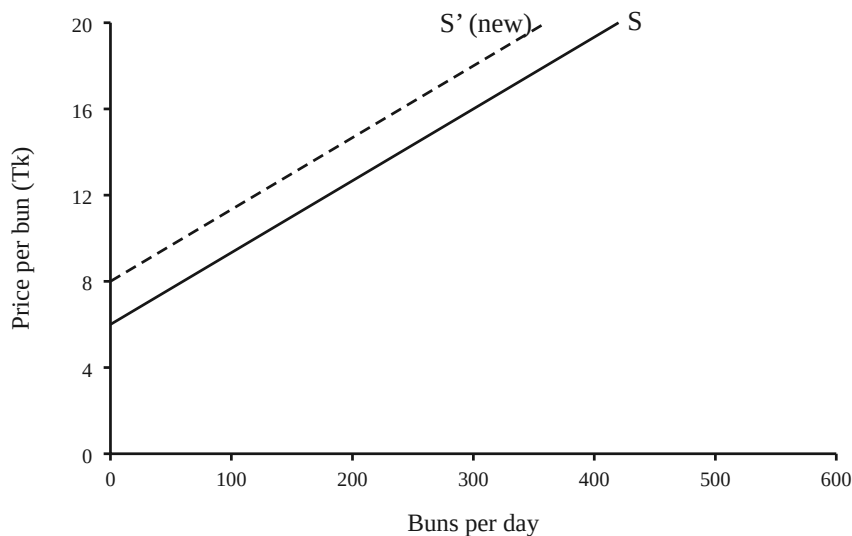


FIG. 9. The supply curve of the neighbourhood bun market. A fall in supply, caused by dearer flour, moves the whole curve to the left.

Two reasons explain the upward slope. The first is profit. A higher price makes each extra unit more profitable, so existing sellers want to sell more, and new sellers are drawn in. The second is cost. Producing more usually costs more for each additional unit, because the baker must run the oven longer, pay for extra labour and buy flour in smaller lots. Chapter X shows this carefully. A seller will produce a unit only if the price covers its extra cost, so a higher price makes more units worth producing.

Rising price gives sellers both the reason and the means to offer more.

Return to the fisherman and the baker. Both want the higher price. The baker can raise the quantity she offers, because her extra cost is low and her oven is idle. The fisherman cannot, because his catch is already landed. The law of supply says that the baker's curve slopes up. It does not say how steeply, and the fisherman's curve, in this

Two sellers, one price rise, different readiness to respond.

one day, is nearly vertical. The last section of this chapter measures the difference.

DETERMINANTS OF SUPPLY

Factors other than price move the whole supply curve.

GO SLOWLY HERE.

Determinants of supply. The things other than price that decide supply.

Determinants of supply: six things other than price.

Dearer inputs lower supply; cheaper inputs and better methods raise it.

Technology. The methods and equipment used to make a good.

Better methods lower cost and raise supply.

Subsidy. Money paid by government to lower a seller's cost.

A tax lowers supply; a subsidy raises it.

Expectations, the number of sellers and the weather also move supply.

Change in quantity supplied. A move along the supply curve, caused by a change in the price of the good.

Change in supply. A shift of the whole supply curve, caused by something other than the price.

Movement along the curve versus shift of the curve.

Case file: dearer flour shifts bun supply left.

Test the rule: is it a movement or a shift?

Determinants shift the curve; only the good's own price moves along it.

The law of supply holds everything but price still. In real life these other things change. When one of them changes, the whole curve moves, and so we need to tell the two kinds of change apart. Think of the family with cows. If the price of milk rises, the family sells more milk, and we move along its curve. If the price of cattle feed rises, the family offers less milk at every price, and the curve itself moves.

The things other than price that decide supply are the **determinants of supply**. The main ones are the price of inputs, technology, taxes and subsidies, expectations, the number of sellers, and the weather. Take them one at a time.

Input prices come first. If flour, wages or rent rise, the cost of each bun rises, and bakeries offer fewer buns at every price. Supply falls. If inputs become cheaper, supply rises.

Technology means the methods and equipment used to make a good. A faster oven or a better mixing machine lowers the cost of each bun, so supply rises. The effect is the opposite of that of dearer inputs.

Taxes and subsidies act on cost too. A tax on each unit sold raises the seller's cost, so supply falls. A **subsidy** means money paid by the government to lower a seller's cost. It acts like a cut in input prices, so supply rises. Bangladeshi farmers receive subsidies on some fertiliser, and the effect on supply is of this kind.

Expectations matter as well. A baker who expects the price of buns to rise next week may hold back some buns today, and today's supply falls. The number of sellers matters because market supply is a sum: a new bakery adds a new schedule to the sum, and supply rises. The weather matters most in farming. A flood or a drought lowers the crop a farmer can offer at every price.

Here is the rule, which parallels the one for demand. A **change in quantity supplied** is a movement along the supply curve, and only a change in the price of the good itself causes it. A **change in supply** is a shift of the whole curve, and a change in any determinant causes it. A rise in supply shifts the curve to the right, so more is offered at every price. A fall in supply shifts it to the left.

Case file, illustrative. The price of flour rises, adding Tk 2 to the cost of each bun. Supply shifts to $Q_s = 30P - 240$, which is 60 buns fewer at every price. At Tk 12 the quantity supplied is now 120 instead of 180, and nothing is offered below Tk 8. Figure 9 shows the shift. Chapter VIII uses this new supply.

Pause and check. In each case, say whether supply shifts or the quantity supplied moves along the curve. (a) The price of buns rises. (b) A new oven cuts the cost of baking. (c) The government taxes each bun sold. The answer is at the back of the book.

Before moving on. Six determinants shift supply: input prices, technology, taxes and subsidies, expectations, the number of sellers and the weather. *You should now be able to say:* "A change in the price of the good itself moves us along the curve, and a change in anything else shifts the curve."

EXCEPTIONAL SUPPLY CURVES

The law of supply holds for most goods, but a few cases seem to break it. These cases give **exceptional supply curves**, which means supply curves that do not slope upward in the usual way.

The first case is a good that cannot be made again in the short time that matters. A fisherman's catch, a day's fresh milk, and a famous painting are examples. The quantity is fixed, so the supply curve is a vertical line, whatever the price. The seller must sell what exists. A rise in price cannot call forth more.

The second case is labour. At a low wage, a higher wage may persuade a worker to work longer hours. At a high wage, the worker may prefer more leisure and work fewer hours. Suppose wages of Tk 50, 80, 110, 140 and 170 an hour bring offers of 6, 8, 9, 8 and 7 hours a day. The curve rises, then bends backward. The third case is farm output. Within a season the crop is fixed by what was sown, so supply does not respond to price. A fourth case, a poor seller who needs a fixed sum of money, may sell more when the price is low, in order to raise the sum. This is rare and is limited to cases of great need.

These cases are narrow. Most goods, for most periods, follow the law of supply. Even the exceptions are explained by the way time and need limit the seller's choices, and the last section of this chapter shows that the time allowed is the main factor in how steep the supply curve is.

Before moving on. The exceptional curves are vertical (fixed quantity), backward bending (labour), and in rare cases falling (a seller in need). *You should now be able to say:* "The law of supply fails where the seller cannot or will not respond to a higher price in the usual way."

ELASTICITY OF SUPPLY

Chapter V measured the reaction of buyers. The same method now measures the reaction of sellers. The fisherman and the baker differ in how strongly their quantity responds. We need a number for that difference.

Elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. In words: elasticity of supply = percentage change in quantity supplied ÷ percentage change in price. In symbols, $E_s = \% \Delta Q_s \div \% \Delta P$. Because both quantity and price move in the same direction, the number is positive, and no sign is dropped. The point formula is as in Chapter V: $E_s = (\Delta Q_s \div \Delta P) \times (P \div Q_s)$.

The five types follow the same pattern as for demand. Supply is elastic when the quantity changes by a larger percentage than the price (E_s above 1), inelastic when by a smaller percentage (below 1), and unit elastic when by an equal percentage (exactly 1). It is perfectly elastic (infinite) when sellers will offer any amount at one price, and perfectly inelastic (zero) when the quantity cannot change. Figure 10 shows the five curves.

A few cases do not follow the law of supply, so their curves have other shapes.

Exceptional supply curve. A supply curve that does not slope upward in the usual way.

Fixed-quantity goods have a vertical supply curve.

Labour can bend backward, and a farmer's crop can be fixed for the season.

The exceptions are limited: most goods follow the law.

Exceptions arise from fixed quantity, from leisure and from need.

How strongly sellers react to price: the mirror of the elasticity of demand.

GO SLOWLY HERE.

Elasticity of supply. Percentage change in quantity supplied divided by percentage change in price.

Elasticity of supply is the percentage change in quantity over that in price.

The same five types appear, with the same dividing line at 1.

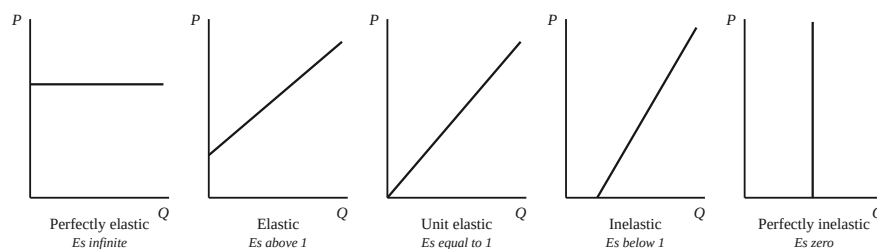


FIG. 10. Five types of elasticity of supply. In each panel, price is on the vertical axis and quantity on the horizontal axis.

The shape of a straight supply curve decides its type, wherever you measure.

Why a line through the origin always gives one.

A straight supply curve has a useful rule, which has no match for straight demand curves. If the line, when extended, meets the price axis above zero, supply is elastic at every point. If it passes through the origin, supply is unit elastic at every point, whatever its slope. If it meets the quantity axis to the right of zero, supply is inelastic at every point. Figure 10 shows all three lines.

The reason is simple. On a line through the origin, quantity is a fixed multiple of price: $Q_s = kP$. Then $P \div Q_s = 1 \div k$, and the slope $\Delta Q_s \div \Delta P$ is k , so $E_s = k \times (1 \div k) = 1$.

ELASTICITY OF SUPPLY ALONG THE BUN SUPPLY CURVE, $Q_s = 30P - 180$

PRICE (TK)	QUANTITY	ELASTICITY
8	60	4.0
10	120	2.5
12	180	2.0
14	240	1.75
16	300	1.6

The bun line cuts the price axis at Tk 6, so supply is elastic everywhere.

Case file: elasticity of supply at Tk 12 and the loaf line.

Three factors decide how readily supply responds: time, spare capacity, ease of storing and switching.

Market period. A time too short to change output.

Short run. A time in which at least one input is fixed.

Long run. A time in which all inputs can change.

Spare capacity and ease of storage or switching raise elasticity.

The table shows the bun supply curve. Its slope is 30 at every point. The line meets the price axis at Tk 6, so the rule says that supply is elastic at every point, and the table agrees: every value is above 1. The value falls toward 1 as the price rises, because the intercept matters less when the price is high. To read a row, find Q_s from the rule, and compute $30 \times P \div Q_s$. At Tk 12: $30 \times 12 \div 180 = 2.0$.

Case file, illustrative. At Tk 12 the elasticity of supply of buns is $30 \times 12 \div 180 = 2.0$. For loaves, supply is $Q_s = 15P$, a line through the origin, so elasticity is 1 at every price. At Tk 20, for example, the quantity is 300 and $15 \times 20 \div 300 = 1$.

Three main factors decide the elasticity of supply. The first is time. Over a very short **market period**, which is a time too short to change output, the quantity is fixed, and supply is perfectly inelastic: the fisherman's day. In the **short run**, a time in which at least one input is fixed, sellers can change variable inputs such as flour and labour, but not the oven, so supply is inelastic or moderately elastic. In the **long run**, a time in which all inputs can change, sellers can build ovens, enter or leave the market, and supply is more elastic. Chapter IX gives these terms their full meaning.

The second factor is spare capacity. A baker with an idle oven can bake more at once, so her supply is elastic. A baker whose oven runs all day cannot, and her supply is inelastic until she buys another. The third factor is how easily a seller can store the good or move inputs to another use. A seller of rice can store it and release it when prices rise, so supply is elastic. A seller of fresh fish cannot, so

supply is inelastic. A farmer who can switch land from jute to rice, or the reverse, has a more elastic supply of each.

Example 1. Given: the price of eggs rises from Tk 50 to Tk 55 a dozen, and a farm raises its offer from 200 to 240 dozen a week. *Find:* the elasticity of supply. *Formula:* $E_s = \% \Delta Q_s \div \% \Delta P$. *Steps.* Price change = $5 \div 50 \times 100 = 10$ per cent. Quantity change = $40 \div 200 \times 100 = 20$ per cent. $E_s = 20 \div 10 = 2.0$. *Answer:* elasticity of supply is 2.0, so supply is elastic. *Check:* the quantity changed by a larger percentage than the price (20 against 10), so the value must exceed 1, and it does.

First example: small clean numbers.

Example 2. Given: the bun market, $Q_s = 30P - 180$. The price rises from Tk 10 to Tk 14. *Find:* the arc elasticity of supply. *Formula:* $E_s = (\Delta Q_s \div \Delta P) \times (\text{average } P \div \text{average } Q_s)$. *Steps.* At Tk 10, $Q_s = 120$. At Tk 14, $Q_s = 240$. $\Delta Q_s = 120$. $\Delta P = 4$. Average $P = 12$. Average $Q_s = 180$. $E_s = (120 \div 4) \times (12 \div 180) = 30 \times 12 \div 180 = 360 \div 180 = 2.0$. *Answer:* the arc elasticity between Tk 10 and Tk 14 is 2.0. *Check:* the point values at the ends are 2.5 and 1.75, and 2.0 lies between them.

Second example: the arc method on the bun market.

Common slips. The slips are those of Chapter V: the wrong starting value, units instead of percentages, and price on top. A new slip is to call a steep supply curve inelastic. Test the curve by the point where it meets an axis, not by its steepness.

Common slips repeat those of Chapter V, with one new one.

Pause and check. At Tk 12 in the bun market, find the elasticity of supply, and compare it with the elasticity of demand from Chapter V. The answer is at the back of the book.

A final test: the same price, two elasticities.

Before moving on. The longer the time allowed, the greater the spare capacity, and the easier the storage and switching, the more elastic supply is. *You should now be able to say:* "A straight supply curve through the origin has an elasticity of one at every point."

Elasticity of supply depends on time, spare capacity and ease of switching.

SUMMARY

Supply is the quantity sellers are able and willing to sell at each price. Market supply adds the supplies of all sellers. The law of supply says that a higher price brings a larger quantity supplied, because of profit and rising cost. Input prices, technology, taxes and subsidies, expectations, the number of sellers and the weather shift the whole curve. A few supply curves are exceptional: vertical for fixed quantities, backward bending for labour. Elasticity of supply is the percentage change in quantity supplied divided by the percentage change in price. A straight supply curve through the origin has an elasticity of one, one that meets the price axis is elastic, and one that meets the quantity axis is inelastic. Time, spare capacity and ease of storage and switching decide the value.

The chapter in a few lines.

THINK IT THROUGH

Two bakeries sell loaves. Bakery X has an idle oven. Bakery Y runs both its ovens all day. The price of loaves rises by 10 per cent. Which bakery will raise its output more today, and why? Which will raise it more after six months, and why might the difference shrink? What would happen to the supply of loaves on a day when no flour could be delivered? Give your reasons in terms of the three factors of this chapter. The answer is at the back of the book.

Two bakeries, one price rise, and the role of spare capacity.

WHERE PEOPLE GO WRONG

A shift and a movement are different events.

Supply is not the stock held.

Steepness is not elasticity here either.

Many students say that supply "rises" when the price rises. When the price of the good rises, the quantity supplied rises, and supply does not change. Supply changes only when a determinant other than price changes.

A second mistake is to confuse supply with the stock a seller holds. Supply is the quantity offered at each price over a period.

A third mistake, as in Chapter V, is to read elasticity from steepness. A steep line that cuts the price axis can be elastic, and a flat line that cuts the quantity axis can be inelastic.

IN HISTORY

A frost in Brazil in 1975 showed how slowly supply can respond.

In July 1975 a severe frost struck the coffee-growing regions of Brazil and destroyed a large part of the crop. The price of coffee on world markets rose sharply over the following year. Higher prices could not call forth a quick increase in supply, because a new coffee tree takes several years to bear fruit. In the market period and the short run, supply was inelastic. Only in the long run, as growers planted new trees, did supply rise and the price ease. The episode is a standard example of the importance of time in supply.

REVIEW QUESTIONS AND PRACTICE

Six review questions test the chapter in new words.

Review questions. (1) Distinguish supply from quantity supplied. (2) How is market supply found from individual supplies? (3) State the law of supply and give two reasons for it. (4) List four determinants of supply, and say which way each shifts the curve. (5) Describe two exceptional supply curves and explain why they arise. (6) State the formula for elasticity of supply, and say what a straight supply curve through the origin tells us.

Items climb from a rule to elasticity and then link to demand.

Practice. Item 1. A supply rule is $Q_s = 20P - 100$. Fill in the quantity supplied at prices of Tk 6, 8, 10, 12 and 14. At what price does supply begin?

Item 2 shows a tax shifting supply.

Item 2. The government taxes sellers Tk 2 on each unit. The new supply rule becomes $Q_s = 20(P - 2) - 100$. Fill in the new quantities at Tk 8, 10, 12 and 14, and say by how much supply has fallen at each price.

Item 3 uses a messier table of three goods.

Item 3. Three goods each rise in price by 10 per cent. (a) Fresh fish: supply rises from 100 to 102 kg. (b) Garments: supply rises from 1,000 to 1,300 units. (c) Rice: supply rises from 10,000 to 11,000 kg. Find the elasticity of supply of each and name the type.

Items 4 and 5 test the through-origin rule and link supply to demand.

Item 4. (a) For $Q_s = 15P$, find the elasticity at Tk 20 and at Tk 40. (b) For $Q_s = 15P + 60$, find the elasticity at Tk 20. (c) For $Q_s = 15P - 60$, find the elasticity at Tk 20. Say which curves are elastic, unit elastic and inelastic.

Item 5 links supply elasticity to demand elasticity.

Item 5. In the bun market, demand is $Q_d = 540 - 30P$ and supply is $Q_s = 30P - 180$. Find the elasticity of each at Tk 12. The price then rises by 5 per cent, to Tk 12.60. Find the new quantity demanded and the new quantity supplied, and say by how much one exceeds the other.

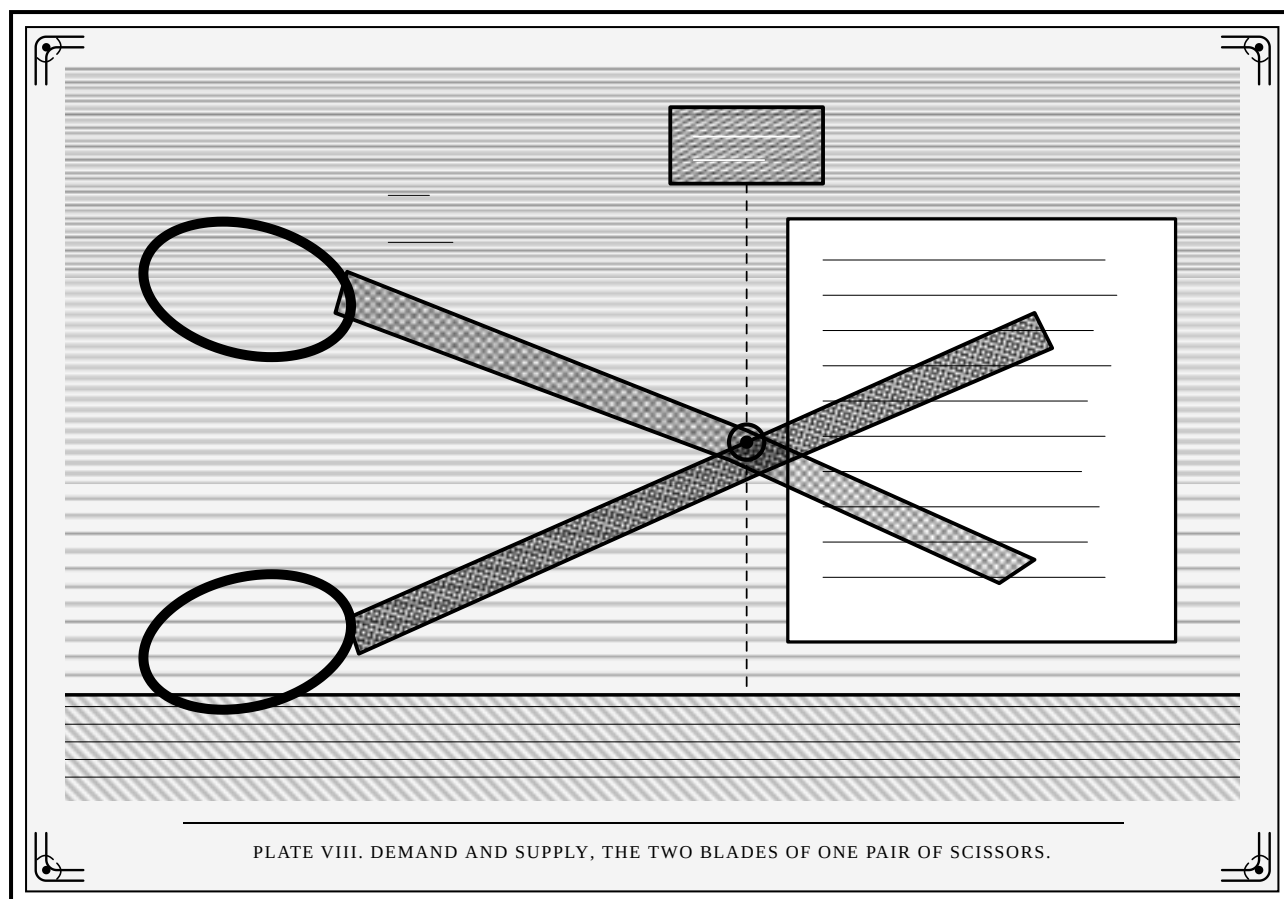


PLATE VIII. DEMAND AND SUPPLY, THE TWO BLADES OF ONE PAIR OF SCISSORS.

CHAPTER VIII

Markets and Equilibrium

Who decides the price, when no one is in charge?



Buyers want low prices and sellers want high ones, yet markets settle on a price at which what is offered matches what is wanted. This chapter brings demand and supply together, shows how the balance is found and restored, how it moves when conditions change, and who gains from it.

A vegetable market's prices fall through the afternoon with no one ordering it.

How does a price find the level that clears a market?

This chapter combines demand, elasticity, surplus and supply.

In economics, a market is any dealing between buyers and sellers, not only a place.

Market. Word in use: in daily life, a place to shop; in economics, buyers and sellers dealing with each other.

A competitive market has many buyers and sellers, none strong enough to set the price.

Competitive market. A market with so many buyers and sellers that none can change the price alone.

Other kinds of market exist and are studied in Chapter XI.

A market connects buyers and sellers of one good; a competitive market has too many to matter singly.

This is the main idea of the chapter, and it repays slow reading.

GO SLOWLY HERE.

Where the quantity wanted equals the quantity offered, the market balances.

In the morning, tomatoes at a town market cost Tk 60 a kilogram. By the evening the same tomatoes cost Tk 40, and almost nothing is left on the stalls. No official has ordered the fall, and no seller has announced a meeting. Each seller has simply watched the buyers, noticed the unsold heap, and lowered the price a little. Each buyer has noticed the lower price and bought a little more. By closing time sellers have sold nearly all they brought, and buyers have found what they wanted.

Here is the puzzle. Buyers act for themselves and so do sellers. Yet the result is a price at which the amount wanted is close to the amount offered. How does this happen? This chapter answers the question with the demand curve of Chapters IV and V and the supply curve of Chapter VII. It also asks what happens when something changes, and who gains from the trade.

You will need demand and its shifts (Chapter IV), consumers' surplus (Chapter VI), and supply and its shifts (Chapter VII).

WHAT A MARKET IS

The word **market** has two meanings. In daily life it means a place where people shop, such as the market at the end of the road. In economics, a market means the dealing between the buyers and sellers of a good, wherever they are. It need not be one place. There is a market for mobile phones, for jute and for taxi rides, and these buyers and sellers may be scattered over a whole country, or linked only by telephone and the internet. What matters is that buyers and sellers are in contact, and that the same good is being bought and sold.

The simplest kind of market is the competitive market. A **competitive market** means a market with so many buyers and sellers that no single one can change the price alone. Each bakery in the neighbourhood bun market is small. If one bakery raised its price a lot, its customers would walk to another. Each customer is also small, and one customer's choice cannot move the price. In such a market the price is set by all buyers and sellers together, and demand and supply explain it.

Not every market is competitive. In some a single seller controls the supply, and in others a few large sellers watch one another. Chapter XI describes these types, and Chapters XII to XIV study them. This chapter treats the competitive market, because it is the clearest case and the basis of the others.

Before moving on. A market is the dealing between buyers and sellers of a good, and a competitive market is one in which no one can set the price alone. You should now be able to say: "A market need not be a place."

EQUILIBRIUM PRICE AND QUANTITY

The meeting of demand and supply is the central idea of this chapter. It can be seen in three ways: in a table, on a graph and in algebra. All three give the same answer, and each helps to understand the other two.

BALANCE IN A TABLE

Take the neighbourhood bun market of the case file. Demand is $Q_d = 540 - 30P$ a day (Chapter IV) and supply is $Q_s = 30P - 180$ (Chapter VII). The table sets them side by side. Read a row: at Tk 10, buyers want 240 buns and sellers offer 120. The last column subtracts quantity supplied from quantity demanded.

DEMAND AND SUPPLY IN THE BUN MARKET, PER DAY

PRICE (TK)	QUANTITY DEMANDED	QUANTITY SUPPLIED	DEMANDED MINUS SUPPLIED
6	360	0	360
8	300	60	240
10	240	120	120
12	180	180	0
14	120	240	-120
16	60	300	-240
18	0	360	-360

Look at the last column. It is positive when the price is low, negative when the price is high, and zero at Tk 12. At that price buyers want 180 buns and sellers offer 180. We say that the market is in **market equilibrium**, which means a state in which the quantity demanded equals the quantity supplied. You met the word *equilibrium* in Chapter III, where a buyer had no reason to change her purchases. The meaning is the same here: nobody has a reason to change. The price at which it happens, Tk 12, is the **equilibrium price**, and the quantity, 180, is the equilibrium quantity.

BALANCE ON A GRAPH

Now draw both curves on one graph, with price on the vertical axis and quantity on the horizontal axis. The demand curve slopes down and the supply curve slopes up, so they must cross, and they cross once. In Figure 11 the crossing is at E, with price Tk 12 and quantity 180. Every other price gives a gap between the curves. At Tk 10 the supply curve is to the left of the demand curve, and at Tk 16 it is to the right. The gaps are the quantities in the last column of the table.

Only at Tk 12 do the two quantities match.

Market equilibrium. A state in which quantity demanded equals quantity supplied.

Equilibrium price. The price at which quantity demanded equals quantity supplied.

On a graph, equilibrium is the point where the two curves cross.

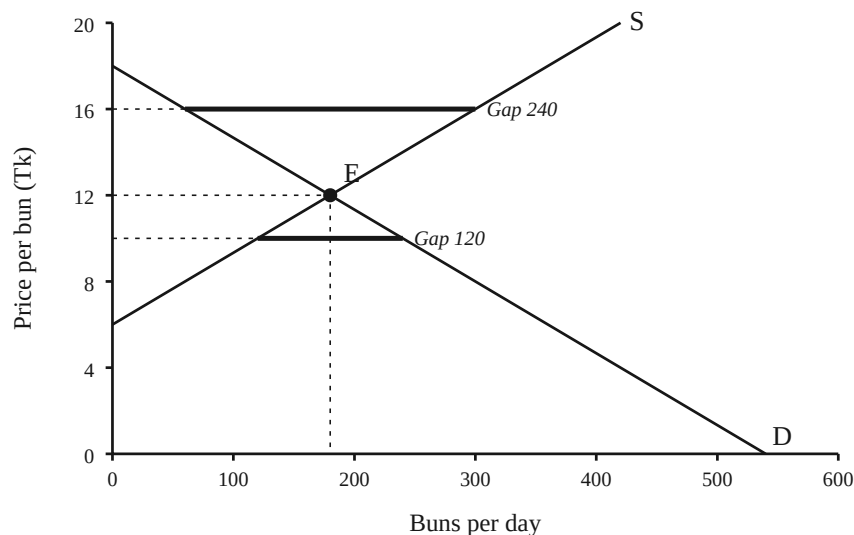


FIG. 11. The bun market in equilibrium at E (Tk 12, 180 buns). At Tk 10 demand exceeds supply by 120; at Tk 16 supply exceeds demand by 240.

Case file, illustrative. The neighbourhood bun market is in equilibrium at a price of Tk 12 and a quantity of 180 buns a day, where $Q_d = 540 - 30P$ equals $Q_s = 30P - 180$.

To find the price algebraically, set Q_d equal to Q_s and solve.

First example: small clean numbers.

Second example: the bun market.

Common slips: the wrong equation, and the idea that equilibrium means fair.

A test of the balance idea.

Equilibrium is the price at which Q_d equals Q_s ; solve by setting the rules equal.

What happens at a wrong price, and why the price moves toward Tk 12.

GO SLOWLY HERE.

Excess demand. The quantity demanded is greater than the quantity supplied at the price.

Excess supply. The quantity supplied is greater than the quantity demanded at the price.

Excess demand pushes the price up, step by step.

Excess supply pushes the price down.

BALANCE IN ALGEBRA

The graph and the table suggest a method. At equilibrium, quantity demanded equals quantity supplied, so we put the two rules equal and solve for the price. Then we put that price into either rule to find the quantity. The first example uses very small numbers, so that the method is the only new thing.

Example 1. Given: demand $Q_d = 10 - P$ and supply $Q_s = 2P - 2$. Find: the equilibrium price and quantity. Formula: equilibrium where $Q_d = Q_s$. Steps. Set $10 - P = 2P - 2$. Add P to both sides: $10 = 3P - 2$. Add 2: $12 = 3P$. Divide by 3: $P = 4$. Quantity: $Q_d = 10 - 4 = 6$, and $Q_s = 2 \times 4 - 2 = 6$. Answer: equilibrium price 4 and quantity 6. Check: both rules give 6 at a price of 4, so the two sides match.

Example 2. Given: $Q_d = 540 - 30P$ and $Q_s = 30P - 180$. Find: the equilibrium price and quantity. Steps. Set $540 - 30P = 30P - 180$. Add $30P$ to both sides: $540 = 60P - 180$. Add 180: $720 = 60P$. Divide by 60: $P = 12$. Quantity: $Q_d = 540 - 30 \times 12 = 540 - 360 = 180$, and $Q_s = 30 \times 12 - 180 = 180$. Answer: the price is Tk 12 and the quantity is 180 buns. Check: the table shows a gap of zero at Tk 12, and the graph shows the crossing at E.

Common slips. Do not set price equal to quantity. Set one quantity rule equal to the other. Do not forget to find the quantity after the price. Do not take "equilibrium" to mean that the price is fair or that everyone is pleased. It means only that the quantities match and nobody can gain by changing the price on his or her own.

Pause and check. In the table, at what price would buyers want 120 buns, and how many would sellers offer at that price? What does the gap tell you? The answer is at the back of the book.

Before moving on. The equilibrium price is where the two curves cross, and where $Q_d = Q_s$. You should now be able to say: "At the equilibrium price, the quantity buyers want equals the quantity sellers offer."

FORCES ABOVE AND BELOW EQUILIBRIUM

The equilibrium price is the one the market tends toward, but a market may begin at any price. Suppose the bun price is Tk 10 on a certain morning. Buyers want 240 buns and bakeries offer 120. There is a gap of 120.

When the quantity demanded is greater than the quantity supplied, there is **excess demand**. When the quantity supplied is greater than the quantity demanded, there is **excess supply**. At Tk 10 there is excess demand of 120 buns. At Tk 16 buyers want 60 and sellers offer 300, so there is excess supply of 240 buns. This book uses these two words for a market imbalance. It keeps the word *surplus* for the gains of buyers and sellers, which Section 8.5 describes.

Follow the market from Tk 10. First, the bakeries sell out early, and some customers go home without buns. Second, some customers offer a little more to be served, and bakeries notice that they can raise the price without losing sales. Third, the price rises. Fourth, as it rises, buyers want fewer buns and bakeries offer more, so the gap narrows. The process continues until the price reaches Tk 12, where the gap is zero and nothing pushes the price further.

At Tk 16 the story runs the other way. Bakeries find unsold buns at the end of the day. They cut prices to sell them. As the price falls, buyers want more and bakeries offer fewer, so the gap narrows until the price is Tk 12. Whether the price starts too high or too low, the market moves toward the equilibrium.

This self-correcting process is the **price mechanism**, which means the way changes in price bring the quantity demanded and the quantity supplied into balance. It also answers the opening puzzle. The tomato sellers did not need an order. Excess supply of tomatoes pushed the price down, and the lower price drew in the buyers.

Case file, illustrative. At a bun price of Tk 10, quantity demanded is 240 and quantity supplied is 120, so there is excess demand of 120. At Tk 16, quantity demanded is 60 and quantity supplied is 300, so there is excess supply of 240.

Sometimes the government does not want the market to find its own price. A **price ceiling** is the highest price that the law allows. A **price floor** is the lowest price that the law allows. Suppose a ceiling of Tk 10 is placed on buns. The ceiling is below the equilibrium price, so it binds. At Tk 10, sellers offer only 120 buns, though buyers want 240. The quantity traded is the smaller of the two, 120, which is less than the 180 that would have been traded without the ceiling. Excess demand of 120 remains. In practice it leads to queues, to sales to favoured customers, and to illegal sales at higher prices. Sellers may also cut quality. A ceiling set above the equilibrium price, such as Tk 14, changes nothing, because the market price of Tk 12 is already below it.

A floor works the other way. A floor of Tk 16 is above the equilibrium price. Buyers want only 60 buns at that price, and sellers offer 300. Excess supply of 240 remains, and only 60 are sold. Governments sometimes set floors to help sellers, such as farmers or workers. The benefit goes only to those who sell, and some sellers cannot find buyers at all.

Before moving on. Excess demand raises the price and excess supply lowers it, until the market balances. A ceiling or a floor that binds stops this. *You should now be able to say:* "A price ceiling below the equilibrium price leaves excess demand, and the quantity traded is the quantity supplied."

Price mechanism. The way changes in price bring quantity demanded and supplied into balance.

A government may hold the price away from equilibrium, with predictable results.

Price ceiling. The highest price allowed by law.

Price floor. The lowest price allowed by law.

A floor above equilibrium leaves goods unsold.

The price mechanism restores balance; controls prevent it.

SHIFTS AND NEW EQUILIBRIA

Chapters IV and VII showed that determinants shift the curves. When a curve shifts, the crossing point moves, and the market finds a new equilibrium. Three steps find the new equilibrium on a graph. First, decide which curve is affected. Second, decide whether it moves right or left. Third, read the new crossing.

Take flour first. A rise in the price of flour adds Tk 2 to the cost of each bun, and supply shifts left, as in Chapter VII. The new supply rule is $Q_s = 30P - 240$. Demand is unchanged. Set $540 - 30P = 30P - 240$. Then $780 = 60P$, so $P = 13$. The quantity is $540 - 30 \times 13 = 150$, and the supply rule agrees: $30 \times 13 - 240 = 150$. The price rises from Tk 12 to Tk 13, and the quantity falls from 180 to 150. Buyers pay Tk 1 more of the Tk 2 cost rise, and the bakeries absorb the rest.

Now take a festival. In Chapter IV, demand rose by 60 buns at every price, to $Q_d = 600 - 30P$. Supply is unchanged at $Q_s = 30P - 180$. Set $600 - 30P = 30P - 180$. Then $780 = 60P$ and $P = 13$. The quantity is $600 - 30 \times 13 = 210$. Both the price and the quantity rise. Notice that the price is Tk 13 in both stories, but the quantity is 150 in one and 210 in the other. The direction of the quantity tells us which curve moved.

Case file, illustrative. After the flour price rise, supply is $Q_s = 30P - 30 \times 8$, that is $30P - 240$, and the new equilibrium is Tk 13 and 150 buns. After the festival rise in demand, to $Q_d = 600 - 30P$, the equilibrium is Tk 13 and 210 buns.

When demand or supply shifts, the equilibrium moves to a new place.

A rise in the cost of flour moves supply left, raising the price and cutting the quantity.

A rise in demand moves demand right, raising both price and quantity.

THE FOUR SINGLE SHIFTS, WITH BUN EXAMPLES

CHANGE	PRICE	QUANTITY	BUN EXAMPLE
Demand rises ($Q_d = 600 - 30P$)	Rises	Rises	Tk 13, 210
Demand falls ($Q_d = 480 - 30P$)	Falls	Falls	Tk 11, 150
Supply rises ($Q_s = 30P - 120$)	Falls	Rises	Tk 11, 210
Supply falls ($Q_s = 30P - 240$)	Rises	Falls	Tk 13, 150

The table gives the four cases and one bun example for each.

The table gives the four cases. Read a row: if demand rises, price and quantity both rise, and the bun market moves to Tk 13 and 210. If supply rises, the price falls but the quantity rises. Check the last row against the flour example above.

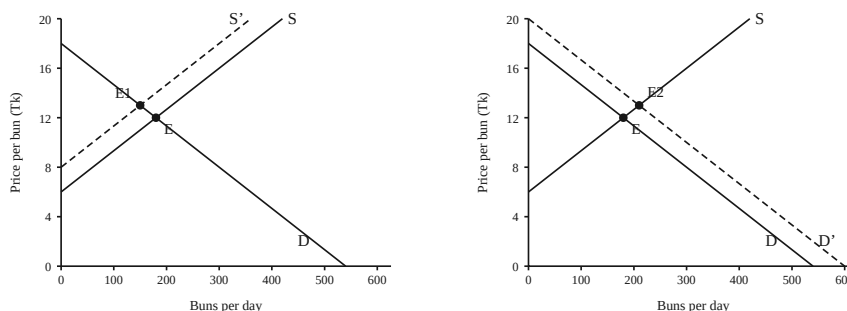


FIG. 12. Two shifts. Left: dearer flour moves supply left, from E to E1 (Tk 13, 150). Right: a festival moves demand right, from E to E2 (Tk 13, 210).

When both curves shift, one of price or quantity is certain, but not both.

What if both curves shift at once? Then one of the two results is certain and the other depends on the sizes. If demand rises (to $600 - 30P$) and supply falls (to $30P - 240$), the price rises for certain, to Tk 14. The quantity stays at 180, but it could rise or fall with other sizes. If demand rises (to $600 - 30P$) and supply rises (to $30P - 120$), the quantity rises for certain, to 240. The price stays at Tk 12, but it could rise or fall with other sizes. Check each case by solving the two rules.

Third example: a shift of supply.

Example 3. Given: demand $Q_d = 540 - 30P$, and supply after a rise in flour prices $Q_s = 30P - 240$. Find: the new equilibrium. Steps. Set $540 - 30P = 30P - 240$, so $780 = 60P$, and $P = 13$. Then $Q = 540 - 30 \times 13 = 150$. Answer: Tk 13 and 150 buns. Check: the supply rule gives $30 \times 13 - 240 = 150$, which equals the demand quantity.

A shift moves the crossing point; read the new one.

Before moving on. A rise in demand lifts price and quantity, a fall in supply lifts the price and lowers the quantity, and so on. You should now be able to say: "To find the effect of a change, decide which curve shifts, in which direction, and read the new crossing."

GAINS FROM TRADE

A trade benefits both sides, and the gains can be measured.

GO SLOWLY HERE.

Producers' surplus. The price a seller receives minus the lowest price the seller would accept.

Why do buyers and sellers trade? Because both gain. Chapter VI measured the buyer's gain as consumers' surplus. The seller's gain has a matching measure.

Producers' surplus means the price a seller receives minus the lowest price the seller would have accepted. The lowest price for each unit is read from the supply curve, because it shows the least that will bring that unit to market. In the bun market, sellers will offer nothing below Tk 6, so the first bun would have been sold for as little as Tk 6. At the market price of Tk 12, a seller of that first bun gains Tk 6 above the minimum. Added

over all units, the gains form a triangle between the price line and the supply curve.

At the equilibrium, 180 buns are sold. Consumers' surplus is the triangle below the demand curve and above the price. Its base is 180 and its height is $18 - 12 = 6$, so its area is $\frac{1}{2} \times 180 \times 6 = \text{Tk } 540$. Producers' surplus is the triangle above the supply curve and below the price. Its base is 180 and its height is $12 - 6 = 6$, so its area is $\frac{1}{2} \times 180 \times 6 = \text{Tk } 540$. Figure 13 shows both.

The two triangles at Tk 12 each have an area of Tk 540.

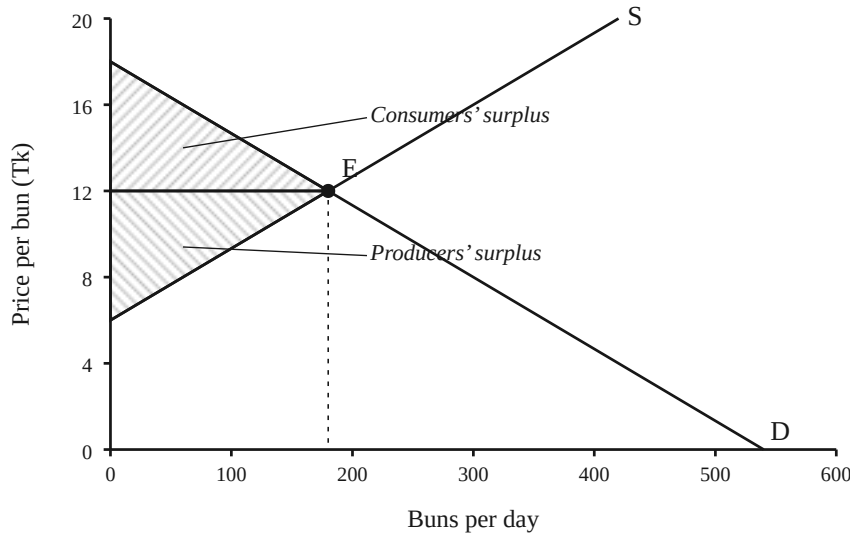


FIG. 13. The gains from trade in the bun market. Consumers' surplus (above the price) and producers' surplus (below the price) are each Tk 540 a day.

The sum of the two is the **total surplus**, which means consumers' surplus plus producers' surplus. Here it is $540 + 540 = \text{Tk } 1,080$ a day. This is the total gain of the market from the trade of 180 buns. Any change in the market that reduces the total surplus is a loss to society, even if some people gain.

Total surplus. Consumers' surplus plus producers' surplus.

Case file, illustrative. At the equilibrium, consumers' surplus is Tk 540 a day, producers' surplus is Tk 540, and total surplus is Tk 1,080.

The two surpluses help to judge public policy. Take a tax of Tk 3 on every bun sold, paid by the bakeries. Supply becomes $Q_s = 30(P - 3) - 180 = 30P - 270$. This is the fourth example.

Surplus lets us measure what a tax does.

Example 4. Given: $Q_d = 540 - 30P$, and after a Tk 3 tax $Q_s = 30P - 270$. Find: the new equilibrium, and the change in the surpluses. Steps. Set $540 - 30P = 30P - 270$. Then $810 = 60P$, so $P = 13.5$ and $Q = 540 - 405 = 135$. Buyers pay Tk 13.50. Bakeries keep $13.50 - 3 = \text{Tk } 10.50$. Tax revenue $= 3 \times 135 = \text{Tk } 405$. Consumers' surplus $= \frac{1}{2} \times 135 \times (18 - 13.5) = \frac{1}{2} \times 135 \times 4.5 = \text{Tk } 303.75$. Producers' surplus $= \frac{1}{2} \times 135 \times (10.5 - 6) = \text{Tk } 303.75$. Answer: total surplus plus tax revenue is $303.75 + 303.75 + 405 = \text{Tk } 1,012.50$, against Tk 1,080 before. Check: the loss is $1,080 - 1,012.50 = \text{Tk } 67.50$, which equals $\frac{1}{2} \times 3 \times 45$, where 45 is the fall in quantity from 180 to 135. Chapter XIII gives this loss a name.

Fourth example: a tax on buns.

The result shows that the buyers pay Tk 1.50 more ($13.50 - 12$) and the bakeries receive Tk 1.50 less ($12 - 10.50$). Both share the tax, though the law taxed only the bakeries. A subsidy works in the opposite way: it lowers the cost, shifts supply right, and raises the surpluses of both sides, though the government pays for it. Trade between countries can be judged the same way, by asking whether the gains to buyers and sellers outweigh the losses to others.

A tax shifts the price by less than the tax, and both sides pay.

Two surpluses add to a total, and policy is judged by its effect on the total.

Before moving on. Consumers' surplus plus producers' surplus makes the total surplus, and a tax or a control reduces it. *You should now be able to say:* "Producers' surplus is what a seller gains above the lowest price the seller would accept."

The chapter in a few lines.

A market is the dealing of buyers and sellers of a good. In a competitive market no one can set the price alone. Market equilibrium is where quantity demanded equals quantity supplied, found in a table, on a graph or by solving $Q_d = Q_s$. For the bun market it is Tk 12 and 180 buns. Excess demand pushes the price up and excess supply pushes it down, which is the price mechanism. A price ceiling below equilibrium leaves excess demand, and a floor above it leaves excess supply. A shift of demand or supply moves the equilibrium: a rise in demand lifts price and quantity, and a fall in supply lifts the price and lowers the quantity. Consumers' surplus and producers' surplus add to total surplus, which a tax reduces.

A festival, a rise in demand, and a price limit.

The market for sweets in a town has demand $Q_d = 2,000 - 4P$ and supply $Q_s = 6P - 1,000$, in kilograms a day, with the price P in taka a kilogram. Find the equilibrium. Before a festival, demand rises by 500 kilograms at every price. Find the new equilibrium. The town council then sets a ceiling at the old price. How many kilograms are traded, and how large is the excess demand? Who gains and who loses from the ceiling, and what may happen in practice? The answer is at the back of the book.

A movement and a shift can be confused.

Students often say that the price rise caused "a fall in demand". A price rise caused by a fall in supply moves us up the demand curve, and demand itself has not changed.

Equilibrium is not fairness, and it is not permanent.

Another mistake is to think that equilibrium means a fair price or one that lasts. It is only the price at which the quantities match, and it moves whenever a determinant changes.

A ceiling that does not bind does nothing.

A third mistake is to think that every ceiling lowers the price. A ceiling above the equilibrium price has no effect.

American petrol queues in 1973-74 showed the cost of a binding ceiling.

In 1971 the United States government placed controls on prices, and later extended limits on the price of petrol. After October 1973, when oil supplies were cut and world prices rose sharply, the controlled price of petrol stayed below the level at which demand matched supply. In the winter of 1973-74 drivers queued for hours at filling stations, and some stations ran out of fuel. Some states allowed purchases only on alternate days, depending on the last digit of the number plate. The episode is a standard example of excess demand under a ceiling.

SUMMARY

THINK IT THROUGH

WHERE PEOPLE GO WRONG

IN HISTORY

SOURCES

Alfred Marshall's *Principles of Economics* (1890) set out the equilibrium of demand and supply. He compared the two to the blades of a pair of scissors: it is idle to ask which blade cuts the paper, because both do. The comparison explains why a price depends on both buyers and sellers.

Marshall's picture of demand and supply as two blades of a pair of scissors.

REVIEW QUESTIONS AND PRACTICE

Review questions. (1) How does the meaning of "market" in economics differ from its meaning in daily life? (2) What is a competitive market? (3) Define equilibrium price and equilibrium quantity, and name three ways to find them. (4) Describe what happens, step by step, when the price is above the equilibrium. (5) Explain what happens when a ceiling is set below the equilibrium price. (6) Name the four single shifts and say what each does to price and quantity.

Six review questions test the chapter in new words.

Practice. Item 1. Demand and supply of notebooks are as follows, with price in taka and quantities per day. Price 20, 30, 40, 50, 60. Demand 100, 80, 60, 40, 20. Supply 20, 40, 60, 80, 100. Find the equilibrium. At Tk 30, what is the gap, and what will happen to the price?

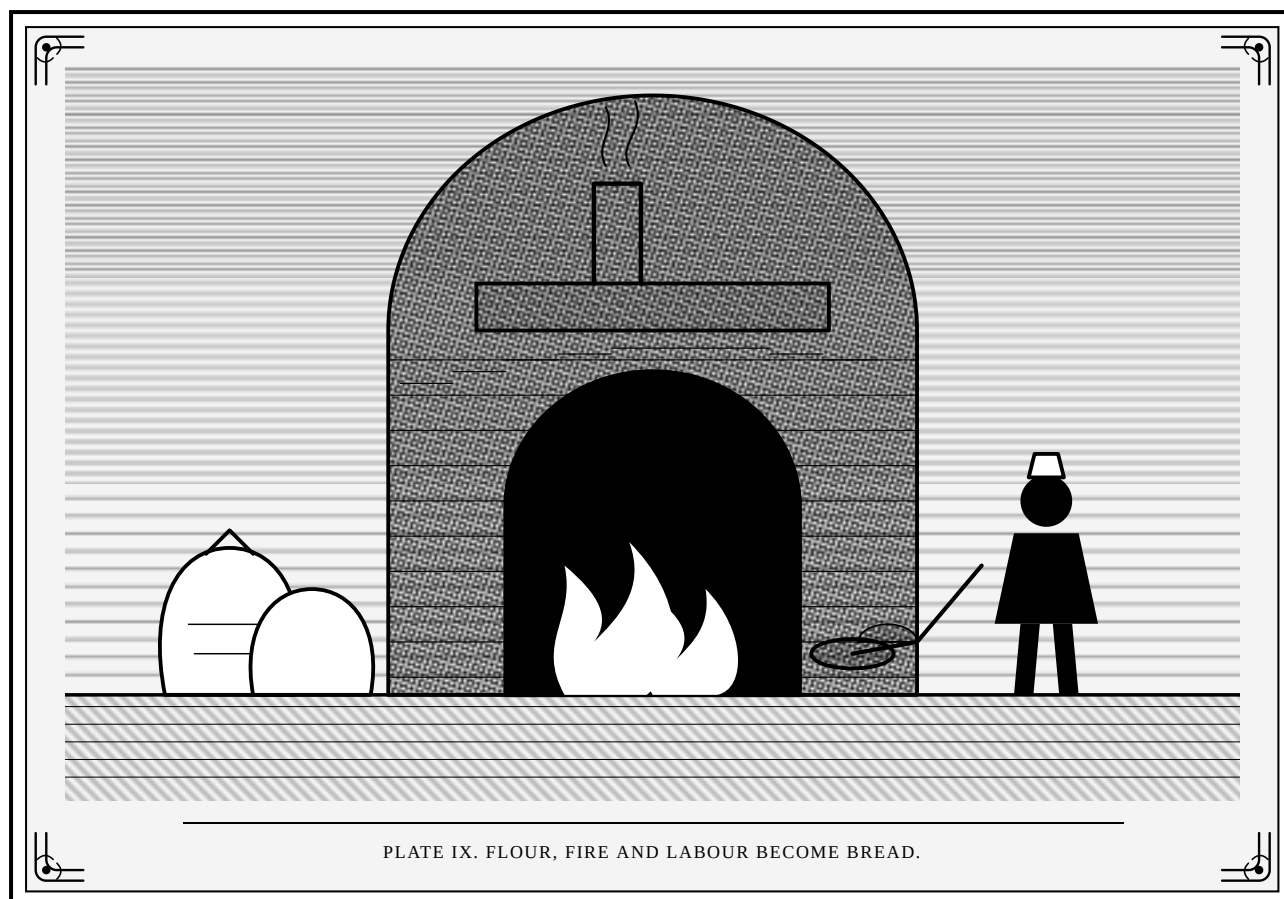
Items climb from schedules to equations, taxes and controls.

Item 2. $Q_d = 120 - 4P$ and $Q_s = 6P - 30$. Find the equilibrium price and quantity.

Item 3. A tax of Tk 5 per unit is placed on sellers in the market of Item 2, so that supply becomes $Q_s = 6P - 60$. Find the new price paid by buyers, the price received by sellers, the quantity, and the tax revenue.

Items 3 and 4 add a tax and a ceiling to the market of Item 2.

Item 4. In the market of Item 2, a ceiling of Tk 12 is set. Find the quantity demanded, the quantity supplied, the excess demand, and the quantity actually traded.



CHAPTER IX

Production

How do flour, fire and labour become bread?



This chapter follows inputs into a workshop and output out of it. It shows why each extra worker may add less than the one before, and what happens when a firm grows all its inputs together. It ends with the choice of the cheapest mix of inputs.

A puzzle: more bakers bring more bread, until they do not.

What you should know before reading on.

GO SLOWLY HERE.

Start at home: inputs go into a kitchen and a meal comes out.

Production means making goods and services.

Production. Making goods and services that satisfy wants.

Inputs go in; output comes out.

Input. Any resource used to make a good or service.

Output. The goods and services that production makes.

Fixed inputs cannot change quickly; variable inputs can.

Fixed input. An input whose amount cannot be changed in the period studied.

Variable input. An input whose amount can be changed in the period studied.

The short run keeps something fixed; the long run does not.

Short run. The period in which at least one input is fixed.

Long run. The period in which every input can be changed.

The production function links inputs to the greatest output.

Production function. The link between amounts of inputs and the greatest output they can make.

The bakery has one oven and one small kitchen. At first the owner works with a single baker, and the loaves come slowly. A second baker arrives and the work speeds up, because one can shape the dough while the other watches the oven. A third baker helps again. Then the kitchen fills. Bakers wait for the oven and block each other's way to the trays. By the seventh baker, no extra bread is made at all. Yet every baker is paid the same wage each day. Why does more labour first help and then stop helping? And what would change if the owner bought a second oven as well? This chapter answers both questions, and in doing so it explains how any firm turns inputs into output.

You will need: the four kinds of resource from Chapter I (land, labour, capital, entrepreneur); the idea of the margin from Chapter III; and the words short run and long run, which Chapter VII previewed. This chapter teaches those two words in full.

PRODUCTION AND THE PRODUCTION FUNCTION

Think of cooking rice at home. You use rice, water, fuel, a pot and your own time. At the end there is a meal. The things you used are the inputs, and the meal is the output. Nobody calls a home kitchen a factory, yet the process is the same one that a bakery follows, only smaller. Keep this picture in mind, because every term in this section can be read in the kitchen first.

Economists give the name **production** to any activity that makes goods or services that satisfy wants. A baker who bakes bread produces a good. A bus driver who carries passengers produces a service. Moving, storing and selling a good also count as production, because each of them makes the good more useful to the buyer.

An **input** is any resource used in production. Chapter I sorted resources into land, labour, capital and entrepreneur. For the bakery, the flour, the oven, the baker's work and the owner's planning are all inputs. The **output** is what comes out. For the bakery, it is loaves. We write output as a number of units in a period, such as 120 loaves a day.

Some inputs are easy to change and some are not. A baker can be hired next week. An oven cannot be built next week. A **fixed input** is an input whose amount cannot be changed in the period we are studying. The oven is fixed. A **variable input** is an input whose amount can be changed in that period. The number of bakers is variable. Notice that "fixed" describes a period of time, not the nature of the thing. Over a longer time, the oven too can be replaced.

This idea gives us two periods. The **short run** is the period in which at least one input is fixed. The **long run** is the period in which every input can be changed. These names do not count days or months. For a bakery the long run may be a few weeks, since it can install another oven in that time. For a steel mill it may be many years. Sections 9.2 and 9.3 are about the short run. Section 9.4 is about the long run.

We now need one rule that joins the two sides. The **production function** is the link between the amounts of inputs used and the greatest output they can make. In words: output depends on inputs. In symbols, $Q = f(L, K)$. Here Q is output, L is labour, K is capital, and f means "depends on". A table that lists the loaves made by one baker, two bakers, three bakers and so on, with one oven, is a production function in its plainest form. The words "greatest output" matter. The function assumes that the firm uses its inputs well and with the methods it knows.

Before moving on. Production turns inputs into output. Some inputs are fixed and some are variable. In the short run at least one input is fixed; in the long run all can change. The production function links inputs to output.

Pause and check. A tailor owns two sewing machines and can hire more tailors at short notice. Which input is fixed, and which is variable? Name one change that she could make only in the long run.

You should now be able to say: production is making goods and services from inputs, and the production function shows the most output that each amount of inputs can make.

Recap of the section.

A check on fixed and variable inputs.

The sentence to test yourself against.

PRODUCT IN THE SHORT RUN

To study the short run, we measure output in three ways. The **total product** is all the output made with a given number of workers. The **average product** is the output per worker. The **marginal product** is the extra output made when one more worker is hired. "Marginal" means "at the edge", as it did for utility in Chapter III: it looks at the last unit added, not at the whole amount.

In words: average product equals total product divided by the number of workers. Marginal product equals the new total product minus the old total product. In symbols, $AP = TP \div L$, and $MP = \text{change in } TP \div \text{change in } L$, where TP is total product. When we add one worker at a time, the change in L is 1, so MP equals the change in TP.

Case file, illustrative. Step 9. The bakery has one oven, which is fixed in the short run, and it hires bakers, which are variable. With one to seven bakers, the loaves made per day are 20, 50, 90, 120, 140, 150 and 150.

Given. One oven. Total product for one to seven bakers: 20, 50, 90, 120, 140, 150, 150. *Find.* The average product and the marginal product at each number of bakers. *Formula.* $AP = TP \div L$; $MP = TP \text{ now} - TP \text{ before}$.

Steps. Take three bakers first. $AP = 90 \div 3 = 30$ loaves per baker. $MP = 90 - 50 = 40$ loaves. For the first baker, "TP before" is zero, because no baker means no loaves: $MP = 20 - 0 = 20$. For four bakers, $AP = 120 \div 4 = 30$ and $MP = 120 - 90 = 30$. We repeat the same two lines for every row. The table shows the result. We have written the average product to one decimal place.

Three ways to measure the output of a variable input.

Total product. All the output made with a given amount of the variable input.

Average product. Output per unit of the variable input.

Marginal product. The extra output from one more unit of the variable input.

The three measures written as formulas.

The bakery's output with one oven, from one to seven bakers.

A worked example: setting out the problem.

A worked example: the arithmetic.

PRODUCT OF THE BAKERY'S KITCHEN, PER DAY, ONE OVEN

BAKERS	TOTAL PRODUCT	AVERAGE PRODUCT	MARGINAL PRODUCT
1	20	20.0	20
2	50	25.0	30
3	90	30.0	40
4	120	30.0	30
5	140	28.0	20
6	150	25.0	10
7	150	21.4	0

Answer. The marginal product rises to 40 with the third baker, then falls to zero with the seventh. The average product rises to 30, then falls. *Check.* The marginal products must add up to the total product. For three bakers, $20 + 30 +$

The answer and a test of it.

Marginal and average product follow a fixed pattern.

40 = 90, which matches the table. For seven bakers, $20 + 30 + 40 + 30 + 20 + 10 + 0 = 150$.

The table shows three rules. First, while marginal product is above zero, total product rises. At the seventh baker, marginal product is zero, so total product stops rising. Second, marginal product pulls the average with it. Think of your scores in a class. A new score above your average raises the average, and a new score below it lowers the average. The third baker adds 40 loaves, which is above the old average of 25, so the average rises to 30. The fifth baker adds 20, below the old average of 30, so the average falls to 28. Third, when the new unit adds exactly the old average, the average does not change. The fourth baker adds 30, the same as the old average of 30, so the average stays at its highest point. Marginal product therefore cuts the average product curve at the top of that curve.

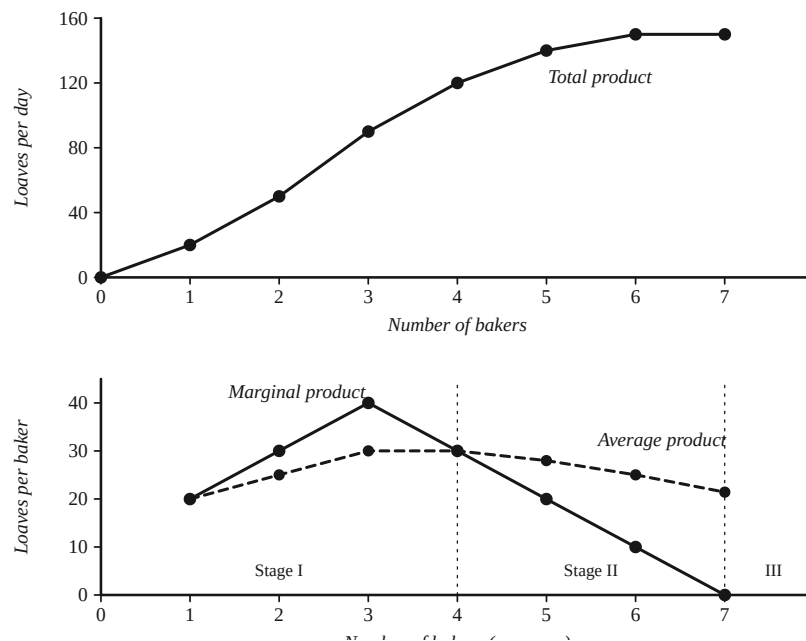


FIG. 14. *The bakery's product: total product above; marginal and average product below. Marginal product cuts average product at its highest point.*

Three slips to avoid.

Common slips. Do not find marginal product by dividing total product by the number of workers; that gives the average. Do not forget that the first marginal product is measured from zero. Do not read marginal product as the output of the whole group: it belongs to the last worker added.

Recap of the section.

Before moving on. Total product is all the output. Average product is output per worker. Marginal product is the extra output of the last worker. When marginal product is above average product, the average rises; when it is below, the average falls.

A check on the arithmetic.

Pause and check. A garage has one workshop, and two mechanics repair 6 cars a week. With three mechanics the garage repairs 10. What is the marginal product of the third mechanic, and what is the average product of three?

THE LAW OF DIMINISHING RETURNS

GO SLOWLY HERE.

A farmer's plot shows the problem in a different setting.

Picture a farmer with one acre of land. One worker can plant, weed and water only part of it, so the land is poorly tended. A second worker lets every task be done on time, and the harvest rises sharply. A third helps again. By the fifth worker, every task is done, and the land can use no more care. The sixth worker

mostly stands and watches. The land, which is the fixed input, has become the limit. The bakery's oven plays the same part in the table above. This pattern is so regular that economists state it as a law.

The **law of diminishing returns** says this: when more and more of a variable input is added to a fixed input, the extra output from each added unit becomes smaller beyond some point. It is a statement of tendency, as Chapter II explained for every economic law. Look at the bakery's marginal product: 20, 30, 40, then 30, 20, 10, 0. The marginal product rises for the first three bakers, because each one can specialise on a task. From the fourth baker onward, it falls, because the oven and the kitchen limit what more bakers can do.¹

Why must the law be true? Suppose it were false, and each extra worker always added as much as the one before. Then a farmer could keep adding workers to one flowerpot and grow all the world's wheat there. Nobody has ever seen this. So the extra output must shrink, because the fixed input has to be shared among more and more units of the variable input.

The law is true only under conditions. First, at least one input is fixed, so we are in the short run. Second, the units of the variable input are alike: the seventh baker is as skilled as the first. Third, the methods of production stay the same. Fourth, we change one input at a time and keep the others unchanged. If any condition fails, the law is not tested by the table. A new oven, for instance, would end the short run and bring us to the long run.

We can divide production into three stages. The first stage runs while average product is still rising. The second stage runs from the top of average product until marginal product reaches zero. The third stage is where marginal product is below zero and total product falls. The table introduces the stages for the bakery. We have not tabulated a stage three, because the case file stops at seven bakers; it would begin if an eighth baker made the kitchen so crowded that output fell.

STAGES OF PRODUCTION IN THE BAKERY

STAGE	WHAT HAPPENS	BAKERY
One	Average product rises	One to four bakers
Two	Marginal product falls to zero; total product still rises	Four to seven bakers
Three	Marginal product is negative; total product falls	More than seven bakers

A sensible firm avoids stage one and stage three. In stage one, more workers still raise output per worker, so the firm should not stop. In stage three, extra workers reduce output, so the firm should never hire them. The firm therefore works in stage two. Where in stage two depends on the wage and on the price of bread. We return to that choice in Chapter XV. For now, note that the seventh baker adds nothing but costs Tk 600, so no owner would hire the seventh.

The law does not say that total output falls; it says that the *increase* in output falls. It does not forbid growth: a better method or a better seed lifts the whole production function, as the history at the end of this chapter shows. And it matters for later chapters. Because each extra worker adds less, each extra loaf costs more to make. That rising cost, studied in Chapter X, is why firms in the short run supply more only at higher prices, as Chapter VII found.

Before moving on. Beyond a point, each extra unit of a variable input adds less output. The law holds when one

The law in one plain statement.

Law of diminishing returns. Beyond some point, each extra unit of a variable input adds less output than the one before.

If the law did not hold, one small plot could feed the world.

The law needs four conditions.

Production passes through three stages as workers are added.

A sensible firm works in stage two.

What the law does not say, and why it matters.

Recap of the section.

¹ Some books call the whole pattern, with its early rise, the law of variable proportions.

A check on the law.

The sentence to test yourself against.

The long run lets every input change together.

Splitting a job into tasks makes workers faster.

Division of labour. Splitting a job into separate tasks, each done by a different worker.

Scale means the size of all inputs together.

Scale of production. The size of the firm, with all its inputs taken together.

Returns to scale. The change in output when all inputs change by the same proportion.

Three cases: output rises faster, equally, or more slowly.

Increasing returns to scale. Output rises by a larger proportion than inputs.

Constant returns to scale. Output rises by the same proportion as inputs.

Decreasing returns to scale. Output rises by a smaller proportion than inputs.

The case file's plan for a larger bakery.

A worked example: setting out the problem.

A worked example: the arithmetic.

input is fixed, units are alike and methods do not change. A firm works in stage two.

Pause and check. In the bakery, which baker is the first to add less than the baker before? Why is the sixth baker still worth considering, even though the marginal product is low?

You should now be able to say: when a fixed input is shared among more and more of a variable input, each extra unit adds less output, and the law needs the short run and unchanged methods.

RETURNS TO SCALE

In the long run, no input is fixed, so the owner can add ovens and bakers together. A new question arises: if all inputs double, does output double? Adam Smith (1776) described a small pin workshop in which ten workers, each given one or two of the separate tasks, made far more pins in a day than the same ten could make working alone.² His example introduces a cause of what follows.

Division of labour means splitting a job into separate tasks, each done by a different worker. In a larger bakery one person mixes, one shapes, one bakes and one packs. Each becomes quick at one task, and little time is lost moving between tasks. This is possible only when the workforce is large enough to share the tasks.

The **scale of production** is the size of the firm, with all its inputs taken together. **Returns to scale** means the change in output when all inputs change by the same proportion. If the owner doubles ovens and bakers, inputs rise by 100 per cent. Returns to scale ask by how much per cent output rises.

There are three cases. With **increasing returns to scale**, output rises by a larger proportion than inputs. With **constant returns to scale**, output rises by the same proportion as inputs. With **decreasing returns to scale**, output rises by a smaller proportion than inputs. A firm can pass through all three as it grows.

Case file, illustrative. Step 9, long run. A plan to enlarge the bakery adds ovens and bakers together, one oven for every two bakers. The daily output at one to five ovens is 50, 110, 180, 240 and 280 loaves. The first row agrees with the short-run table: one oven and two bakers make 50 loaves.

Given. The output at scale 1 to 5 above. *Find.* The kind of returns to scale at each step. *Formula.* Per cent change = $(\text{new} - \text{old}) \div \text{old} \times 100$, for inputs and for output. Compare the two.

Steps. From scale 1 to 2, inputs rise by $(2 - 1) \div 1 \times 100 = 100$ per cent. Output rises by $(110 - 50) \div 50 \times 100 = 120$ per cent. Output rises faster: increasing returns. From 2 to 3, inputs rise 50 per cent and output $(180 - 110) \div 110 \times 100 = 63.6$ per cent: increasing. From 3 to 4, inputs rise 33.3 per cent and output $(240 - 180) \div 180 \times 100 = 33.3$ per cent: constant. From 4 to 5,

² Smith's figures for the workshop are widely quoted; this book leaves them out and keeps the point.

inputs rise 25 per cent and output $(280 - 240) \div 240 \times 100 = 16.7$ per cent: decreasing.

RETURNS TO SCALE IN THE BAKERY'S PLAN

SCALE (OVENS)	OUTPUT PER DAY	RISE IN INPUTS	RISE IN OUTPUT	RETURNS TO SCALE
1	50	—	—	—
2	110	100.0%	120.0%	Increasing
3	180	50.0%	63.6%	Increasing
4	240	33.3%	33.3%	Constant
5	280	25.0%	16.7%	Decreasing

Answer. Returns increase up to three ovens, are constant from three to four, and decrease after four. *Check.* Output per scale unit (output $\div$ ovens) is 50, 55, 60, 60 and 56. It rises, stays level, then falls, which agrees with the verdicts. The causes differ. Increasing returns come from division of labour and from machines that work well only when large. Constant returns arise when the firm can repeat what it already does. Decreasing returns arise when a larger firm is harder to manage, and instructions travel slowly and are less clear.

Returns to scale and diminishing returns answer different questions. The comparison below shows how. Read each row across: the left column gives the feature, and the other two columns say how each idea treats it.

DIMINISHING RETURNS AND RETURNS TO SCALE

FEATURE	DIMINISHING RETURNS	RETURNS TO SCALE
Period	Short run	Long run
Inputs changed	One input; others fixed	All inputs together
What is measured	Marginal product of the added unit	Total output against total inputs
Main cause	A fixed input is shared more thinly	Division of labour; later, management strain

Common slips. Do not call a firm's falling marginal product "decreasing returns to scale"; that is diminishing returns. Do not change only the number of bakers and call it a change of scale. A firm can show increasing returns to scale and diminishing returns together, because they belong to different periods.

Before moving on. Returns to scale compare the rise in output with the rise in all inputs together. They may increase, stay constant or decrease. Diminishing returns concern one input in the short run.

LEAST-COST COMBINATION

A farmer can grow the same crop with many workers and one power tiller, or with fewer workers and two tillers. Both ways give the same harvest, but they cost different amounts. Which mix is cheapest? This section answers by two lines and a rule. It extends the idea of the budget line from Chapter VI.

The answer, a check, and the causes behind each stage.

Two ideas that are often confused.

Three slips to avoid.

Recap of the section.

ADVANCED SECTION; YOU MAY DEFER IT.

Many mixes of inputs can make the same output.

The isoquant shows equal-output mixes; the isocost line shows equal-cost mixes.

Isoquant. A line joining all mixes of two inputs that make the same output.

Isocost line. A line joining all mixes of two inputs that cost the same money.

The cheapest mix has the lowest cost for the chosen output.

Least-cost combination. The mix of inputs that makes a given output at the lowest cost.

A worked example: setting out the problem.

A worked example: the arithmetic and the answer.

An **isoquant** is a line joining all mixes of two inputs that make the same output. The word means "equal quantity". An **isocost line** is a line joining all mixes of two inputs that cost the same money. A bigger budget gives an isocost line farther from the corner. The cheapest mix for a given output lies where the isoquant touches the lowest possible isocost line.

The **least-cost combination** is the mix of inputs that makes a given output at the lowest cost. We find it by working out the cost of each mix on the isoquant and choosing the smallest.

Given. A small farm can make one season's rice with these mixes of labour-days and tiller-days: (24, 2), (16, 3), (12, 4), (8, 6), (6, 8). A labour-day costs Tk 500 and a tiller-day costs Tk 1,500. **Find.** The cheapest mix. **Formula.** Cost = labour-days $\times$ Tk 500 + tiller-days $\times$ Tk 1,500.

Steps. The mix (24, 2) costs $12,000 + 3,000 =$ Tk 15,000. The mix (16, 3) costs $8,000 + 4,500 =$ Tk 12,500. The mix (12, 4) costs $6,000 + 6,000 =$ Tk 12,000. The mix (8, 6) costs $4,000 + 9,000 =$ Tk 13,000. The mix (6, 8) costs $3,000 + 12,000 =$ Tk 15,000. **Answer.** The cheapest mix is 12 labour-days and 4 tiller-days, at Tk 12,000. The figure shows the isoquant as a smooth curve, and the isocost line for Tk 12,000, which touches it at that mix.

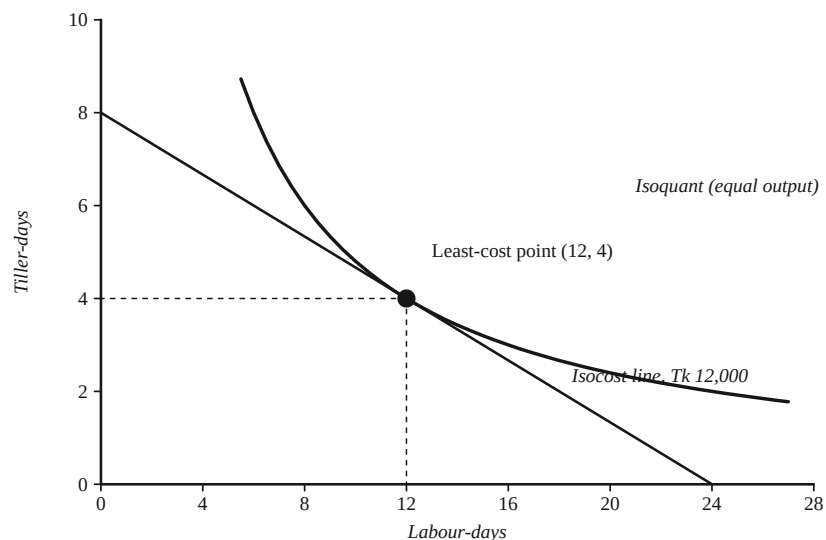


FIG. 15. The isoquant touches the lowest isocost line (Tk 12,000) at 12 labour-days and 4 tiller-days.

The rule behind the answer.

Equal-ratio rule. At least cost, each input gives the same extra output for each taka spent on it.

Check. Moving from (16, 3) to (12, 4) adds one tiller-day, costing Tk 1,500, and saves four labour-days, worth Tk 2,000, so it saves Tk 500. Moving on to (8, 6) adds two tiller-days, costing Tk 3,000, and saves four labour-days, worth Tk 2,000, so it costs Tk 1,000 more. We stop at (12, 4). This is the **equal-ratio rule**: at least cost, the extra output for each taka spent on labour equals the extra output for each taka spent on the tiller. In symbols, MP of labour $\div$ price of labour = MP of tiller $\div$ price of tiller. If the two ratios differ, the firm moves spending to the input that gives more output per taka. Chapter III used the same idea for the consumer.

Common slips. Do not compare the number of units; compare the cost of each mix. Do not choose the cheapest input alone: the cheapest mix may use a dearer

Slips and a recap.

input, if it saves enough of the other. *Before moving on.* The isoquant shows equal output, the isocost line shows equal cost, and the least-cost mix is where they touch.



SUMMARY

Production turns inputs into output, and the production function links the two. In the short run one input is fixed. Marginal product first rises and then falls, and the law of diminishing returns states the fall. In the long run all inputs change; returns to scale may be increasing, constant or decreasing. A firm finds its cheapest mix of inputs where the isoquant touches the lowest isocost line.

The chapter in a few lines.

THINK IT THROUGH

A bakery owner says: "Average product is highest with four bakers, so four is the right number to hire." Using the table in section 9.2, explain whether the owner is right. What else must the owner know before deciding?

A claim to test against the product table.

WHERE PEOPLE GO WRONG

Error: "Diminishing returns means total output falls." *Correction:* only the extra output falls; total output still rises until marginal product is zero. *Error:* "Marginal and average product are the same." *Correction:* marginal is the last unit; average is the total shared by all units. *Error:* "Diminishing returns and decreasing returns to scale are one idea." *Correction:* the first concerns one input in the short run, the second all inputs in the long run. *Error:* "Short run means a few months." *Correction:* it means that at least one input is fixed, however long that lasts.

Four common errors, each corrected.

IN HISTORY

In 1966 the International Rice Research Institute in the Philippines released a short-stemmed rice variety called IR8. With enough water and fertiliser, it gave far higher yields from the same land than traditional varieties. This did not break the law of diminishing returns. It moved the production function upward: the same land and labour now made more rice. Later extra fertiliser still added less and less, as the law says. The lesson is that methods are one of the conditions of the law, and new methods can lift output for a long time.

A new seed lifted the whole production function.

SOURCES

Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), opens with the division of labour in pin-making. David Ricardo and Edward West each published essays in 1815 that set out diminishing returns on land, which later writers extended to all inputs.

Where the ideas first appeared.

REVIEW QUESTIONS AND PRACTICE

Review questions. 1. Define production and give one example of a good and one of a service. 2. What is the difference between a fixed and a variable input? 3. Define the production function. 4. State the law of diminishing returns and give

Six review questions.

its conditions. 5. Name the three kinds of returns to scale. 6. Define an isoquant and an isocost line, and state the equal-ratio rule.

Practice item 1: marginal product from total product.

Practice 1. A tailoring shop has two sewing machines. With one to six tailors, it makes 8, 20, 33, 42, 47 and 47 shirts a day. (a) Find the marginal product of each tailor. (b) Which tailor is the first to add less than the one before? (c) Between which numbers of tailors does total product stop rising?

Practice item 2: average product and its link to marginal product.

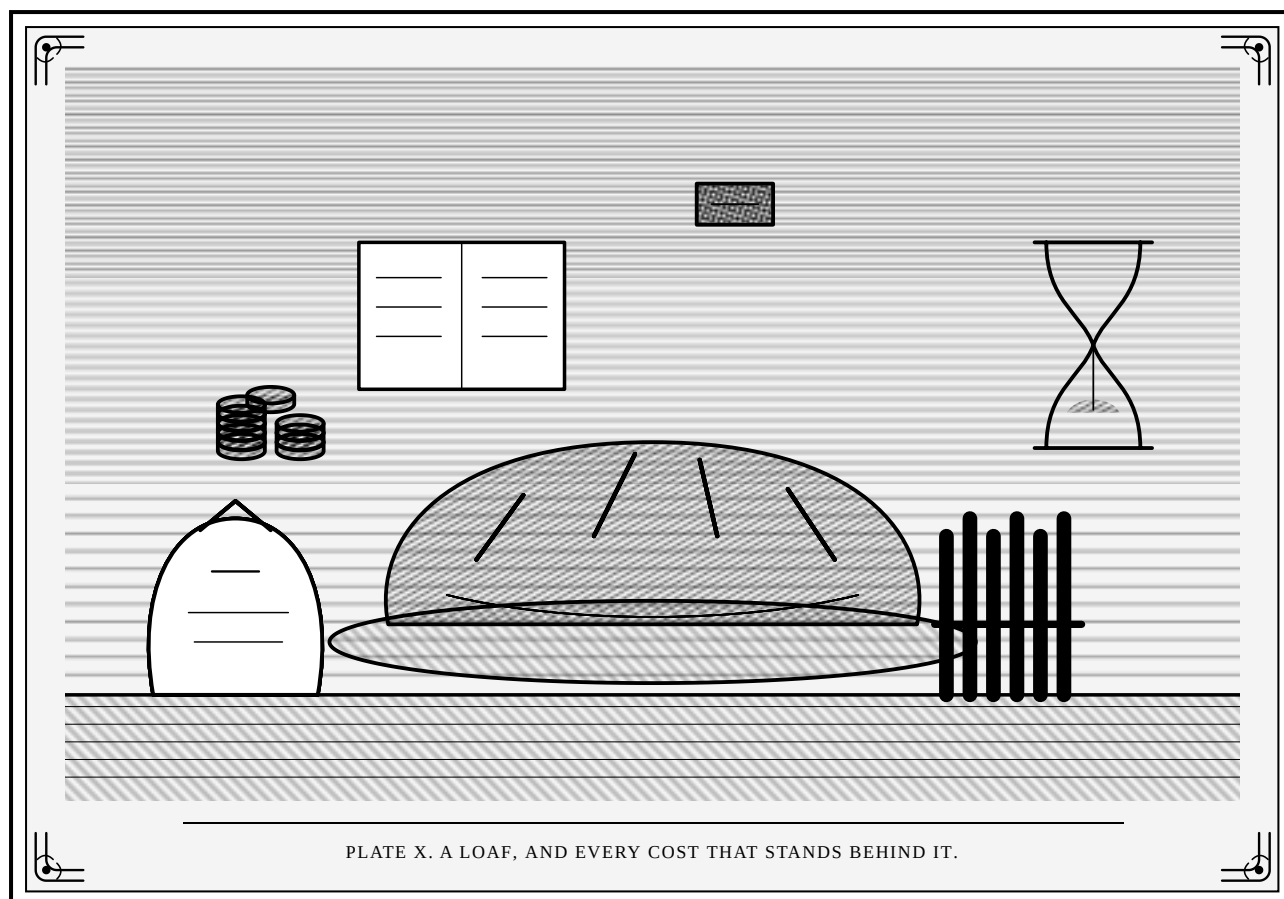
Practice 2. Using the shop in Practice 1, find the average product at each number of tailors, to one decimal place. At which numbers does the marginal product lie above the average, and what happens to the average there?

Practice item 3: classify returns to scale.

Practice 3. A printing firm adds machines and workers together. At scales 1, 2, 3 and 4 it makes 30, 66, 99 and 120 pages a minute. Find the per cent rise in inputs and in output at each step, and state the returns to scale.

Practice item 4: least cost at new prices.

Practice 4. The farm in section 9.5 uses the same mixes, but the labour-day now costs Tk 800 and the tiller-day Tk 1,500. Find the cost of each mix, the cheapest mix, and the saving against the mix that was cheapest before.



CHAPTER X

Cost

What does a loaf really cost?



This chapter counts every cost of production, including those that no one is paid in cash. It builds the cost curves of a firm from its product table, shows why average cost is shaped like a U, and asks what happens to cost when the firm grows.

A puzzle: a shop can show a profit and still be a poor bargain.

What you should know before reading on.

Word in use. Cost. In daily life, the price you pay for a thing. In economics, the value of what you give up to get it.

Cost means the value of what is given up.

Cost. The value of what is given up to make or get something.

Some costs are paid in money; others are not.

Explicit cost. A cost paid in money to someone outside the firm.

Implicit cost. The cost of using resources the owner already has; no money is paid.

Accountants count explicit costs; economists count all costs.

Accounting cost. The explicit costs recorded in the firm's books.

Economic cost. All costs of the firm, explicit and implicit together.

A cost already paid and lost cannot guide a new choice.

Sunk cost. A cost already paid that cannot be recovered.

A normal profit is the owner's own opportunity cost.

Normal profit. The least profit that keeps an entrepreneur in the business.

Recap of the section.

At the end of its first full month, the bakery's books show that it paid rent, flour, wages and the electricity bill. They show no wage for the owner, who works every day, and no interest on the savings that were used to open the shop. A friend studies the books and asks a hard question. "You left a job that paid Tk 40,000 a month, and your savings would have earned Tk 3,000 a month in a bank. Do your books count these?" They do not. So is the shop's profit a true profit? To answer, we must decide what counts as a cost. Then we must see how cost changes as the bakery bakes more loaves.

You will need: opportunity cost from Chapter I, and the bakery's product table from Chapter IX (total, average and marginal product, and the law of diminishing returns).

COST AND OPPORTUNITY COST

Chapter I taught that every choice gives something up. We can now put a money value on the loss. A **cost** is the value of what is given up to make or get something. For the bakery owner, the best alternative to opening the shop was to keep the job and deposit the savings: Tk 40,000 + Tk 3,000 = Tk 43,000 a month. That sum is the opportunity cost of using her time and savings in the shop. To measure any opportunity cost, find the best alternative and write down its money value.

Costs come in two kinds. An **explicit cost** is a cost paid in money to someone outside the firm. The bakery's rent of Tk 25,000 a month is explicit, as are flour and wages. An **implicit cost** is the cost of using resources the owner already has, where no money is paid. The owner's own time, valued at Tk 40,000, and her own savings, valued at Tk 3,000, are implicit costs. A family that farms its own land has an implicit cost too: the rent the land could have earned if it had been let to others.

The two kinds of cost give two totals. The **accounting cost** is the explicit costs recorded in the firm's books. The **economic cost** is all costs, explicit and implicit together. In words: economic cost equals explicit cost plus implicit cost. For the bakery, the implicit cost is Tk 43,000 a month, which is about Tk 1,433 a day in a month of 30 days. The books leave this out, so accounting cost is smaller than economic cost.

A **sunk cost** is a cost already paid that cannot be recovered. Suppose a tailor pays Tk 5,000 for a licence that cannot be refunded or sold. The money is gone whatever she does next. When she decides whether to open the shop this month, the Tk 5,000 should not weigh on the choice, because no choice can bring it back. Only costs that still depend on the decision belong in it.

The owner of a firm gives up something too, so her reward is also a cost. **Normal profit** is the least profit that keeps an entrepreneur in the business. It equals what her time and money could earn in the best other use: for the bakery, Tk 43,000 a month. Economists therefore count normal profit as a cost. A firm that earns only normal profit is doing as well as the owner's best alternative. A firm that earns less will, in time, lose its owner. Chapter XVI returns to this when it compares economic and accounting profit.

Before moving on. Cost is the value of what is given up. Explicit costs are paid in money; implicit costs are not. Economic cost counts both; accounting cost counts only the explicit kind. Normal profit is a cost, because it is the owner's opportunity cost. A sunk cost cannot be recovered and should not guide a new choice.

Pause and check. A farmer works her own land, using family labour, and pays only for seed and fertiliser. Name two implicit costs.

A check on implicit costs.

KINDS OF COST

Costs also differ in how they behave when output changes. This difference is the base of everything that follows in this chapter. We use the short run of Chapter IX, in which one input (the oven and the shop) is fixed.

A **fixed cost** is a cost that does not change with output. The shop rent is paid even on a day when no loaf is baked. A **variable cost** is a cost that changes with output. Flour, fuel and bakers' wages all rise when more bread is made. The **total cost** is fixed cost plus variable cost. In symbols, $TC = FC + VC$.¹

Some costs stay the same whatever the output; others change.

Fixed, variable and total cost.

Fixed cost. Cost that does not change with output.

Variable cost. Cost that changes with output.

Total cost. Fixed cost plus variable cost.

An older name for a similar idea.

Older books use the name **supplementary cost**. It means overhead cost, the cost that does not change with output. In this book we say fixed cost for the same idea. When you meet "supplementary cost" in another book, read it as fixed cost.

Supplementary cost. An older name for overhead cost, which does not change with output.

Cost can be seen in two ways. **Money cost** is cost measured in money. **Real cost** is the effort and sacrifice of the people who supply the inputs: the baker's tiring work and the saver's waiting. Money cost is what we can count in a table, so the rest of this chapter uses it. Real cost explains why people want a wage at all.

Cost can be seen as money or as sacrifice.

Money cost. Cost measured in money, such as taka paid.

Real cost. The effort and sacrifice of the people who supply the inputs.

Case file, illustrative. Step 10. Fixed cost is Tk 1,000 a day, which is Tk 30,000 a month (rent Tk 25,000 and other costs Tk 5,000) divided by 30 days. A baker costs Tk 600 a day, and materials cost Tk 20 for each loaf. The table counts explicit costs only. The owner's implicit cost of Tk 43,000 a month is not in it.

The bakery's cost table follows from the product table.

Given. The product table of Chapter IX: 1 to 6 bakers make 20, 50, 90, 120, 140 and 150 loaves. $FC = \text{Tk } 1,000$ a day; baker Tk 600 a day; materials Tk 20 a loaf. *Find.* Variable cost and total cost at each output. *Formula.* $VC = \text{bakers} \times \text{Tk } 600 + \text{loaves} \times \text{Tk } 20$; $TC = FC + VC$.

A worked example: setting out the problem.

Steps. Take 90 loaves, made by three bakers. $VC = 3 \times 600 + 90 \times 20 = 1,800 + 1,800 = \text{Tk } 3,600$. $TC = 1,000 + 3,600 = \text{Tk } 4,600$. Repeat for each row. *Answer.* The table shows the result. *Check.* TC minus FC must equal VC in every row. For 150 loaves, $7,600 - 1,000 = 6,600$, which is $6 \times 600 + 150 \times 20 = 3,600 + 3,000$.

A worked example: the arithmetic and the answer.

COSTS OF THE BAKERY, PER DAY (TAKA)

LOAVES	BAKERS	FIXED COST	VARIABLE COST	TOTAL COST
20	1	1,000	1,000	2,000
50	2	1,000	2,200	3,200
90	3	1,000	3,600	4,600
120	4	1,000	4,800	5,800
140	5	1,000	5,800	6,800
150	6	1,000	6,600	7,600

¹ Another common symbol is $TC = TFC + TVC$, where F and V stand for fixed and variable.

Recap of the section.

A check on kinds of cost.

GO SLOWLY HERE.

Start with pens: an average and a new unit.

Four measures of cost per loaf.

Average fixed cost. Fixed cost divided by output.

Average variable cost. Variable cost divided by output.

Average cost. Total cost divided by output; the cost of one unit on average.

Marginal cost. The extra cost of making one more unit.

A worked example: setting out the problem.

A worked example: the arithmetic.

Before moving on. Fixed cost does not change with output; variable cost does; total cost is their sum. Money cost is counted in taka. Real cost is the sacrifice behind it. The cost table follows from the product table.

Pause and check. In the table, what is the total cost of 50 loaves, and what part of it is fixed? What is the total cost if the bakery bakes nothing?

AVERAGE AND MARGINAL COST

Buy four pens for Tk 40. The cost of each pen on average is Tk 10. Now buy a fifth pen for Tk 20. The total is Tk 60, and the average becomes Tk 12. The new pen cost more than the old average, so it pulled the average up. Keep this small case in mind: an extra unit that costs more than the average raises the average, and an extra unit that costs less lowers it. A firm's cost curves follow the same rule.

The **average fixed cost** is fixed cost divided by output. The **average variable cost** is variable cost divided by output. The **average cost** is total cost divided by output. The **marginal cost** is the extra cost of making one more unit. In symbols: $AFC = FC \div Q$; $AVC = VC \div Q$; $AC = TC \div Q = AFC + AVC$; $MC = \text{change in } TC \div \text{change in } Q$. Here Q is output.

Given. The cost table of section 10.2. *Find.* AFC, AVC, AC and MC at each output. *Formula.* As above.

Steps. Take 90 loaves. $AFC = 1,000 \div 90 = \text{Tk } 11.11$. $AVC = 3,600 \div 90 = \text{Tk } 40.00$. $AC = 4,600 \div 90 = \text{Tk } 51.11$. MC is measured from the row before, where TC was Tk 3,200 at 50 loaves: $MC = (4,600 - 3,200) \div (90 - 50) = 1,400 \div 40 = \text{Tk } 35$. The first row has no row before it, so we measure from zero output, where TC equals the fixed cost of Tk 1,000: $MC = (2,000 - 1,000) \div (20 - 0) = \text{Tk } 50$.

COST PER LOAF AT THE BAKERY (TAKA)

LOAVES	AFC	AVC	AC	MC
20	50.0	50.0	100.0	50
50	20.0	44.0	64.0	40
90	11.1	40.0	51.1	35
120	8.3	40.0	48.3	40
140	7.1	41.4	48.6	50
150	6.7	44.0	50.7	80

The answer, and two checks on it.

Marginal cost is the mirror image of marginal product.

Answer. Average cost falls from Tk 100 to Tk 48.3 at 120 loaves, then rises. Marginal cost falls to Tk 35 at 90 loaves, then rises sharply. *Check.* In every row AC equals AFC plus AVC, apart from a difference of Tk 0.1 caused by rounding. The gap between AC and AVC is AFC, and it shrinks as output grows.

Why does marginal cost first fall and then rise? The product table answers. Each step in the table adds one baker, who costs Tk 600, and the baker's marginal product tells how many loaves that Tk 600 buys. So $MC = (\text{Tk } 600 \div MP) + \text{Tk } 20$ for materials. For the third baker, MP is 40 and $MC = 15 + 20 = \text{Tk } 35$. For the sixth baker, MP is 10 and $MC = 60 + 20 = \text{Tk } 80$. When marginal product rises, marginal cost falls. When marginal product falls, marginal cost

risers. The law of diminishing returns is therefore the cause of rising marginal cost.

The pen case gives the rule. When marginal cost is below average cost, average cost falls. When marginal cost is above average cost, average cost rises. So the marginal cost curve cuts the average cost curve at the lowest point of the average cost curve. The table agrees. At 120 loaves, MC is Tk 40, below AC of Tk 48.3, so AC is still falling. At 140 loaves, MC is Tk 50, above AC of Tk 48.6, so AC has begun to rise. The true lowest point lies between 120 and 140 loaves. Among the rows of the table, the lowest average cost is at 120 loaves. The same rule links marginal cost to average variable cost.

The rule linking average and marginal cost.

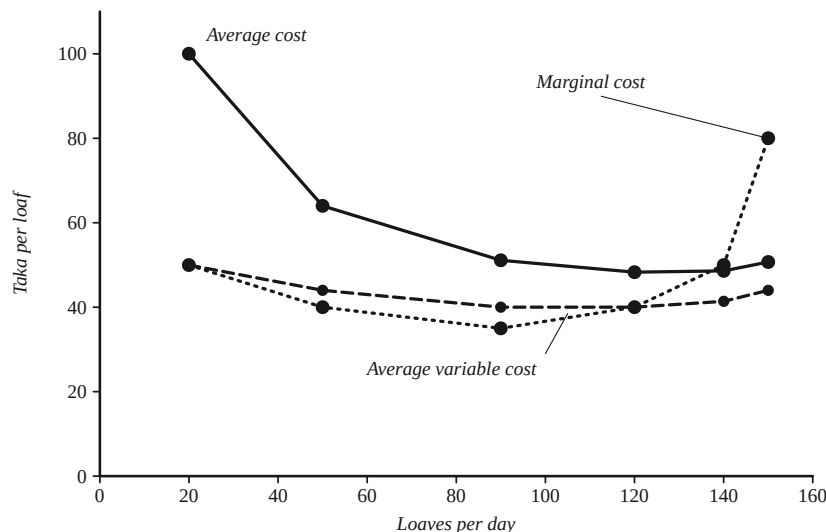


FIG. 16. Cost curves drawn through the points of the bakery's cost table. Marginal cost cuts average cost near its lowest point.

Average cost is U-shaped because two forces act on it. First, the fixed cost is spread over more loaves, so AFC falls all the time. This force dominates at low output. Second, diminishing returns raise AVC after some output. At small output, the first force wins and AC falls. At large output, the second wins and AC rises. Since AFC keeps falling after AVC has started to rise, the lowest point of AC lies at a larger output than the lowest point of AVC.

Why average cost has the shape of a U.

We can also read the totals together. Fixed cost is a flat line. Variable cost and total cost have the same shape, and the gap between them is always the fixed cost. Marginal cost is the rate at which total cost rises. Average cost is total cost shared among the loaves. If you know any one of these curves, you can find the others by the formulas above.

How the totals and the averages connect.

Common slips. Do not divide by the wrong output: AC divides by the output of its own row. Do not measure marginal cost from the same row; use the row before. Do not forget that total cost is not zero when output is zero: it equals the fixed cost. Do not say that MC and AC are lowest at the same output; MC is lowest earlier.

Four slips to avoid.

Before moving on. AFC falls all the time. AVC, AC and MC fall and then rise. MC rises first because marginal product falls. MC cuts AC and AVC at their lowest points.

Recap of the section.

You should now be able to say: when marginal cost is below average cost, average cost falls; when it is above, average cost rises; so the two curves meet where average cost is lowest.

The sentence to test yourself against.

SHORT-RUN AND LONG-RUN COST CURVES

In the long run the firm may choose its plant.

The long-run curve is the lowest cost over all plant sizes.

Long-run average cost. The lowest average cost of each output when every input can change.

The curves above belong to the short run, with one oven. In the long run, the owner can choose any size of plant: one oven, two or five. Each plant size has its own short-run average cost curve. A small plant has low cost at small output but high cost at large output, because its fixed oven is crowded. A large plant costs more at small output, because its fixed costs are spread over few loaves, but less at large output.

The **long-run average cost** is the lowest average cost of each output when every input can change. For each output, the owner picks the plant that makes it most cheaply. The long-run curve therefore runs along the lower edge of all the short-run curves. For this reason it is called an envelope: it wraps around them from below. In the long run there are no fixed costs, since every input can be changed.

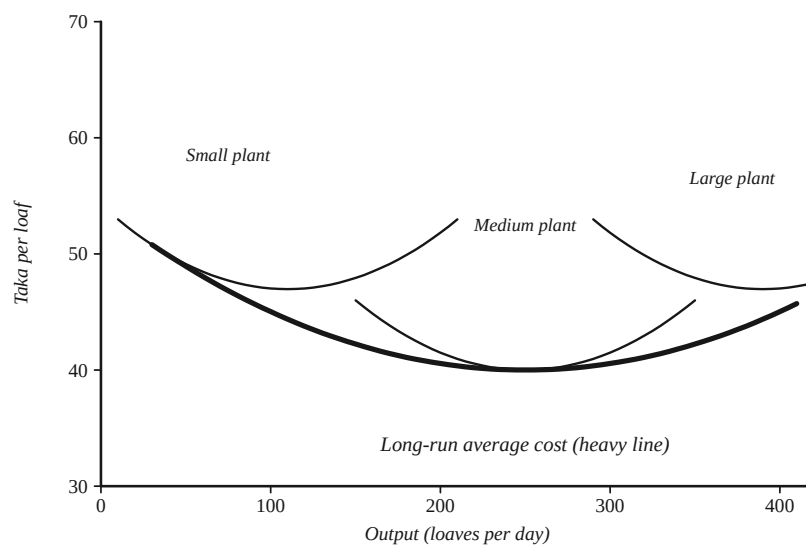


FIG. 17. Short-run average cost curves for three plant sizes, and the long-run average cost curve that wraps around them.

Where the curves touch.

Each short-run curve touches the long-run curve at one point and nowhere lies below it. The firm can always choose that plant, so the long run can never cost more than any one plant. Only for the medium plant, at the bottom of the long-run curve, is the touching point also the lowest point of the short-run curve. For smaller plants the long-run curve is falling, so the touching point is on the falling side of the short-run curve. For larger plants it is on the rising side. The long-run curve is also flatter than the short-run curves, because in the long run the firm can adjust every input.

A check on the envelope.

Pause and check. Why can the long-run average cost of any output never be higher than the short-run average cost of the same output with one particular plant?

ECONOMIES AND DISECONOMIES OF SCALE

What makes cost per loaf fall as the firm grows.

Economies of scale. Average cost falls as the scale of the firm grows.

When the long-run curve falls, the firm enjoys **economies of scale**, which means that average cost falls as the scale grows. There are several sources. Technical economies come from bigger machines and division of labour. Managerial economies come from hiring a specialist for each task. Marketing economies come from buying flour in bulk and spreading the cost of advertising over more

loaves. Financial economies come from borrowing at lower rates, which banks offer to large firms. Risk-bearing economies come from selling several products, so a loss on one can be met from another.

Economies may also be *internal* or *external*. Internal economies arise inside one firm, as above. External economies arise outside the firm, when the whole industry grows. For example, when many bakeries gather in one town, trained bakers, flour merchants and repair shops settle nearby. All the bakeries then enjoy lower costs, though no single one has grown.

When the long-run curve rises, the firm suffers **diseconomies of scale**, which means that average cost rises as the scale grows. The chief cause is management strain: orders pass through many people, delays grow, and it is harder to see waste. Buying very large amounts of an input may also raise its price. The bottom of the long-run curve is the scale at which cost is lowest. Between falling and rising parts, there is often a flat stretch where average cost hardly changes.

How does this link to returns to scale in Chapter IX? Returns to scale compare output with inputs, without prices. Economies of scale compare cost with output, in taka. If input prices do not change, the two move together. In the bakery's plan, output per scale unit was 50, 55, 60, 60 and 56. If each unit of scale costs the same, the cost per loaf falls from scale 1 to 3, is level from 3 to 4, and rises at 5. That agrees with increasing, constant and decreasing returns to scale. The two differ when input prices change, as when bulk buying lowers the price of flour.

Before moving on. Economies of scale make average cost fall as the firm grows; diseconomies make it rise. Sources of economies include technical, managerial, marketing, financial and risk-bearing gains. Returns to scale describe quantities; economies of scale describe cost.

Some economies arise outside the firm.

Growth beyond a point raises cost again.

Diseconomies of scale. Average cost rises as the scale of the firm grows.

Costs and returns to scale are linked but not the same.

Recap of the section.



SUMMARY

Cost is the value of what is given up. Economic cost counts explicit and implicit costs; accounting cost counts only explicit ones; normal profit is a cost. In the short run, total cost is fixed plus variable cost. Average cost is U-shaped, and marginal cost cuts it at its lowest point. In the long run the firm chooses its plant, and the long-run average cost curve wraps around the short-run curves. It falls with economies of scale and rises with diseconomies.

The chapter in a few lines.

THINK IT THROUGH

A family sells rice worth Tk 200,000 in a year. It pays Tk 140,000 for seed, fertiliser, hired labour and water. The family's own labour could earn Tk 40,000 elsewhere, and its land could be let for Tk 30,000. Find the accounting profit and the economic profit. What does the sign of the second tell the family?

A problem on implicit cost.

WHERE PEOPLE GO WRONG

Error: "Fixed cost is zero when output is zero." *Correction:* fixed cost must still be paid; only variable cost is zero. *Error:* "Money already spent should guide the choice." *Correction:* a sunk cost cannot be recovered, so it should not. *Error:* "Marginal cost and average cost are lowest at the same output." *Correction:* marginal cost reaches its lowest point first and cuts average cost at the average's

Four common errors, each corrected.

lowest point. *Error:* "Unpaid resources have no cost." *Correction:* the owner's time and savings have an opportunity cost, which economists count.

IN HISTORY

A fall in cost made a car cheap.

The Ford Motor Company began making the Model T in 1908. In 1913 it introduced a moving assembly line at its Highland Park factory, and each worker repeated one task as the car moved past. Output rose, and the cost per car fell. Ford passed part of the saving to buyers: the price fell from about 850 US dollars in 1908 to about 260 dollars in the mid-1920s. The story shows economies of scale at work, from division of labour and from large output spread over a single design.

SOURCES

Where the ideas first appeared.

Alfred Marshall, *Principles of Economics* (1890), discussed real cost and money cost. Jacob Viner's article of 1931 on cost curves and supply curves set out the long-run curve as an envelope of short-run curves.

REVIEW QUESTIONS AND PRACTICE

Six review questions.

Review questions. 1. Define cost and say how an opportunity cost is measured. 2. Distinguish explicit and implicit costs. 3. Why do economists count normal profit as a cost? 4. Define fixed cost and variable cost, with one example of each. 5. State the rule linking average cost and marginal cost. 6. What is the long-run average cost curve, and why is it called an envelope?

Practice item 1: build the cost columns from a new table.

Practice 1. A tailoring shop has fixed cost Tk 600 a day. A tailor costs Tk 300 a day, and thread and cloth cost Tk 10 for each shirt. With 1 to 5 tailors it makes 8, 20, 33, 42 and 47 shirts a day. Find the variable cost, total cost, average cost and marginal cost at each output. Measure the first marginal cost from zero output.

Practice item 2: the lowest average cost.

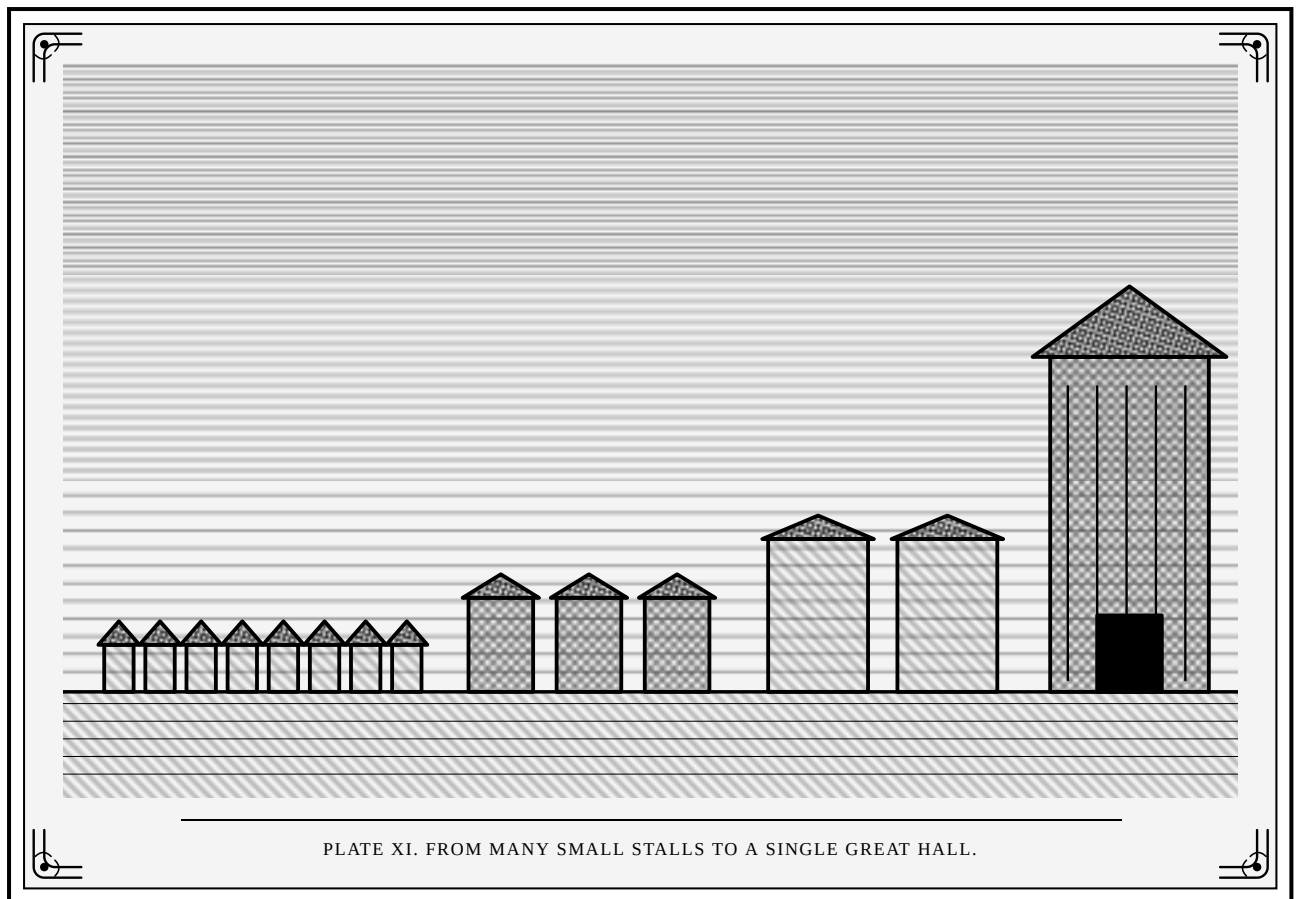
Practice 2. Using Practice 1, find the average variable cost at each output. At which output is average cost lowest among the rows, and at which is average variable cost lowest?

Practice item 3: the average-marginal rule.

Practice 3. Without calculating, say whether average cost is rising, falling or level: (a) AC Tk 50 and MC Tk 40; (b) AC Tk 50 and MC Tk 50; (c) AC Tk 50 and MC Tk 60. Then check your rule on two rows of Practice 1.

Practice item 4: implicit costs of a different firm.

Practice 4. Anwar runs a tea stall in a shop that he owns. He works every day and used his own savings to buy the equipment. List three implicit costs of the stall and say how each could be valued.



CHAPTER XI

Market Structures and the Goal of the Firm

Who sets the price, and what does the seller want?



This chapter sorts markets by the number of sellers, the kind of product and the ease of entry. It introduces revenue, shows how a seller who takes the price differs from one who sets it, and states the rule that a firm follows when it chooses its output.

A puzzle: one seller cannot choose the price, another can.

What you should know before reading on.

A market has features that shape how its sellers behave.

Market structure. The features of a market: the sellers, the product and the ease of entry.

The first feature: the number of sellers.

The second feature: whether products are alike.

Homogeneous product. A product that buyers see as identical from every seller.

Differentiated product. A product that buyers see as different from rivals' products.

The third feature: how easily new sellers can enter.

Barrier to entry. Anything that stops new sellers from entering a market.

The fourth feature: how far sellers can choose the price.

Recap of the section.

A check on the four features.

Four main types arise from the four features.

The bakery sells a loaf at Tk 60. If the owner asks Tk 65, the neighbours buy from the many other bakeries, and her sales fall to almost nothing. Now think of the only pharmacy on a small island. Its owner can raise the price of a medicine, and the islanders must still buy. Both are sellers, and both want a good profit, yet one must accept the price and the other can choose it. What makes the difference? The answer lies in the shape of the market, and it decides how each firm picks its price and its output. This chapter describes the shapes, measures the money a firm receives, and states the rule for choosing output.

You will need: the demand curve and the market (Chapters IV and VIII); total revenue and elasticity (Chapter V); and average and marginal cost (Chapter X).

WHAT MARKET STRUCTURE MEANS

Chapter VIII defined a market as buyers and sellers dealing with each other. Markets differ in how they are built. **Market structure** means the features of a market that decide how its sellers behave: how many sellers there are, what kind of product they sell, how easily new sellers can enter, and how much control sellers have over price. We study these features one by one.

The first feature is the number of sellers. A town may have thousands of vegetable sellers, or a few large firms may supply all its cement, or one firm may supply its gas. The fewer the sellers, the more each seller's choice affects the price and the rivals.

The second feature is the kind of product. A **homogeneous product** is a product that buyers see as identical from every seller; a kilogram of coarse rice of one grade is an example. A **differentiated product** is a product that buyers see as different from rivals' products, because of brand, quality or place; the milk bun of one bakery is an example. When products are alike, buyers care only about price. When they are different, a buyer may pay more for a favourite.

The third feature is ease of entry. A **barrier to entry** is anything that stops new sellers from entering a market. A law that allows only one licensed seller is a barrier. So is a very large cost of starting, or control of the only source of a key input. Where entry is free, high profits attract new sellers and bring prices down. Where entry is blocked, high profits can last.

The fourth feature is control over price. It follows from the other three. Many sellers of a homogeneous product, with free entry, leave each seller no control. One seller behind a barrier has a great deal. Between these two lie the other cases, which the next section names.

Before moving on. Market structure is the number of sellers, the kind of product, the ease of entry and the control over price. A homogeneous product is the same from every seller; a differentiated product is not. A barrier to entry blocks new sellers.

Pause and check. A town has one railway station and only one licensed taxi stand. Which of the four features does this describe, and what is the effect on control over price?

THE MAIN TYPES

Combining the features gives four main types. We state each type briefly here. Chapters XII to XIV study them in turn.

Perfect competition is a market with many sellers, one homogeneous product, free entry and exit, and full knowledge of prices by buyers and sellers. **Pure competition** is a market with the first three of these only: it has no element of monopoly, but knowledge and the movement of inputs may be imperfect. Pure competition is the wider idea; perfect competition adds the last two conditions.¹ A seller in either market has no control over price. A village rice market with many growers and buyers comes close.

Monopoly is a market with one seller of a product that has no close substitute. The seller has strong control over price, if entry is blocked. Later chapters name the kinds: a *pure monopoly* is a sole seller with no substitute, and a *bi-lateral monopoly* is one seller facing one buyer (Chapter XIII).

Monopolistic competition is a market with many sellers of products that differ from each other, and with free entry. Restaurants in one town are an example. Each restaurant is the only seller of its own dishes, so it has some control over price, but rivals are close, so the control is small.

Oligopoly is a market with only a few sellers. Each firm is so large that its price and output change what its rivals do. A market served by three or four large firms, such as the market for mobile phone service, is an example.

The table sets the types side by side. Read each row across: the first column names the type, and the others give its features.

THE MAIN MARKET STRUCTURES

TYPE	SELLERS	PRODUCT	ENTRY	CONTROL OVER PRICE
Perfect competition	Very many	Homogeneous	Free	None
Monopoly	One	No close substitute	Blocked	Strong
Monopolistic competition	Many	Differentiated	Free	Small
Oligopoly	Few	Either kind	Hard	Considerable, but rivals reply

Before moving on. Perfect competition has many sellers and no control over price. Monopoly has one seller. Monopolistic competition has many sellers with different products. Oligopoly has a few sellers.

REVENUE

To choose its output, a firm must know how much money it receives. This section measures that money in three ways. You already know the first from Chapter V.

Perfect and pure competition: many sellers, no control over price.

Perfect competition. A market with many sellers, one homogeneous product, free entry and full knowledge.

Pure competition. Competition with no element of monopoly: many sellers, an identical product and free entry.

Monopoly: one seller.

Monopoly. A market with one seller of a product that has no close substitute.

Monopolistic competition: many sellers, differing products.

Monopolistic competition. A market with many sellers of products that differ from each other.

Oligopoly: a few sellers.

Oligopoly. A market with only a few sellers.

The four types compared in a table.

Recap of the section.

The money a firm receives is the starting point for its choice.

¹ Books differ: some use "pure" and "perfect" as names for the same market. This book keeps them apart, as the table above does.

Total, average and marginal revenue.

Average revenue. The revenue received for each unit sold.

Marginal revenue. The extra revenue from selling one more unit.

A seller who must accept the market price.

Price taker. A seller who must accept the price set by the market.

The bakery as a price taker.

A worked example: setting out the problem.

A worked example: the arithmetic, the answer and a check.

The total revenue is the price times the quantity sold. The **average revenue** is the revenue received for each unit sold; it equals total revenue divided by quantity, which is the price. The **marginal revenue** is the extra revenue from selling one more unit. In symbols, $TR = P \times Q$, $AR = TR \div Q = P$, and $MR = \text{change in } TR \div \text{change in } Q$.

A **price taker** is a seller who must accept the price set by the market. The seller can sell any amount at that price, but nothing at a higher one. For such a seller, each extra loaf brings the same price, so marginal revenue equals price. Average revenue also equals price. The seller's demand curve is a horizontal line.

Case file, illustrative. Step 11. The loaf price of Tk 60 is given by the market. For the bakery, average revenue and marginal revenue are both Tk 60 at every output.

Given. Price Tk 60 a loaf. Outputs of 20, 50, 90, 120, 140 and 150 loaves. *Find.* Total, average and marginal revenue. *Formula.* $TR = P \times Q$; $AR = TR \div Q$; $MR = \text{change in } TR \div \text{change in } Q$.

Steps. At 20 loaves, $TR = 60 \times 20 = \text{Tk } 1,200$ and $AR = 1,200 \div 20 = \text{Tk } 60$. From 20 to 50 loaves, TR rises from 1,200 to 3,000, so $MR = (3,000 - 1,200) \div 30 = \text{Tk } 60$. Repeat for each row. *Answer.* The table shows that AR and MR are both Tk 60 throughout. *Check.* The marginal revenue of one loaf is the price of that loaf, so $MR = P$.

REVENUE OF THE BAKERY AS A PRICE TAKER (TAKA)

LOAVES	PRICE (AR)	TOTAL REVENUE	MARGINAL REVENUE
20	60	1,200	60
50	60	3,000	60
90	60	5,400	60
120	60	7,200	60
140	60	8,400	60
150	60	9,000	60

A seller who sets the price.

Price maker. A seller who can set the price, within what buyers will pay.

The one-seller bun market.

A worked example: setting out the problem.

A worked example: the arithmetic and the answer.

A **price maker** is a seller who can set the price, within what buyers will pay. Such a seller faces a falling demand curve, as Chapter IV found for the market. To sell more, the seller must lower the price. And the lower price applies to every unit, not only the extra one. This is why marginal revenue is below the price.

Case file, illustrative. Step 11. Suppose one seller served the whole neighbourhood bun market of Chapter IV. Its demand is $P = 18 - Q \div 30$. Then $TR = 18Q - Q \times Q \div 30$, and $MR = 18 - Q \div 15$.

Given. $P = 18 - Q \div 30$. *Find.* Price, total revenue and marginal revenue at 0, 60, 120, 180, 240 and 300 buns. *Formula.* $TR = P \times Q$; $MR = \text{change in } TR \div \text{change in } Q$.

Steps. At 60 buns, $P = 18 - 60 \div 30 = \text{Tk } 16$, and $TR = 16 \times 60 = \text{Tk } 960$. At 120 buns, $P = 14$ and $TR = 1,680$. The step from 60 to 120 adds $1,680 - 960 = \text{Tk } 720$ over 60 buns, so $MR = 720 \div 60 = \text{Tk } 12$. Repeat for each row. *Answer.* The table shows that MR falls faster than price, and is zero from 240 to 300 buns. *Check.* The formula $MR = 18 - Q \div 15$ at the middle of the first step (30 buns) gives $18 - 2 = \text{Tk } 16$, which equals the first step in the table.

REVENUE OF A ONE-SELLER BUN MARKET (TAKA)

BUNS	PRICE (AR)	TOTAL REVENUE	MARGINAL REVENUE
0	18	0	—
60	16	960	16
120	14	1,680	12
180	12	2,160	8
240	10	2,400	4
300	8	2,400	0

The price taker's marginal revenue stays at the price, but the price maker's falls below the price and can reach zero. Zero marginal revenue is the peak of total revenue, which is at 270 buns and Tk 9, where Chapter V found unit elasticity. So a price maker who wants more revenue stops raising output at the point of unit elasticity.

Common slips. Do not say that marginal revenue equals price for every seller; it does only for a price taker. Do not forget that a price cut applies to all units. Do not take total revenue as profit: it takes no account of cost.

Before moving on. Total revenue is price times quantity; average revenue is the price; marginal revenue is the extra revenue of one more unit. A price taker has MR equal to price. A price maker has MR below price.

The two sellers compared, and the link to elasticity.

Three slips to avoid.

Recap of the section.

THE PROFIT RULE AND OTHER GOALS

Profit is revenue minus cost. In symbols, $\text{profit} = \text{TR} - \text{TC}$. **Profit maximisation** means choosing output to make profit as large as possible. It is the usual goal in the study of the firm, because it gives a clear rule. Chapter X showed that cost can mean accounting cost or economic cost; the table of this chapter uses the cost table of Chapter X, which counts explicit costs.

Given. The bakery sells at Tk 60, and the cost table of Chapter X. *Find.* The output with most profit. *Steps.* At 140 loaves, $\text{TR} = 8,400$ and $\text{TC} = 6,800$, so profit = Tk 1,600. At 150 loaves, $\text{TR} = 9,000$ and $\text{TC} = 7,600$, so profit = Tk 1,400. The table shows every row. *Answer.* Profit is largest at 140 loaves. *Check.* Profit rises until 140 and then falls, so 140 is the peak among these outputs.

What a firm wants decides what it does.

Profit. Revenue minus cost.

Profit maximisation. Choosing output to make profit as large as possible.

Profit can be found row by row.

PROFIT OF THE BAKERY AT A PRICE OF TK 60 (TAKA)

LOAVES	TOTAL REVENUE	TOTAL COST	PROFIT
20	1,200	2,000	-800
50	3,000	3,200	-200
90	5,400	4,600	800
120	7,200	5,800	1,400
140	8,400	6,800	1,600
150	9,000	7,600	1,400

The profit rule gives a quicker way. Add one more unit if it brings in more than it costs, that is, if marginal revenue is above marginal cost. Stop when marginal revenue is no longer above marginal cost. The firm's best output is therefore where marginal revenue equals marginal cost, or the last unit that satisfies $\text{MR} \geq \text{MC}$. In the bakery, MR is Tk 60 for every loaf. The marginal costs of the steps in Chapter X are Tk 50, 40, 35, 40, 50 and 80. The step to 140 loaves has MC of

The rule that gives the same answer without a long table.

Tk 50, below Tk 60, so it adds profit. The step to 150 has MC of Tk 80, above Tk 60, so it reduces profit. The firm stops at 140, as the table shows.

Why the rule works.

The rule works because each step is a small decision. If the last loaf adds more to revenue than to cost, producing it raises profit. If it adds less, producing it lowers profit. At the best output, no further step can raise profit. Chapter XII uses this rule for a firm in perfect competition, and Chapter XIII for a monopoly.

Firms may seek other goals.

Some firms seek goals other than profit. A manager paid by the amount sold may prefer a high revenue to a high profit. The one-seller bun market shows the difference. Revenue is largest at 270 buns, where marginal revenue is zero, but profit is largest where marginal revenue equals the higher marginal cost. A firm may also seek growth, to become larger than its rivals, or safety, to survive in a risky market. Berle and Means (1932) noted that in large firms the managers who decide are often not the owners who want profit, which opens the way for such goals. In this book, unless we say otherwise, a firm maximises profit.

Recap of the section.

Before moving on. Profit is revenue minus cost. The best output is where marginal revenue equals marginal cost. Firms may seek sales, growth or safety, but profit maximisation is our standard goal.

A check on the profit rule.

Pause and check. At 90 loaves the bakery's price is Tk 60 and the marginal cost of the last step is Tk 35. Should it make more? What does the answer say about MR and MC?

HOW STRUCTURE SHAPES PRICING

The same costs can lead to different prices in different markets.

Suppose four bakeries have the same cost table, but each sells in a different market. Their choices differ, because of what each can do about the price. This section sets out what each type of seller can and cannot do.

The perfect competitor can choose only output.

In perfect competition, the baker cannot choose the price. She chooses only the output, by the profit rule, and the market sets the price at Tk 60. If she raises her price, she loses all her buyers. If she lowers it, she gives up revenue and gains nothing. Chapter XII follows this firm.

The monopolist chooses both, but is limited by demand.

A monopolist can choose the price or the output, but not both independently, because buyers decide how much they will take at each price. Choosing a higher price means selling less. The profit rule applies, with marginal revenue below price. The result is a higher price and a smaller output than under competition. Chapter XIII shows this.

The monopolistic competitor has a little room.

A firm in monopolistic competition can set its own price within a narrow range, because buyers like its product. If its price rises too far, buyers move to the many rivals. The bakery's branded milk bun is an example. Chapter XIV follows it.

The oligopolist must watch its rivals.

A firm in an oligopoly cannot decide alone. If it cuts its price, rivals may cut theirs, and nobody gains. If it raises its price, rivals may keep theirs, and it loses buyers. Each firm's choice depends on rivals' replies. Chapter XIV also explains this.

Recap of the section.

Before moving on. A price taker chooses only output. A monopolist faces demand and sets price and output together. A monopolistic competitor has a narrow range of price. An oligopolist watches its rivals.



SUMMARY

Market structure is the number of sellers, the kind of product, the ease of entry and the control over price. The main types are perfect competition, monopoly, monopolistic competition and oligopoly. A price taker has marginal revenue equal to price; a price maker has marginal revenue below price. A profit-maximising firm chooses the output where marginal revenue equals marginal cost. Some firms seek other goals.

The chapter in a few lines.

THINK IT THROUGH

A firm can sell 100 units and earn Tk 20,000 in revenue at a cost of Tk 15,000, or sell 140 units and earn Tk 23,000 at a cost of Tk 18,800. Find the profit in each case. A manager paid by revenue prefers which? Compare the extra revenue and the extra cost of the 40 extra units, and say what this shows about marginal revenue and marginal cost.

A choice between more sales and more profit.

WHERE PEOPLE GO WRONG

Error: "All sellers can set the price." *Correction:* only sellers with market power can; a price taker cannot. *Error:* "Marginal revenue always equals price." *Correction:* only for a price taker. *Error:* "Highest revenue means highest profit." *Correction:* profit also depends on cost. *Error:* "A monopoly is a large firm." *Correction:* monopoly means one seller, however large or small.

Four common errors, each corrected.

IN HISTORY

By the late nineteenth century, a few large firms in the United States controlled whole industries such as oil and railways. Many people feared that these firms would set prices without rivals. On 2 July 1890, Congress passed the Sherman Act, the first major law against agreements that restrained trade and against monopolising a market. It showed that societies treat market structure as a matter of public concern. Bangladesh also has a competition law, the Competition Act 2012, which is enforced by a Competition Commission. Chapter XIII returns to it.

A law was written to protect competition.

SOURCES

Edward Chamberlin, *The Theory of Monopolistic Competition* (1933), and Joan Robinson, *The Economics of Imperfect Competition* (1933), separated the cases between perfect competition and monopoly. Adolf Berle and Gardiner Means, *The Modern Corporation and Private Property* (1932), raised the question of who controls the large firm.

Where the ideas first appeared.

REVIEW QUESTIONS AND PRACTICE

Review questions. 1. What is market structure, and what are its four features? 2. Distinguish a homogeneous from a differentiated product. 3. Name the four main types of market and give one feature of each. 4. Why is marginal revenue below price for a price maker? 5. State the profit rule. 6. Name two goals of a firm other than profit.

Six review questions.

Practice 1. Name the type of market in each case: (a) paddy at a rural market with hundreds of growers and buyers; (b) one firm that supplies piped gas to a

Practice item 1: classify five markets.

district, and the law forbids others; (c) three firms that make most of a country's cement; (d) twenty tailors in a town, each with a different style; (e) the only bus company on a remote route, with a licence needed to run buses.

Practice item 2: a price taker's revenue.

Practice 2. A potato farmer sells at Tk 40 a kilogram, and the price does not change with her sales. Find total revenue and marginal revenue at 10, 20, 30 and 40 kg.

Practice item 3: a price maker's revenue.

Practice 3. A craftsman sells hand-printed cloth. He can sell 10 metres at Tk 80, 20 at Tk 60, 30 at Tk 40 and 40 at Tk 20 a metre. Find total revenue and marginal revenue at each quantity, measuring the first from zero. At which quantity is total revenue greatest, and what is marginal revenue there?

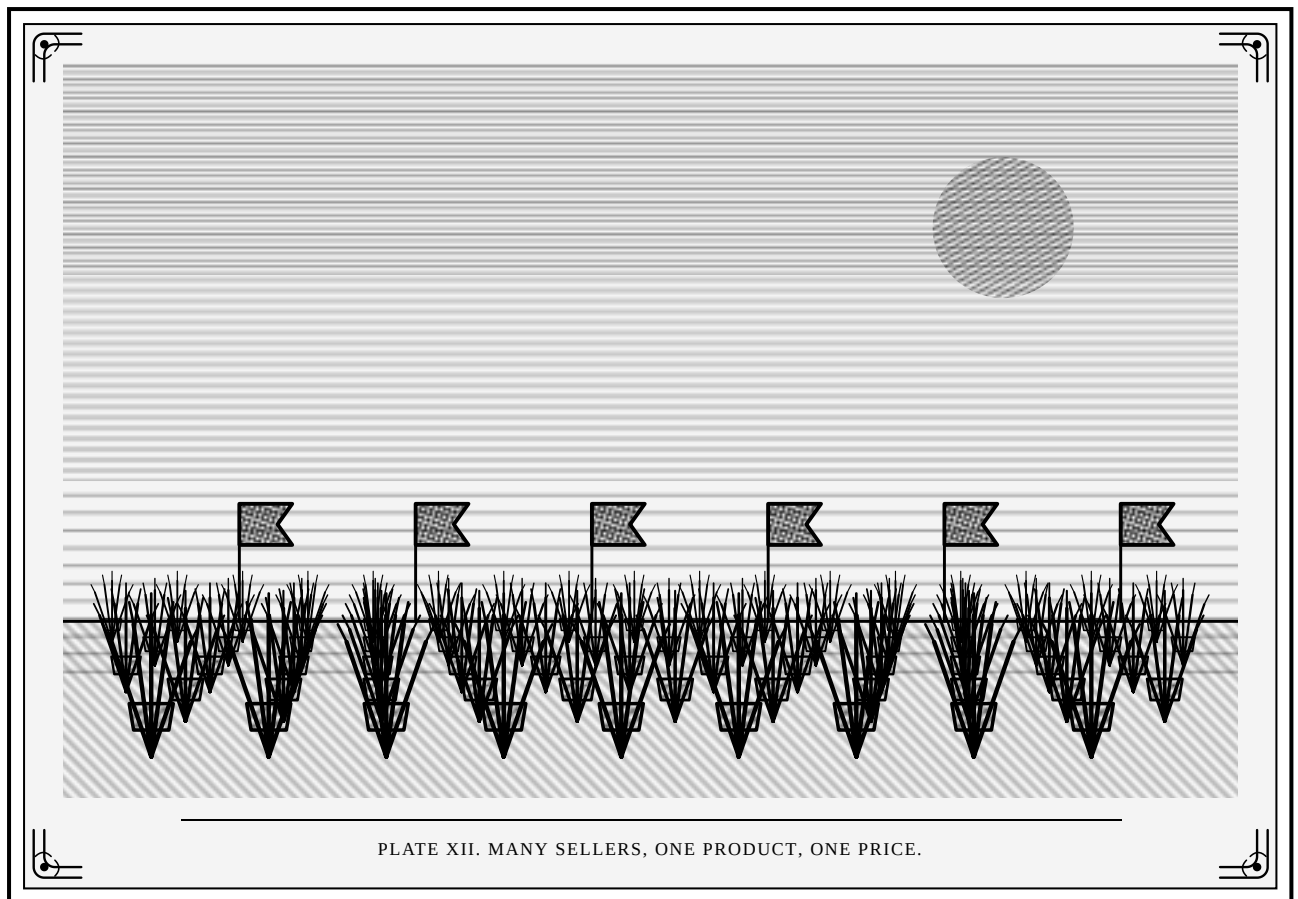


PLATE XII. MANY SELLERS, ONE PRODUCT, ONE PRICE.

CHAPTER XII

Perfect Competition

When no seller can move the price, what is left to choose?



This chapter follows a firm that must accept the market price. It shows how the firm picks its output, when it should keep producing at a loss and when it should close, how the industry adds up its firms, and how entry and exit bring profit to its normal level in the long run.

A puzzle: a firm that cannot choose its price still has a hard decision.

What you should know before reading on.

Five conditions make a market perfectly competitive.

Perfect competition asks for more than pure competition.

Because of these conditions, each firm is a price taker.

Perfectly competitive firm. A firm in perfect competition, which must accept the market price.

Industry. All the firms that make the same product.

Recap of the section.

The price depends on how much time sellers have.

In the market period, supply cannot change.

In the short run, firms change variable inputs only.

Early one morning the bakery learns that the market price of a loaf has fallen. The owner cannot argue with the price, and she cannot change it. She can decide only how many loaves to bake. Should she bake as many as before, bake fewer, or close the shop until the price rises? Each answer costs money. In perfect competition, the output choice is the whole of the firm's decision, so it must be made with care. This chapter gives the rule, tests it as the price falls, and then asks what happens when many firms are drawn into, or driven out of, the market.

You will need: the profit rule and the price taker from Chapter XI; the cost table of Chapter X (average, average variable and marginal cost); and equilibrium from Chapter III.

FEATURES

Chapter XI named perfect competition. We now list its conditions, because every result in this chapter depends on them. First, there are very many sellers and buyers, each so small that none can change the price. Second, the product is homogeneous, so buyers do not care which seller supplies it. Third, entry and exit are free: a new firm can join, and an old firm can leave, at any time. Fourth, buyers and sellers know the price asked everywhere. Fifth, inputs can move freely from one use to another.

Pure competition needs only the first three conditions, as Chapter XI said. Perfect competition adds full knowledge and free movement of inputs. If buyers did not know prices, a seller might charge more than rivals and still sell. So full knowledge is what makes the single price possible. No market meets all five conditions, but some come close: markets for grain, jute, vegetables and foreign currency.

A **perfectly competitive firm** is a firm in perfect competition, and it must accept the market price. The **industry** means all the firms that make the same product; the bakeries of the neighbourhood are the bread industry. The industry as a whole, through its supply and the buyers' demand, decides the price. The firm then faces this price as a given.

Before moving on. Perfect competition needs many sellers, a homogeneous product, free entry and exit, full knowledge and free movement of inputs. Its firms are price takers. The industry is all the firms that make the product.

PRICE DETERMINATION

The market price is set by demand and supply, as Chapter VIII showed. But supply depends on how much time sellers have to change their output, as Chapter VII explained. We therefore look at three periods.

In the market period, output is already made and cannot be changed. The loaves baked this morning are all that exist. Supply is then a fixed amount, and price depends on demand alone. A strong demand raises the price; a weak demand lowers it. A seller of perishable goods, such as fish, must sell today whatever the price.

In the short run, firms can change their variable inputs, but the plant is fixed (Chapter IX). Supply can rise as firms hire more bakers, but only up to the limit of their ovens. Price is set where demand meets this supply. In the long run, firms can change all inputs, and new firms can enter, so supply can change much more. We study that case in section 12.5.

Once the industry has set the price, the single firm faces that price. It can sell any amount at Tk 60, and nothing at Tk 61. So its demand curve is a horizontal line at Tk 60. The firm's marginal revenue equals its price, as Chapter XI showed. The firm's own output is too small to move the industry's price, in the way that one drop of water does not change the level of a lake.

Before moving on. Price is set by the industry. In the market period supply is fixed. In the short run it can rise within the limits of plants. The firm faces a horizontal demand curve at the market price.

The firm's demand curve is a horizontal line.

Recap of the section.

SHORT-RUN EQUILIBRIUM OF THE FIRM

Chapter III defined equilibrium as a state with no reason to change. The **short-run equilibrium** of a firm is the output at which it has no reason to change its output, with its plant fixed. For a profit-maximising firm, this is where marginal revenue equals marginal cost. For a price taker, marginal revenue equals the price, so the condition is price equals marginal cost. If price were above marginal cost, one more loaf would add profit. If price were below marginal cost, one fewer loaf would add profit.

Case file, illustrative. Step 12. The loaf market price is Tk 60. The bakery's best output is 140 loaves a day. Its revenue is Tk 8,400, its cost is Tk 6,800, and its profit is Tk 1,600 a day.

Given. Price Tk 60. Marginal cost of the steps to 20, 50, 90, 120, 140 and 150 loaves: Tk 50, 40, 35, 40, 50 and 80. Total cost: 2,000; 3,200; 4,600; 5,800; 6,800; 7,600. *Find.* The equilibrium output and the profit. *Formula.* Produce while MC is not above P; profit = $P \times Q - TC$.

Steps. The steps to 20, 50, 90, 120 and 140 loaves each have MC of Tk 50 or less, which is below the price. The step to 150 has MC of Tk 80, above the price, so the bakery stops at 140. Profit = $60 \times 140 - 6,800 = 8,400 - 6,800 = \text{Tk } 1,600$. *Answer.* 140 loaves, with a profit of Tk 1,600. *Check.* At 120 loaves the profit is $7,200 - 5,800 = \text{Tk } 1,400$, and at 150 it is $9,000 - 7,600 = \text{Tk } 1,400$. Both are lower.

Chapter X said that normal profit is the owner's opportunity cost and counts as a cost. Profit above that level is **supernormal profit**. A firm earns supernormal profit when its revenue covers all its costs, including normal profit, with something left. We return to this in section 12.5, because the profit in the table counts explicit costs only.

Now let the price fall. The bakery follows the same rule at each price. At Tk 48, the steps to 120 loaves have MC of Tk 40 or less, but the step to 140 has MC of Tk 50, above the price. So the output is 120 loaves. Revenue is $48 \times 120 = \text{Tk } 5,760$, and cost is Tk 5,800, so the result is a loss of Tk 40. At Tk 40, the output is 120 or 90; revenue is Tk 4,800 against a variable cost of Tk 4,800. At Tk 30, no step has MC below 30 except none: the first step costs Tk 50, so the firm should produce nothing.

The **break-even point** is the price at which revenue just covers total cost. For the bakery it is the lowest average cost in the table, about Tk 48, reached at 120 loaves. Below it the firm makes a **loss**, which is revenue below total cost. The **shut-down point** is the price at which revenue covers only variable cost. For the bakery it is the lowest average variable cost, Tk 40. Between Tk 40 and Tk 48, the firm should keep producing, because its revenue covers all its variable cost and part of its fixed cost. Below Tk 40, it should close, because each loaf then adds more to cost than to revenue.

GO SLOWLY HERE.

A firm is in equilibrium when it has no reason to change its output.

Short-run equilibrium. The output at which a firm has no reason to change, with its plant fixed.

Case file: the bakery at a price of Tk 60.

A worked example: setting out the problem.

A worked example: the arithmetic, the answer and a check.

A profit above the normal level is called supernormal.

Supernormal profit. Profit above normal profit.

What happens as the price falls.

Two key prices: break-even and shut-down.

Break-even point. The price at which revenue just covers total cost.

Shut-down point. The price at which revenue covers only variable cost.

A worked example: a firm decides between producing and closing.

Given. Fixed cost Tk 1,000 a day. *Find.* The loss at Tk 48, Tk 40 and Tk 30, if the bakery produces and if it closes. *Steps.* If it closes, it loses its fixed cost, Tk 1,000, at every price. At Tk 48 it produces 120 loaves and loses Tk 40, which is far less than Tk 1,000. At Tk 40 it loses $4,800 - 5,800 = -\text{Tk } 1,000$ at 120 loaves, the same as closing. At Tk 30 it would lose more than Tk 1,000 by producing, so it closes. *Answer.* It produces at Tk 48, is indifferent at Tk 40, and closes at Tk 30. *Check.* The rule is to compare the loss when producing with the fixed cost, and choose the smaller.

The table gives the best output and the result at each price.

THE BAKERY'S CHOICE AT FOUR PRICES (TAKA A DAY)

PRICE	BEST OUTPUT	REVENUE	TOTAL COST	PROFIT OR LOSS
60	140	8,400	6,800	1,600
48	120	5,760	5,800	-40
40	120	4,800	5,800	-1,000
30	0 (close)	0	1,000	-1,000

Three slips to avoid.

Common slips. Do not close the firm as soon as it makes a loss: compare the loss with the fixed cost. Do not use average cost for the choice of output; use marginal cost. Do not forget that fixed cost must be paid even when the firm closes, in the short run.

Recap of the section.

Before moving on. A price taker is in equilibrium where price equals marginal cost. At the break-even price, revenue covers all cost. At the shut-down price, it covers only variable cost. Between these two prices, the firm produces at a loss; below the shut-down price, it closes.

The sentence to test yourself against.

You should now be able to say: a price taker produces where price equals marginal cost, and in the short run it closes only when the price is below average variable cost.

SHORT-RUN EQUILIBRIUM OF THE INDUSTRY

The firm's marginal cost curve is its supply curve.

The rule above tells us the output of the firm at each price. It is therefore the firm's supply curve: the part of the marginal cost curve above the shut-down price. At Tk 40, the bakery supplies about 120 loaves; at Tk 50, 140; at Tk 80, 150. For a price between these points, the firm supplies the output of the last point whose marginal cost does not exceed the price. At Tk 60, it supplies 140.

The industry's supply is the sum of its firms' supplies.

To find the industry supply, add the supplies of all firms at each price. Suppose the industry has 100 firms like the bakery. Then at Tk 40 the industry supplies 12,000 loaves; at Tk 50, 14,000; at Tk 80, 15,000. The table shows the sums. This is the same addition as for market demand in Chapter IV.

SUPPLY OF ONE BAKERY AND OF AN INDUSTRY OF 100 BAKERIES

PRICE (TK)	ONE FIRM (LOAVES)	INDUSTRY (LOAVES)
40	120	12,000
50	140	14,000
80	150	15,000

Industry equilibrium sets the price that the firm takes.

The industry is in short-run equilibrium where industry supply equals industry demand. That gives the price. Each firm then takes that price and applies the rule of section 12.3. If demand is such that the price is Tk 60, each of the 100 bakeries produces 140 loaves, and the industry produces 14,000. If demand falls, the price falls, and the firms move down their marginal cost curves, so the industry

output falls. In the short run, the number of firms is fixed, so firms may earn supernormal profit, or make a loss.

Before moving on. The firm's supply curve is its marginal cost curve above the shut-down price. The industry supply is the sum of its firms' supplies. Demand and industry supply set the price.

Recap of the section.

LONG-RUN EQUILIBRIUM

The short run allows no change in the number of firms. In the long run, **entry** is a new firm joining the industry, and **exit** is an old firm leaving it. Entry is free in perfect competition. The **long-run equilibrium** is a state in which no firm wants to enter or leave the industry, and no firm wants to change its plant.

In the long run, firms may enter or leave.

Long-run equilibrium. A state in which no firm wants to enter or leave the industry, and no firm wants to change its plant.

Supernormal profit draws new firms in.

If firms earn supernormal profit, new firms are attracted. The industry supply increases, because there are more firms. The supply curve moves to the right, so the price falls. Profit falls with it. Entry continues until supernormal profit is zero. If firms make losses, some leave. Industry supply moves to the left, and the price rises. Exit continues until the losses have gone.

Here we must use the cost idea of Chapter X with care. The Tk 1,600 a day of profit counts only explicit costs. The owner's implicit cost is Tk 43,000 a month, which is about Tk 1,433 a day. So the bakery's economic profit at Tk 60 is about $1,600 - 1,433 = \text{Tk } 167$ a day, which is a small supernormal profit. It is enough to attract entry.

The bakery's profit counts explicit costs only.

Given. Explicit cost per day as in the cost table; implicit cost Tk 1,433 a day. *Find.* The long-run price at which economic profit is zero. *Steps.* Add the implicit cost to total cost at each output, and divide by output. At 120 loaves, $(5,800 + 1,433) \div 120 = \text{Tk } 60.28$. At 140 loaves, $(6,800 + 1,433) \div 140 = \text{Tk } 58.81$. At 150 loaves, $(7,600 + 1,433) \div 150 = \text{Tk } 60.22$. *Answer.* The lowest is Tk 58.81, at 140 loaves, so entry pushes the price down to about Tk 59. *Check.* At Tk 58.81, revenue is $58.81 \times 140 = \text{Tk } 8,233$, which equals the total of explicit and implicit cost.

The price falls until economic profit is zero.

In long-run equilibrium, each firm earns only normal profit. The price equals the lowest average cost, counting normal profit as a cost, and it also equals marginal cost, because the firm produces where $P = MC$. Each firm therefore produces at the lowest cost possible with its plant, and buyers pay the lowest price that covers it. This is why economists value competition: it brings the price to the cost of production.

The long-run position: price equals the lowest average cost.

Real markets do not always reach this position. Costs and demand keep changing, so the firms are always adjusting. Entry takes time, and some firms are more efficient than others. The result is a tendency, as Chapter II described for every economic law.

Cautions about the result.

Before moving on. Supernormal profit draws entry; losses drive exit. In the long run, price equals the lowest average cost and each firm earns normal profit. Price also equals marginal cost.

Recap of the section.



SUMMARY

A perfectly competitive firm takes the price and chooses only its output. In the short run it produces where price equals marginal cost. It closes if the price falls below average variable cost. Its marginal cost curve is its supply curve, and the

The chapter in a few lines.

industry supply is the sum of these. In the long run, entry and exit bring the price down to the lowest average cost, so each firm earns only normal profit.

THINK IT THROUGH

A firm in a loss: produce or close?

A tailoring shop, in a perfectly competitive market, has a fixed cost of Tk 600 a day. At a price of Tk 45 a shirt, its best output is 42 shirts, with a total cost of Tk 2,220. Find the revenue and the loss. Compare this loss with the loss if the shop closed. Should it produce or close, and why?

WHERE PEOPLE GO WRONG

Four common errors, each corrected.

Error: "A firm in a loss should close at once." *Correction:* in the short run it should continue if revenue covers variable cost. *Error:* "A firm sets its price by adding a profit to its cost." *Correction:* a price taker accepts the price. *Error:* "In the long run, firms earn no profit at all." *Correction:* they earn normal profit, which counts as a cost. *Error:* "A firm should produce where average cost is lowest." *Correction:* it should produce where price equals marginal cost.

IN HISTORY

Grading made a crop a single product.

The Chicago Board of Trade was founded in 1848 as a meeting place for grain merchants. Before long, merchants agreed on standard grades of grain, so that a buyer in one city could purchase grain from a seller in another without seeing it. Once the grain was graded, a bushel of one grade was the same from every seller. Standard grades made grain a homogeneous product, which is one of the conditions of perfect competition. The case shows how a market can be built toward the conditions.

REVIEW QUESTIONS AND PRACTICE

Six review questions.

Review questions. 1. List the conditions of perfect competition. 2. Why is a perfectly competitive firm's demand curve horizontal? 3. State the condition for the firm's short-run equilibrium. 4. Define break-even point and shut-down point. 5. How is the industry's supply found from the firms' supplies? 6. Explain how entry and exit lead to long-run equilibrium.

Practice item 1: output at a given price.

Practice 1. A tailoring shop has fixed cost Tk 600. Its marginal costs for steps to 8, 20, 33, 42 and 47 shirts are Tk 47.5, 35, 33.1, 43.3 and 70, and its total costs are Tk 980, 1,400, 1,830, 2,220 and 2,570. At a price of Tk 55, find the best output and the profit.

Practice item 2: break-even and shut-down.

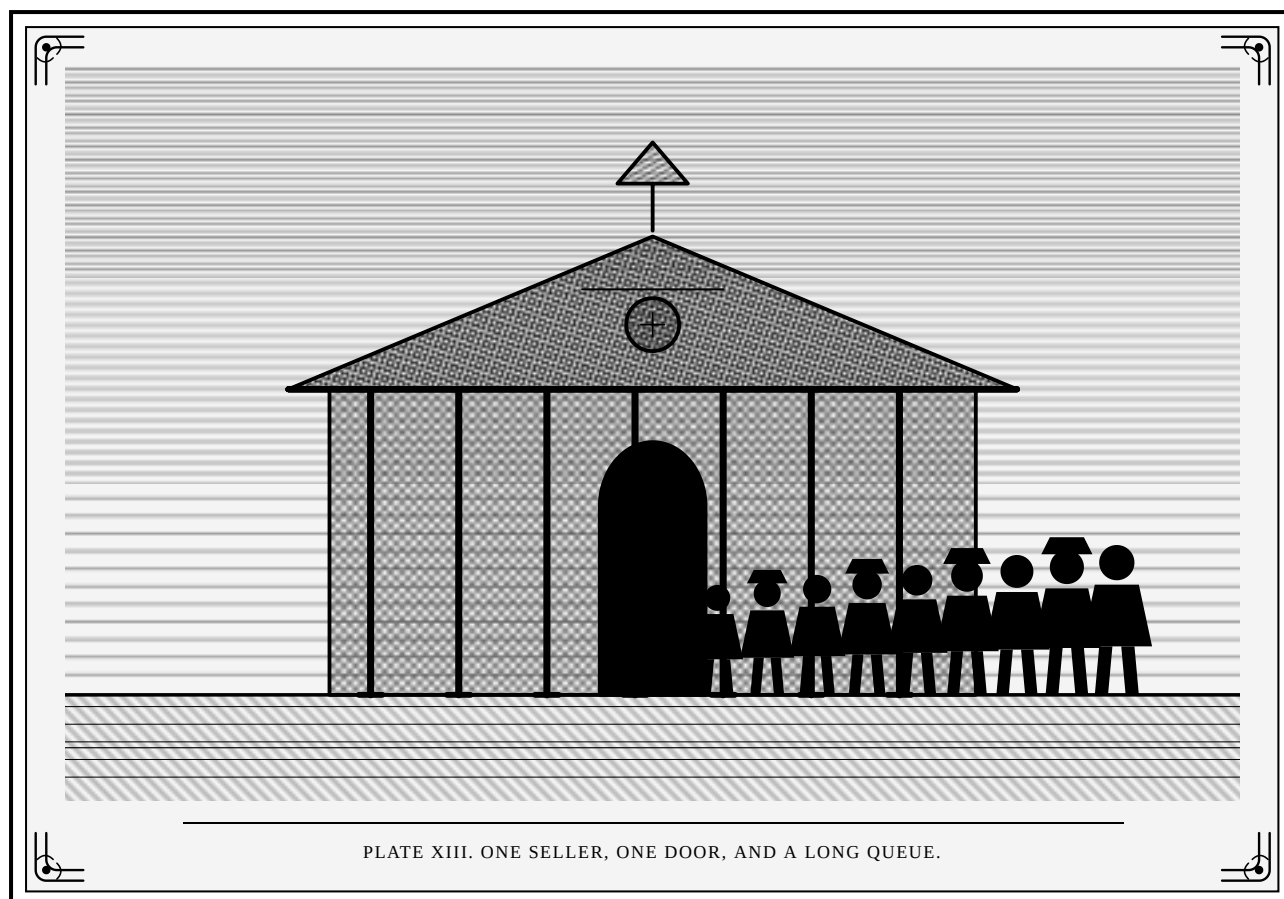
Practice 2. For the shop in Practice 1, the average costs are Tk 122.5, 70, 55.5, 52.9 and 54.7, and the average variable costs are Tk 47.5, 40, 37.3, 38.6 and 41.9. Find the break-even price and the shut-down price, and say what the shop does at Tk 45 and at Tk 36.

Practice item 3: industry supply.

Practice 3. One hundred shops like the one in Practice 1 sell in one market. Find the industry output at a price of Tk 55 and at Tk 38.

Practice item 4: long-run adjustment.

Practice 4. Each shop's owner has an implicit cost of Tk 70 a day. At a price of Tk 55, find a shop's economic profit. What will happen to the number of shops, and to the price? Estimate the long-run price by adding Tk 70 to the total cost at each output and dividing by output.



CHAPTER XIII

Monopoly

What happens when buyers have nowhere else to go?



This chapter studies the market with a single seller. It shows how such a seller chooses a price and an output, what buyers lose when compared with competition, how a seller may charge different buyers different prices, and how public bodies try to control the power of a sole seller.

A town with one ferry shows the question: how does a sole seller choose a price?

The ferry owner chooses the fare; the baker of Chapter XII had to accept one.

A second question follows: does the owner's best fare also serve the town?

You need the profit rule of Chapter XI, the shapes of Chapter XII and surplus from Chapter VIII.

Chapter XI named monopoly; here we state it exactly and ask where it comes from.

A pure monopoly has one seller and no close substitute.

Pure monopoly. A sole seller of a good that has no close substitute.

Real sellers have more or less power, so pure monopoly is a limiting case.

Monopoly power. The ability to set a price above marginal cost without losing all buyers.

Monopoly power comes from barriers that keep new sellers out.

Law can create a monopoly through patents and licences.

Patent. A legal right to be the only seller of an invention for a fixed number of years.

Legal monopoly. A monopoly created by law, such as a patent or a sole licence.

A wide river divides a market town from the villages on the far bank. There is no bridge. The nearest crossing is a day away by road. One family owns the only ferry. The owner could ask Tk 100 from each passenger. Few people would pay, and the boat would cross almost empty. The owner could ask Tk 2, and the boat would be full, but the day's takings would be small. Somewhere between these two fares lies the best one. How does the owner find it?

Notice what is missing. There is no rival to undercut the owner and no market price to accept. In Chapter XII the baker took the loaf price of Tk 60 from the market and chose only the output. The ferry owner chooses the fare itself. That freedom is a form of power. It has limits, because every rise in the fare drives some passengers away.

A second question follows from the first. The fare that is best for the owner may not be best for the town. Some villagers who would gladly pay Tk 10, and whose crossing would cost the owner far less, may stay at home. Both sides lose from that missed crossing. This chapter explains how the sole seller chooses, what is lost, and what can be done about it.

You will need three earlier ideas. First, the profit rule of Chapter XI: a firm raises output while marginal revenue is above marginal cost. Second, the competitive result of Chapter XII, which we use as a standard of comparison. Third, consumers' and producers' surplus from Chapter VIII, because the cost to buyers is measured in surplus.



NATURE AND KINDS

In Chapter XI we met monopoly as a market with one seller. We now state the idea more exactly.

A **pure monopoly** is a market with one seller and no close substitute for its good. The ferry is close to this case. A traveller who does not use it must spend a day on the road, and that is a poor substitute. The word *close* matters. A seller of a good that buyers can replace easily has little power, even if it is the only seller.

Pure monopoly is rare. Most sellers face some substitutes. So economists speak of degrees of **monopoly power**: the ability of a seller to set a price above marginal cost without losing all its buyers. The ferry owner has a great deal of it. A baker in a street with ten bakeries has very little.

Where does monopoly power come from? From barriers to entry, which we met in Chapter XI. Four sources are common. A firm may own the only supply of a key input, such as the one mine in a region. A firm may hold a legal right, as described below. A firm may be so large that no rival could match its low cost. Or buyers may have no wish to switch, because the seller's good is the only one that works with their equipment.

Law is a strong barrier. A **patent** is a legal right to be the only seller of an invention for a fixed period, usually many

years.¹ A **legal monopoly** is any monopoly that the law creates, whether by patent, by sole licence, or by giving one firm the exclusive right to serve a district. The reason is usually to reward effort or to guarantee a service.

Scale can also create a monopoly. Think of water pipes in a city. If two firms each laid pipes along every road, the cost per household would double. One network is cheapest. A **natural monopoly** is a market where one firm can serve all buyers at a lower cost than two or more firms could. Electricity grids and railways are common examples. No law is needed to create it. Cost itself does so.

A market can also have one buyer. A **monopsony** is a market with a single buyer. Imagine a town whose only large employer is one factory. Workers there have nowhere else to sell their labour, so the factory can pay a lower wage than a competitive market would give. When a single seller meets a single buyer, we have a **bilateral monopoly**. A mine that sells all its coal to one power station is an example. No rule fixes the price. It lies somewhere between the lowest price the seller will accept and the highest price the buyer will pay, and bargaining settles it.

Before moving on. A pure monopoly has one seller and no close substitute. Monopoly power is a matter of degree. It comes from owned inputs, patents and licences, or scale. A monopsony is one buyer; a bilateral monopoly joins one seller to one buyer.

Pause and check. A firm holds a patent on a medicine. Another firm sells a different medicine that treats the same illness equally well. Is the first firm a pure monopoly? (Answer at the back.)

Some goods are cheapest to supply through one firm, which gives a natural monopoly.

Natural monopoly. A market in which one firm can supply the whole demand at lower cost than two or more.

The buying side can also have a single member, called a monopsony.

Monopsony. A market with a single buyer.

Bilateral monopoly. A market with one seller facing one buyer.

Before moving on: sole seller, no close substitute, power from barriers.

Pause and check: test the word close in the definition.

DEMAND, REVENUE AND EQUILIBRIUM

We take this section in four steps. The first step is about demand. A competitive baker faces a flat demand line at the market price, because any single baker is too small to move it. A monopolist is different. It is the whole market. So the demand curve it faces is the market demand curve, and that curve slopes down. To sell more, the monopolist must lower its price.

The second step explains why marginal revenue is lower than price. Start with an everyday case. A sole mango seller sells 10 kg at Tk 50 a kilogram. To sell 11 kg she must lower the price to Tk 48, and she cannot lower it for the eleventh kilogram alone, because buyers would simply wait for the lower price. So all 11 kg sell at Tk 48. Revenue rises from Tk 500 to Tk 528. The extra revenue from the last kilogram is only Tk 28, far below its price of Tk 48.

The gain from the new kilogram is its price, Tk 48. The loss is Tk 2 on each of the ten kilograms that were selling at Tk 50. That loss is Tk 20. So the net gain is Tk 48 minus Tk 20, which is Tk 28. This is the whole reason that marginal revenue (the extra revenue from one more unit) is below price for a seller with a downward demand curve. Compare Chapter XII: the price taker lowers no price to sell more, so its marginal revenue equals its price.

Case file, illustrative. Suppose one seller served the whole neighbourhood bun market of Chapter IV. Its demand is the market demand, $P = 18 - Q/30$, where P is the price in taka and Q is the number of buns a day. Total revenue is price times quantity, so $TR = 18Q - Q^2/30$. Marginal revenue is $MR = 18 - Q/15$. Marginal cost is $MC = 6 + Q/30$, the same cost line as the supply curve of Chapter VII.

GO SLOWLY HERE: FOUR STEPS, ONE AT A TIME.

The sole seller's demand curve is simply the market demand curve.

Everyday bridge: a lone mango seller who lowers the price must lower it for all mangoes.

Marginal revenue is below price because the lower price applies to every unit sold.

Case file: tabulating the one-seller bun market.

¹ Under international trade rules the usual period is 20 years from the date of the application. The exact rules differ by country.

Reading the table: each row gives price, revenue, and the two margins.

In the table, read each row from left to right. Price comes from the demand line. Total revenue is price times quantity. Marginal revenue and marginal cost are taken from their formulas at that output. Marginal revenue falls twice as fast as price. At 90 buns the price is Tk 15 but marginal revenue is only Tk 12.

THE ONE-SELLER BUN MARKET, PER DAY (CASE FILE, ILLUSTRATIVE)

BUNS (Q)	PRICE (TK)	TOTAL REVENUE (TK)	MARGINAL REVENUE (TK)	MARGINAL COST (TK)
0	18	0	18	6
30	17	510	16	7
60	16	960	14	8
90	15	1,350	12	9
120	14	1,680	10	10
150	13	1,950	8	11
180	12	2,160	6	12

The rule for a straight demand line: same starting point, twice the slope.

The pattern in the table gives a rule. When demand is a straight line, marginal revenue is also a straight line. It starts at the same price on the vertical axis and falls twice as fast. Demand $P = 18 - Q/30$ has slope $1/30$, so marginal revenue has slope $2/30$, which is $1/15$. That is why $MR = 18 - Q/15$. The marginal revenue line reaches zero at 270 buns, where total revenue is highest.

Marginal revenue is zero where demand has unit elasticity, so the seller avoids the inelastic range.

This links to Chapter V. At 270 buns the price is Tk 9, and we found there that elasticity is exactly one. Beyond that point marginal revenue is negative: selling more would reduce total revenue. A seller who also pays to make each bun would never go that far. So a monopolist always operates on the elastic part of its demand curve, where elasticity is above one.

Step three: the profit rule still applies, so output is set where MR equals MC.

The third step is the choice of output. The profit rule of Chapter XI does not change. If marginal revenue is above marginal cost, one more unit adds more to revenue than to cost, so output should rise. If marginal revenue is below marginal cost, output should fall. Profit is highest where the two are equal. In the table the two meet at 120 buns, where both are Tk 10.

Step four: the price is read from the demand curve, not from the cost curve.

The fourth step is the price, and it is where beginners often slip. The rule $MR = MC$ gives the output only. To find the price, go up from that output to the demand curve and read the price buyers will pay. At 120 buns the demand line gives Tk 14. The price is above marginal cost (Tk 10). That gap is the monopolist's power at work. In a competitive market the price equals marginal cost, as Chapter XII showed.

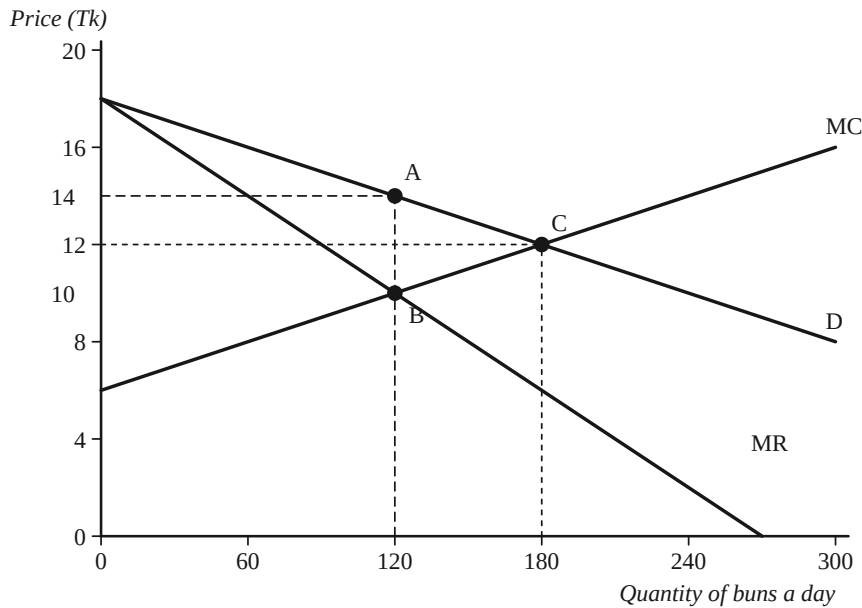


FIG. 18. The one-seller bun market. The seller sets output where marginal revenue meets marginal cost (B), then reads the price from the demand curve (A). Competition would give point C.

Worked example. Given: demand $P = 18 - Q/30$ and marginal cost $MC = 6 + Q/30$. Find: the profit-maximising output and price. **Formula:** in words, marginal revenue equals marginal cost; in symbols, $MR = MC$. We first need the MR line. Here are the steps.

Worked example: finding the monopoly output and price by algebra.

STEPS FOR THE MONOPOLY OUTPUT

STEP	WORKING
1. Marginal revenue: same intercept, double slope	$MR = 18 - Q/15$
2. Set MR equal to MC	$18 - Q/15 = 6 + Q/30$
3. Move the terms in Q to one side	$18 - 6 = Q/15 + Q/30$
4. Combine, using a common denominator of 30	$12 = 2Q/30 + Q/30 = 3Q/30 = Q/10$
5. Solve for Q	$Q = 12 \times 10 = 120$ buns
6. Read the price from the demand line	$P = 18 - 120/30 = 18 - 4 = \text{Tk } 14$

Answer: the seller offers 120 buns a day at Tk 14 each. **Check:** at 120 buns, $MR = 18 - 120/15 = 18 - 8 = 10$, and $MC = 6 + 120/30 = 6 + 4 = 10$, so the two agree. Total revenue is $14 \times 120 = \text{Tk } 1,680$, which matches the table. Variable cost is the area under the marginal cost line, 960, so revenue above variable cost is Tk 720. This equals the producers' surplus of the next section. Fixed cost is not given, so we cannot say more about profit.

The check: MR and MC agree at 120, and revenue is price times quantity.

Common slips. First, reading the price from the marginal cost line instead of the demand line. Second, writing the MR slope as $1/30$ instead of $1/15$. Third, mixing units: P is in taka, Q in buns a day. Fourth, expecting a supply curve. A monopolist has none. It does not respond to a given price. It chooses price and output together from its demand and cost.

Common slips: using the cost line for price, or forgetting to double the slope.

Before moving on. The monopolist faces the market demand curve. Its marginal revenue is below price. It picks the output where MR equals MC, then reads the price from the demand curve.

Before moving on: demand is the market's, MR is below price, price is read from demand.

Pause and check: should the seller raise output from 90 to 120 buns, and then from 120 to 150?

You should now be able to say why a sole seller's price exceeds marginal cost.

We compare the one-seller market with the competitive market using the same demand and cost.

Case file: comparing the competitive and one-seller results.

The table sets the two markets side by side.

Buyers lose 300, the seller gains 180, and 120 vanishes.

Deadweight loss is the surplus that vanishes when output is cut.

Deadweight loss. The surplus lost to everyone when output falls below the competitive level.

Pause and check. Use the table. Moving from 90 to 120 buns, total revenue rises from Tk 1,350 to Tk 1,680. What is the extra revenue per bun? Marginal cost over this step is between Tk 9 and Tk 10. Should the seller go ahead? Repeat for 120 to 150 buns. (Answer at the back.)

You should now be able to say: "A sole seller sets output where marginal revenue equals marginal cost, and then charges the highest price buyers will pay for that output, which is above marginal cost."

COMPARED WITH COMPETITION

Now use the same demand and the same cost lines to compare two worlds. In the first, many bakers compete and the price settles where demand equals supply. In the second, one seller serves the whole market. Because the buyers and the costs are the same, any difference comes from the market structure alone.

Case file, illustrative. Under competition (Chapter VIII), $P = 18 - Q/30$ meets $P = 6 + Q/30$ at $Q = 180$ and $P = \text{Tk } 12$. Under one seller, $Q = 120$ and $P = \text{Tk } 14$. Consumers' surplus is 540 under competition. Under the single seller it is the triangle between the demand line and the price of Tk 14, which is $\frac{1}{2} \times 120 \times (18 - 14) = \text{Tk } 240$. Producers' surplus is revenue less the area under marginal cost: $1,680 - 960 = \text{Tk } 720$.

The table sets the results side by side. The first column is the competitive market, the second the one-seller market.

COMPETITION AND ONE SELLER COMPARED, PER DAY (CASE FILE, ILLUSTRATIVE)

MEASURE	COMPETITIVE MARKET	ONE SELLER
Output (buns)	180	120
Price (Tk)	12	14
Consumers' surplus (Tk)	540	240
Producers' surplus (Tk)	540	720
Deadweight loss (Tk)	0	120
Total surplus (Tk)	1,080	960

Look at the changes. Buyers lose Tk 300 of surplus. The seller gains only Tk 180. The difference, Tk 120, is gained by nobody. Part of what buyers lose is a plain transfer: they pay Tk 2 more on each of 120 buns, which is Tk 240, and the seller receives it. The rest disappears.

This vanished amount is the **deadweight loss**: the surplus lost to everyone when output falls below the competitive level. Consider the 60 buns between 120 and 180. Each is worth to a buyer between Tk 12 and Tk 14. Each would cost the seller between Tk 10 and Tk 12 to make. So each is worth more than it costs, yet none is made, because the seller sees that selling them would force down the price on the other 120 buns. The gain from these trades is lost.

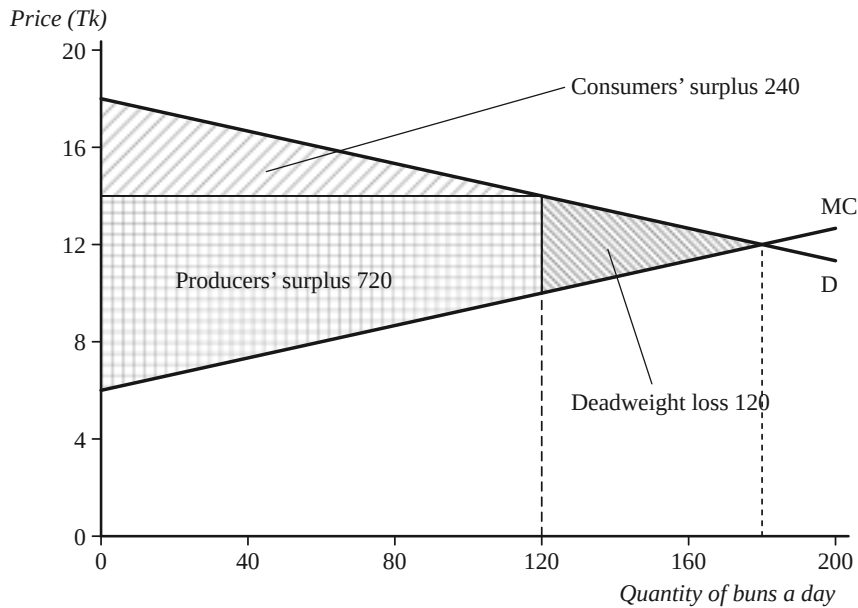


FIG. 19. *Surplus under one seller. The loss falls on buyers and, in part, on the seller.*

In the figure, the hatched triangle above the price is consumers' surplus (240). The crossed area between the price and marginal cost is producers' surplus (720). The small dark triangle is the deadweight loss (120). Its corners are the monopoly point, the point on marginal cost directly below it, and the competitive point.

A fair judgement does not stop there. Monopoly is not always worse in every way. A natural monopoly may have much lower cost than several small firms, and then one firm may give cheaper service even after allowing for its power. A patent gives the inventor a reward that may make the invention possible. On the other hand, a sole seller faces little pressure to keep its costs low. So the result depends on the case, and public policy tries to separate the harmful cases from the useful ones.

Seen from the buyer's seat, monopoly means a higher price and fewer goods. Seen from the owner's seat, it means a higher price and a larger share of the surplus. Seen from the town's seat, it means a smaller total.

PRICE DISCRIMINATION

Suppose the ferry owner could charge students less than office workers. Would that be wise? Often it is, and many sellers do it.

Price discrimination means selling the same good at different prices to different buyers, where the difference is not due to cost. A cinema charges less for afternoon shows. A doctor charges richer patients more. A bus company offers lower fares to children. The key words are *same good*: if the price differs because the cost differs, as with a longer journey, that is not discrimination.

Three conditions are needed. The seller must have some power over price, so it is not a price taker. The seller must be able to keep the groups apart, so that people who pay the low price cannot resell to those who pay the high price. And the groups must differ in how strongly they react to price, which means different elasticities. A cinema separates groups by showtime and by student card.

The figure shows the three areas; the lost triangle lies between the monopoly and the competitive output.

A fair judgement also counts what monopoly may bring.

Viewpoint: buyer and owner see the same market differently.

The ferry owner might do even better by charging different people different fares.

Price discrimination means different prices for the same good.

Price discrimination. Selling the same good at different prices to different buyers, for reasons not based on cost.

Three conditions must hold for the plan to work.

The rule: charge more where demand is less elastic, so that MR is equal in each market.

Case file: a small cinema shows two prices for one good.

The rule for the seller follows from the profit rule. Treat each group as its own market. In each market set marginal revenue equal to marginal cost. Since marginal cost is the same, marginal revenue ends up equal in both markets. The group whose demand is less elastic gets the higher price. We show this with a worked case.

Case file, illustrative. This example uses a small town cinema, not the bakery. Cost per seat filled is Tk 10, whatever the show. The evening show draws working adults, with demand $P = 30 - Q/10$, where Q is seats sold. The afternoon show draws students, with demand $P = 20 - Q/20$. Fixed costs are left out.

STEPS FOR THE DISCRIMINATING SELLER

STEP	WORKING
1. Marginal revenue in each market: same intercept, double slope	Evening: $30 - Q/5$; afternoon: $20 - Q/10$
2. Evening: set $MR = MC$	$30 - Q/5 = 10$, so $Q = 100$ seats
3. Evening price	$P = 30 - 100/10 = \text{Tk } 20$
4. Afternoon: set $MR = MC$	$20 - Q/10 = 10$, so $Q = 100$ seats
5. Afternoon price	$P = 20 - 100/20 = \text{Tk } 15$
6. Profit with two prices	$(20 - 10) \times 100 + (15 - 10) \times 100 = 1,000 + 500 = \text{Tk } 1,500$
7. Compare one price, Tk 17	Evening $300 - 10 \times 17 = 130$; afternoon $400 - 20 \times 17 = 60$; $(17 - 10) \times 190 = \text{Tk } 1,330$

The two-price plan beats any single price, and the higher price falls on the less elastic group.

Answer: the cinema charges Tk 20 in the evening and Tk 15 in the afternoon, and earns Tk 1,500. *Check:* the best single price (about Tk 16.67) would earn only about Tk 1,333, so two prices do better. Elasticity at the chosen points confirms the rule: in the evening it is $10 \times 20/100 = 2.0$, and in the afternoon $20 \times 15/100 = 3.0$. The group that reacts less strongly to price, the evening audience, pays more.

Effects are mixed: some buyers lose, others gain.

The effects on buyers are mixed. Compare a single price of Tk 17. Evening buyers now pay Tk 20 instead of Tk 17, so they lose. Students pay Tk 15 instead of Tk 17, so they gain. Total seats sold are 200, against 190. In the extreme case of *perfect price discrimination*, where each buyer pays the most he or she would pay, the seller takes all consumers' surplus, but output reaches the competitive level and no deadweight loss remains.

CONTROLLING MONOPOLY

Governments use three tools: regulation, competition law and public ownership.

Regulation sets rules on price or conduct; a ceiling can raise output.

Regulation. Rules set by a public body on a firm's prices or conduct.

Because monopoly can harm buyers, governments act. Three tools are common.

Regulation means rules set by a public body on a firm's prices or conduct. The usual rule is a price ceiling. A well-set ceiling can even raise output. Return to the one-seller bun market. Left alone, the seller offers 120 buns at Tk 14. Suppose the price may not exceed Tk 12. The seller can sell up to 180 buns at that price, so marginal revenue is a flat Tk 12 up to 180 buns, while marginal cost reaches Tk 12 at exactly 180 buns. Output rises to the competitive level. A ceiling set too low, however, can bring a loss that drives the firm out, so regulators set it with care. For a natural monopoly, a price equal to marginal cost may not cover total cost, so a higher price is allowed.

Competition law forbids abuse of power, such as agreements to fix prices, and can stop a merger that would create a dominant firm. In Bangladesh the Competition Act 2012 set up a Competition Commission for this purpose (Chapter II). The third tool is public ownership, where the state owns the firm. Railways and water supply are often run this way. Its risk is that a public firm may also face little pressure to keep costs low. Each tool has costs, so the choice depends on the market.

Competition law and public ownership are the other two tools.



SUMMARY

A monopoly has one seller. Its power comes from barriers: inputs, law or scale. The monopolist's demand curve is the market's, and marginal revenue lies below price, so it sets MR equal to MC and reads the price from demand. In the bun market this gives 120 buns at Tk 14, against 180 at Tk 12 under competition. Buyers lose more than the seller gains, and the difference is deadweight loss (Tk 120). A seller may charge different groups different prices if it can keep them apart. Governments respond by regulation, competition law and public ownership.

Summary of the chapter in six sentences.

THINK IT THROUGH

A ferry has no rival. Its marginal cost is Tk 10 a passenger. Demand is $P = 40 - Q/5$, where Q is passengers a day. (a) Find the owner's output and fare. (b) Find the output if the fare equalled marginal cost, and the deadweight loss from the owner's choice. (c) The authority sets a fare ceiling of Tk 18. What happens to output? Does the owner gain or lose? Explain why a ceiling can raise output here, when Chapter VIII showed ceilings often reduce it. (Hint: look at what the ceiling does to marginal revenue.)

A ferry problem with a fare ceiling.

WHERE PEOPLE GO WRONG

"A monopolist can charge any price it likes." It can choose its price, but buyers then choose how much to buy. A high price loses buyers. "The monopolist reads its price from the marginal cost line." It reads the price from the demand line. "Monopoly profit is the same as deadweight loss." Profit goes to the seller; deadweight loss goes to nobody. "Price discrimination is always unfair." It can bring in buyers who would otherwise be left out, as the student audience shows, though it can also raise prices for the less sensitive group.

Four common mistakes about monopoly.

IN HISTORY

Standard Oil was founded in Ohio in 1870. By the 1880s it controlled most of the oil refining in the United States, and railways gave it special rates that rivals could not match. The Sherman Act of 2 July 1890 made such monopolising unlawful (Chapter XI). On 15 May 1911 the United States Supreme Court ruled that Standard Oil was an unlawful monopoly and ordered it divided into separate companies. Economists still debate how much the break-up helped buyers, but the case shows a state using competition law to dissolve a monopoly.

The break-up of Standard Oil in 1911.

REVIEW QUESTIONS AND PRACTICE

Review questions on the chapter.

REVIEW QUESTIONS

NO.	QUESTION
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- | | |
|---|---|
| 1 | Why does a pure monopoly still face a downward-sloping demand curve? |
| 2 | Explain, with a numerical example, why marginal revenue is below price. |
| 3 | Name three sources of monopoly power and give one example of each. |
| 4 | Describe deadweight loss in words and show it on a diagram. |
| 5 | State the three conditions for price discrimination. |
| 6 | Compare regulation with public ownership as ways of controlling a natural monopoly. |
-

Practice problems, from simple to combined.

PRACTICE

ITEM	PROBLEM
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- | | |
|---|---|
| 1 | A sole seller faces demand $P = 50 - 2Q$. Find the marginal revenue line; marginal revenue and total revenue at $Q = 5$; and the output at which marginal revenue is zero. |
| 2 | Demand is $P = 80 - 2Q$ and marginal cost is $MC = 20 + Q$. Find the sole seller's output and price. Then find the output and price under competition. |
| 3 | Using item 2, find consumers' surplus under each market and the deadweight loss from monopoly. |
| 4 | A hill bus operator has marginal cost Tk 4. Adults' demand is $P = 20 - Q/5$ and children's is $P = 12 - Q/10$. Find the fare for each group and the profit. Check which group's demand is less elastic. |
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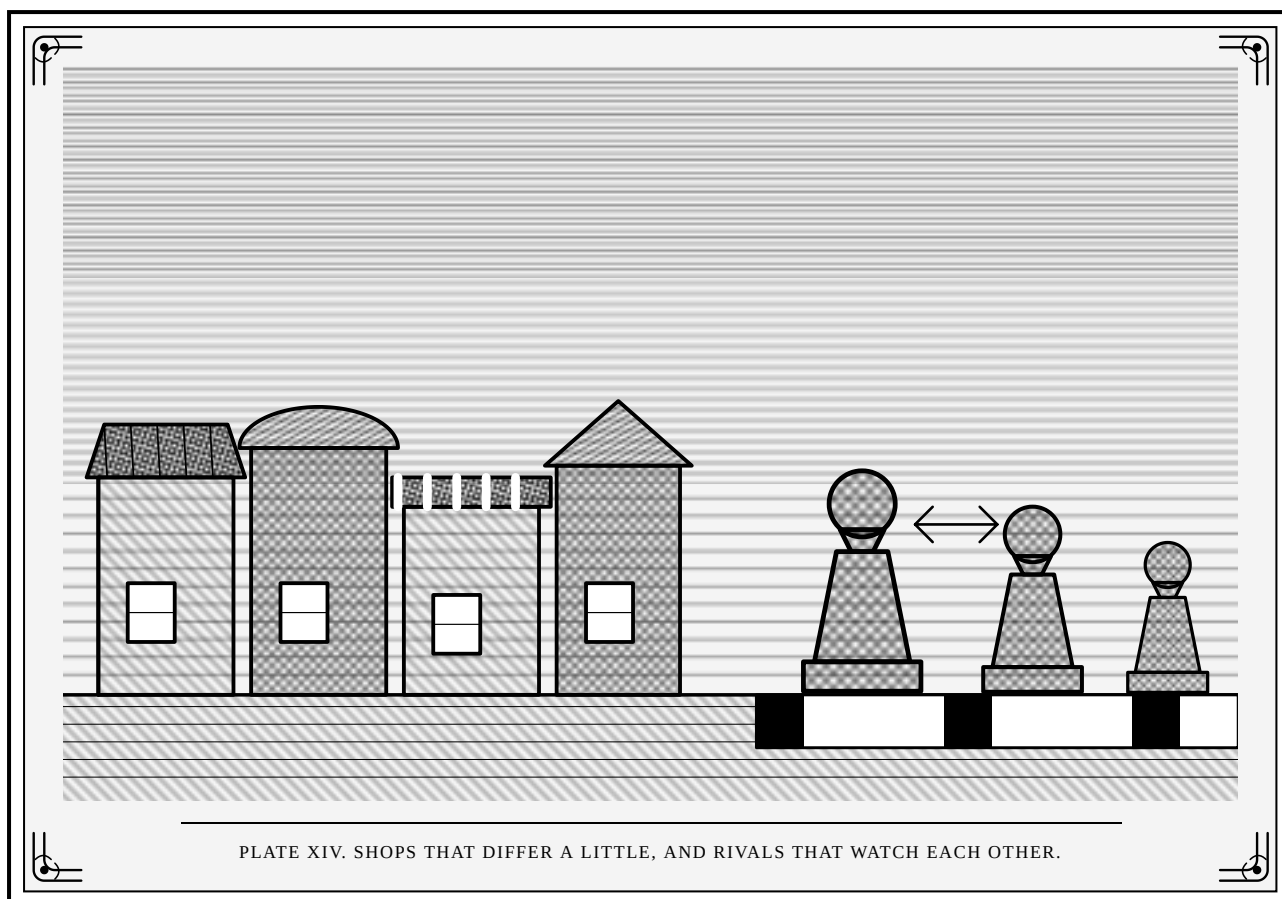


PLATE XIV. SHOPS THAT DIFFER A LITTLE, AND RIVALS THAT WATCH EACH OTHER.

CHAPTER XIV

Monopolistic Competition and Oligopoly

Many sellers, each a little different; or a few, each watching the others.



This chapter studies the two markets that lie between perfect competition and monopoly. In one, many firms sell goods that differ a little. In the other, a few firms watch each other closely. It shows how price and output are set in each, and then compares all four kinds of market.

Two puzzles: why small differences matter, and why a few big sellers rarely cut prices.

The second puzzle concerns markets with a few big sellers.

The two puzzles share one answer: the seller is neither alone nor lost in a crowd.

You need the profit rule, the cost curves and the shapes of Chapters XI to XIII.

Walk down a busy lane in any Bangladeshi town and count the bakeries. There may be a dozen. No two sell exactly the same bun. One bakes early and sells warm. One uses more butter. One stands beside the bus stop. Their prices are close, yet not equal. A baker who charges Tk 1 more than the neighbour does not lose every customer. Why not? And yet no baker on that lane grows rich for long. Why not?

Now move to a different market: mobile phone service. A few large firms supply nearly everyone. When one firm announces a cheaper plan, the others often reply within days. When one firm raises its price, rivals often leave theirs unchanged. Yet prices in such markets do not fall every week. What holds them?

Both puzzles have one root. In each market the seller is neither alone, as the ferry owner was in Chapter XIII, nor lost in a crowd of identical sellers, as the baker was in Chapter XII. This chapter studies the two middle cases.

You will need the profit rule of Chapter XI, the cost curves of Chapter X, and the equilibrium ideas of Chapters XII and XIII. We use them again here, with one change: now the seller's demand curve slopes down, but rivals are close.



THE IDEA

Monopolistic competition mixes a little monopoly with plenty of competition.

The name shows the mix: monopoly because of the difference, competition because of the rivals.

Many everyday markets fit this description.

Case file: the bakery adds a branded milk bun.

Before moving on: many sellers, differentiated goods, free entry.

Take the lane of bakeries again. Each baker has a small monopoly over a bun that buyers think is a little different. But each baker also has many rivals whose buns are close substitutes. Monopolistic competition is the market that results.¹ It has three features. First, there are many sellers. Second, the sellers' goods are differentiated: they are alike, but not the same. Third, entry is free, so new sellers can join when profits tempt them.

The name explains the mix. The word *monopoly* points to each firm's power over its own variety: it is the sole seller of that variety. The word *competition* points to the many close rivals and the free entry. Two books of 1933 set out the idea: Edward Chamberlin, *The Theory of Monopolistic Competition*, and Joan Robinson, *The Economics of Imperfect Competition*. Before them, textbooks treated markets as either perfectly competitive or monopolistic.

Many everyday markets in Bangladesh fit. Tailors, restaurants, hair salons, private tutors and pharmacies each have many sellers who differ in place, service or quality. Compare the market with the earlier two. A perfectly competitive seller faces a flat demand line, since all goods are alike. A monopolist faces the whole market demand. A monopolistic competitor sits in between: its demand slopes down, but because rivals are close, buyers react strongly to its price. That demand is elastic.

Case file, illustrative. The bakery adds a branded milk bun. Three rival bakeries in the neighbourhood already sell their own buns, so the milk bun competes closely with several others, but it is the bakery's own variety. The figures used in 14.3 are fixed here: demand for the milk bun before any rival copies it is $P = 20 - Q/10$, where Q is the number sold a day; total cost for the milk bun line is $TC = 60 + 8Q + Q^2/20$, with all costs counted, implicit costs included. These are separate from the loaf figures of Chapter X.

Before moving on. Monopolistic competition has many sellers, goods that differ a little, and free entry. Each firm has a downward demand curve, which is elastic because rivals are close.

¹ Economists call any market in which sellers face downward-sloping demand *imperfect competition*. That includes monopoly, monopolistic competition and oligopoly.

PRODUCT DIFFERENTIATION AND SELLING COSTS

Why does a customer pass a cheaper bakery and walk on to another? Because the goods are not quite alike in the buyer's eyes.

Product differentiation means making a good different from rivals' goods, in fact or in the buyer's mind. The difference may be real: more butter, a better fit, a faster repair. It may be one of place: a shop close to home. It may be one of service: free delivery, longer hours. It may even be a matter of belief, when the goods are alike but the buyer thinks otherwise. Whatever its source, the difference makes some buyers loyal, and loyalty gives the seller some power over price.

Sellers use a brand to fix the difference in the buyer's mind. A **brand** is a name or sign that identifies one seller's good. The milk bun, sold in a printed paper bag with the bakery's name, is a brand. A brand helps a buyer remember the good and find it again. It also lets the seller build a reputation that is worth money.

Sellers also spend money to tell buyers about the difference. **Selling cost** means money spent to promote sales, such as signboards, flyers, advertisements and free samples. It differs from the cost of production, which is the cost of making the good. A Tk 2,000 signboard is a selling cost. The flour in the bun is a production cost. Selling costs push the demand curve to the right, since more buyers learn about the good. They also add to the seller's average cost.

Advertising does two jobs. It informs: a buyer learns that the warm bun is on sale at 7 a.m. It persuades: a buyer comes to prefer one bun to another because of what an advertisement says. Economists disagree about which job matters more. The information is useful. The persuasion can be wasteful if every seller advertises and no one gains in the end.

Sellers in these markets often compete in ways other than price. **Non-price competition** is rivalry through quality, brand, service or advertising, not through price. A baker who adds a new filling is competing in this way. Such competition gives buyers more variety, but it costs money, and that cost is paid by buyers through prices.

Before moving on. Product differentiation creates loyalty. A brand fixes the difference in the mind. Selling cost is money spent to promote sales. Non-price competition is rivalry by quality or service.

Pause and check. The bakery pays Tk 3,000 for a new signboard and Tk 400 for flour. Which is a selling cost and which a production cost? (Answer at the back.)

Differences make buyers loyal; loyalty gives the seller some power.

Product differentiation makes buyers see one seller's good as different.

Product differentiation. Making a good different from rivals' goods, in fact or in the buyer's mind.

A brand names a seller's good and fixes the difference in the buyer's mind.

Brand. A name or sign that identifies one seller's good and sets it apart from others.

Selling costs are paid to win buyers, not to make goods.

Selling cost. Money spent to promote a seller's goods, such as advertising.

Advertising informs and persuades; both effects can occur at once.

Non-price competition. Rivalry among sellers through quality, brand, service or advertising, not through price.

Non-price competition is rivalry through quality, service and brand.

Before moving on: differences, brands and selling costs.

Pause and check: a signboard and the flour.

THE THEORY

Now we set out how the monopolistic competitor chooses. Its demand curve slopes down, as a monopolist's does, so its marginal revenue lies below price (Chapter XIII). It follows the same profit rule: it raises output until marginal revenue equals marginal cost. The new element is entry, which acts in the long run.

Worked example. Given: milk bun demand $P = 20 - Q/10$ and $TC = 60 + 8Q + Q^2/20$, so $MC = 8 + Q/10$. Find: output, price and profit. Formula: $MR = MC$; profit = total revenue – total cost. The steps follow.

The firm's demand slopes down but is elastic; the profit rule gives its output.

Worked example one: the short run.

SHORT-RUN EQUILIBRIUM OF THE MILK BUN LINE (CASE FILE, ILLUSTRATIVE)

STEP	WORKING
1. Marginal revenue: same intercept, double slope	$MR = 20 - Q/5$
2. Set MR equal to MC	$20 - Q/5 = 8 + Q/10$
3. Collect terms	$12 = Q/5 + Q/10 = 3Q/10$
4. Solve for Q	$Q = 12 \times 10/3 = 40$ buns
5. Price from the demand line	$P = 20 - 40/10 = \text{Tk } 16$
6. Revenue and cost	$TR = 16 \times 40 = 640$; $TC = 60 + 320 + 80 = 460$
7. Profit	$640 - 460 = \text{Tk } 180$ a day

The answer and its check: the profit equals the price-minus-average-cost gap times output.

Profit attracts entry: rivals copy the milk bun, and each firm's demand shifts left.

Worked example two: after entry, demand is $P = 14 - Q/10$ and the firm just breaks even.

Answer: the bakery sells 40 milk buns a day at Tk 16 and earns Tk 180 a day above all costs. *Check:* average cost at 40 is $460/40 = \text{Tk } 11.50$, and $(16 - 11.50) \times 40 = 180$. This is supernormal profit, a profit above normal (Chapter XII).

A profit of Tk 180 does not go unnoticed. Entry is free, so rivals bake their own milk buns. Each new variety takes buyers from the others. For our bakery, demand falls: at every price, fewer buns sell. The demand curve shifts left. It also becomes more elastic, because there are now more close substitutes. Entry continues as long as a profit remains. It stops when the demand curve just touches the average cost curve, so that price equals average cost and profit is zero.

Worked example. Given: after entry, demand is $P = 14 - Q/10$; cost is unchanged. *Find:* output, price and profit. *Formula:* $MR = MC$ again.

LONG-RUN EQUILIBRIUM OF THE MILK BUN LINE (CASE FILE, ILLUSTRATIVE)

STEP	WORKING
1. New marginal revenue	$MR = 14 - Q/5$
2. Set MR equal to MC	$14 - Q/5 = 8 + Q/10$
3. Collect terms	$6 = 3Q/10$, so $Q = 20$ buns
4. Price	$P = 14 - 20/10 = \text{Tk } 12$
5. Average cost at 20 buns	$AC = 60/20 + 8 + 20/20 = 3 + 8 + 1 = \text{Tk } 12$
6. Compare	$P = AC = 12$, so profit = 0

The check: the AC curve has the same slope as demand at 20 buns, so the curves just touch.

Answer: 20 buns at Tk 12, with no profit above normal. *Check:* total revenue $12 \times 20 = 240$ and total cost $60 + 160 + 20 = 240$. The slope of the average cost curve at 20 buns is $-60/400 + 1/20 = -0.10$, the same as the slope of demand, so the two curves just touch.

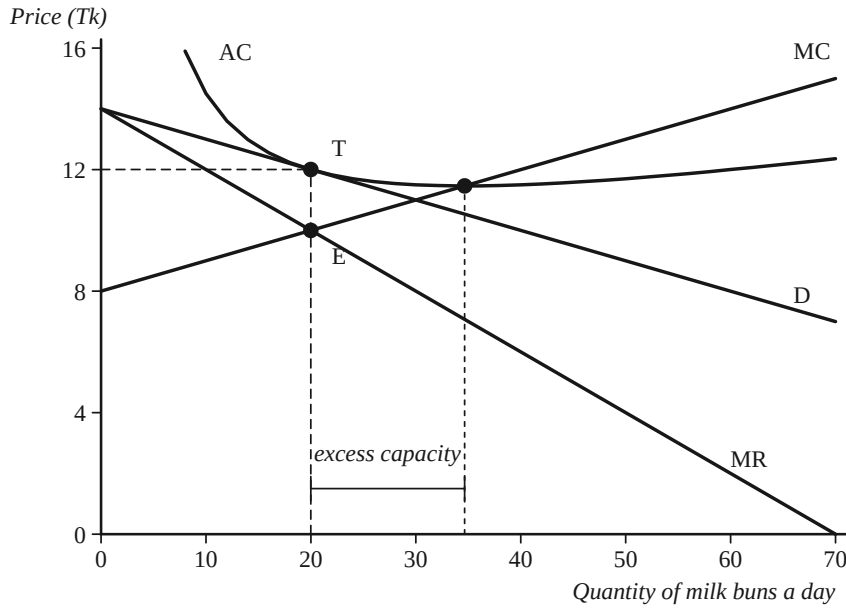


FIG. 20. Long-run equilibrium of the milk bun line. Demand touches average cost at T; output is 20, below the output of lowest average cost.

In the figure, the demand line touches the average cost curve at T, directly above the point E where marginal revenue equals marginal cost. A firm that earns zero supernormal profit still earns *normal profit*, since all costs, including the owner's time and money, have been counted (Chapter X). It does not close. If profits had turned into losses, firms would leave, and demand for those who remain would shift right again.

There is a second result. The lowest average cost is at 34.6 buns, where average cost is about Tk 11.46. The firm makes only 20. The gap, about 15 buns, is **excess capacity**: output below the level of lowest average cost. The firm is not idle. Its equipment could produce more at lower cost per bun, but the firm cannot sell more at a price that covers cost, because its demand curve slopes down.

Compare this with perfect competition (Chapter XII). A competitive firm ends at the lowest point of average cost, and price equals marginal cost. Here price (Tk 12) is above marginal cost (Tk 10), and output is below the lowest-cost level. Some economists call this waste. Others reply that buyers value the variety: they could have a single standard bun made at lower cost, but they prefer a choice. The truth depends on how much buyers value the variety.

Before moving on. The firm sets MR equal to MC. Profit draws in new firms until demand just touches average cost. Then profit above normal is zero and capacity is unused.

Pause and check. In the long run the milk bun line earns zero profit above normal. Should the owner close it? Explain in one sentence. (Answer at the back.)

The figure shows tangency: price equals average cost, so entry stops.

Excess capacity: the firm makes less than the output of lowest average cost.

Excess capacity. Output that lies below the level at which average cost is lowest.

Compared with perfect competition, price is higher than marginal cost but buyers gain variety.

Before moving on: marginal revenue meets marginal cost; entry removes profit; excess capacity remains.

Pause and check: does a firm with zero profit close?

Oligopoly has few sellers and high barriers, so each firm's choice affects the others.

Interdependence. Each firm's best choice depends on the choices of its rivals.

Interdependence means no single rule fixes the price.

Price war. A series of price cuts, each rival cutting below the last.

A price war leaves every firm worse off.

Kinked demand curve. A demand curve with a bend at the present price.

Kinked demand: rivals follow a cut but not a rise.

Case file: a kinked-demand variant of the milk bun.

The kink gives a break in marginal revenue, so costs can change without a change in price.

OLIGOPOLY

Now the second puzzle. Oligopoly is a market with a few sellers, and its key feature is **interdependence**: each firm's best choice depends on what rivals do. Barriers to entry keep new firms out, so the few stay few. Mobile phone service and air travel on a busy route are examples. A firm that cuts its price cannot treat the market as given, because rivals may cut too, and the gain disappears.

Because of interdependence, economists have several models of oligopoly, not one. We study the four main ideas: price war, kinked demand, collusion and price leadership.

A **price war** is a series of price cuts in which each rival cuts below the last. Everyone's profit falls. Firms fear this, and so they often avoid price cuts and compete in other ways, such as brand, service and advertising.

The **kinked demand curve** is a simple model of this fear. A firm believes that if it raises its price, rivals will not follow, so it loses many buyers: demand is elastic above the present price. If it cuts its price, rivals will follow, so it gains little: demand is inelastic below. The demand curve has a bend at the present price.

Case file, illustrative. A variant of the milk bun. Suppose the four bakeries are the only sellers of branded buns and each sells at Tk 16, with this bakery selling 40 a day. Above Tk 16, demand is $P = 20 - Q/10$, because rivals hold their prices. Below Tk 16, demand is $P = 24 - Q/5$, because rivals match. Marginal cost at 40 buns is taken as Tk 10; only the demand lines are borrowed from the milk bun above.

The bend creates a break in the marginal revenue line. Above the kink, marginal revenue is $20 - Q/5$, which is Tk 12 at 40 buns. Below the kink, it is $24 - 2Q/5$, which is Tk 8 at 40 buns. So between Tk 8 and Tk 12 there is a vertical gap. Marginal cost at 40 buns is Tk 10, inside the gap. So the firm has no reason to change its output or price. If marginal cost rose to Tk 11 or fell to Tk 9, nothing would change.

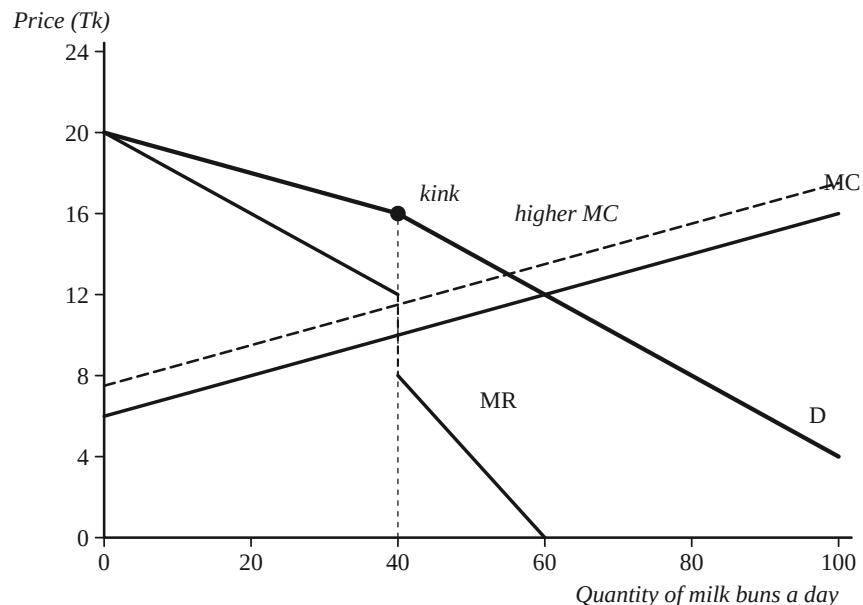


FIG. 21. The kinked demand curve. Marginal revenue breaks at 40 buns; any marginal cost cutting the gap leaves price and output unchanged.

The model explains why prices in oligopoly are often sticky. It does not explain how the price of Tk 16 came about in the first place. It also fits poorly when all firms face the same cost rise and all raise prices together. So it is one idea, not a full theory.²

Rivals may avoid risk by agreeing. **Collusion** means an agreement among rivals to fix a price or limit output. A group of firms that do this openly is a **cartel**. If the firms agree, they behave like one monopolist and share its profit. But each member is tempted to cheat. By selling slightly more at the agreed price, one firm gains, while the others lose. If all cheat, the cartel breaks down. Many countries forbid cartels by law (Chapter XIII).

A quieter route is **price leadership**: one firm, often the largest, announces a price and the others follow it. No agreement is signed. The rivals simply learn that following is safer than leading. Price leadership can give much the same result as a cartel.

Before moving on. Oligopoly has few sellers and interdependence. Four ideas describe the outcome: price war, kinked demand, collusion and price leadership. No single rule fixes the price.

Pause and check. In the kinked case, marginal cost at 40 buns rises from Tk 10 to Tk 11. What happens to output and price? (Answer at the back.)

The model explains why prices stay stuck but not how the price was set.

Collusion. An agreement among rivals to fix a price or limit output.

Cartel. A group of firms that agree together on prices or output.

Collusion turns rivals into a single seller, but each member is tempted to cheat.

Price leadership. One firm in the market sets the price and the others follow.

Price leadership is a quiet way to agree without meeting.

Before moving on: interdependence, price war, kinked demand, collusion, price leadership.

Pause and check: what happens if marginal cost rises from Tk 10 to Tk 11 at 40 buns in the kinked case?

PRICE DETERMINATION COMPARED

We now have all four markets. Their differences can be shown in a table. Read each column as one kind of market and each row as one feature. The last row gives an example from the case file or from daily life.

We now compare how price is set in the four kinds of market.

THE FOUR MARKETS COMPARED

FEATURE	PERFECT COMPETITION	MONOPOLY	MONOPOLISTIC COMPETITION	OLIGOPOLY
Number of sellers	Very many	One	Many	A few
Product	Identical	One, no close substitute	Differentiated	Identical or differentiated
Entry	Free	Blocked	Free	Hard
Control over price	None	Large	Some	Some, shared with rivals
Price against marginal cost	Equal	Above	Above	Above
Long-run profit above normal	None	Can persist	None	Can persist
Example	Loaves at Tk 60 (Chapter XII)	One-seller bun market	The milk bun	Mobile phone service

The way price is set also differs. In perfect competition, market demand and supply set the price, and each firm takes it. In monopoly, the seller sets output

Each market has its own way of setting the price.

² The kinked demand idea was published by Paul Sweezy in 1939, and independently by Hall and Hitch in the same year.

where marginal revenue equals marginal cost and reads the price from demand. In monopolistic competition, the firm uses the same rule, but entry then brings price down to average cost. In oligopoly, the price depends on the replies of rivals, so no single rule exists.

The bakery's own markets show all four ideas on one street.

Case file, illustrative. Place the bakery's products on this scale. Loaves sell at the market price of Tk 60 (perfect competition). If one seller served the bun market, the price would be Tk 14 against Tk 12 under competition (monopoly). The milk bun settles at Tk 12 with no profit above normal (monopolistic competition). In the kinked variant, price holds at Tk 16 (oligopoly).

Before moving on: the four markets differ in power over price and in long-run profit.

Before moving on. As we move from perfect competition to monopoly, control over price rises. Price lies above marginal cost in every market except perfect competition. Entry decides whether profit lasts.



SUMMARY

A short summary of the chapter.

Monopolistic competition has many sellers of differentiated goods and free entry. Each firm's demand slopes down and is elastic. In the short run it earns a profit; entry removes it, leaving price equal to average cost and some excess capacity. Oligopoly has a few sellers and interdependence, so firms may fight price wars, face a kinked demand, collude, or follow a leader. The four markets differ in power over price and in how long profit lasts.

THINK IT THROUGH

Two rival bakeries and a price cut.

Two bakeries sell the same milk bun at Tk 16, and each now earns Tk 240 a day. Each can hold the price or cut it to Tk 14. The table shows the daily profit for each pair of choices.

DAILY PROFIT (TK), BAKERY A FIRST AND BAKERY B SECOND (ILLUSTRATIVE)

	B HOLDS AT 16	B CUTS TO 14
A holds at 16	240, 240	120, 300
A cuts to 14	300, 120	160, 160

The questions ask what each firm will do and why the outcome is poor.

Suppose each decides alone. (a) What is each firm's best choice if the other holds? If the other cuts? (b) What is the result, and how does it compare with both holding? (c) Name two things that could keep both at Tk 16. (d) Why might a written agreement to hold be unlawful?

WHERE PEOPLE GO WRONG

Four common mistakes about these markets.

"Differentiated means monopoly." The firm has power over its own variety only; rivals are close. *"Zero profit means the firm is failing."* It earns normal profit, which covers the owner's own time and money. *"Excess capacity means idle workers."* It means output below the level of lowest cost. *"The kinked curve shows how price is chosen."* It shows only why price tends to stay put.

IN HISTORY

OPEC, a cartel of oil exporting countries, began on 14 September 1960.

On 14 September 1960, representatives of five oil exporting countries met in Baghdad and founded the Organization of the Petroleum Exporting Countries.

Its aim was to coordinate the oil policies of its members. In later years it tried to hold prices up by agreeing on output quotas. Members have often produced more than their quotas, and this shows the weakness of a cartel: each member gains by cheating. The case shows both the power of a group of sellers and the difficulty of keeping them together.

SOURCES

Chamberlin (1933) and Robinson (1933) wrote the first full theories of markets between competition and monopoly. They are the basis of this chapter. Sweezy (1939) and Hall and Hitch (1939) set out the kinked demand curve. These sources are listed with full details at the back of the book.

Where the ideas of this chapter began.

REVIEW QUESTIONS AND PRACTICE

REVIEW QUESTIONS

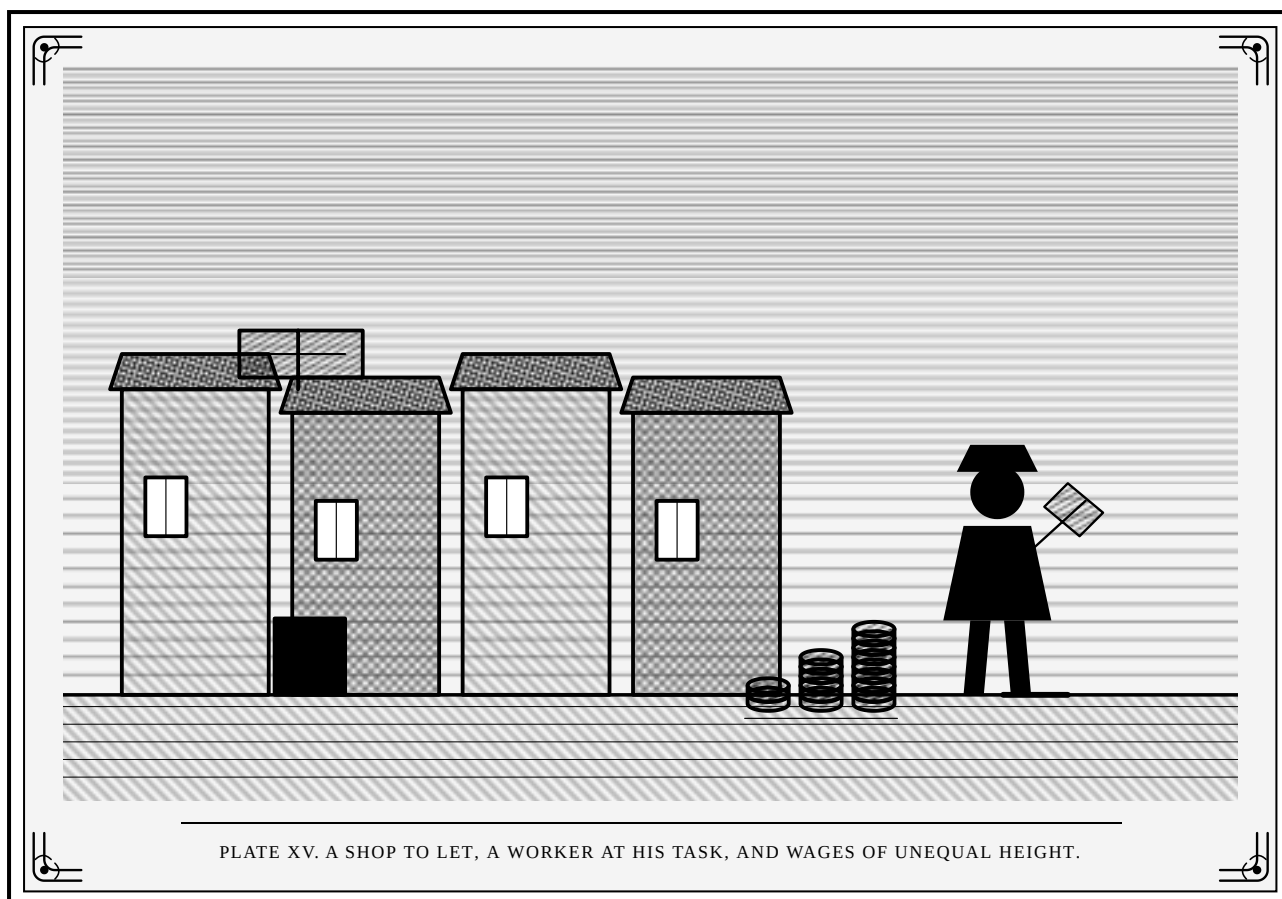
Review questions on the chapter.

NO.	QUESTION
1	What is product differentiation, and how can a seller achieve it?
2	Distinguish selling costs from production costs, with one example of each.
3	Why is a monopolistic competitor's demand curve downward-sloping and elastic?
4	Explain, with a diagram, why a firm in monopolistic competition ends up with excess capacity.
5	Explain why the kinked demand curve makes prices sticky in oligopoly.
6	Compare the four kinds of market on control over price and long-run profit.

PRACTICE

Practice problems, from simple to combined.

ITEM	PROBLEM
1	Name the market structure for each: (a) a wheat farmer selling in a large wholesale market; (b) the only water supplier in a city; (c) one of thirty tailors in a bazaar; (d) one of two airlines on a route; (e) a tea stall among twelve with its own blend.
2	A salon has demand $P = 100 - 3Q$ and $TC = 100 + 20Q + Q^2$, so $MC = 20 + 2Q$. Find output, price and profit.
3	Entry moves the salon's demand to $P = 60 - 3Q$. Find output, price and profit. Find the output of lowest average cost and the excess capacity. (Hint: average cost is lowest where it equals marginal cost.)
4	Kinked demand: above Tk 50, $P = 60 - Q/10$; below, $P = 70 - Q/5$. Marginal cost is a constant. Find output and price when it is Tk 32, and when it is Tk 45.



CHAPTER XV

Rent and Wages

Two payments, one question: what is a shop, or a pair of hands, worth?



*A bakery pays Tk 25,000 for a shop and Tk 600 a day for a baker.
This chapter shows how the market sets such payments for land
and for labour, why they differ from place to place and person to
person, and what a payment really buys.*

A bakery owner must pay for a shop and for a baker, and wonders how such prices arise.

Two equal shops carry different prices, and the chapter must explain the gap.

Two workers with equal hours earn unequal pay, and the market explains it.

Rent and wages are prices of inputs, not of finished goods.

A baker is wanted for the loaves, so the demand for bakers depends on loaf demand.

Factor market. A market where inputs are bought and sold.

The two terms: the market for inputs and the price paid in it.

Demand for an input exists only because buyers want the goods the input makes.

Derived demand. Demand for an input that comes from the demand for the good it makes.

The owner compares what one more worker adds to revenue with what that worker costs.

Marginal revenue product. The extra revenue a firm gains from one more unit of an input.

The name for the extra revenue is marginal revenue product.

GO SLOWLY HERE: THIS RULE IS USED FOR THE WHOLE CHAPTER.

The hiring rule says: hire another worker while the worker adds at least as much as the wage.

One clean example fixes the method.

Marginal revenue product usually falls as more workers are hired, which is why the firm stops.

The supply of an input also matters, and it differs between land and labour.

Look at two shops in one town. One stands on a quiet lane and costs Tk 25,000 a month. The other stands on the main road and costs Tk 40,000. The land under them is the same size. Why does the second owner ask Tk 15,000 more?

Now look at two workers. One weaves cloth in a mill. The other repairs the mill's machines. The second earns far more, though both work eight hours. Nobody sets these payments by a rule of fairness. The market sets them, and this chapter shows how.

We study rent and wages together because both are prices of inputs. In Chapters VIII to XIV we studied the prices of goods. Now we turn to the prices of the things used to make them.

15.1 PRICING THE FACTORS

A bakery buys flour, a shop and the work of bakers. Each purchase happens in a market, but this market is not the one where loaves are sold. It is the market for inputs.

A **factor market** is a market where inputs such as land and labour are bought and sold. The price paid in it is the **factor price**. Rent is the price of land, and a wage is the price of labour. (Chapter XVI treats the prices of capital and of the entrepreneur.)

Nobody wants a baker for his own sake. The owner wants a baker because customers want loaves. We therefore say that the demand for bakers is a **derived demand**. It is derived from, that is, it follows from, the demand for loaves. If customers stopped buying loaves, the bakery would stop hiring bakers.

How many bakers should the owner hire? In Chapter IX we met marginal product, the extra output from one more worker. The owner now asks a money question: how much extra revenue does that output bring?

This extra revenue is the **marginal revenue product** of the input. For a seller who must accept the market price (Chapter XII), it equals marginal product multiplied by the price of the good. When the firm also pays for materials, we subtract that cost, and the result is the extra money the worker truly brings in.

The rule is simple. Hire one more worker while the marginal revenue product is at least the wage. Stop when it falls below the wage. A worker who adds Tk 500 to revenue and costs Tk 400 raises profit by Tk 100. A worker who adds Tk 300 and costs Tk 400 lowers it by Tk 100.

Given. One more worker adds 10 units of output. Each unit sells at Tk 50. Materials cost nothing extra. The wage is Tk 400 a day. *Find.* Should the firm hire the worker? *Formula.* Marginal revenue product = marginal product x price. *Steps.* 10 units x Tk 50 = Tk 500. Compare Tk 500 with Tk 400. *Answer.* Hire: the worker adds Tk 100 a day to profit. *Check.* If the wage were Tk 600, the same worker would cost Tk 100 more than he adds, so the firm would not hire.

The marginal revenue product falls as the firm hires more. The reason is diminishing returns (Chapter IX): each added worker adds less output. So the demand curve for an input slopes downward, like the demand curve for a good. A lower wage means that more workers are worth hiring.

The other side of the market is supply. More people will work in a trade if it pays more, so the supply of labour to one trade rises with the wage. Land is different. A society cannot make more land when rent rises. Next we see what this means.

15.2 RENT

In daily speech, *rent* means the money paid to use a house, shop or machine. In economics the word has a second, sharper meaning. **Economic rent** is the part of a payment to an input that is above the least payment needed to keep that input in use. It is a surplus: more than the owner needs to be paid.



David Ricardo wrote about rent in 1817.¹ He asked a plain question. Farmers pay rent for good land. Why do they pay nothing for the worst land they use? His answer rests on two facts. Land differs in fertility and in position, and the amount of land is limited.

Suppose three plots of land, each one acre, are used to grow rice. The same work and seed on each plot cost Tk 6,000, and this sum includes the farmer's normal profit. The plots differ in fertility. Plot A is the best, plot B is average, and plot C is the poorest that anyone farms.

THREE GRADES OF LAND, ONE ACRE EACH (ILLUSTRATIVE)

PLOT	HARVEST (SACKS)	VALUE AT TK 300 A SACK	COST (TK)	RENT (TK)
A	40	12,000	6,000	6,000
B	30	9,000	6,000	3,000
C	20	6,000	6,000	0

How do we find the price of Tk 300? Rice from plot C costs Tk 6,000 for 20 sacks, which is Tk 300 a sack. Farmers will use plot C only if the price covers that cost. So the price settles at Tk 300. Plot C just pays its way and earns no rent. It is the *no-rent land*, the margin of use.

Plots A and B cost the same to farm but give more rice. Their extra return is **differential rent**: rent that comes from a difference. Plot A gives 20 more sacks than plot C, worth $20 \times \text{Tk } 300 = \text{Tk } 6,000$. Plot B gives 10 more sacks, worth Tk 3,000. These are the rents in the last column.

Now let the demand for rice grow. Farmers must bring in a fourth plot, D, which gives only 10 sacks. It will be used only if the price rises to Tk 600 a sack (Tk 6,000 divided by 10). At that price every rent rises. Plot A earns $40 \times 600 - 6,000 = \text{Tk } 18,000$. Plot B earns Tk 12,000 and plot C earns Tk 6,000. Plot D now earns none.

Given. A plot gives 35 sacks. The worst plot in use gives 20 sacks. Both cost Tk 6,000 to farm. *Find.* The rent of the better plot. *Formula.* Rent = harvest $\times$ price - cost, where price = cost of worst plot / its harvest. *Steps.* Price = $6,000 / 20 = \text{Tk } 300$. Value = $35 \times 300 = \text{Tk } 10,500$. Rent = $10,500 - 6,000$. *Answer.* Rent is Tk 4,500. *Check.* The extra 15 sacks over the worst plot are worth $15 \times 300 = \text{Tk } 4,500$. The two methods agree.

Ricardo drew a bold conclusion. The price of rice is set by the cost on the worst land. So rent is *the result* of a high price, not its cause. For the farming community as a whole, he said, a landlord's demand for rent does not raise the price of rice.

The theory has limits. Few lands are truly free of rent. Land can be used for other crops, so a farmer who rents it must pay at least what another crop would

In daily speech rent is a payment for a house; in economics it means something wider.

Economic rent. A payment to an input above the least needed to keep it in use.

Ricardo's question is why fertile land earns a surplus while poor land earns none.

GO SLOWLY HERE: THE THREE-GRADE TABLE IS THE HEART OF THE THEORY.

Ricardo explained rent by differences in the land itself.

Three plots of equal size and cost give different harvests.

The price is fixed by the poorest plot in use, which earns no rent.

Differential rent. The extra return of better land over the worst land in use.

Better plots earn a differential rent.

Rent follows the price: when demand brings poorer land into use, every rent rises.

A short example shows the steps for a plot of one's own.

Ricardo's famous claim: rent does not make the price, the price makes the rent.

The theory has limits that later writers pointed out.

¹ The source is chapter 2 of his *Principles*; see Sources at the end of this chapter.

Contract rent is what the tenant actually pays, and it may differ from economic rent.

Contract rent. The sum agreed between owner and user for the use of a thing.

The location of a shop shows differential rent in a town.

The modern view says any input in fixed supply can earn economic rent.

Man-made things earn a temporary surplus that economists call quasi-rent.

Quasi-rent. A temporary surplus earned by a man-made asset whose supply is fixed in the short run.

A wage is the pay for work, and what matters is what it can buy.

Wage. The payment made for the use of labour.

Money wages may rise while buying power falls.

Nominal wage. The wage measured in money.

Real wage. What the wage can buy, shown by dividing the money wage by a price index.

Price index. A number showing the average level of prices compared with a base period set at 100.

A real wage is found by dividing the money wage by the index and multiplying by 100.

pay. For one farmer, then, rent is a real cost. Also, land can be improved by drainage and fertiliser, and part of what looks like rent is a return on such work. Still, the central idea survives: a payment fixed by demand alone, on an input that cannot be increased.

Contract rent is the sum that the owner and user agree for the use of a thing, such as the Tk 25,000 paid for the lane shop. It may include more than economic rent. It can contain interest on a building the owner put up, and pay for the owner's upkeep. Economic rent is only the surplus part.

Return to the two shops. The main road brings more customers, so the same bakery would earn more there. The extra earning is the reason the owner can ask more. The gap of Tk 15,000 is a differential rent, a rent of position instead of fertility.

The modern view widens the idea. Economic rent is the payment to any input minus its *transfer earnings*, the least payment that would keep it in its present use. A singer paid Tk 100,000 for an evening, whose best other offer is Tk 10,000, earns an economic rent of Tk 90,000. If the supply of an input is fixed and cannot move elsewhere, all its payment is rent. When supply can move easily, rent is small.

A bakery's oven is made by people, so more ovens can be built in time. In the short run, however, the number of ovens is fixed. If loaf prices are high, the oven earns a surplus above its running costs. This is **quasi-rent**, *quasi* meaning "almost": it looks like rent but lasts only until new ovens appear. In the bakery of Chapter XII, revenue is Tk 8,400 a day and variable cost is Tk 5,800, so the fixed assets earn Tk 2,600 a day. After paying their Tk 1,000 fixed cost, Tk 1,600 remains. Other bakers will notice this, build ovens, and the surplus will shrink.²

15.3 WAGES: MEANING

A **wage** is the payment made for the use of labour. In everyday speech some people say "wages" for hourly or daily pay and "salary" for monthly pay. Economics uses *wage* for all of them. Pay may be by time (so much a day) or by piece (so much for each shirt sewn).

The **nominal wage** is the wage in money: Tk 600 a day. The **real wage** is what that money can buy. If prices double and the nominal wage stays the same, the worker is poorer. To compare, we use a **price index**, a number that shows the average level of prices against a base of 100.

Given. A wage is Tk 500 a day. Prices rise so that the index goes from 100 to 125. *Find.* The real wage in old prices. *Formula.* Real wage = nominal wage / price index x 100. *Steps.* 500 / 125 = 4. 4 x 100 = 400. *Answer.* The real wage is Tk 400 in old prices. *Check.* Prices rose by a quarter, so the same money buys one-fifth less (400 is four-fifths of 500).

² The term comes from Alfred Marshall's *Principles of Economics* (1890). Quasi-rent is often called *rent of a man-made asset*.

Real wages also include non-money gains: free meals, housing, shorter hours. Two jobs with the same money pay may differ in real reward. We use this idea in section 15.5.

A second reading of the real wage includes non-money benefits.

15.4 HOW WAGES ARE DETERMINED

In a competitive market for labour, the wage is set as the price of any good is set in Chapter VIII. Employers demand labour, and workers supply it. Where the two quantities are equal, we have equilibrium in the labour market.

The wage in a trade comes from the demand for and supply of labour.

The demand for labour is the marginal revenue product curve of section 15.1. Each employer hires up to the point where the extra worker adds as much as the wage. At a high wage few workers are worth hiring; at a low wage many are. This is *marginal productivity theory*: the wage tends to equal the marginal revenue product of the last worker hired.

Employers demand labour according to what a worker adds.

The supply of labour to one trade rises with the wage, because workers move in from poorer trades. Supply in a trade also depends on the cost and time of learning the skill. Where the wage settles, the quantity of labour wanted equals the quantity offered.

Workers supply more labour to a trade when it pays more.

A rise in the demand for loaves raises the demand for bakers and pushes the wage up. More people learning to bake raises the supply of bakers and pushes it down. These are the same shifts we saw in Chapter VIII, applied to labour.

A change in either side moves the wage.

A **trade union** is a group of workers who act together. In **collective bargaining** the union deals with employers for all its members, instead of each worker making his own deal. A strong union may win a wage above the market level, though employers may then hire fewer workers. A strong employer, one of few buyers of labour, may hold wages below the level of marginal product. The result depends on who has more bargaining power.

Bargaining can move a wage away from the market value.

Trade union. A group of workers who act together on pay and conditions.

Collective bargaining. Talks in which a union deals with employers on behalf of its members.

A **minimum wage** is the lowest wage that the law permits. It is a price floor (Chapter VIII). If the floor lies below the market wage, it changes nothing. If it lies above, workers who keep jobs earn more, but employers may want fewer workers, so some may lose work. How large that loss is depends on how strongly employers cut hiring, a question of elasticity (Chapter V). Careful studies of real cases have found different results, which is why economists still debate it.³

A minimum wage is a floor on the price of labour.

Minimum wage. The lowest wage that the law permits an employer to pay.

15.5 WHY WAGES DIFFER

In one economy a surgeon, a bus driver and a field worker all earn different wages. If labour were one uniform thing, competition would equalise pay. It is not. Workers differ, and so do jobs. Adam Smith named several of the causes in 1776.

The same market forces give different wages to different jobs.

A **compensating difference** is extra pay that makes up for the unpleasant side of a job. A worker in a dangerous shipbreaking yard, or one who works at night, is paid more than one with a safe daytime job. Without the extra, few would take the work. The wage evens out the total reward between jobs.

Hard, risky or unpleasant work pays more.

Compensating difference. Extra pay given for work that is unpleasant, risky or insecure.

Training costs money and years. A student who spends five years learning to be an engineer gives up earnings meanwhile, and may also pay fees. The wage after training must repay this, or no one would train. So skilled work

Skill that takes long and costly training must be paid for.

³ Studies of minimum wages in different countries and periods have found effects on jobs ranging from small to large; the outcome depends on how high the floor is set.

Workers in different grades do not compete with each other.

Non-competing groups. Groups of workers whose skills differ so much that one cannot easily move into the other.

Bargaining power, place and other causes also play a part.

pays more than unskilled work. Fewer people also have the skill, so supply is small.

A fitter cannot become a doctor next month. Workers form **non-competing groups**, layers of labour with little movement between them. Each layer has its own demand and supply, so each has its own wage. The gap between layers lasts for as long as training or entry is difficult.

Other causes add to the list. Strong unions can lift pay in a trade. Wages differ between regions where living costs differ. Some jobs offer secure work for life and others do not. The table collects the main causes.

MAIN CAUSES OF WAGE DIFFERENCES

CAUSE	EFFECT ON WAGE	EXAMPLE
Risk or discomfort	Higher	Night shift in a factory
Cost and time of training	Higher	Engineer or doctor
Scarce skill	Higher	Machine fitter in a mill
Strong bargaining power	Higher	Workers in a strong union
Easy entry by many workers	Lower	Unskilled day labour

WORKED CASE: THE BAKERY HIRES

The bakery uses these ideas to choose a shop and a number of bakers.

The hiring rule is applied to the product table of Chapter IX.

With a wage of Tk 600 a day, the owner stops at the fifth baker.

A rise in prices lowers the buying power of a fixed wage.

Case file, illustrative. The owner can rent a shop on the lane for Tk 25,000 a month or a main-road shop for Tk 40,000. The difference is Tk 15,000 a month. The extra rent is justified only if the main road brings extra revenue of more than Tk 15,000 a month.

Case file, illustrative. The bakery bakes loaves at Tk 60. Materials cost Tk 20 for each loaf, so each loaf adds Tk 40 after materials. The product table of Chapter IX shows the marginal product of each baker. The fourth baker adds 30 loaves, worth Tk 1,200. The fifth adds 20 loaves: $20 \times (60 - 20) = \text{Tk } 800$. The sixth adds 10 loaves: $10 \times 40 = \text{Tk } 400$.

Case file, illustrative. The wage is Tk 600 a day. The fifth baker adds Tk 800 and costs Tk 600, so hiring raises profit by Tk 200. The sixth adds Tk 400 and costs Tk 600, so hiring would lower profit by Tk 200. The bakery therefore hires five bakers. *Check.* The sixth baker still adds loaves, yet the owner does not hire him, because he adds less than he costs.

Case file, illustrative. Suppose prices rise by 5%, so the index is 105. The nominal wage of Tk 600 is unchanged, but the real wage is $600 / 1.05 = \text{Tk } 571.43$ in old prices. The baker is poorer by about Tk 28.57 a day, though the pay slip shows no change.

SUMMARY

The chapter's main points in one place.

Rent and wages are prices of inputs, set in factor markets. Demand for an input is derived from demand for the good, and a firm hires while marginal revenue product is at least the price of the input. Economic rent is a surplus above the least payment needed; Ricardo explained it by differences in land, with the no-rent land fixing the price. Contract rent is the sum agreed; quasi-rent is a temporary surplus on man-made assets. A wage is the payment for labour; the real wage is the nominal wage divided by a price index. Wages are set by demand and supply, and by bargaining, and they differ because jobs and workers differ.

THINK IT THROUGH

A jute mill pays weavers Tk 15,000 a month and machine fitters Tk 27,000. The mill's owner says the gap is unfair, since both work the same hours. Using ideas from this chapter, give two reasons why the gap exists and one reason why it may shrink. Would a rule that fixed both wages at Tk 21,000 solve the problem?

A question to apply the chapter.

WHERE PEOPLE GO WRONG

"Rent is what a tenant pays." This is contract rent. Economic rent is only the surplus part of it.

Four common mistakes, each corrected.

"High rent makes the price of rice high." In Ricardo's theory the cause runs the other way: the price, set by the worst land, creates the rent.

The second mistake, with its correction.

"A higher money wage means a better life." Not if prices rise faster. Compare real wages.

The third mistake, with its correction.

"A minimum wage always helps workers" or "always harms them" are both too simple. The effect depends on the level of the floor and on how employers react.

The fourth mistake, with its correction.

IN HISTORY

After the collapse of the Rana Plaza building on 24 April 2013, wages in the garment industry became a matter of national debate. A wage board set a new minimum for garment workers in 2013, and raised it in 2018 and again in 2023, to Tk 12,500 a month. Workers argued that rising prices had cut the real value of their pay. Owners argued that higher pay would raise their costs against foreign rivals. Both sides used the ideas of this chapter: real wage, marginal productivity and bargaining power.

A true account from Bangladesh.

SOURCES

David Ricardo, *On the Principles of Political Economy and Taxation* (1817), chapter 2 "On Rent". Adam Smith, *The Wealth of Nations* (1776), book 1, chapter 10, on wage differences.

Where to read more.

REVIEW QUESTIONS AND PRACTICE

Review. (1) Why is the demand for an input called a derived demand? (2) State the rule for hiring one more worker. (3) What is no-rent land, and why does it fix the price? (4) Distinguish economic rent from contract rent. (5) Why do real wages differ from nominal wages? (6) Name three causes of wage differences.

Short questions on the whole chapter.

Practice 1. Three plots of jute land each cost Tk 9,000 to farm and give 60, 45 and 30 bales. Find the price per bale and the rent of each plot. Then let a fourth plot of 22.5 bales be brought into use, and find the new rents.

Four problems with new data.

Practice 2. A worker earns Tk 18,000 a month. The price index rises from 100 to 108. Find the real wage. If the nominal wage rises 5%, does the real wage rise or fall?

Real wage problem.

Practice 3. A sewing unit adds 12, 10, 8, 6 and 4 shirts with the first to fifth worker. A shirt sells at Tk 90 and its materials cost Tk 40. Find the marginal revenue product of each worker. How many workers will the unit hire at a wage of Tk 350? At Tk 250?

Hiring problem.

Choice of shop problem.

Practice 4. A shopkeeper pays Tk 30,000 a month for a shop. A better corner shop costs Tk 44,000 and would give Tk 18,000 more in monthly revenue after other costs. Should she move? State the rule you use.

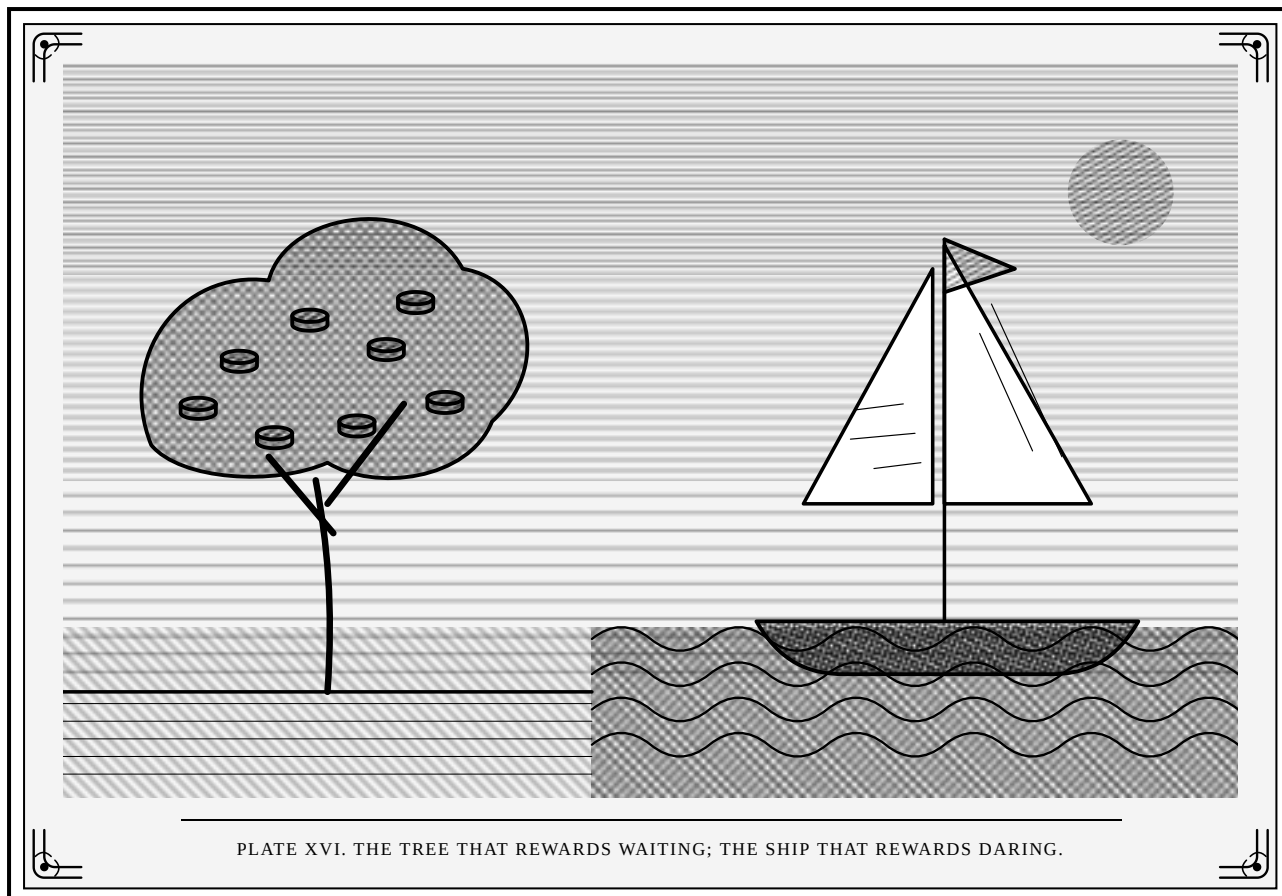


PLATE XVI. THE TREE THAT REWARDS WAITING; THE SHIP THAT REWARDS DARING.

CHAPTER XVI

Interest and Profit

The reward for waiting, and the reward for daring.



A bank lends money and asks for more back. A baker risks savings and may gain or lose. This chapter shows why money earns interest, how the rate is set, what profit really is, and why it appears at all.

A lender is paid for waiting, and an owner for running a business; the chapter asks why.

The same figure looks large to the books and small to the friend.

A second puzzle: why is a price paid for the use of money.

Interest and profit complete the four rewards of production.

In daily speech interest means curiosity; in economics it means a payment for money.

Interest. The payment made for the use of money for a period of time.

A short example shows the arithmetic of simple interest.

Why does money earn interest? Because the lender gives up something.

Time preference. The tendency to value present goods more than the same goods in the future.

A second reason is that money can be used to make more goods.

Two forms of interest: what the lender receives, and what is left for waiting alone.

Gross interest. The whole payment a lender receives, including pay for risk and for the cost of lending.

Net interest. The part of gross interest that pays only for waiting.

The rate of interest is a price, so demand and supply set it.

The first view: the rate balances the demand for and supply of loans.

Loanable funds. Money offered for lending and money sought for borrowing.

At the end of one month the baker's books show Tk 48,000 of profit. A friend who works in an office says, "You have earned almost nothing. I earn Tk 40,000 and do not risk my savings." Who is right?

A second puzzle sits beside it. The baker wants a second oven and asks a bank for Tk 400,000. The bank agrees, but it will ask for more than Tk 400,000 back. Why should anyone pay for the use of money? Money does not bake bread.

These two puzzles lead to the last two rewards in our list of inputs. In Chapter XV we met rent for land and wages for labour. Here we study interest for capital and profit for the entrepreneur.

16.1 INTEREST

In daily speech, *interest* means curiosity: "an interest in cricket". In economics, **interest** is the payment made for the use of money for a period of time. A borrower pays it; a lender receives it. It is usually stated as a percentage of the sum for a year, called the *rate of interest*.

Given. A loan of Tk 400,000 at 12% a year, for one year. *Find.* The interest. *Formula.* Interest = sum lent x rate x years. *Steps.* 400,000 x 0.12 x 1 = 48,000. *Answer.* The interest is Tk 48,000 a year, which is Tk 4,000 a month. *Check.* Ten per cent of 400,000 is 40,000, and a further 2% is 8,000, so the total is 48,000.

The answer begins with **time preference**, the tendency to value goods in the present more than the same goods in the future. Most people would rather have Tk 1,000 today than Tk 1,000 a year from now. To give up money today, a lender must be paid. The Austrian economist Eugen von Böhm-Bawerk made this idea central to the theory of interest.¹

A second reason concerns the borrower. Money lets the baker buy an oven that makes more loaves. If the oven earns Tk 60,000 a year, the baker can afford to pay part of that sum for the loan. The borrower's gain from using the money sets the highest interest he would pay. The lender's wish to be paid for waiting sets the lowest he would accept.

The total payment is **gross interest**. It contains more than pay for waiting. Suppose a lender charges 14%. Of this, 8% pays for waiting, 4% covers the chance that the borrower will not repay, and 2% covers the cost of handling the loan. The 8% left for waiting is **net interest**.

16.2 HOW INTEREST RATES ARE DETERMINED

Interest is the price of using money, so we can apply demand and supply. Economists give two views. They look at different sides of the same market and should be read together.

In the **loanable funds** view, the rate balances the money offered for lending against the money sought for borrowing. Firms borrow to build and buy. Households borrow to buy homes. The government borrows to pay for projects. Savers, and banks creating credit, supply the funds. A higher rate makes borrowers want less and lenders offer more.

¹ Böhm-Bawerk wrote *Capital and Interest* in the 1880s; the reference is in Sources.

LOANABLE FUNDS AT DIFFERENT RATES (TK MILLION, ILLUSTRATIVE)

RATE OF INTEREST (%)	FUNDS SOUGHT	FUNDS OFFERED	POSITION
6	90	20	Excess demand
8	70	35	Excess demand
10	50	50	Equilibrium
12	30	65	Excess supply
14	10	80	Excess supply

At 6% borrowers want Tk 90 million but lenders offer only Tk 20 million. Competition among borrowers pushes the rate up. At 14% lenders offer more than borrowers want, and the rate is pushed down. The balance comes at 10%, where both are Tk 50 million. A rise in saving moves the supply to the right and lowers the rate. A rise in business hope moves the demand to the right and raises it.

John Maynard Keynes proposed the second view in 1936. Holding cash earns no interest, yet people hold it. They need it for daily purchases, for emergencies, and sometimes because they expect to buy something cheaper later. This wish to hold cash is **liquidity preference**. To persuade people to part with cash, the rate of interest must be high enough. The supply of money is set by the central bank, here Bangladesh Bank, and the rate settles where the demand for cash equals that supply.

The two views agree on the main point: interest is a price, and it changes when demand or supply changes. They differ on which changes matter most. For a beginner, the loanable funds view is the easier first step.

The **nominal interest rate** is the rate stated in the loan. The **real interest rate** is the rate after removing the rise in prices. As a rule, real rate is about nominal rate minus the rate of price rise. A loan at 12% when prices rise 7% a year pays a real rate of about 5%.² If prices rose by 12%, the lender would gain nothing in real terms.

Given. Nominal rate 12%; prices rise 7% a year. *Find.* The real rate. *Formula.* Real rate = nominal rate - rate of price rise (approximately). *Steps.* 12 - 7 = 5. *Answer.* About 5% a year. *Check.* A lender who is repaid Tk 112 for each Tk 100 lent can buy only about 5% more goods than before.

16.3 PROFIT

In everyday talk, profit is what is left after paying the bills. An accountant calls this **accounting profit**: revenue minus explicit costs, the costs actually paid. An economist asks for more. **Economic profit** is revenue minus *all* costs, including implicit costs, the costs of owned resources that nobody pays in cash (Chapter X). The owner's own time and money are resources, and they could have earned money elsewhere.

The case file of Chapter I gives the bakery's figures. Revenue is Tk 252,000 a month and explicit cost is Tk 204,000. Accounting profit is therefore Tk 48,000. The

A table shows the demand and supply of loans at five rates.

The rate settles where funds sought equal funds offered.

The second view: people choose how to hold their wealth, in cash or in loans.

Liquidity preference. The wish to hold wealth in cash instead of in forms that earn interest.

Both views agree that the rate is a price set by two sides.

A nominal rate hides the effect of rising prices.

Nominal interest rate. The rate of interest stated in money, before allowing for rising prices.

Real interest rate. The rate of interest after the effect of rising prices is removed.

Check the real-rate rule with an example.

Profit is the difference between what a firm earns and what it costs.

Accounting profit. Revenue minus explicit costs, as recorded in the firm's books.

Economic profit. Revenue minus all costs, including implicit costs.

The bakery's month shows both kinds of profit.

² The exact value is $(1.12 / 1.07) - 1$, which is about 4.7%; the simple subtraction is a good guide when rates are small.

owner's forgone pay is Tk 40,000 and forgone interest is Tk 3,000, both implicit costs.

ACCOUNTING AND ECONOMIC PROFIT OF THE BAKERY, ONE MONTH (ILLUSTRATIVE)

ITEM	TK
Revenue	252,000
Explicit costs	204,000
Accounting profit	48,000
Less owner's forgone pay	40,000
Less forgone interest on savings	3,000
Economic profit	5,000

The friend in the opening puzzle was partly right: much of the profit pays for the owner's own time and money.

Normal profit is the minimum reward that keeps an entrepreneur in the business.

Now the opening puzzle has an answer. Of the Tk 48,000, Tk 43,000 only repays the owner for what he gave up. The friend was almost right: only Tk 5,000 is a gain beyond the owner's best alternative.

In Chapter X we called the owner's minimum reward *normal profit*. It is a cost of production, because a firm that cannot pay it will lose its entrepreneur. When economic profit is above zero, the firm earns **supernormal profit** (Chapter XII) in addition to normal profit. When economic profit is below zero, the owner would do better elsewhere.

16.4 WHY PROFIT OCCURS

Economists give several answers to the question of why profit exists.

The oldest answer: profit pays for taking a risk.

Knight's view: the reward is for facing what cannot be calculated.

Uncertainty. A situation in which the chances of outcomes cannot be known.

Schumpeter's view: profit comes from doing something new.

Innovation. The introduction of a new product, method or market.

Another view treats the entrepreneur as one more input with a marginal product.

If competition drives economic profit to zero in the long run (Chapter XII), why does profit appear at all? Economists have given five main answers. Each explains a different part of the picture.

The first is the *risk-bearing* view. Every business may fail. The owner puts money in, and may lose it. Profit is the reward for taking this risk. An American economist, Frederick Hawley, put this view in 1893. Many risks, such as fire, can be insured, and an insurance premium is simply a cost. So the pay for risk alone cannot explain all profit.

Frank Knight drew a sharp line in 1921.³ A risk whose chances can be counted, like fire, can be insured. **Uncertainty** is a situation in which the chances cannot be known: will customers like a new product? Nobody can insure against this. The entrepreneur bears it, and profit is the reward. In Knight's view profit is the reward of uncertainty, not of risk.

Joseph Schumpeter wrote in 1911 that profit comes from **innovation**, the introduction of something new. A baker who first sells a milk bun, or who first delivers by bicycle, earns more than the rest, until others copy. Competition then removes the gain, and the next innovator earns a new profit. On this view profit is temporary but never disappears for long.

The *marginal productivity* view treats the entrepreneur like any input. His reward equals the extra output his skill adds. The idea has a weakness: there is usually one entrepreneur in a firm, so it is hard to say what one *extra* entrepreneur would add. It is still useful, because it links profit to the value the entrepreneur creates.

³ Knight's book, *Risk, Uncertainty and Profit*, is listed in Sources.

Not every profit is earned. A **windfall profit** comes by chance. If floods destroy a neighbour's stock, a baker with sacks of flour in store may sell at a much higher price. And a monopoly (Chapter XIII) may earn profit through market power, by keeping price above cost. The table sets out the theories side by side.

FIVE VIEWS OF PROFIT

VIEW	SOURCE OF PROFIT	MAIN THINKER	WEAKNESS
Risk-bearing	Taking risk	Hawley, 1893	Insurable risks are only a cost
Uncertainty	Facing the unknowable	Knight, 1921	Hard to separate from risk
Innovation	New products and methods	Schumpeter, 1911	Not all profit is innovative
Marginal productivity	Entrepreneur's own product	Early marginalists	One entrepreneur; no "extra"
Windfall or market power	Chance or monopoly	Various	Not a reward for effort

No single view is complete. A farmer whose crop depends on rain fits the uncertainty view. A firm with a new product fits the innovation view. A firm with a sole licence fits the market-power view. Good analysis picks the theory that fits the case.

Some profit arises by chance, and some from market power.

Windfall profit. Unexpected profit that comes by chance, not by the owner's effort or skill.

No single view is complete; each fits some cases.

16.5 THE WHOLE PICTURE

We can now put the four rewards of production together. Chapter I named four kinds of resource. Each has its own reward and its own price.

Four inputs, four rewards: this is the summary of Chapters XV and XVI.

THE FOUR INPUTS AND THEIR REWARDS

INPUT	REWARD	WHERE STUDIED
Land	Rent	Chapter XV
Labour	Wages	Chapter XV
Capital	Interest	Chapter XVI
Entrepreneur	Profit	Chapter XVI

The rewards depend on one another. Revenue pays rent, wages and interest first, and what remains is profit. Profit can be negative: the entrepreneur bears the loss. That is why profit is called the *residual* reward. A rise in wages, for example, lowers profit unless the price rises.

The rewards depend on one another: a firm must pay the first three before profit remains.

Microfinance is the supply of small loans and other financial services to poor people. Bangladesh is the home of a famous example. The Grameen project began in 1976 with small loans to villagers near Chittagong, and became the Grameen Bank by law in 1983.⁴ Poor borrowers could not offer a bank anything to secure a loan, and small loans cost much to manage. In the loanable-funds view, funds did not reach them. Microfinance brought the lenders and these borrowers together. Debate continues over the rates charged and the risk of too much debt.

Microfinance applies the theory of interest to poor borrowers.

Microfinance. The supply of small loans and other financial services to poor people.

⁴ Dates as commonly given for the Grameen project and the law of 1983; see the verify list.

WORKED CASE: THE SECOND OVEN

The bakery borrows to buy a second oven and sees what interest does to profit.

The loan adds a new explicit cost.

The oven is worth buying only if it earns more than it costs in interest.

The chapter's main points in one place.

A question to apply the chapter.

Four common mistakes, each corrected.

The second mistake, with its correction.

The third mistake, with its correction.

The fourth mistake, with its correction.

A true account from Bangladesh.

Case file, illustrative. In a month of 30 days the bakery bakes 140 loaves a day. Revenue is Tk 252,000 and cost is Tk 204,000 (Tk 6,800 x 30). Accounting profit is Tk 48,000. The owner's forgone pay is Tk 40,000 and forgone interest is Tk 3,000, so economic profit is Tk 5,000 a month.

Case file, illustrative. The owner borrows Tk 400,000 for a second oven at 12% a year. Interest is Tk 48,000 a year, or Tk 4,000 a month. This is an explicit cost, paid in cash. Before the oven earns anything, accounting profit falls to 48,000 - 4,000 = Tk 44,000, and economic profit falls to 5,000 - 4,000 = Tk 1,000.

Case file, illustrative. The oven is worth buying only if it adds more than Tk 4,000 a month to revenue, after its running costs. *Check.* If it added Tk 3,000, the net effect would be a fall of Tk 1,000, and economic profit would drop from Tk 5,000 to Tk 4,000. The rule is to compare the gain with the interest.

SUMMARY

Interest is the payment for the use of money. It arises from time preference and from the use of money to produce more. Gross interest includes pay for risk and cost; net interest pays for waiting. The rate is set by loanable funds, or by liquidity preference and the supply of money; the real rate is about the nominal rate minus the rate of price rise. Accounting profit counts explicit costs only; economic profit counts all costs, and normal profit is the minimum reward of the entrepreneur. Profit is explained by risk, uncertainty, innovation, marginal productivity, windfall and market power. Rent, wages, interest and profit are the four rewards of production.

THINK IT THROUGH

A family shop earns an accounting profit of Tk 84,000 a year. The owner could earn Tk 72,000 working for another firm, and the Tk 480,000 invested in the shop could earn 5% in a bank. Find the economic profit. Should the owner close the shop? What would change your answer?

WHERE PEOPLE GO WRONG

"Interest is only paid on borrowed money." An owner's own money also has an interest cost, the interest forgone. This is an implicit cost.

"A high nominal rate is always a good return." If prices rise faster, the real rate is low or negative.

"Accounting profit is the true profit." It leaves out implicit costs. A firm with accounting profit may have an economic loss.

"Profit is only a reward for hard work." Profit may come from risk, uncertainty, innovation, chance or market power.

IN HISTORY

In 1976 the economist Muhammad Yunus lent small sums to villagers near Chittagong who had been unable to borrow from banks. Many repaid. The project grew, and in 1983 it became the Grameen Bank. The idea spread to many countries. Its lesson for this chapter is that interest and risk depend

on who can reach the loanable funds, and that new ways of lending can change that.

SOURCES

Frank Knight, *Risk, Uncertainty and Profit* (1921). Joseph Schumpeter, *The Theory of Economic Development* (1911). Eugen von Böhm-Bawerk, *Capital and Interest* (1884 to 1889). John Maynard Keynes, *The General Theory of Employment, Interest and Money* (1936).

Where to read more.

REVIEW QUESTIONS AND PRACTICE

Review. (1) What two reasons explain why interest is paid? (2) Distinguish gross from net interest. (3) How does the loanable funds view set the rate? (4) What is liquidity preference? (5) Distinguish accounting from economic profit. (6) Name three explanations of profit.

Short questions on the whole chapter.

Practice 1. A tailor earns revenue of Tk 360,000 a year. Paid costs are Tk 250,000. His forgone pay is Tk 70,000, and his Tk 200,000 of savings in the shop could earn 5%. Find the accounting profit and the economic profit.

Four problems with new data.

Practice 2. A loan of Tk 250,000 at 9% a year runs for two years, with simple interest. Find the interest paid. If prices rise 6% a year, find the approximate real rate.

Interest and real-rate problem.

Practice 3. Which view of profit best fits each case? (a) A shop gains when a flood closes a rival's shop. (b) A trader opens the first mobile payment service in a market. (c) A farmer sells grain without knowing next season's rain.

A problem on choosing a theory.

Practice 4. Funds sought at rates of 5%, 7% and 9% are Tk 80, 60 and 40 million. Funds offered are Tk 30, 50 and 70 million. Treat the figures as straight lines. Find the balancing rate and quantity. What happens at 5%?

A problem on loanable funds.

GLOSSARY OF KEY TERMS

Accounting cost. The explicit costs recorded in the firm's books. (*Chapter X, page 84*)

Accounting profit. Revenue minus explicit costs, as recorded in the firm's books. (*Chapter XVI, page 135*)

Arc elasticity. Elasticity between two points, using average price and average quantity. (*Chapter V, page 38*)

Average cost. Total cost divided by output; the cost of one unit on average. (*Chapter X, page 86*)

Average fixed cost. Fixed cost divided by output. (*Chapter X, page 86*)

Average product. Output per unit of the variable input. (*Chapter IX, page 75*)

Average revenue. The revenue received for each unit sold. (*Chapter XI, page 94*)

Average variable cost. Variable cost divided by output. (*Chapter X, page 86*)

Bandwagon effect. Demand rises as more people buy. (*Chapter IV, page 32*)

Barrier to entry. Anything that stops new sellers from entering a market. (*Chapter XI, page 92*)

Bilateral monopoly. A market with one seller facing one buyer. (*Chapter XIII, page 107*)

Brand. A name or sign that identifies one seller's good and sets it apart from others. (*Chapter XIV, page 117*)

Break-even point. The price at which revenue just covers total cost. (*Chapter XII, page 101*)

Budget line. A line of all the baskets that cost exactly the whole income. (*Chapter VI, page 49*)

Capital. Made things used to make other goods. (*Chapter I, page 3*)

Cardinal. Measured in numbers. (*Chapter III, page 21*)

Cartel. A group of firms that agree together on prices or output. (*Chapter XIV, page 121*)

Ceteris paribus. All other things held unchanged. (*Chapter II, page 14*)

Change in demand. A shift of the whole demand curve, caused by something other than price. (*Chapter IV, page 30*)

Change in quantity demanded. A move along the demand curve, caused by a change in the price. (*Chapter IV, page 30*)

Change in quantity supplied. A move along the supply curve, caused by a change in the price of the good. (*Chapter VII, page 58*)

Change in supply. A shift of the whole supply curve, caused by something other than the price. (*Chapter VII, page 58*)

Choice. Picking one use of limited resources. (*Chapter I, page 4*)

Collective bargaining. Talks in which a union deals with employers on behalf of its members. (*Chapter XV, page 129*)

Collusion. An agreement among rivals to fix a price or limit output. (*Chapter XIV, page 121*)

Compensating difference. Extra pay given for work that is unpleasant, risky or insecure. (*Chapter XV, page 129*)

Competitive market. A market with so many buyers and sellers that none can change the price alone. (*Chapter VIII, page 64*)

Complementary goods. Goods used together. (*Chapter IV, page 31*)

Constant returns to scale. Output rises by the same proportion as inputs. (*Chapter IX, page 78*)

Consumer. A person who uses a good. (*Chapter III, page 23*)

Consumers' surplus. What a buyer would be willing to pay, minus what is actually paid. (*Chapter VI, page 46*)

Consumption. Using a good to meet a want. (*Chapter III, page 23*)

Contract rent. The sum agreed between owner and user for the use of a thing. (*Chapter XV, page 128*)

Cost. The value of what is given up to make or get something. (*Chapter X, page 84*)

Cross elasticity. Percentage change in demand for one good divided by percentage change in the price of another. (*Chapter V, page 41*)

Deadweight loss. The surplus lost to everyone when output falls below the competitive level. (*Chapter XIII, page 110*)

Decreasing returns to scale. Output rises by a smaller proportion than inputs. (*Chapter IX, page 78*)

Demand. In daily life, a strong request or order. In economics, the quantity buyers are able and willing to buy at each price. (*Chapter IV, page 28*)

Demand curve. The line drawn from a demand schedule. (*Chapter IV, page 28*)

Demand schedule. A table of prices and quantities. (*Chapter IV, page 28*)

Derived demand. Demand for an input that comes from the demand for the good it makes. (*Chapter XV, page 126*)

Determinants of demand. The things other than price that decide demand. (*Chapter IV, page 30*)

Determinants of supply. The things other than price that decide supply. (*Chapter VII, page 58*)

Differential rent. The extra return of better land over the worst land in use. (*Chapter XV, page 127*)

Differentiated product. A product that buyers see as different from rivals' products. (*Chapter XI, page 92*)

Diseconomies of scale. Average cost rises as the scale of the firm grows. (*Chapter X, page 89*)

Division of labour. Splitting a job into separate tasks, each done by a different worker. (*Chapter IX, page 78*)

Economic cost. All costs of the firm, explicit and implicit together. (*Chapter X, page 84*)

Economic law. A statement of a regular tendency. (*Chapter II, page 14*)

Economic model. A simple picture of how something works. (*Chapter II, page 14*)

Economic profit. Revenue minus all costs, including implicit costs. (*Chapter XVI, page 135*)

Economic rent. A payment to an input above the least needed to keep it in use. (*Chapter XV, page 127*)

Economic system. The way a society answers what, how and for whom. (*Chapter I, page 7*)

Economics. The study of choice under scarcity. (*Chapter I, page 6*)

Economies of scale. Average cost falls as the scale of the firm grows. (*Chapter X, page 88*)

Elastic. Quantity changes by a larger percentage than price. (*Chapter V, page 37*)

Elasticity. How strongly one thing reacts when another thing changes. (*Chapter V, page 36*)

- Elasticity of supply.** Percentage change in quantity supplied divided by percentage change in price. (*Chapter VII, page 59*)
- Entrepreneur.** A person who starts and runs a business and bears its risk. (*Chapter I, page 3*)
- Equal-ratio rule.** At least cost, each input gives the same extra output for each taka spent on it. (*Chapter IX, page 80*)
- Equi-marginal rule.** Spend so that marginal utility per taka is equal across goods. (*Chapter III, page 24*)
- Equilibrium.** A state of balance with no reason to change. (*Chapter III, page 24*)
- Equilibrium price.** The price at which quantity demanded equals quantity supplied. (*Chapter VIII, page 65*)
- Exceptional supply curve.** A supply curve that does not slope upward in the usual way. (*Chapter VII, page 59*)
- Excess capacity.** Output that lies below the level at which average cost is lowest. (*Chapter XIV, page 119*)
- Excess demand.** The quantity demanded is greater than the quantity supplied at the price. (*Chapter VIII, page 66*)
- Excess supply.** The quantity supplied is greater than the quantity demanded at the price. (*Chapter VIII, page 66*)
- Explicit cost.** A cost paid in money to someone outside the firm. (*Chapter X, page 84*)
- Factor market.** A market where inputs are bought and sold. (*Chapter XV, page 126*)
- Factors of production.** Land, labour, capital and entrepreneur. (*Chapter I, page 2*)
- Firm.** An organisation that makes goods for sale. (*Chapter II, page 12*)
- Fixed cost.** Cost that does not change with output. (*Chapter X, page 85*)
- Fixed input.** An input whose amount cannot be changed in the period studied. (*Chapter IX, page 74*)
- Free good.** A good that exists in unlimited amount. (*Chapter I, page 3*)
- Giffen good.** A good whose demand falls when its price falls. (*Chapter IV, page 32*)
- Good.** In daily life, good means nice or kind. In economics, a good is a thing that satisfies a want. (*Chapter I, page 2*)
- Gross interest.** The whole payment a lender receives, including pay for risk and for the cost of lending. (*Chapter XVI, page 134*)
- Homogeneous product.** A product that buyers see as identical from every seller. (*Chapter XI, page 92*)
- Household.** People who live together and spend together. (*Chapter II, page 12*)
- Implicit cost.** The cost of using resources the owner already has; no money is paid. (*Chapter X, page 84*)
- Income (of a consumer).** The money a buyer has to spend in a period. (*Chapter VI, page 49*)
- Income effect.** The change in buying power after a price change. (*Chapter IV, page 30*)
- Income elasticity.** Percentage change in quantity demanded divided by percentage change in income. (*Chapter V, page 41*)
- Increasing returns to scale.** Output rises by a larger proportion than inputs. (*Chapter IX, page 78*)

- Indifference curve.** A line joining baskets of two goods that give equal satisfaction. (*Chapter VI, page 48*)
- Indifference map.** A set of indifference curves, each higher curve preferred to the lower. (*Chapter VI, page 49*)
- Indifference schedule.** A table of baskets of two goods that give equal satisfaction. (*Chapter VI, page 48*)
- Individual demand.** The demand of one buyer. (*Chapter IV, page 29*)
- Individual supply.** The supply of one seller. (*Chapter VII, page 56*)
- Industry.** All the firms that make the same product. (*Chapter XII, page 100*)
- Inelastic.** Quantity changes by a smaller percentage than price. (*Chapter V, page 37*)
- Inferior good.** A good whose demand falls as income rises. (*Chapter IV, page 31*)
- Innovation.** The introduction of a new product, method or market. (*Chapter XVI, page 136*)
- Input.** Any resource used to make a good or service. (*Chapter IX, page 74*)
- Interdependence.** Each firm's best choice depends on the choices of its rivals. (*Chapter XIV, page 120*)
- Interest.** The payment made for the use of money for a period of time. (*Chapter XVI, page 134*)
- Isocost line.** A line joining all mixes of two inputs that cost the same money. (*Chapter IX, page 80*)
- Isoquant.** A line joining all mixes of two inputs that make the same output. (*Chapter IX, page 80*)
- Kinked demand curve.** A demand curve with a bend at the present price. (*Chapter XIV, page 120*)
- Labour.** Human work of the body or the mind. (*Chapter I, page 2*)
- Land.** Gifts of nature, such as soil and water. (*Chapter I, page 2*)
- Law of demand.** A higher price brings a smaller quantity demanded, other things equal. (*Chapter IV, page 30*)
- Law of diminishing marginal utility.** Each extra unit adds less satisfaction than the one before. (*Chapter III, page 21*)
- Law of diminishing returns.** Beyond some point, each extra unit of a variable input adds less output than the one before. (*Chapter IX, page 77*)
- Law of supply.** A higher price brings a larger quantity supplied, other things equal. (*Chapter VII, page 57*)
- Least-cost combination.** The mix of inputs that makes a given output at the lowest cost. (*Chapter IX, page 80*)
- Legal monopoly.** A monopoly created by law, such as a patent or a sole licence. (*Chapter XIII, page 106*)
- Liquidity preference.** The wish to hold wealth in cash instead of in forms that earn interest. (*Chapter XVI, page 135*)
- Loanable funds.** Money offered for lending and money sought for borrowing. (*Chapter XVI, page 134*)
- Long run.** A time in which all inputs can change. (*Chapter VII, page 60*)
- Long-run average cost.** The lowest average cost of each output when every input can change. (*Chapter X, page 88*)
- Long-run equilibrium.** A state in which no firm wants to enter or leave the industry, and no firm wants to change its plant. (*Chapter XII, page 103*)
- Luxury.** A good people can do without when money is short. (*Chapter V, page 40*)

- Macroeconomics.** The study of the whole economy. (*Chapter II, page 12*)
- Margin.** The edge: the next or last unit. (*Chapter III, page 20*)
- Marginal cost.** The extra cost of making one more unit. (*Chapter X, page 86*)
- Marginal product.** The extra output from one more unit of the variable input. (*Chapter IX, page 75*)
- Marginal rate of substitution.** The units of one good given up for one more unit of the other. (*Chapter VI, page 49*)
- Marginal revenue.** The extra revenue from selling one more unit. (*Chapter XI, page 94*)
- Marginal revenue product.** The extra revenue a firm gains from one more unit of an input. (*Chapter XV, page 126*)
- Marginal utility.** The extra satisfaction from one more unit. (*Chapter III, page 20*)
- Market.** Word in use: in daily life, a place to shop; in economics, buyers and sellers dealing with each other. (*Chapter VIII, page 64*)
- Market demand.** The sum of all buyers' demand. (*Chapter IV, page 29*)
- Market equilibrium.** A state in which quantity demanded equals quantity supplied. (*Chapter VIII, page 65*)
- Market period.** A time too short to change output. (*Chapter VII, page 60*)
- Market structure.** The features of a market: the sellers, the product and the ease of entry. (*Chapter XI, page 92*)
- Market supply.** The sum of the supplies of all sellers. (*Chapter VII, page 56*)
- Microeconomics.** The study of single households, firms and markets. (*Chapter II, page 12*)
- Microfinance.** The supply of small loans and other financial services to poor people. (*Chapter XVI, page 137*)
- Minimum wage.** The lowest wage that the law permits an employer to pay. (*Chapter XV, page 129*)
- Money cost.** Cost measured in money, such as taka paid. (*Chapter X, page 85*)
- Monopolistic competition.** A market with many sellers of products that differ from each other. (*Chapter XI, page 93*)
- Monopoly.** A market with one seller of a product that has no close substitute. (*Chapter XI, page 93*)
- Monopoly power.** The ability to set a price above marginal cost without losing all buyers. (*Chapter XIII, page 106*)
- Monopsony.** A market with a single buyer. (*Chapter XIII, page 107*)
- Natural monopoly.** A market in which one firm can supply the whole demand at lower cost than two or more. (*Chapter XIII, page 107*)
- Necessity.** A good people need and cannot easily do without. (*Chapter V, page 40*)
- Net interest.** The part of gross interest that pays only for waiting. (*Chapter XVI, page 134*)
- Nominal interest rate.** The rate of interest stated in money, before allowing for rising prices. (*Chapter XVI, page 135*)
- Nominal wage.** The wage measured in money. (*Chapter XV, page 128*)
- Non-competing groups.** Groups of workers whose skills differ so much that one cannot easily move into the other. (*Chapter XV, page 130*)
- Non-price competition.** Rivalry among sellers through quality, brand, service or advertising, not through price. (*Chapter XIV, page 117*)
- Normal good.** A good whose demand rises with income. (*Chapter IV, page 31*)

- Normal profit.** The least profit that keeps an entrepreneur in the business. (*Chapter X, page 84*)
- Normative statement.** A statement of what ought to be. (*Chapter II, page 15*)
- Oligopoly.** A market with only a few sellers. (*Chapter XI, page 93*)
- Opportunity cost.** The best alternative given up. (*Chapter I, page 4*)
- Ordinal.** Ranked as first, second, third. (*Chapter III, page 21*)
- Output.** The goods and services that production makes. (*Chapter IX, page 74*)
- Patent.** A legal right to be the only seller of an invention for a fixed number of years. (*Chapter XIII, page 106*)
- Percentage change.** The change divided by the starting value, times 100. (*Chapter V, page 36*)
- Perfect competition.** A market with many sellers, one homogeneous product, free entry and full knowledge. (*Chapter XI, page 93*)
- Perfectly competitive firm.** A firm in perfect competition, which must accept the market price. (*Chapter XII, page 100*)
- Perfectly elastic.** Buyers take any amount at one price and nothing above it. (*Chapter V, page 37*)
- Perfectly inelastic.** Quantity demanded does not change at all when price changes. (*Chapter V, page 37*)
- Point elasticity.** Elasticity at one point of the demand curve, for a very small price change. (*Chapter V, page 38*)
- Positive statement.** A statement that can be tested by facts. (*Chapter II, page 15*)
- Price ceiling.** The highest price allowed by law. (*Chapter VIII, page 67*)
- Price discrimination.** Selling the same good at different prices to different buyers, for reasons not based on cost. (*Chapter XIII, page 111*)
- Price effect.** The change in a buyer's best basket when the price of a good changes. (*Chapter VI, page 50*)
- Price elasticity of demand.** Percentage change in quantity demanded divided by percentage change in price. (*Chapter V, page 36*)
- Price floor.** The lowest price allowed by law. (*Chapter VIII, page 67*)
- Price index.** A number showing the average level of prices compared with a base period set at 100. (*Chapter XV, page 128*)
- Price leadership.** One firm in the market sets the price and the others follow. (*Chapter XIV, page 121*)
- Price maker.** A seller who can set the price, within what buyers will pay. (*Chapter XI, page 94*)
- Price mechanism.** The way changes in price bring quantity demanded and supplied into balance. (*Chapter VIII, page 67*)
- Price taker.** A seller who must accept the price set by the market. (*Chapter XI, page 94*)
- Price war.** A series of price cuts, each rival cutting below the last. (*Chapter XIV, page 120*)
- Producers' surplus.** The price a seller receives minus the lowest price the seller would accept. (*Chapter VIII, page 68*)
- Product differentiation.** Making a good different from rivals' goods, in fact or in the buyer's mind. (*Chapter XIV, page 117*)
- Production.** Making goods and services that satisfy wants. (*Chapter IX, page 74*)
- Production function.** The link between amounts of inputs and the greatest output they can make. (*Chapter IX, page 74*)

Profit. Revenue minus cost. (*Chapter XI, page 95*)

Profit maximisation. Choosing output to make profit as large as possible. (*Chapter XI, page 95*)

Pure competition. Competition with no element of monopoly: many sellers, an identical product and free entry. (*Chapter XI, page 93*)

Pure monopoly. A sole seller of a good that has no close substitute. (*Chapter XIII, page 106*)

Quantity demanded. The amount bought at one price. (*Chapter IV, page 28*)

Quantity supplied. The amount sellers offer for sale at one price. (*Chapter VII, page 56*)

Quasi-rent. A temporary surplus earned by a man-made asset whose supply is fixed in the short run. (*Chapter XV, page 128*)

Real cost. The effort and sacrifice of the people who supply the inputs. (*Chapter X, page 85*)

Real interest rate. The rate of interest after the effect of rising prices is removed. (*Chapter XVI, page 135*)

Real wage. What the wage can buy, shown by dividing the money wage by a price index. (*Chapter XV, page 128*)

Regulation. Rules set by a public body on a firm's prices or conduct. (*Chapter XIII, page 112*)

Resource. Anything used to make goods. (*Chapter I, page 2*)

Returns to scale. The change in output when all inputs change by the same proportion. (*Chapter IX, page 78*)

Scale of production. The size of the firm, with all its inputs taken together. (*Chapter IX, page 78*)

Scarcity. Too little exists to satisfy all wants. (*Chapter I, page 3*)

Selling cost. Money spent to promote a seller's goods, such as advertising. (*Chapter XIV, page 117*)

Service. Work done for others, such as teaching. (*Chapter I, page 2*)

Short run. A time in which at least one input is fixed. (*Chapter VII, page 60*)

Short-run equilibrium. The output at which a firm has no reason to change, with its plant fixed. (*Chapter XII, page 101*)

Shut-down point. The price at which revenue covers only variable cost. (*Chapter XII, page 101*)

Snob effect. Demand falls as more people buy. (*Chapter IV, page 32*)

Subsidy. Money paid by government to lower a seller's cost. (*Chapter VII, page 58*)

Substitute goods. Goods used in place of each other. (*Chapter IV, page 31*)

Substitution effect. Switching to a cheaper good. (*Chapter IV, page 30*)

Sunk cost. A cost already paid that cannot be recovered. (*Chapter X, page 84*)

Supernormal profit. Profit above normal profit. (*Chapter XII, page 101*)

Supplementary cost. An older name for overhead cost, which does not change with output. (*Chapter X, page 85*)

Supply. The quantity sellers are able and willing to sell at each price. (*Chapter VII, page 56*)

Supply curve. The line that plots a supply schedule. (*Chapter VII, page 56*)

Supply schedule. A table of prices and the quantities offered at each. (*Chapter VII, page 56*)

Technology. The methods and equipment used to make a good. (*Chapter VII, page 58*)

Time preference. The tendency to value present goods more than the same goods in the future. (*Chapter XVI, page 134*)

Total cost. Fixed cost plus variable cost. (*Chapter X, page 85*)

Total product. All the output made with a given amount of the variable input. (*Chapter IX, page 75*)

Total revenue. Price multiplied by the quantity sold. (*Chapter V, page 42*)

Total surplus. Consumers' surplus plus producers' surplus. (*Chapter VIII, page 69*)

Total utility. All the satisfaction from the units consumed. (*Chapter III, page 20*)

Trade union. A group of workers who act together on pay and conditions. (*Chapter XV, page 129*)

Uncertainty. A situation in which the chances of outcomes cannot be known. (*Chapter XVI, page 136*)

Unit elastic. Quantity and price change by equal percentages. (*Chapter V, page 37*)

Util. A made-up unit of satisfaction. (*Chapter III, page 20*)

Utility. In daily life, the usefulness of a tool. In economics, the satisfaction a person gets from a good. (*Chapter III, page 20*)

Variable cost. Cost that changes with output. (*Chapter X, page 85*)

Variable input. An input whose amount can be changed in the period studied. (*Chapter IX, page 74*)

Wage. The payment made for the use of labour. (*Chapter XV, page 128*)

Want. A wish for something. (*Chapter I, page 2*)

Welfare. The well-being of people. (*Chapter II, page 13*)

Willingness to pay. The most a buyer will give for one unit of a good. (*Chapter VI, page 46*)

Windfall profit. Unexpected profit that comes by chance, not by the owner's effort or skill. (*Chapter XVI, page 137*)

ABOUT THIS BOOK

A first course in microeconomics for the reader who is meeting the subject for the first time. Starting from a household budget and a neighbourhood bakery, the book builds the subject step by step: choice and cost, demand and supply, elasticity, production and cost curves, the four market structures, and the payments of rent, wages, interest and profit.

Examples are drawn from daily life in Bangladesh and priced in taka. Side notes, worked cases, and practice at the end of every chapter lead the reader from the first idea to the last.

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